

- 3.** A charged particle at rest is subjected to an external magnetic field $\vec{B}$. If no other force acts on the particle, then it will :
- (a) accelerate along $\vec{B}$
 - (b) remain at rest
 - (c) move in a circular path
 - (d) move with a constant speed along $\vec{B}$

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(b) remain at rest

5. A current carrying loop is placed in a uniform magnetic field. The torque acting on it does ***not*** depend upon the :

- | | |
|----------------------|-------------------------|
| (a) magnetic field | (b) current in the loop |
| (c) area of the loop | (d) shape of the loop |

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(d) Shape of loop

- 6.** Two identical coils carrying equal currents are held concentric with their planes perpendicular to each other. The ratio of the magnitude of magnetic field at the centre of one coil to that of the resultant magnetic field at the centre is :

(a) $\frac{1}{\sqrt{2}}$

(b) $\sqrt{2}$

(c) $\frac{1}{2}$

(d) 2

- 6.** Two identical coils carrying equal currents are held concentric with their planes perpendicular to each other. The ratio of the magnitude of magnetic field at the centre of one coil to that of the resultant magnetic field at the centre is :

(a) $\frac{1}{\sqrt{2}}$

(b) $\sqrt{2}$

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(a) $\frac{1}{\sqrt{2}}$

7. An electron is revolving around a nucleus in a circular orbit of radius r with a constant speed v . The magnetic moment associated with the circulating current is :

(a) $\frac{evr}{4}$

(b) $4\ evr$

(c) $2\ evr$

(d) $\frac{evr}{2}$

7. An electron is revolving around a nucleus in a circular orbit of radius r with a constant speed v . The magnetic moment associated with the circulating current is :

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- 32.** (a) (i) Two long parallel straight conductors carrying current I_1 and I_2 are kept r distance apart in air. Obtain an expression for the force per unit length on one conductor due to the magnetic field produced by the other conductor. Hence, define one ampere. Under what condition will the force between the conductors be attractive in nature ?
- (ii) A closely wound circular coil of radius 6.28 cm, having 50 turns carries a steady current of 4 A. Find the magnetic field at the centre of the coil. How will the magnitude of the magnetic field be affected if the radius of the coil is halved keeping other factors same ?

(i) Obtaining expression of force/length	1½
Definition of one ampere	1
Condition of force to be attractive	½
(ii) Calculation of magnetic field	1½
Effect on magnitude of magnetic field when radius is halved	½

- (b) (i) A particle of mass m and charge q , at the origin moves with a velocity $\vec{v}$ in xy -plane making an angle θ with x -axis. It is subjected to a uniform magnetic field $\vec{B}$ along x -axis. Justify that the particle will move in a helical path. Hence, obtain expression for the radius of the helix.
- (ii) A long straight wire kept horizontally carries a current of 4 A from west to east direction. Find the direction and magnitude of magnetic field $\vec{B}$ at a point 20 cm below the wire.

(i) Consider two long parallel conductors P and Q, distance r apart carrying currents I_1 and I_2 respectively. magnetic field of P at a point in the position of Q $B_1 = \frac{\mu_o I_1}{2\pi r} \text{ (i)}$ Magnetic force on Q at length l $F_{QP} = B_1 I_2 l = \frac{\mu_o I_1 I_2}{2\pi r} l \text{ (ii)}$ $f_2 = \frac{F_{QP}}{l} = \frac{\mu_o I_1 I_2}{2\pi r} \text{ (iii)}$ Similarly force per unit length of P due to magnetic field of Q $f_1 = \frac{F_{PQ}}{l} = \frac{\mu_o I_1 I_2}{2\pi r} \text{ (iv)}$ From eq. (ii) and (iv) $f_1 = f_2 = f = \frac{\mu_o I_1 I_2}{2\pi r} \text{ (v)}$ Definition of one ampere One ampere is the value of the steady current which when maintained in each of the two very long straight parallel conductors of negligible cross section, placed one meter apart in vaccum, would produce on each of these conductors a force equal to 2×10^{-7} newton per meter of length. The force between the two conductors will be attractive when they carry current in the same direction.		(ii) Magnetic field at the centre of the current carrying coil $B = N \frac{\mu_o I}{2a}$ a= 6.28 cm N = 50 I = 4A $B = \frac{50 \times 4\pi \times 10^{-7} \times 4}{2 \times 6.28 \times 10^{-2}}$ $= 4 \times 10^{-3} \text{ T}$ When radius of the coil is halved i.e. $\frac{a}{2}$ then magnetic field becomes two times. $B = 8 \times 10^{-3} \text{ T}$
	1/2	1/2
	1/2	1/2
	1/2	
	1	
	1/2	

18. *Assertion (A) :* A current carrying square loop made of a wire of length L is placed in a magnetic field. It experiences a torque which is greater than the torque on a circular loop made of the same wire carrying the same current in the same magnetic field.

Reason (R) : A square loop occupies more area than a circular loop, both made of wire of the same length.

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Reason (R) : A square loop occupies more area than a circular loop, both made of wire of the same length.

(d) Assertion (A) is false and Reason (R) is also false.

- 23.** A particle of charge q and mass m starts moving with uniform velocity $v_0 \hat{i}$. Specify the direction of magnetic field which should be set up in the region so that the particle moves (a) straight undeviated, and (b) in a circle. Justify your answers.

2

Specification of the direction of magnetic field for particle to move

- | | |
|-------------------------|---|
| (a) straight undeviated | 1 |
| (b) in a circle | 1 |

(a) For straight undeviated path, magnetic field should be set up parallel or antiparallel to velocity vector

$$\vec{F} = q(\vec{v} \times \vec{B})$$

Alternatively: angle between $\vec{v}$ and $\vec{B}$, $\theta = 0^\circ$, or $\theta = 180^\circ$
 $\theta = 0^\circ$, or $\theta = 180^\circ$

(b) If charge particle moves in circle magnetic field should be setup perpendicular to the velocity vector.

$$\vec{F} = q(\vec{v} \times \vec{B}) \quad \vec{v} \perp \vec{B}$$

1/2

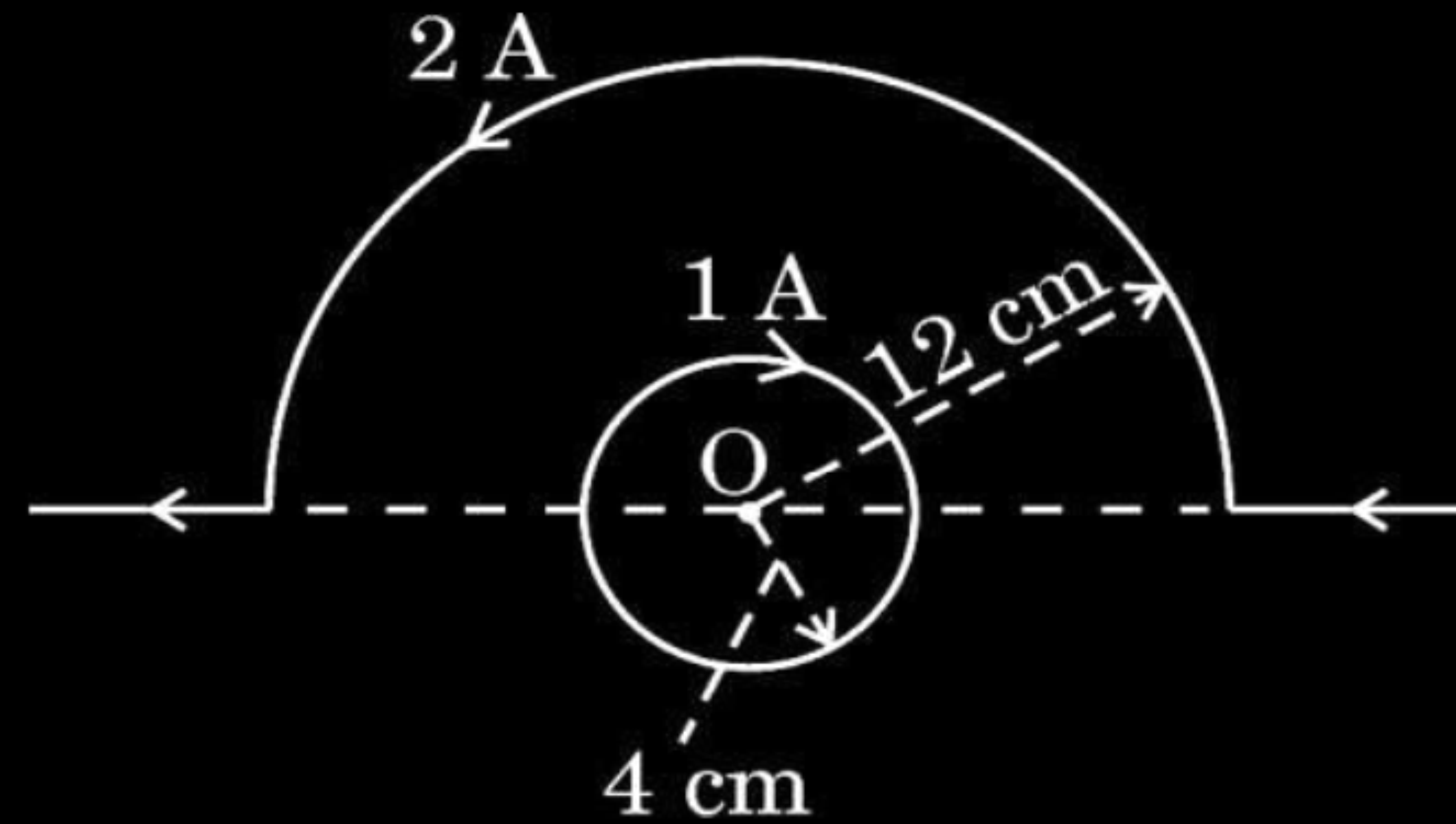
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- 27.** A current carrying circular loop and a straight wire bent partly in the form of a semicircle are placed as shown in the figure. Find the magnitude and direction of net magnetic field at point O.

3



Magnetic field due to inner circle at O,

$$\begin{aligned} B_1 &= \frac{\mu_0 I}{2r} \\ &= \frac{2\pi \times 10^{-7} \times 1}{4 \times 10^{-2}} \\ &= \frac{\pi}{2} \times 10^{-5} \text{ T} \quad \text{into the page} \end{aligned}$$

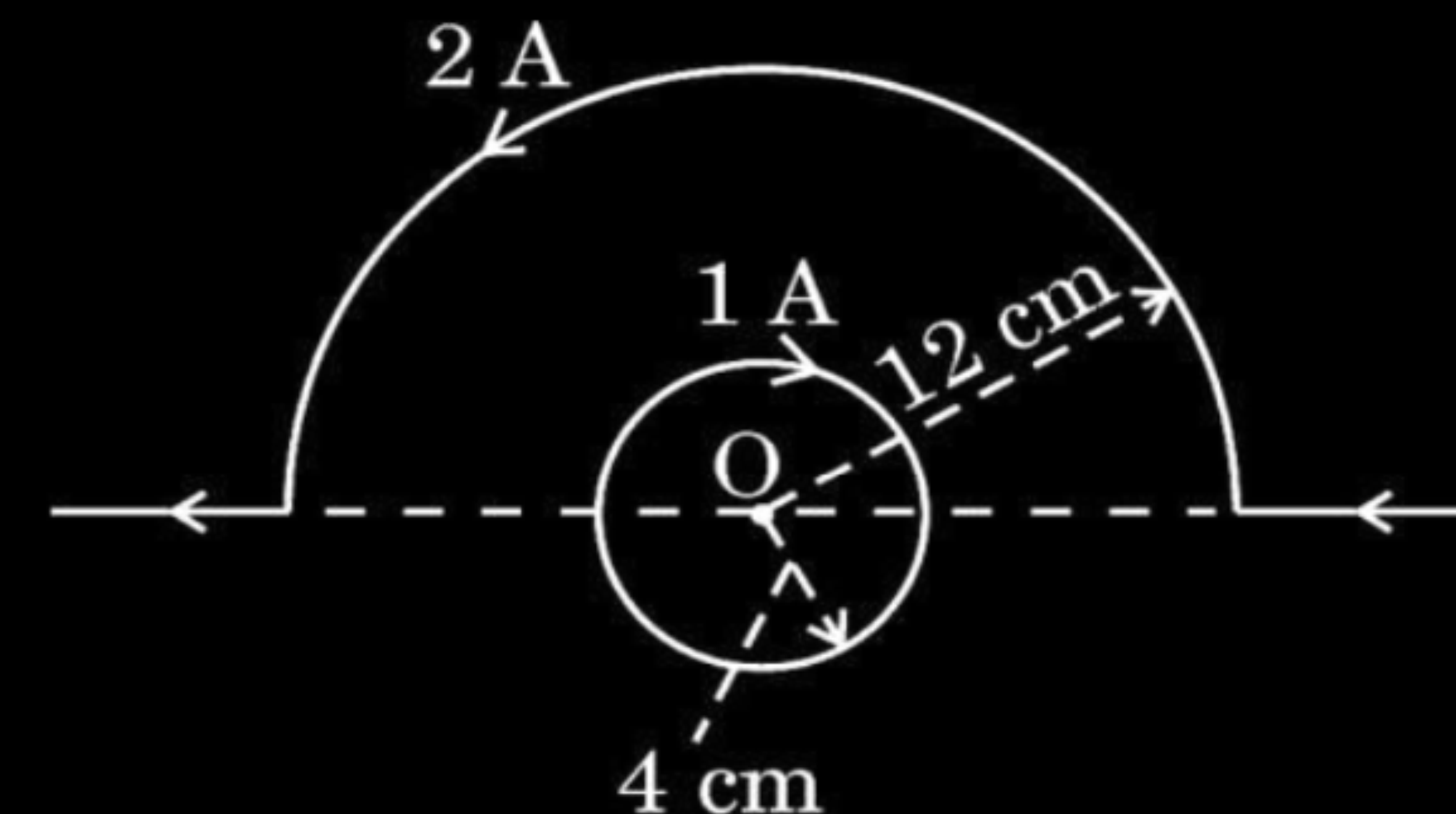
Magnetic field due to semi- circle at O,

$$\begin{aligned} B_2 &= \frac{\mu_0 I}{2r} \cdot \frac{1}{2} \\ &= \frac{2\pi \times 10^{-7} \times 2}{12 \times 10^{-2}} \\ &= \frac{\pi}{6} \times 10^{-5} \text{ T} \quad \text{out of the page} \end{aligned}$$

$$B_{net} = B_1 - B_2$$

$$= \left(\frac{\pi}{2} \times 10^{-5} - \frac{\pi}{6} \times 10^{-5} \right) T \approx 1.0 \times 10^{-5} T$$

Direction of the net magnetic field is into the page, since $B_1 > B_2$.



7. The ratio of the magnitude of magnetic moment to that of angular momentum, of an electron revolving around the nucleus in a hydrogen atom is :

(a) Zero

(b) $9.27 \times 10^{-24} \text{ C/kg}$

(c) $8.8 \times 10^{10} \text{ C/kg}$

(d) $6.6 \times 10^{-12} \text{ C/kg}$

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(c) $8.8 \times 10^{10} \text{ C/kg}$

29. Two infinitely long parallel conductors A and B, separated by 2 cm, carry 6 A and 2 A currents in opposite directions respectively. Find :

- (a) the net magnetic field at a point midway between A and B.
- (b) the force acting per unit length on B.

3

(a) Calculation of magnetic field due to current in A and B	$\frac{1}{2} + \frac{1}{2}$
Net magnetic field	1
(b) Force per unit length	1

(a) $B_1 = \frac{\mu_0 I_1}{2\pi r} = \frac{2 \times 10^{-7} \times 6}{1 \times 10^{-2}}$ $= 1.2 \times 10^{-4} \text{ T}$ into the page	$\frac{1}{2}$
$B_2 = \frac{\mu_0 I_2}{2\pi r} = \frac{2 \times 10^{-7} \times 2}{1 \times 10^{-2}}$ $= 0.4 \times 10^{-4} \text{ T}$ into the page	$\frac{1}{2}$
$\therefore B_1$ and B_2 are in same direction.	
$B_{net} = B_1 + B_2$ $= 1.2 \times 10^{-4} \text{ T} + 0.4 \times 10^{-4} \text{ T}$	
$= 1.6 \times 10^{-4} \text{ T}$ into the page	$\frac{1}{2} + \frac{1}{2}$
(b) Magnitude of magnetic force per unit length	
$\frac{F}{l} = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{r}$	$\frac{1}{2}$
$= \frac{2 \times 10^{-7} \times 6 \times 2}{2 \times 10^{-2}} = 1.2 \times 10^{-4} \text{ N/m}$	$\frac{1}{2}$

2. An electron enters a uniform magnetic field with speed v . It describes a semicircular path and comes out of the field. The final speed of the electron is :

(a) Zero

(b) v

(c) $\frac{v}{2}$

(d) $2v$

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(a) Zero

(b) v

(c) $\frac{v}{2}$

(d) $2v$

(b) v

- 26.** Two circular loops A and B, each of radius 3 m, are placed coaxially at a distance of 4 m. They carry currents of 3 A and 2 A in opposite directions respectively. Find the net magnetic field at the centre of loop A.

3

Calculation of magnetic field due to loop A	1
Calculation of magnetic field due to loop B	1
Net magnetic field	1

$B_1 = \frac{\mu_0 I}{2r}$	
$B_1 = \frac{4\pi \times 10^{-7} \times 3}{2 \times 3} = 2\pi \times 10^{-7} T = 6.28 \times 10^{-7} T$	1
$B_2 = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}$	
$= \frac{2\pi \times 10^{-7} \times 2 \times 9}{(3^2 + 4^2)^{3/2}}, \text{ opposite to } B_1$	
$B_2 = \frac{36\pi \times 10^{-7}}{125} T = 0.9 \times 10^{-7} T$	1
$B_{net} = B_1 - B_2$	
$= 6.28 \times 10^{-7} - 0.9 \times 10^{-7}$	
$B_{net} = 5.38 \times 10^{-7} T$	1

18. *Assertion (A) :* When radius of a circular loop carrying a steady current is doubled, its magnetic moment becomes four times.

Reason (R): The magnetic moment of a circular loop carrying a steady current is proportional to the area of the loop.

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Reason (R): The magnetic moment of a circular loop carrying a steady current is proportional to the area of the loop.

(a) Both Assertion (A) and Reason(R) are true and Reason (R) is the correct explanation of the Assertion (A).

- 22.** A wire of length L is bent round in the form of (i) a square, and then (ii) an equilateral triangle. If current I is passed through each of them, find the ratio of magnetic moment of the square loop to that of the triangle.

For square loop; $L = 4a \Rightarrow a = \frac{L}{4}$	
Area of square loop $= \frac{L^2}{16}$	$\frac{1}{2}$
For equilateral triangle; $L = 3a \Rightarrow a = \frac{L}{3}$	
Area of equilateral triangle $= \frac{\sqrt{3}}{4} a^2$	$\frac{1}{2}$
$= \frac{\sqrt{3}}{4} \left(\frac{L^2}{9} \right) = \frac{\sqrt{3}L^2}{36}$	
Magnetic moment $\propto$ Area of the loop.	$\frac{1}{2}$
$\Rightarrow \frac{m_s}{m_t} = \frac{L^2}{16} \times \frac{36}{\sqrt{3}L^2} = \frac{3\sqrt{3}}{4}$	$\frac{1}{2}$

- 33.** (a) (i) Write the principle and explain the working of a moving coil galvanometer. A galvanometer as such cannot be used to measure the current in a circuit. Why ?
- (ii) Why is the magnetic field made radial in a moving coil galvanometer ? How is it achieved ?

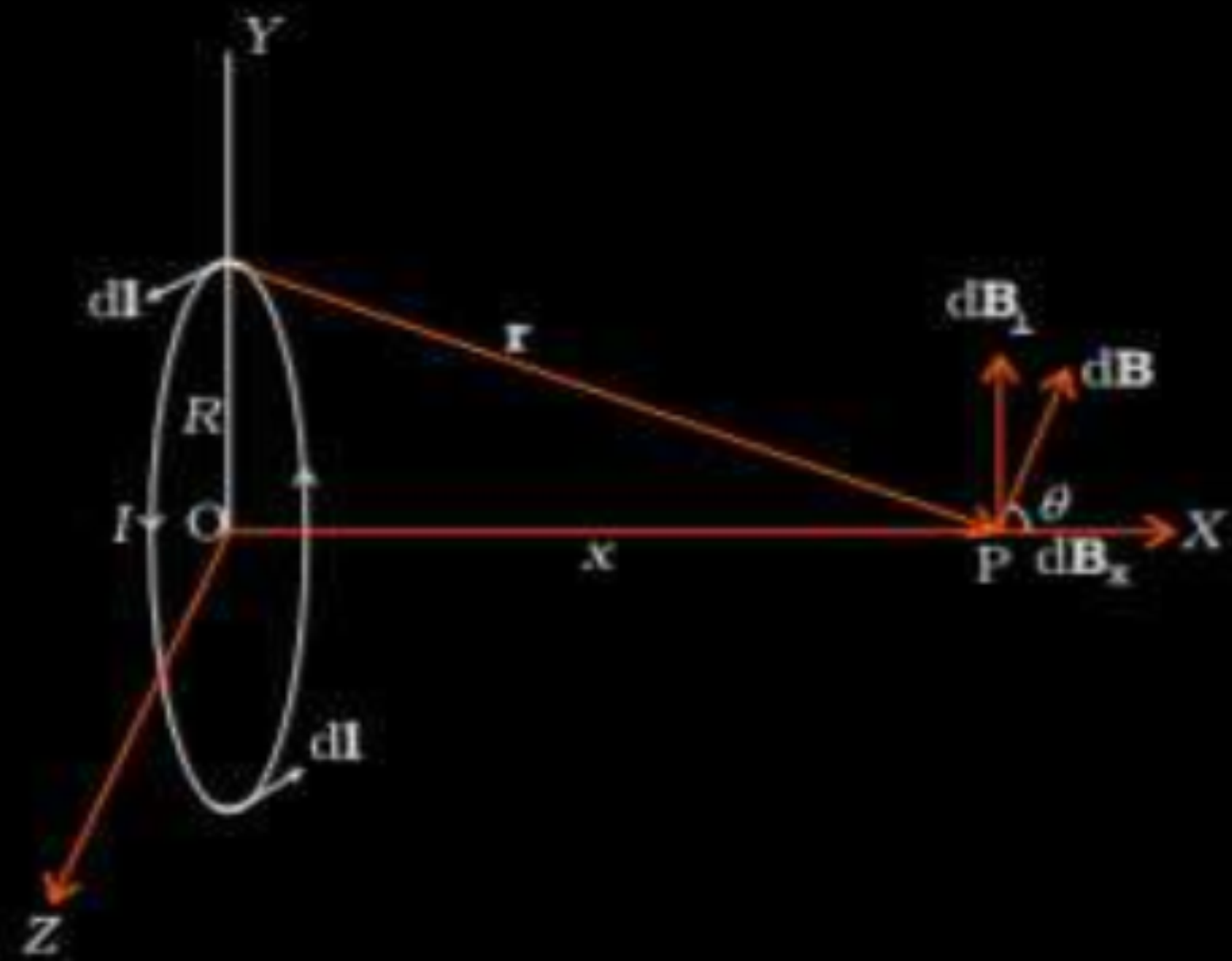
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(i) For a moving coil galvanometer	
Principle	1
Working	1
Reason it cannot be used as such	1
(ii) Reason for radial field	1
How radial field is achieved	1

(i) Principle – When a rectangular loop carrying current I is placed in a uniform magnetic field, it experiences a torque. Working:- When a current flows through the coil of a galvanometer, a torque acts on it. $\tau = NiAB \sin \theta$ For radial magnetic field; $\sin \theta = 1$ The spring provides a counter or restoring toqrue $k\phi$. $k\phi = NiAB$ In equilibrium; $\phi = \left(\frac{NAB}{k} \right) i$	1
Galvanometer cannot be used as such to measure current because: -It has large resistance and hence will change the value of current in the circuit. -It is a sensitive device. (Any one of the above)	1
(ii) The magnetic field is made radial in a moving coil galvanometer so that the magnetic dipole moment ($\vec{m}$) is always perpendicular to the magnetic field ($\vec{B}$) Hence, $\sin \theta = 1$ always Alternatively: The magnetic field is made radial in a moving coil galvanometer to make the scale linear.	1
It is achieved by using curved magnetic poles. Alternatively:-By using soft iron cylindrical core.	1

- (b) (i) Derive an expression for magnetic field on the axis of a current carrying circular loop.

(i)



$$dB = \frac{\mu_0}{4\pi} \frac{I |d\mathbf{l} \times \mathbf{r}|}{r^3}$$

$$= \frac{\mu_o idl \sin 90^\circ}{4\pi (x^2 + R^2)}$$

$dB_{\perp}$ cancels out.

$$\text{Net B} = \int dB_x = \int dB \cos \theta$$

$$= \frac{\mu_o}{4\pi} \int \frac{idl}{(x^2 + R^2)} \times \frac{R}{(x^2 + R^2)^{1/2}}$$

$$= \frac{\mu_o iR}{4\pi (x^2 + R^2)^{3/2}} \int dl$$

$$= \frac{\mu_o iR}{4\pi (x^2 + R^2)^{3/2}} (2\pi R)$$

$$\mathbf{B} = B_x \hat{\mathbf{i}} = \frac{\mu_o I R^2}{2(x^2 + R^2)^{3/2}} \hat{\mathbf{i}}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

10. In a moving coil galvanometer, the deflecting torque τ acting on the coil is related to the current I flowing through it as :

(a) $\tau \propto I^3$

(b) $\tau \propto I^2$

(c) $\tau \propto I$

(d) $\tau \propto \sqrt{I}$

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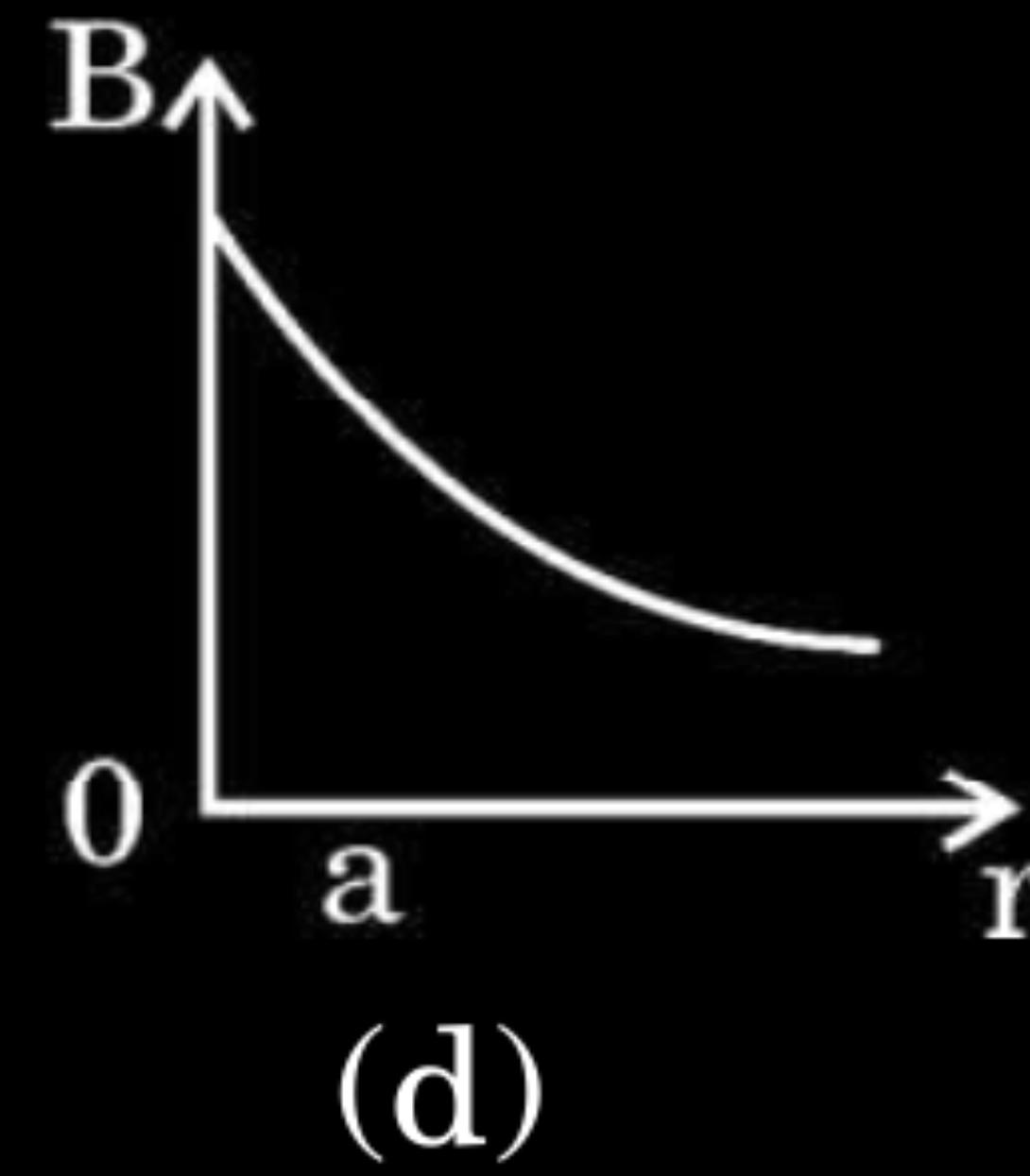
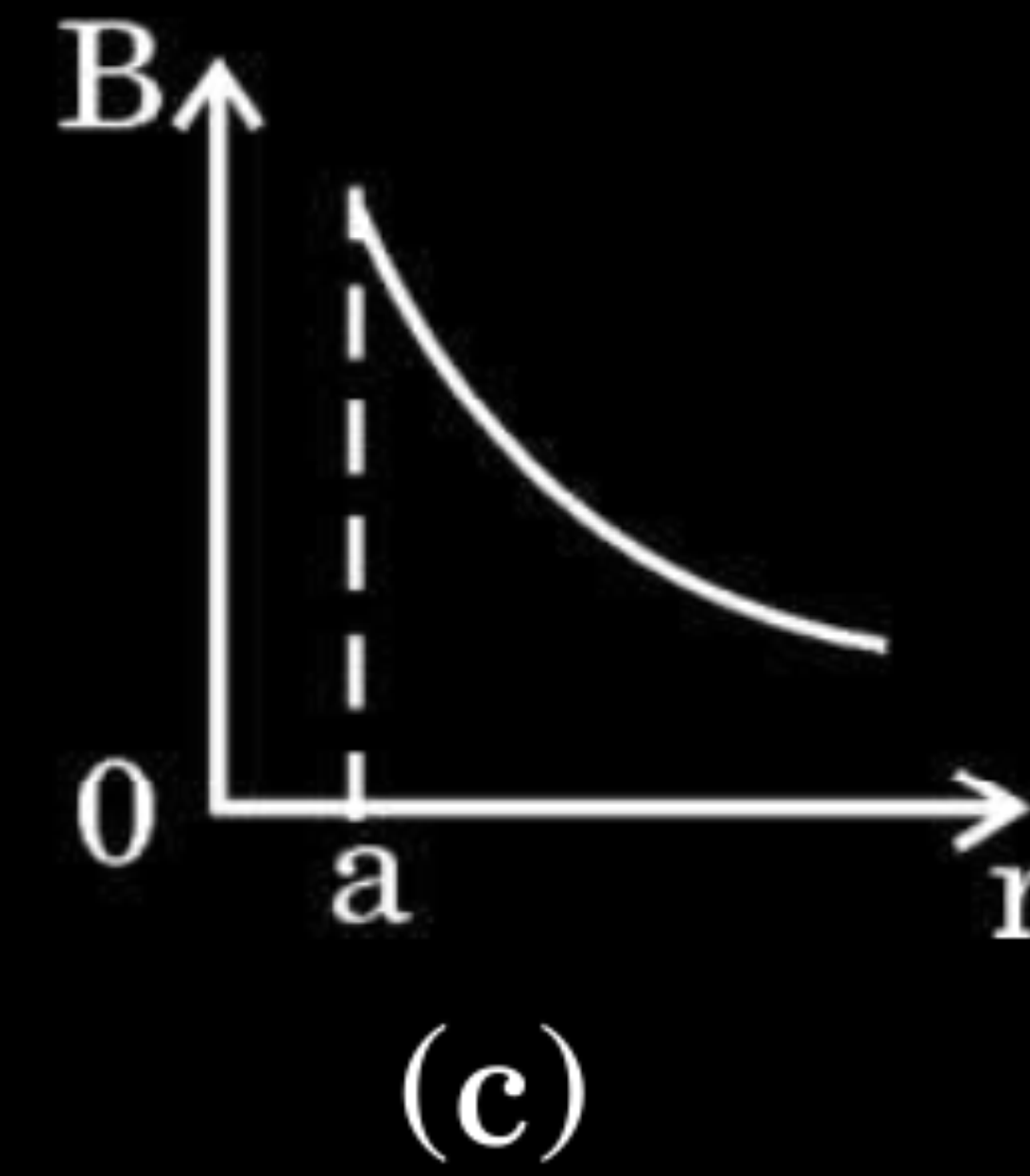
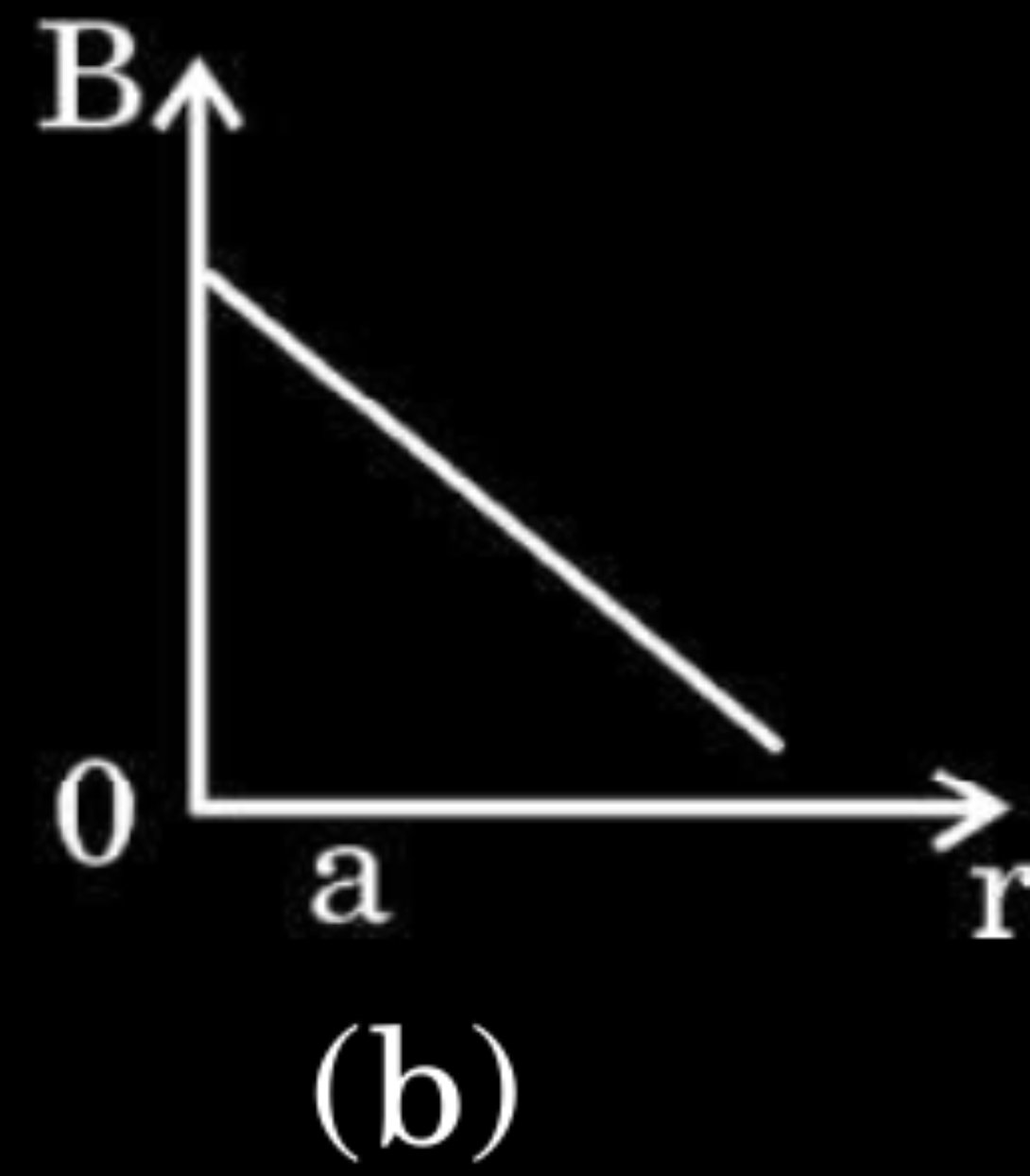
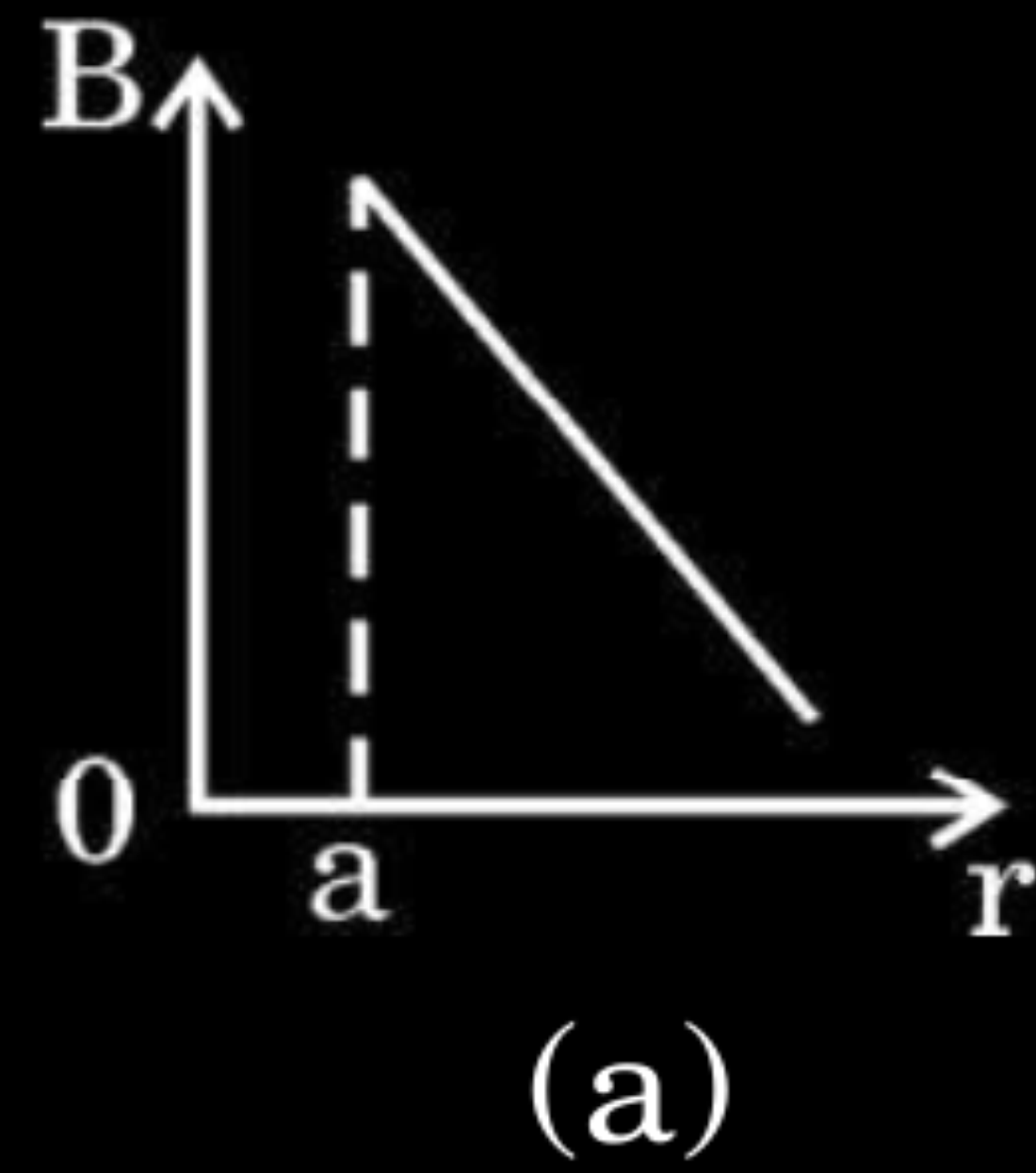
(b) $\tau \propto I^2$

(c) $\tau \propto I$

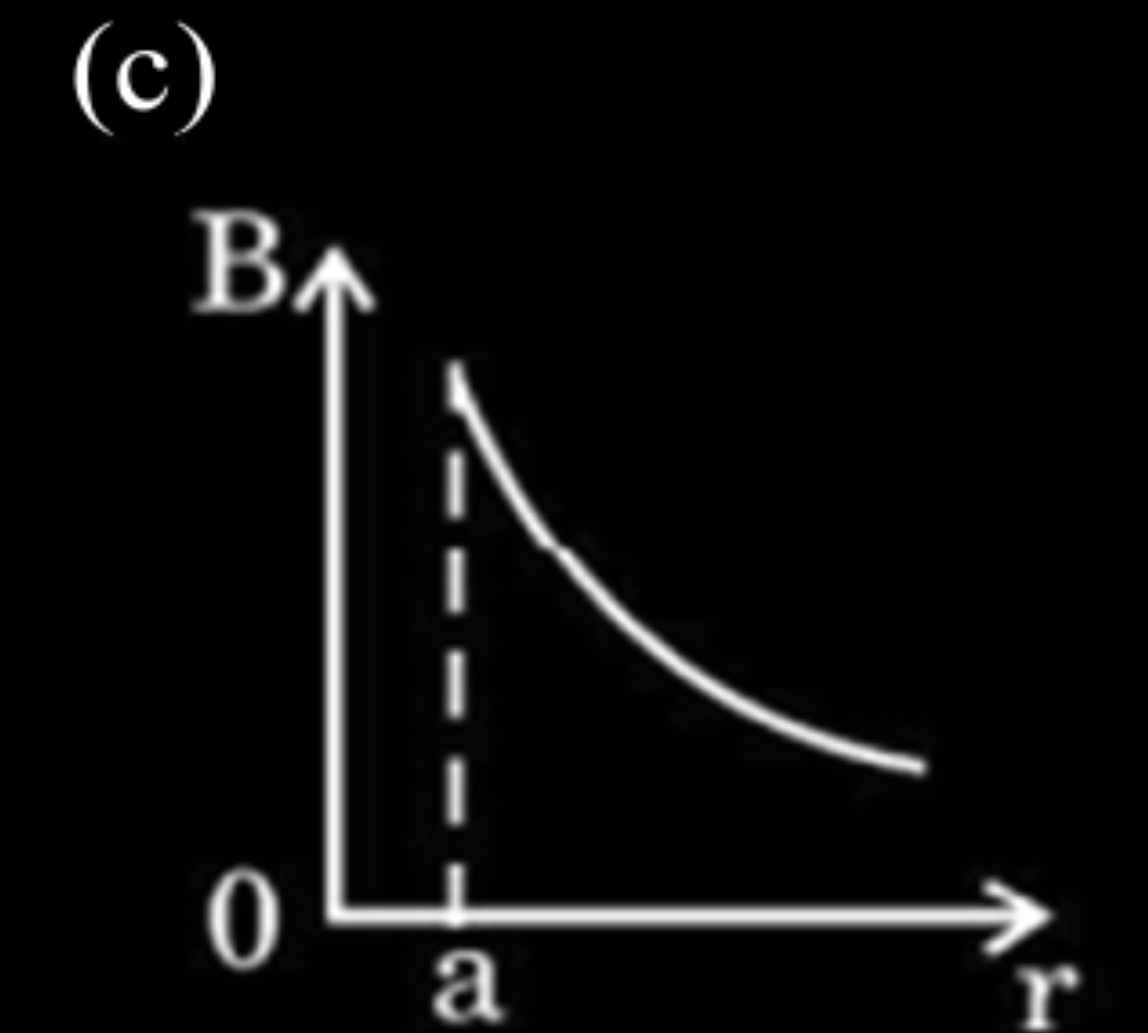
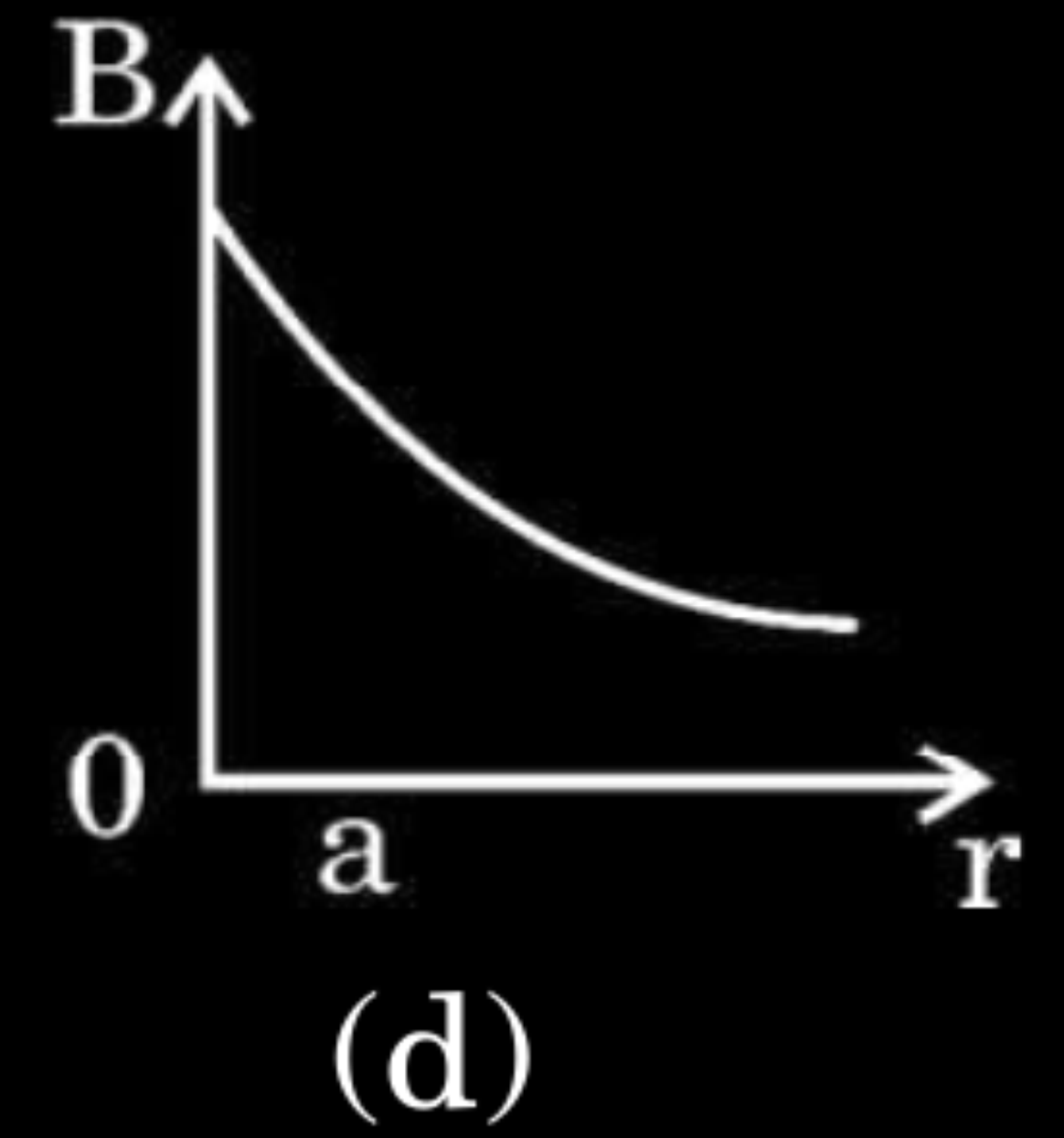
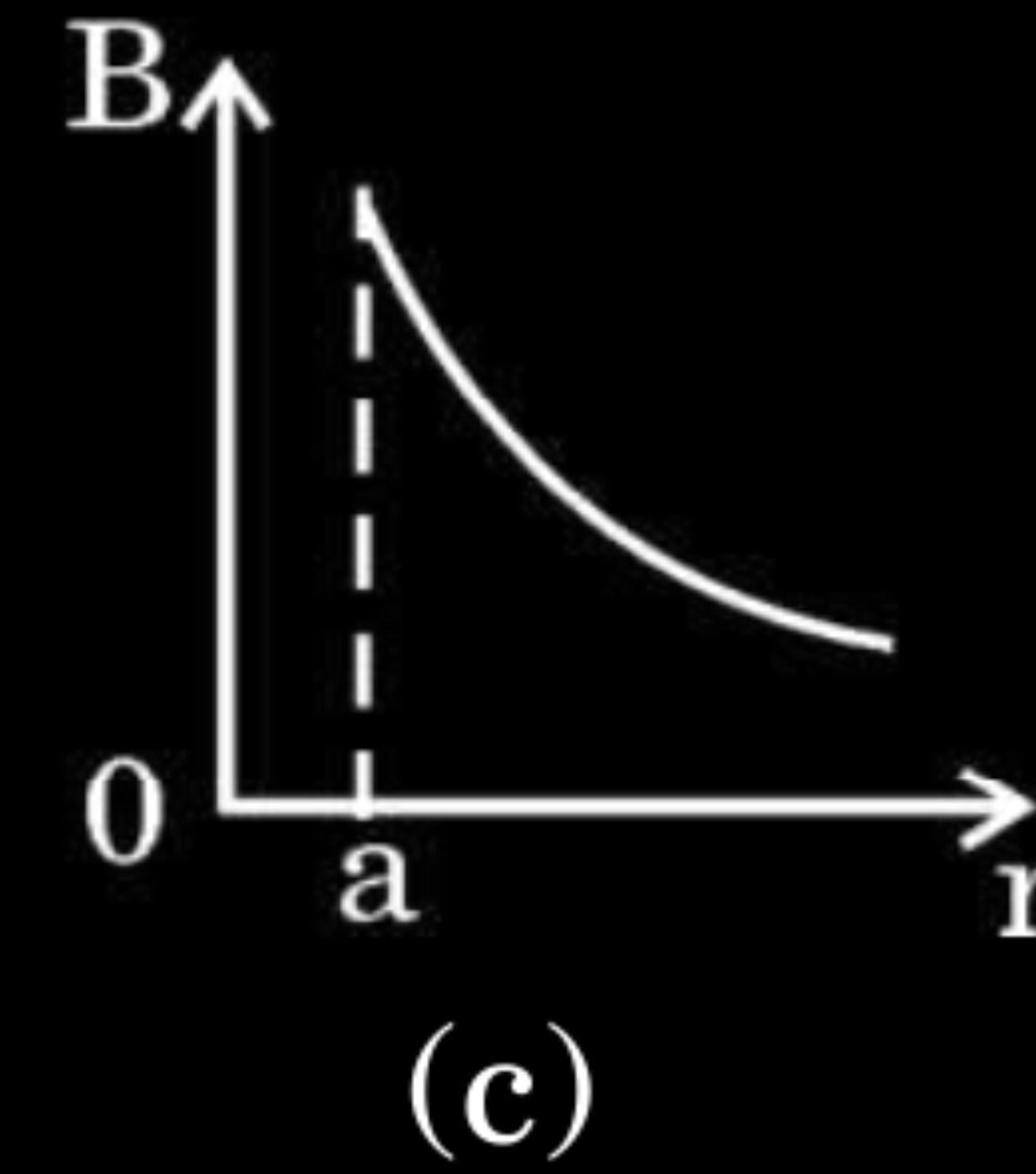
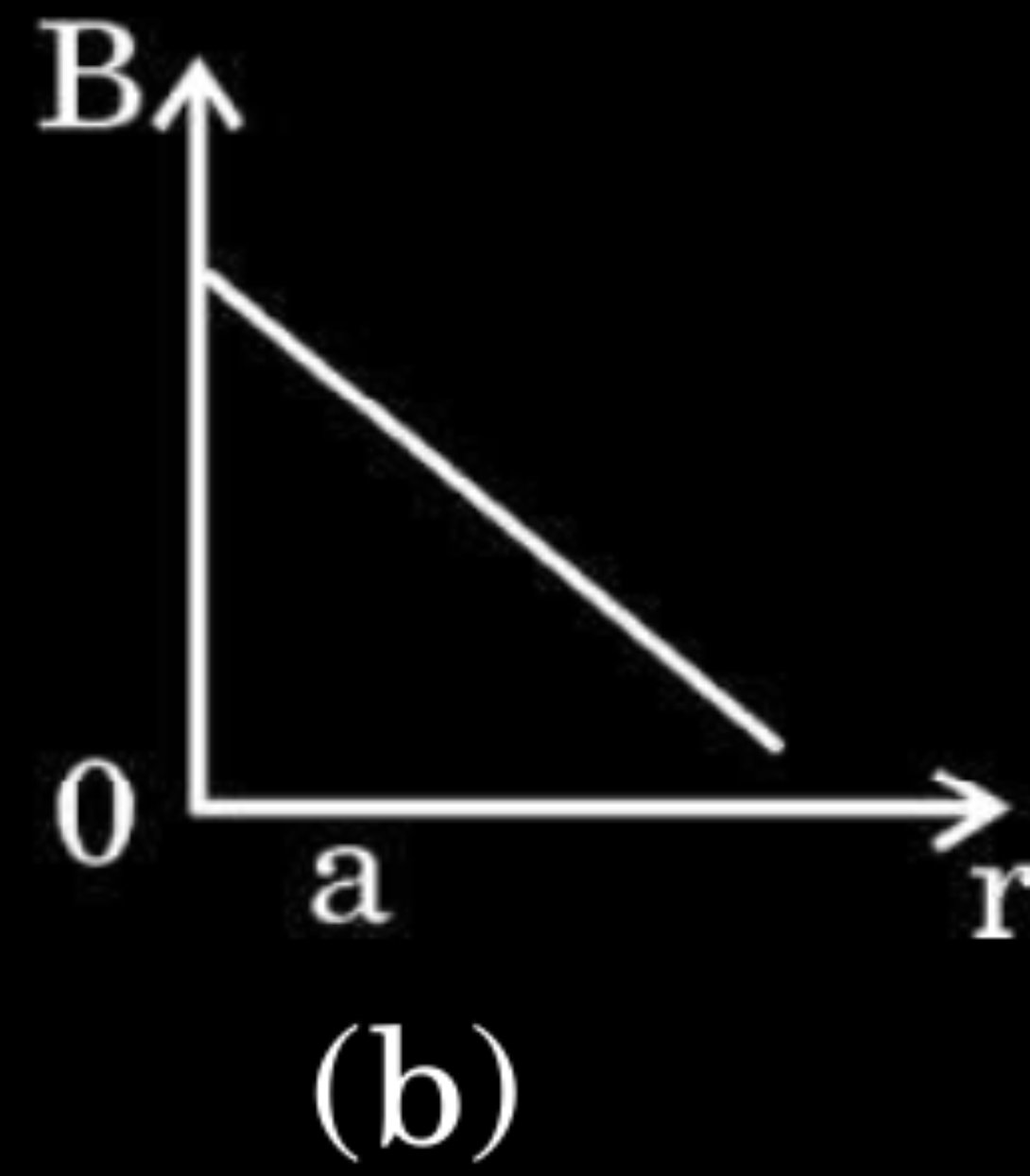
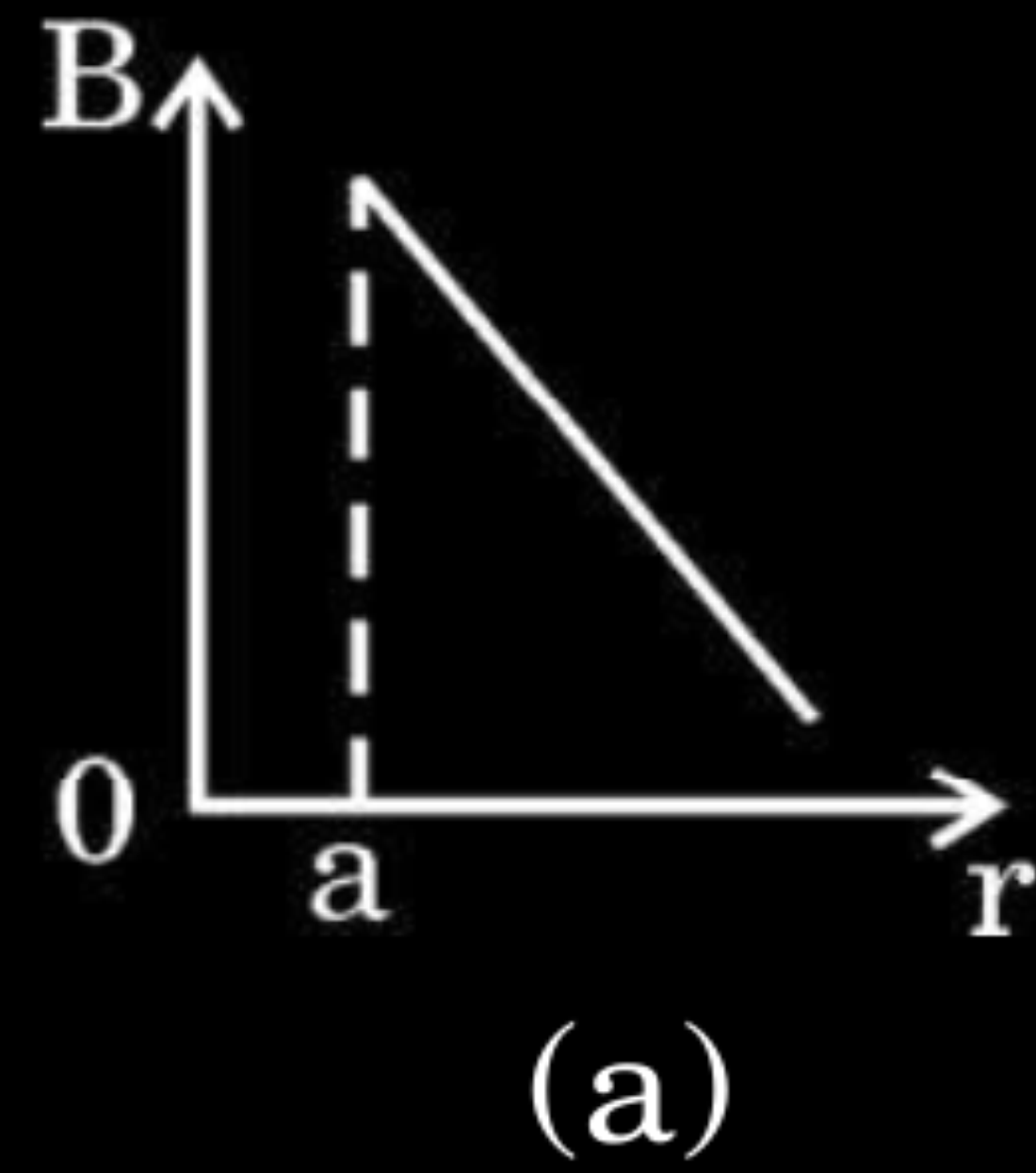
(d) $\tau \propto \sqrt{I}$

(c) $\tau \propto I$

15. Which of the following graphs correctly represents the variation of the magnitude of the magnetic field outside a straight infinite current carrying wire of radius 'a', as a function of distance 'r' from the centre of the wire ?



15. Which of the following graphs correctly represents the variation of the magnitude of the magnetic field outside a straight infinite current carrying wire of radius 'a', as a function of distance 'r' from the centre of the wire ?



- 23.** Two wires of equal lengths are shaped in the form of a square loop and a circular loop. Both loops are suspended in a uniform magnetic field. Prove that for the same current, the circular loop will experience larger torque. 2

Let the length of the wire be l .

For square loop; $l = 4a \Rightarrow a = \frac{l}{4}$

Area of square loop $= a^2 = \frac{l^2}{16}$

---(i)

$\frac{1}{2}$

For Circular loop; $l = 2\pi r \Rightarrow r = \frac{l}{2\pi}$

Area of circular loop(A_c) $= \pi r^2 = \frac{l^2}{4\pi}$

----(ii)

$\frac{1}{2}$

Torque acting on the loop (τ) $\propto A$

$\frac{1}{2}$

$\therefore A_c > A_s \quad \therefore \tau_c > \tau_s$

$\frac{1}{2}$

5. A particle of mass m and charge q moving with a uniform velocity $\vec{v} = v_{0x} \hat{i} + v_{0y} \hat{j}$ enters a region with a magnetic field $\vec{B} = B_0 \hat{j}$. After some time, an electric field $\vec{E} = E_0 \hat{j}$ is also switched on in the region. The resulting path described by the particle will be :
- (a) a circle in x-z plane
 - (b) a parabola in x-y plane
 - (c) a helix with constant pitch
 - (d) a helix with increasing pitch

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- (a) a circle in x-z plane
 - (b) a parabola in x-y plane
 - (c) a helix with constant pitch
 - (d) a helix with increasing pitch

(d) a helix with increasing pitch.

- 19.** A wire of length l is in the form of a circular loop A of one turn. This loop is reshaped into loop B of three turns. Find the ratio of the magnetic fields at the centres of loop A and loop B for the same current through them.

$B = \frac{\mu_o NI}{2r}$	1/2
$l = 2\pi r \text{ ; } 2\pi r_A = 3(2\pi r_B)$	
$r_B = \frac{r_A}{3}$	1/2
$\frac{B_A}{B_B} = \frac{\mu_o N_A I}{2r_A} \times \frac{2r_B}{\mu_o N_B I}$	1/2
$\frac{B_A}{B_B} = \frac{1}{9}$	1/2

- 21.** An alpha particle is projected with velocity $\vec{v} = (3.0 \times 10^5 \text{ m/s}) \hat{i}$ into a region in which magnetic field $\vec{B} = [(0.4 \text{ T}) \hat{i} + (0.3 \text{ T}) \hat{j}]$ exists. Calculate the acceleration of the particle in the region. $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors along x, y and z axis respectively and charge to mass ratio for alpha particle is $4.8 \times 10^7 \text{ C/kg}$.

$$\vec{F} = q(\vec{v} \times \vec{B})$$

1/2

$$= q(3 \times 10^5 \hat{i} \times (0.4 \hat{i} + 0.3 \hat{j})) N$$

$$\vec{F} = q(0.9 \times 10^5 \hat{k}) N$$

1/2

$$\vec{F} = m \vec{a} = q(0.9 \times 10^5 \hat{k}) N$$

$$\vec{a} = \frac{q}{m} (0.9 \times 10^5 \hat{k}) \text{ ms}^{-2}$$

1/2

$$= 4.8 \times 10^7 \times 0.9 \times 10^5 \hat{k} \text{ ms}^{-2}$$

$$= 4.32 \times 10^{12} \hat{k} \text{ ms}^{-2}$$

1/2

Note: Deduct 1/2 mark if a student does not mention the direction of acceleration.

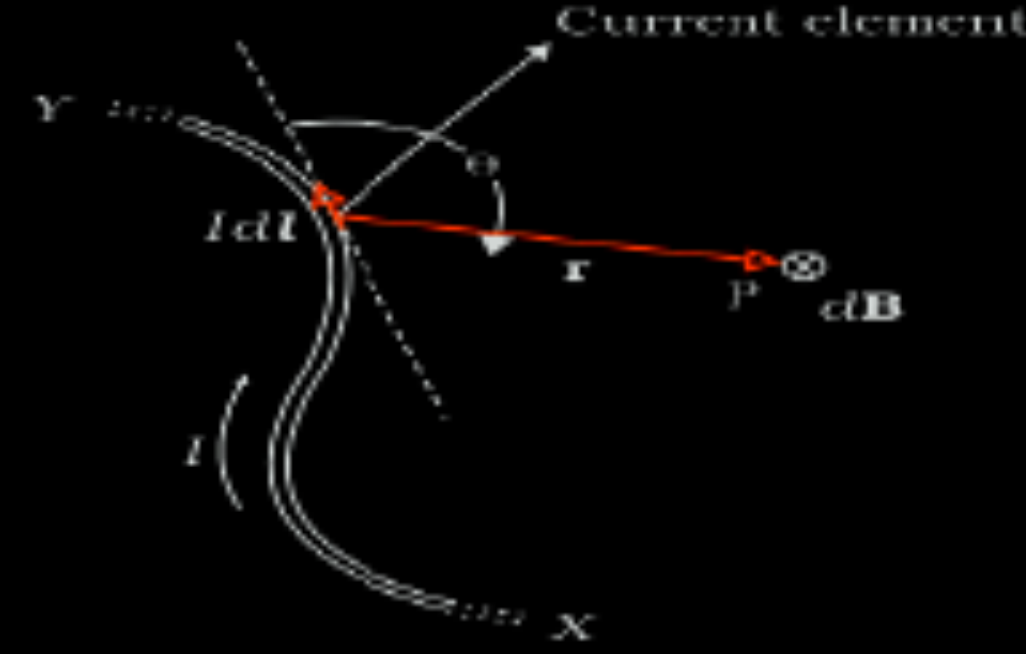
- 33.** (a) (i) State Biot-Savart's law for the magnetic field due to a current carrying element. Use this law to obtain an expression for the magnetic field at the centre of a circular loop of radius 'a' and carrying a current 'I'. Draw the magnetic field lines for a current loop indicating the direction of magnetic field.
- (ii) An electron is revolving around the nucleus in a circular orbit with a speed of 10^7 m s^{-1} . If the radius of the orbit is 10^{-10} m , find the current constituted by the revolving electron in the orbit.

i) Statement of Biot-Savart's law	1
Expression for magnetic field	2
Diagram for magnetic field lines	$\frac{1}{2}$
ii) Finding current by revolving electron	$1 \frac{1}{2}$

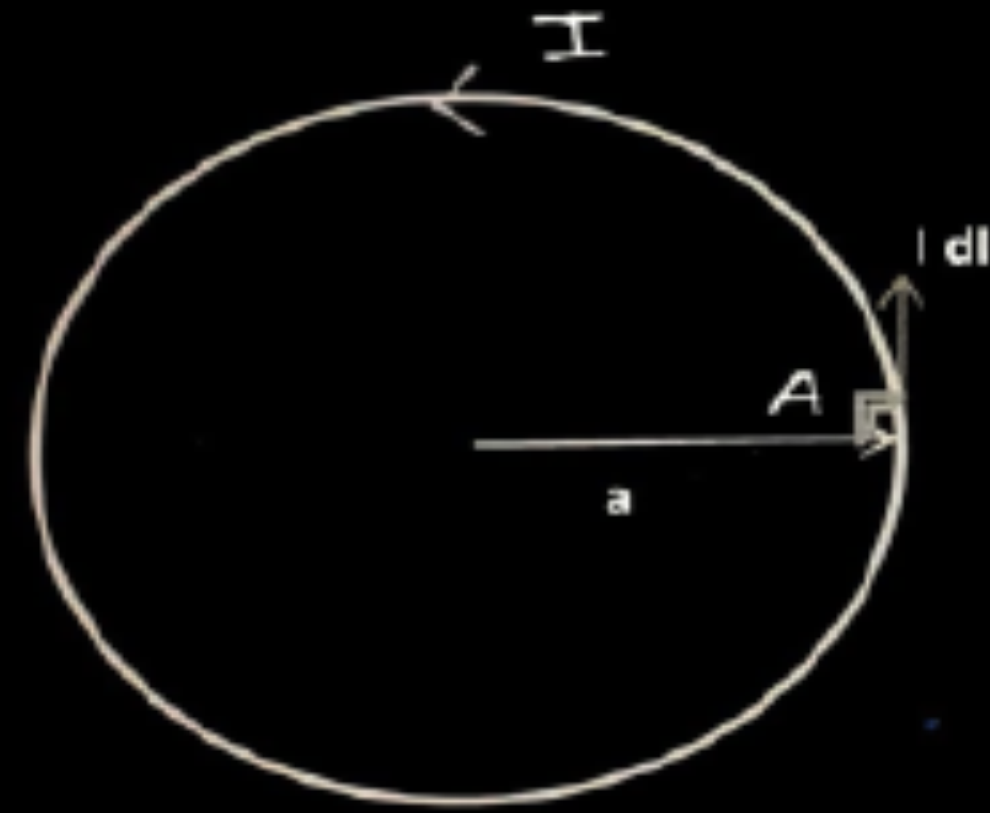
(i)
The magnetic field at a point due to a current carrying element is proportional to magnitude of current, element length and inversely proportional to the square of the distance from the element.

$$\vec{dB} = \frac{\mu_0}{4\pi} I \frac{d\vec{l} \times \vec{r}}{r^3}$$

$$|dB| = \frac{\mu_0}{4\pi} \frac{Idl \sin \theta}{r^2}$$



Consider a circular coil of radius a carrying current I.



According to Biot-Savart's law

$$|dB| = \frac{\mu_0}{4\pi} \frac{Idl \sin \theta}{r^2}$$

At point A $I \vec{dl} \perp \vec{a}$
 $\therefore \theta = 90^\circ, \sin 90^\circ = 1$

$$\text{Hence } Db = \frac{\mu_0}{4\pi} \frac{Idl}{a^2}$$

Magnetic field at centre

$$B = \int_0^{2\pi a} dB = \int_0^{2\pi a} \frac{\mu_0}{4\pi} \frac{Idl}{a^2}$$

$$B = \frac{\mu_0}{4\pi} \times \frac{I}{a^2} \times 2\pi a$$

$$B = \frac{\mu_0 I}{2a}$$

1

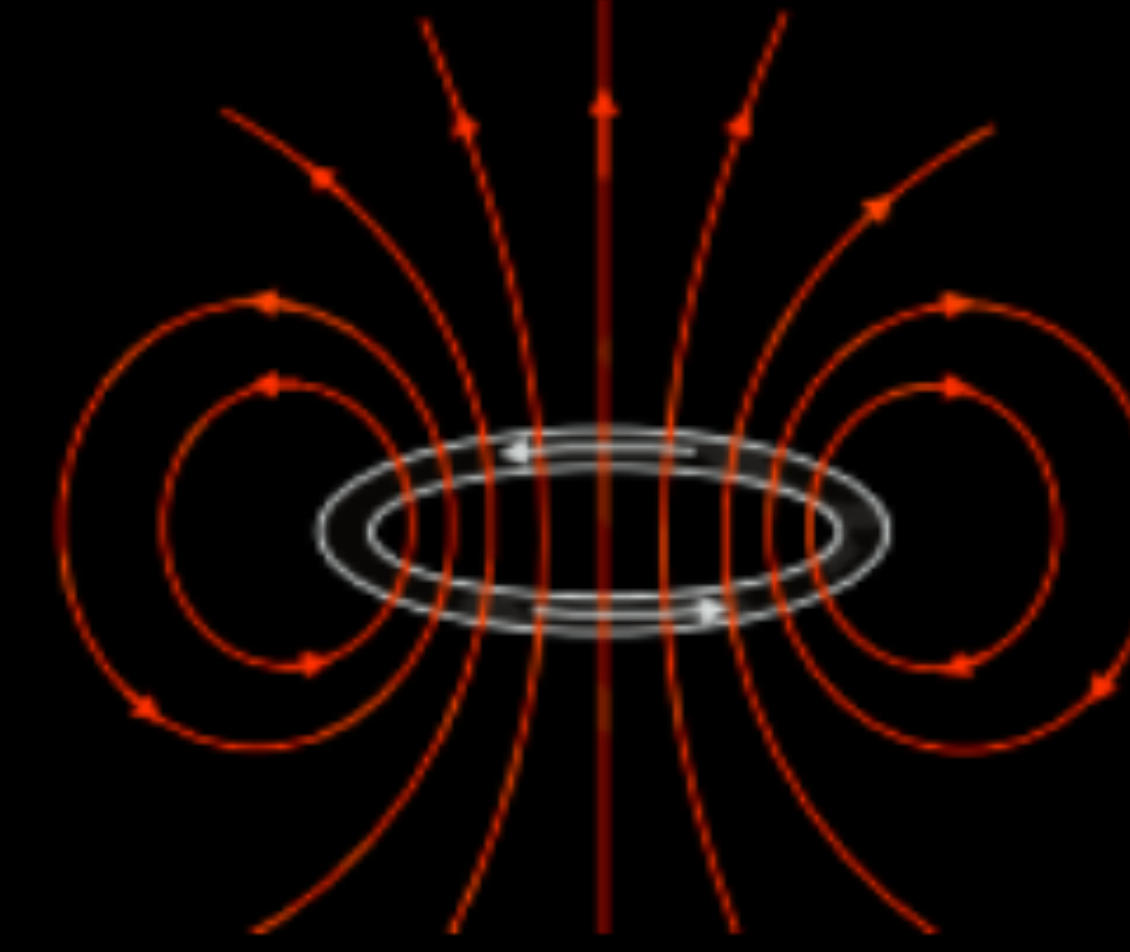
1/2

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1/2

1/2

Note: Give full credit of 2 marks if a student derives the expression for magnetic field at the axis of the loop and then puts distance of point as 0 from the centre.



ii) $q=e$, $v=10^7 \text{ ms}^{-1}$, $r=10^{-10} \text{ m}$

$$i = \frac{q}{T}$$

$$= \frac{qv}{2\pi r}$$

$$= \frac{ev}{2\pi r}$$

$$= \frac{1.6 \times 10^{-19} \times 10^7}{2 \times \pi \times 10^{-10}}$$

$$= \frac{0.8}{\pi} \times 10^{-2} \text{ A}$$

$$= 0.255 \times 10^{-2} \text{ A} = 2.55 \text{ Ma}$$

1/2

1/2

1/2

1/2

- (b) (i) Derive an expression for the force acting on a current carrying straight conductor kept in a magnetic field. State the rule which is used to find the direction of this force. Give the condition under which this force is (1) maximum, and (2) minimum.
- (ii) Two long parallel straight wires A and B are 2.5 cm apart in air. They carry 5.0 A and 2.5 A currents respectively in opposite directions. Calculate the magnitude of the force exerted by wire A on a 10 cm length of wire B.

i) Derivation of expression for force	2
Statement of Rule	$\frac{1}{2}$
Conditions for maximum and minimum force	$\frac{1}{2} + \frac{1}{2}$
ii) Calculation of magnitude of force	$1 \frac{1}{2}$

Consider a rod of uniform cross sectional area A and length l . Let the number density of mobile charge carriers in it be n .

Thus the total number of mobile charge carriers in it is $n l A$.

For steady current I , drift velocity of electrons $\vec{v}_d$, in the presence of external magnetic field $\vec{B}$, the force on these carriers is

$$\vec{F} = n l A q (\vec{v}_d \times \vec{B})$$

$$= \left[\vec{j}Al \right] \times \vec{B}$$

$$= I(\vec{l} \times \vec{B})$$

Where $nq\vec{v}_d$ is current density ($\vec{j}$) and $|\vec{j}A|$ is current (I)

Fleming's left hand Rule: If forefinger, middle finger and thumb are stretched in mutually perpendicular directions, such that forefinger indicates the direction of magnetic field, middle finger indicates the direction of current in the conductor, then thumb indicates the direction of force on the conductor.

Or

Right Hand Thumb Rule : If the fingers of right hand are made to rotate from $\vec{l}$ to $\vec{B}$ through angle θ , the thumb points in the direction of force on the current carrying conductor.

Condition for maximum force $\theta = 90^\circ$

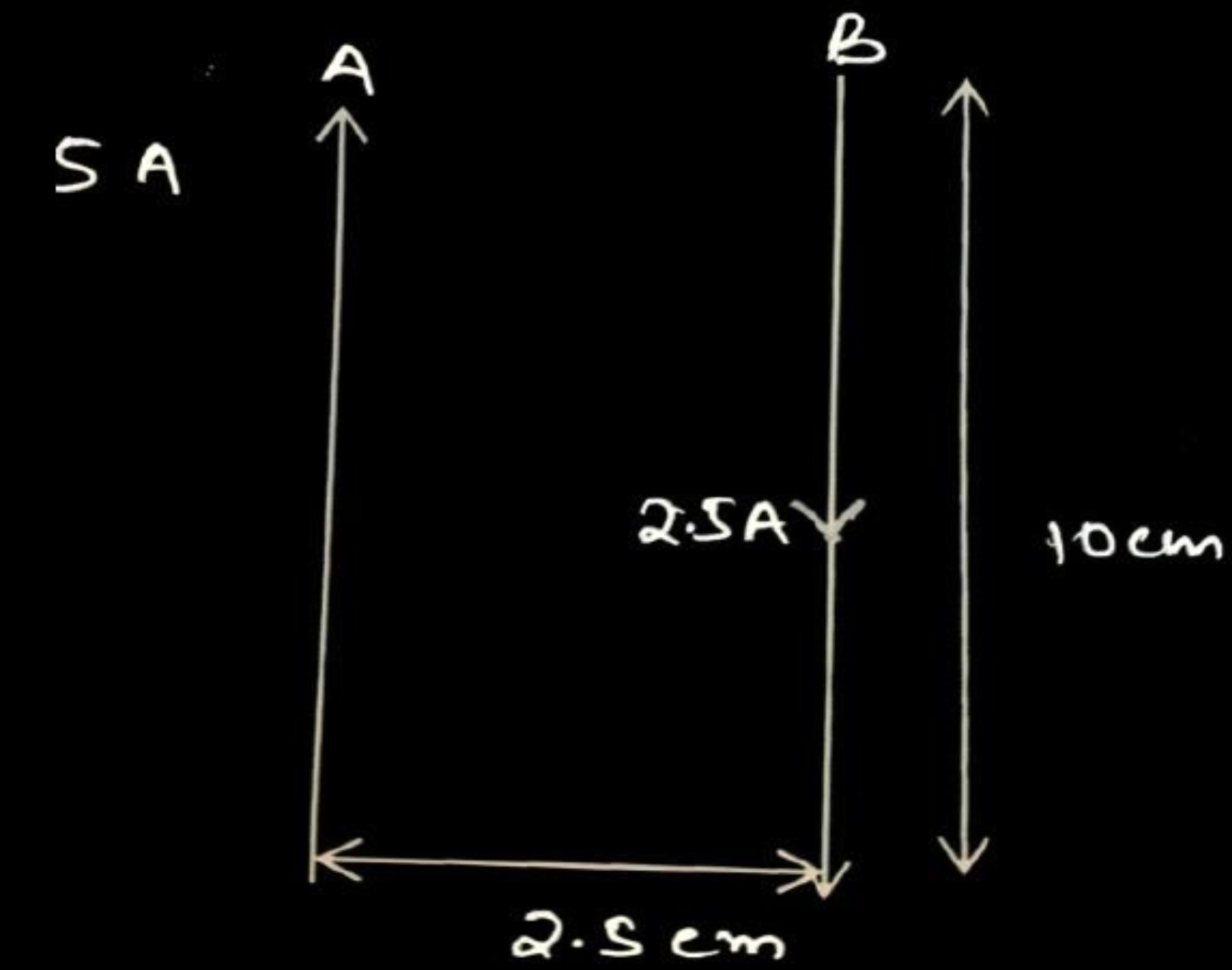
$$|\vec{F}| = I l B \sin \theta = I l B$$

Condition for minimum force $\theta = 0^\circ$ or 180°

$$\left| \overrightarrow{F} \right| = 0$$

 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

ii)



$$F = \frac{\mu_o}{4\pi} \frac{2I_1 I_2}{d} l$$

$$= \frac{10^{-7} \times 2 \times 5 \times 2.5}{2.5 \times 10^{-2}} \times 10 \times 10^{-2} N$$

$$= 10^{-5} \text{ N}$$

 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

1. A long straight wire of radius 'a' carries a steady current 'I'. The current is uniformly distributed across its area of cross-section. The ratio of magnitude of magnetic field $\vec{B}_1$ at $\frac{a}{2}$ and $\vec{B}_2$ at distance 2a is

1

(a) $\frac{1}{2}$

(b) 1

(c) 2

(d) 4

1. A long straight wire of radius 'a' carries a steady current 'I'. The current is uniformly distributed across its area of cross-section. The ratio of magnitude of magnetic field $\vec{B}_1$ at $\frac{a}{2}$ and $\vec{B}_2$ at distance 2a is

1

(a) $\frac{1}{2}$

(b) 1

(c) 2

(d) 4

24. Briefly explain why and how a galvanometer is converted into an ammeter.

2

Explanation of conversion of galvanometer into an ammeter

- Why 1
- How 1

24. Briefly explain why and how a galvanometer is converted into an ammeter. 2

• Due to very high sensitivity Alternatively It has large resistance and hence will change the value of current in circuit.	1
• A galvanometer can be converted into an ammeter of desired range by connecting a shunt of proper value across its coil.	1

Explanation of conversion of galvanometer into an ammeter	
• Why	1
• How	1

13. Beams of electrons and protons move parallel to each other in the same direction. They

1

- (a) attract each other.
- (b) repel each other.
- (c) neither attract nor repel.
- (d) force of attraction or repulsion depends upon speed of beams.

13. Beams of electrons and protons move parallel to each other in the same direction. They

1

- (a) attract each other.
- (b) repel each other.
- (c) neither attract nor repel.
- (d) force of attraction or repulsion depends upon speed of beams.

(b) Repel each other

2. Two horizontal thin long parallel wires, separated by a distance r carry current I each in the opposite directions. The net magnetic field at a point midway between them, will be

1

- (A) zero
- (B) $\left(\frac{\mu_0 I}{2\pi r}\right)$ vertically downward
- (C) $\left(\frac{2\mu_0 I}{r}\right)$ vertically upward
- (D) $\left(\frac{\mu_0 I}{\pi r}\right)$ vertically downward

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- (D) $\left(\frac{\mu_0 I}{\pi r}\right)$ vertically downward

No option is correct.

[Award one mark to each student]

18. **Assertion (A)** : The deflecting torque acting on a current carrying loop is zero when its plane is perpendicular to the direction of magnetic field.

1

Reason (R) : The deflecting torque acting on a loop of magnetic moment $\vec{m}$ in a magnetic field $\vec{B}$ is given by the dot product of $\vec{m}$ and $\vec{B}$.

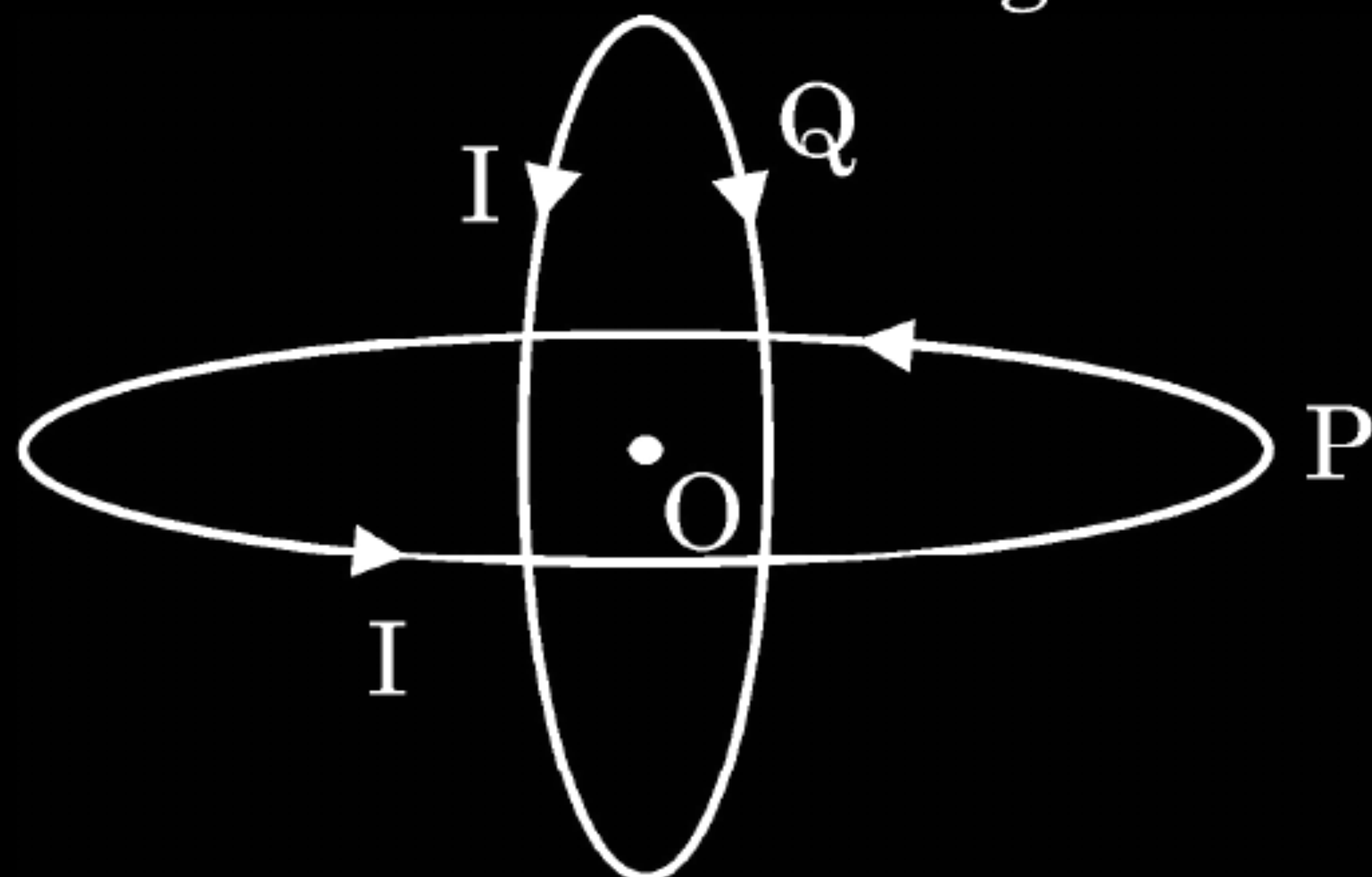
18. **Assertion (A)** : The deflecting torque acting on a current carrying loop is zero when its plane is perpendicular to the direction of magnetic field.

1

Reason (R) : The deflecting torque acting on a loop of magnetic moment $\vec{m}$ in a magnetic field $\vec{B}$ is given by the dot product of $\vec{m}$ and $\vec{B}$.

(C) Assertion (A) is true and Reason (R) is false.

23. (a) Two identical circular loops P and Q, each of radius R carrying current I are kept in perpendicular planes such that they have a common centre O as shown in the figure.



Find the magnitude and direction of the net magnetic field at point O. **2**

(a) Calculation of magnitude of field

(2)

$$B_1 = \frac{\mu_0 I}{2R}$$

$$B_2 = \frac{\mu_0 I}{2R}$$

$$B = \sqrt{B_1^2 + B_2^2}$$

$$B = \sqrt{2 \left(\frac{\mu_0 I}{2R} \right)^2}$$

$$B = \sqrt{2} \frac{\mu_0 I}{2R}$$

Note: Since direction of current is shown clockwise as well as anti-clockwise in the vertical loop. So no marks is assigned for direction of magnetic field.

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- (b) A long straight conductor kept along $X'X$ axis, carries a steady current I along $+x$ direction. At an instant t , a particle of mass m and charge q at point (x, y) moves with a velocity $\vec{v}$ along $+y$ direction. Find the magnitude and direction of the force on the particle due to the conductor.

2

(b) Calculation of magnitude of force
Direction

$(1\frac{1}{2})$
 $(\frac{1}{2})$

$$\vec{F} = q(\vec{V} \times \vec{B})$$

$$\frac{1}{2}$$

$$B = \frac{\mu_o I}{2\pi y} \text{ and } \theta = 90^\circ$$

$$\frac{1}{2}$$

$$F = \frac{\mu_o qvI}{2\pi y}$$

$$\frac{1}{2}$$

Force is along + X axis

$$\frac{1}{2}$$

Alternatively

$$\vec{F} = \frac{\mu_o qvI}{2\pi y} \hat{i}$$

29. (a) Briefly describe how the current sensitivity of a moving coil galvanometer can be increased.
- (b) A galvanometer shows full scale deflection for current I_g . A resistance R_1 is required to convert it into a voltmeter of range $(0 - V)$ and a resistance R_2 to convert it into a voltmeter of range $(0 - 2V)$. Find the resistance of the galvanometer.

3

- | | |
|---|-----|
| (a) Methods to increase current sensitivity of moving coil galvanometer | (1) |
| (b) Calculation of resistance of galvanometer | (2) |

(a) Current sensitivity of moving coil galvanometer can be increased by (1) increasing number of turns in the coil ($I_s \propto N$) (2) increasing magnetic field strength ($I_s \propto B$) (3) using a material having lesser value of torsional constant $\left(I_s \propto \frac{1}{k} \right)$ (4) increasing area of cross-section of the coil (Note- Any one of the above)	1
(b) $R = \frac{V}{I_g} - G$ For range (0–V) $R_1 = \frac{V}{I_g} - G \text{-----(1)}$ For range (0-2V) $R_2 = \frac{2V}{I_g} - G \text{-----(2)}$ On solving equations (1) and (2) $G = R_2 - 2R_1$	$\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

2. Two long parallel wires kept 2 m apart carry 3A current each, in the same direction. The force per unit length on one wire due to the other is

1

(A) $4.5 \times 10^{-5} \text{ Nm}^{-1}$, attractive

(B) $4.5 \times 10^{-7} \text{ N/m}$, repulsive

(C) $9 \times 10^{-7} \text{ N/m}$, repulsive

(D) $9 \times 10^{-5} \text{ N/m}$, attractive

23. (a) Write the expression for the Lorentz force on a particle of charge q moving with a velocity $\vec{v}$ in a magnetic field $\vec{B}$. When is the magnitude of this force maximum ? Show that no work is done by this force on the particle during its motion from a point $\vec{r}_1$ to point $\vec{r}_2$. **2**

Expression for Lorentz force	($\frac{1}{2}$)
Condition for maximum magnitude of force	($\frac{1}{2}$)
Showing no work done in moving the charge from point $\vec{r}_1$ to $\vec{r}_2$	(1)

$$\vec{F}_m = q(\vec{v} \times \vec{B})$$

$$F_m = q v B \sin \theta \quad (\because \vec{v} \perp \vec{B})$$

Force is maximum for $\theta = 90^\circ$

As magnetic force always acts in a direction perpendicular to the velocity vector, hence no work is done by this force on the particle during motion.

Alternatively $W = \vec{F} \cdot (\vec{r}_2 - \vec{r}_1)$

$$= F \left| \vec{r}_2 - \vec{r}_1 \right| \cos 90^\circ$$

$$= 0$$

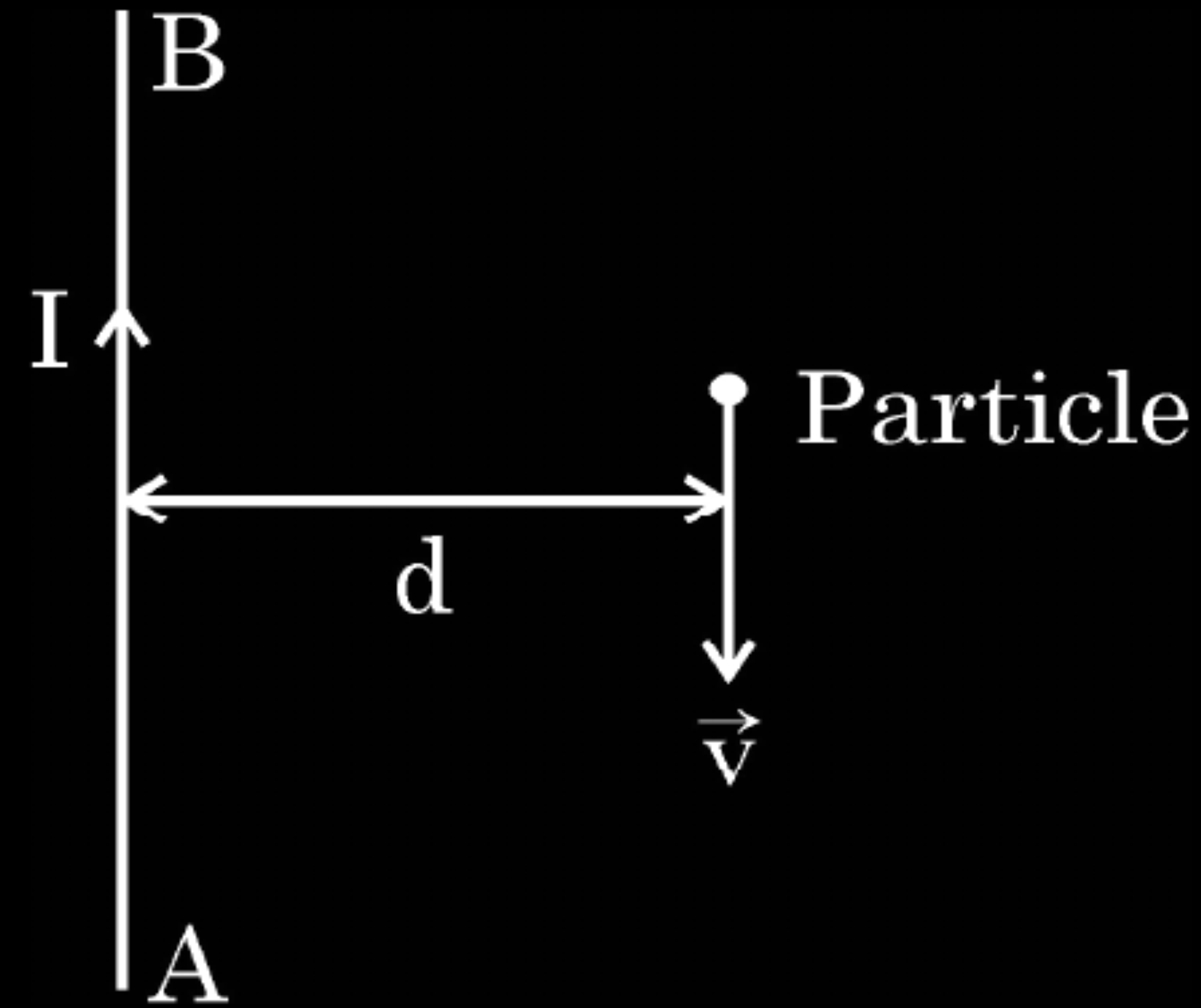
$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$1$$

- (b) A long straight wire AB carries a current I . A particle (mass m and charge q) moves with a velocity $\vec{v}$, parallel to the wire, at a distance d from it as shown in the figure. Obtain the expression for the force experienced by the particle and mention its directions.

2



(b)

Obtaining the expression for force experienced
Direction

(1 1/2)
(1/2)

The magnetic field produced by current carrying conductor AB

$$B = \frac{\mu_0 I}{2\pi d}, \text{ directed into the plane of paper.}$$

1/2

Force experienced by charged particle

$$\vec{F}_m = q(\vec{v} \times \vec{B})$$

1/2

$$\therefore \vec{v} \perp \vec{B}$$

$$F_m = q v B \sin 90^\circ$$

$$F_m = q v \times \frac{\mu_0 I}{2\pi d}$$

1/2

Force is repulsive/acts towards right / away from the conductor.

1/2

29. (a) Briefly describe how the current sensitivity of a moving coil galvanometer can be increased.
- (b) A galvanometer shows full scale deflection for current I_g . A resistance R_1 is required to convert it into a voltmeter of range $(0 - V)$ and a resistance R_2 to convert it into a voltmeter of range $(0 - 2V)$. Find the resistance of the galvanometer.

3

(a) Methods to increase current sensitivity of moving coil galvanometer

(1)

(b) Calculation of resistance of galvanometer

(2)

(a) Current sensitivity of moving coil galvanometer can be increased by

(1) increasing number of turns in the coil ($I_s \propto N$)

(2) increasing magnetic field strength ($I_s \propto B$)

(3) using a material having lesser value of torsional constant $\left(I_s \propto \frac{1}{k} \right)$

(4) increasing area of cross-section of the coil

(**Note-** Any one of the above)

1

(b) $R = \frac{V}{I_g} - G$

$\frac{1}{2}$

For range (0–V)

$R_1 = \frac{V}{I_g} - G$ ----- (1)

$\frac{1}{2}$

For range (0-2V)

$R_2 = \frac{2V}{I_g} - G$ ----- (2)

$\frac{1}{2}$

On solving equations (1) and (2)

$G = R_2 - 2R_1$

$\frac{1}{2}$

5. Two long straight parallel conductors carrying steady currents I_1 and I_2 in the same direction are separated by a distance d . The force exerted by one conductor on unit length of the other is :

(a) $\frac{\mu_0 I_1 I_2}{4\pi d}$, attractive

(b) $\frac{\mu_0 I_1 I_2}{4\pi d}$, repulsive

(c) $\frac{\mu_0 I_1 I_2}{2\pi d}$, attractive

(d) $\frac{\mu_0 I_1 I_2}{2\pi d}$, repulsive

5. Two long straight parallel conductors carrying steady currents I_1 and I_2 in the same direction are separated by a distance d . The force exerted by one conductor on unit length of the other is :

(a) $\frac{\mu_0 I_1 I_2}{4\pi d}$, attractive

(b) $\frac{\mu_0 I_1 I_2}{4\pi d}$, repulsive

(c) $\frac{\mu_0 I_1 I_2}{2\pi d}$, attractive

(d) $\frac{\mu_0 I_1 I_2}{2\pi d}$, repulsive

(c) $\frac{\mu_o I_1 I_2}{2\pi d}$, attractive

- 6.** A long solenoid of length L and number of turns N carrying a current I produces a magnetic field B along its axis. If the current in the solenoid is doubled and its number of turns halved, the magnetic field along its axis will be :

(a) $2B$

(b) $4B$

(c) $\frac{B}{2}$

(d) B

6. A long solenoid of length L and number of turns N carrying a current I produces a magnetic field B along its axis. If the current in the solenoid is doubled and its number of turns halved, the magnetic field along its axis will be :

(a) $2B$

(b) $4B$

(c) $\frac{B}{2}$

(d) B

(d) B

- 32.** (a) (i) A charged particle with speed v_0 enters a region in which a strong and non-uniform magnetic field exists everywhere. It comes out of the region following a complicated trajectory without suffering any collision in the region. Would its final speed v be equal to its initial speed v_0 ? Justify your answer.
- (ii) In a region, a uniform magnetic field of $6 \times 10^{-4} \text{ T}$ is maintained. An electron enters the field with a speed of $3 \times 10^6 \text{ ms}^{-1}$ normal to the field. Find radius of its path. Also calculate its energy in eV.

5

(i) Answer	1
Justification	1
(ii) Calculation of radius	1 ½
Calculation of energy in eV	1 ½

(i) Yes, the final speed v would be equal to its initial speed V_o .

Because the magnetic force is perpendicular to the direction of motion of the charged particle.

(ii) $r = \frac{mv}{qB}$

$$= \frac{9.1 \times 10^{-31} \times 3 \times 10^6}{1.6 \times 10^{-19} \times 6 \times 10^{-4}}$$

$$= 2.8 \text{ cm}$$

$$\text{K.E} = \frac{1}{2}mv^2$$

$$= \frac{\frac{1}{2} \times 9.1 \times 10^{-31} \times (3 \times 10^6)^2}{1.6 \times 10^{-19}} \text{ eV}$$

$$= 25.59 \text{ eV}$$

1

1

$\frac{1}{2}$

1

$\frac{1}{2}$

1

- (b) (i) Derive an expression for the force acting on a conductor carrying current I in a magnetic field $\vec{B}$.
- (ii) A long horizontal conductor carries a current of 30 A, in east to west direction. What are the magnitude and direction of the magnetic field due to the current 1.0 m below the conductor ?

5

(i) Deriving expression for force on a current carrying wire

3

(ii) Finding the magnitude and direction of magnetic field

1+1

(i) Consider a wire of length l , area of cross-section A and having number density of free electrons n and carrying electric current I . Total number of charge carriers in the wire = nIA . If $\vec{v}_d$ is drift velocity of electrons, magnetic force experience by the electrons is $\vec{F} = -(nIA)e\vec{v}_d \times \vec{B}$ As $I\vec{l} = -enIA\vec{v}_d$ $\vec{F} = I(\vec{l} \times \vec{B})$	$\frac{1}{2}$
(ii) $B = \frac{2\mu_o I}{4\pi r}$	$\frac{1}{2}$
$B = \frac{2 \times 10^{-7} \times 30}{1} = 0.6 \times 10^{-5} T$	$\frac{1}{2}$
Magnetic field is directed towards SOUTH	1

17. *Assertion (A) :* A proton and an alpha particle having same kinetic energy are moving in circular paths in a uniform magnetic field. The radii of their circular paths will be equal.

Reason (R) : The centripetal force required to move a charged particle in a circle does not depend on the magnitude of the magnetic field.

17. *Assertion (A) :* A proton and an alpha particle having same kinetic energy are moving in circular paths in a uniform magnetic field. The radii of their circular paths will be equal.

Reason (R) : The centripetal force required to move a charged particle in a circle does not depend on the magnitude of the magnetic field.

(c) Assertion (A) is true, but Reason (R) is false.

- 29.** (a) It is not advisable to use a galvanometer as such to measure current directly. Why ?
- (b) Why should the value of resistance connected in parallel to a galvanometer be low ?
- (c) Is the reading shown by an ammeter in a circuit less than or more than the actual value of current flowing in the circuit ? Why ?

- 29.** (a) It is not advisable to use a galvanometer as such to measure current directly. Why ?
- (b) Why should the value of resistance connected in parallel to a galvanometer be low ?
- (c) Is the reading shown by an ammeter in a circuit less than or more than the actual value of current flowing in the circuit ? Why ?

3

(a) Explanation	1
(b) Explanation	1
(c) Answer	$\frac{1}{2}$
Explanation	$\frac{1}{2}$

(a) It will not measure accurate value of current because its high resistance will affect the current in the circuit.	1
(b) To reduce the galvanometer resistance a small resistance is connected in parallel.	1
(c) It is less than the actual value of current because it has some resistance	1

2. An electron with velocity $\vec{v} = (v_x \hat{i} + v_y \hat{j})$ moves through a magnetic field $\vec{B} = (B_x \hat{i} - B_y \hat{j})$. The force $\vec{F}$ on the electron is : (e is the magnitude of its charge)

(a) $-e (v_x B_y - v_y B_x) \hat{k}$

(b) $e (v_x B_y - v_y B_x) \hat{k}$

(c) $-e (v_x B_y + v_y B_x) \hat{k}$

(d) $e (v_x B_y + v_y B_x) \hat{k}$

2. An electron with velocity $\vec{v} = (v_x \hat{i} + v_y \hat{j})$ moves through a magnetic field $\vec{B} = (B_x \hat{i} - B_y \hat{j})$. The force $\vec{F}$ on the electron is : (e is the magnitude of its charge)

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(b) $e (v_x B_y - v_y B_x) \hat{k}$

(c) $-e (v_x B_y + v_y B_x) \hat{k}$

(d) $e (v_x B_y + v_y B_x) \hat{k}$

(d) $e(v_x B_y + v_y B_x)k$

17. *Assertion (A) :* The magnetic field produced by a current carrying solenoid is independent of its length and cross-sectional area.

Reason (R): The magnetic field inside the solenoid is uniform.

17. *Assertion (A) :* The magnetic field produced by a current carrying solenoid is independent of its length and cross-sectional area.

Reason (R): The magnetic field inside the solenoid is uniform.

(b) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).

17. *Assertion (A) :* A planar loop of irregular shape carrying current is subjected to a magnetic field acting perpendicular to the plane of the loop. If the wire is flexible, the loop takes a circular shape.

Reason (R) : The force acting on each point of a current carrying loop, in a magnetic field perpendicular to its plane, is radially outward.

17. *Assertion (A) :* A planar loop of irregular shape carrying current is subjected to a magnetic field acting perpendicular to the plane of the loop. If the wire is flexible, the loop takes a circular shape.

Reason (R) : The force acting on each point of a current carrying loop, in a magnetic field perpendicular to its plane, is radially outward.

(a) Both Assertion (A) and Reason (R) are true, but Reason (R) is the correct explanation of the Assertion (A)

2. Two insulated concentric coils, each of radius R , placed at right angles to each other, carry currents I and $\sqrt{3} I$ respectively. The magnitude of the net magnetic field at their common centre will be

(A) $\frac{\mu_o I}{R}$

(B) $\frac{\mu_o I}{2R}$

(C) $\frac{\mu_o I}{4R}$

(D) $\frac{2\mu_o I}{R}$

2. Two insulated concentric coils, each of radius R , placed at right angles to each other, carry currents I and $\sqrt{3} I$ respectively. The magnitude of the net magnetic field at their common centre will be

(A) $\frac{\mu_o I}{R}$

(B) $\frac{\mu_o I}{2R}$

(C) $\frac{\mu_o I}{4R}$

(D) $\frac{2\mu_o I}{R}$

(A) $\frac{\mu_0 I}{R}$

4. A galvanometer of resistance $100\ \Omega$ gives full scale deflection for a current of $1.0\ \text{mA}$. It is converted into an ammeter of range $(0 - 1\text{A})$. The resistance of the ammeter will be close to

(A) $0.1\ \Omega$

(B) $0.8\ \Omega$

(C) $1.0\ \Omega$

(D) $10\ \Omega$

4. A galvanometer of resistance $100\ \Omega$ gives full scale deflection for a current of $1.0\ \text{mA}$. It is converted into an ammeter of range $(0 - 1\text{A})$. The resistance of the ammeter will be close to

(A) $0.1\ \Omega$

(B) $0.8\ \Omega$

(C) $1.0\ \Omega$

(D) $10\ \Omega$

(A) $0.1\ \Omega$

16. **Assertion (A)** : The energy of a charged particle moving in a magnetic field does not change.

Reason (R) : It is because the work done by the magnetic force on the charge moving in a magnetic field is zero.

16. **Assertion (A)** : The energy of a charged particle moving in a magnetic field does not change.

Reason (R) : It is because the work done by the magnetic force on the charge moving in a magnetic field is zero.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is correct explanation of Assertion (A).

17. Derive an expression for magnetic force $\vec{F}$ acting on a straight conductor of length L carrying current I in an external magnetic field $\vec{B}$. Is it valid when the conductor is in zig-zag form ? Justify.

Deriving an expression for magnetic force	$1\frac{1}{2}$
Validity and Justification for zig-zag form conductor	$\frac{1}{2}$

Total number of mobile charge carriers in a conductor of length L , cross-sectional area A and number density of charge carriers n :

$$= nLA$$

Force acting on the charge carriers in external magnetic field $\vec{B}$

$$\vec{F} = (nAL) q \vec{v}_d \times \vec{B} \quad \text{-----}(1)$$

Where $\vec{v}_d$ is the drift velocity of the charge carriers

Current flowing

$$I = v_d q n A$$

$$I \vec{L} = \vec{v}_d q n A L \quad \text{-----}(2)$$

On solving equation (1) and (2)

$$\vec{F} = I(\vec{L} \times \vec{B})$$

Yes, because this force can be calculated by considering zig-zag conductor as a collection of linear strips ($d\vec{l}$) and summing them vectorically.

22. (b) (i) State and explain Ampere's circuital law.
- (ii) Two long straight parallel wires separated by 20 cm, carry 5 A and 10 A current respectively, in the same direction. Find the magnitude and direction of the net magnetic field at a point midway between them.

(i) Statement and explanation of Ampere's circuital law	1
(ii) Finding magnitude and direction of magnetic field	2

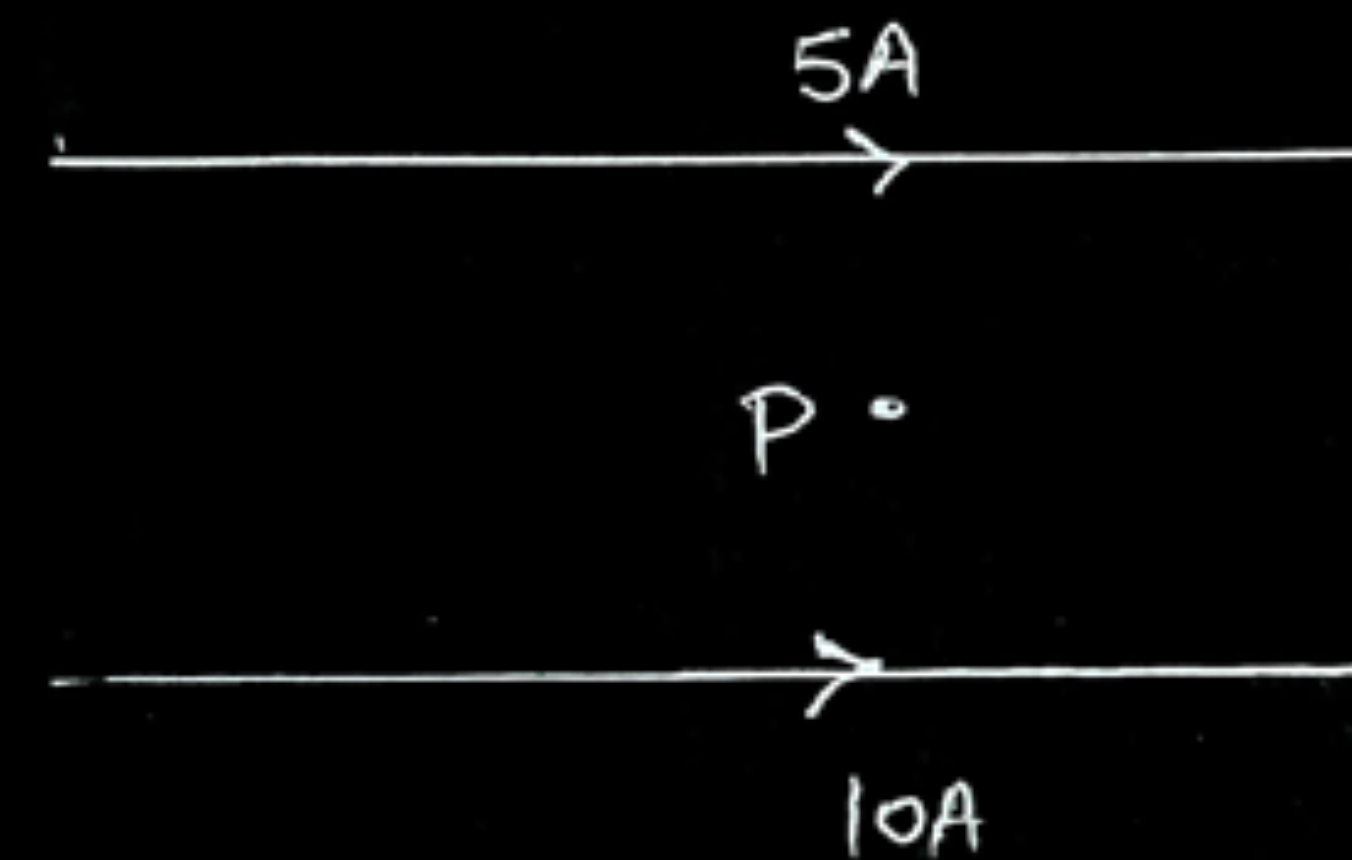
Line integral of magnetic field over a closed loop in vacuum is equal to μ_0 times the total current passing through the loop.

Alternatively

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I$$

The integral in this expression is over a closed loop coinciding with the boundary of the surface.

(ii)



$$B = \frac{\mu_0 I}{2\pi r}$$

Net magnetic field $B = B_2 - B_1$

$$B = \frac{\mu_0 \times 10^2}{20\pi} [10 - 5]$$

$$B = \frac{4\pi \times 10^{-7} \times 10^2 \times 5}{20\pi}$$

$$B = 10^{-5} T$$

Along the direction of magnetic field produced by the conductor carrying current 10A.

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

27. An electron is moving with a velocity $\vec{v} = \left(3 \times 10^6 \frac{\text{m}}{\text{s}}\right)\hat{i}$. It enters a region of magnetic field $\vec{B} = (91 \text{ mT})\hat{k}$.

- (a) Calculate the magnetic force $\vec{F}_B$ acting on electron and the radius of its path.
- (b) Trace the path described by it.

(a) Calculation for magnetic force and radius	2
(b) Tracing the path	1

$$(a) \vec{F}_B = q(\vec{v} \times \vec{B})$$

$$= -1.6 \times 10^{-19} \left[(3 \times 10^6 \hat{i}) \times (91 \times 10^{-3} \hat{k}) \right]$$

$$= 1.6 \times 10^{-19} \left[3 \times 10^6 \times 91 \times 10^{-3} \right] \hat{j}$$

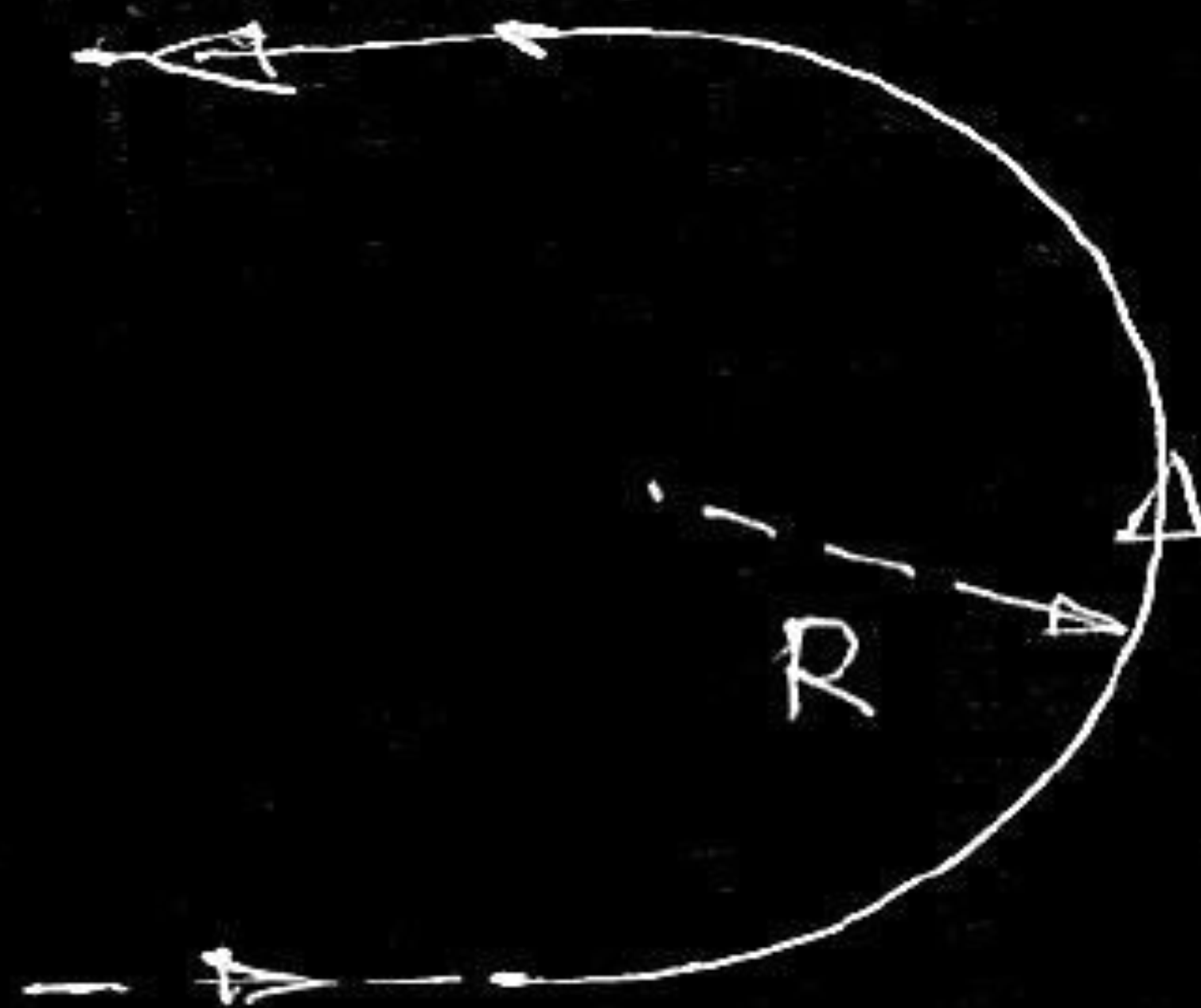
$$= 4.368 \times 10^{-14} \hat{j} \text{ N}$$

$$r = \frac{mv}{qB}$$

$$r = \frac{9.1 \times 10^{-31} \times 3 \times 10^6}{1.6 \times 10^{-19} \times 91 \times 10^{-3}} \text{ m}$$

$$r = 1.875 \times 10^{-4} \text{ m}$$

(b) Anticlockwise circular path



$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1

1. An ammeter and a voltmeter are connected in series to a battery. Their readings are noted as 'A' and 'V' respectively. If a resistor is connected in parallel with the voltmeter, then
 - (A) A will increase, V will decrease.
 - (B) A will decrease, V will increase.
 - (C) Both A and V will decrease.
 - (D) Both A and V will increase.

1. An ammeter and a voltmeter are connected in series to a battery. Their readings are noted as 'A' and 'V' respectively. If a resistor is connected in parallel with the voltmeter, then
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 - (D) Both A and V will increase.

(A) A will increase, V will decrease

4. A galvanometer shows full scale deflection for a current I_g . If a shunt of resistance S_1 is connected to the galvanometer, it gets converted into an ammeter of range $(0 - I)$. When resistance of the shunt is made S_2 , its range becomes $(0 - 2I)$. Then $\left(\frac{S_1}{S_2}\right)$ is

(A) $\frac{I + I_g}{I - I_g}$

(B) $\frac{I - I_g}{I + I_g}$

(C) $\frac{2I - I_g}{I - I_g}$

(D) $\frac{I - I_g}{2I - I_g}$

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(B) $\frac{I - I_g}{I + I_g}$

(C) $\frac{2I - I_g}{I - I_g}$

(D) $\frac{I - I_g}{2I - I_g}$

(C) $\frac{2I - I_g}{I - I_g}$

2. A particle of mass m and charge q describes a circular path of radius R in a magnetic field. If its mass and charge were $2m$ and $\frac{q}{2}$ respectively, the radius of its path would be

(A) $\frac{R}{4}$

(B) $\frac{R}{2}$

(C) $2R$

(D) $4R$

2. A particle of mass m and charge q describes a circular path of radius R in a magnetic field. If its mass and charge were $2m$ and $\frac{q}{2}$ respectively, the radius of its path would be

(A) $\frac{R}{4}$

(B) $\frac{R}{2}$

(C) $2R$

(D) $4R$

(D) $4R$

4. A galvanometer of resistance $50\ \Omega$ is converted into a voltmeter of range $(0 - 2\text{V})$ using a resistor of $1.0\ \text{k}\Omega$. If it is to be converted into a voltmeter of range $(0 - 10\ \text{V})$, the resistance required will be
- | | |
|---------------------------|---------------------------|
| (A) $4.8\ \text{k}\Omega$ | (B) $5.0\ \text{k}\Omega$ |
| (C) $5.2\ \text{k}\Omega$ | (D) $5.4\ \text{k}\Omega$ |

4. A galvanometer of resistance $50\ \Omega$ is converted into a voltmeter of range $(0 - 2\text{V})$ using a resistor of $1.0\ \text{k}\Omega$. If it is to be converted into a voltmeter of range $(0 - 10\ \text{V})$, the resistance required will be
- (A) $4.8\ \text{k}\Omega$ (B) $5.0\ \text{k}\Omega$
(C) $5.2\ \text{k}\Omega$ (D) $5.4\ \text{k}\Omega$

(C) $5.2\ \text{k}\Omega$

5. Two long straight parallel conductors A and B, kept at a distance r , carry current I in opposite directions. A third identical conductor C, kept at a distance $\left(\frac{r}{3}\right)$ from A carry current I_1 in the same direction as in A. The net magnetic force on unit length of C is

1

(A) $\frac{3\mu_0 I I_1}{2\pi r}$, towards A

(B) $\frac{3\mu_0 I I_1}{2\pi r}$, towards B

(C) $\frac{3\mu_0 I I_1}{4\pi r}$, towards A

(D) $\frac{3\mu_0 I I_1}{4\pi r}$, towards B

5. Two long straight parallel conductors A and B, kept at a distance r , carry current I in opposite directions. A third identical conductor C, kept at a distance $\left(\frac{r}{3}\right)$ from A carry current I_1 in the same direction as in A. The net magnetic force on unit length of C is

1

(A) $\frac{3\mu_0 I I_1}{2\pi r}$, towards A

(B) $\frac{3\mu_0 I I_1}{2\pi r}$, towards B

(C) $\frac{3\mu_0 I I_1}{4\pi r}$, towards A

(D) $\frac{3\mu_0 I I_1}{4\pi r}$, towards B

No option is correct, award 1 mark.

8. A galvanometer of resistance $100\ \Omega$ is converted into an ammeter of range $(0 - 1\ \text{A})$ using a resistance of $0.1\ \Omega$. The ammeter will show full scale deflection for a current of about

(A) $0.1\ \text{mA}$

(B) $1\ \text{mA}$

(C) $10\ \text{mA}$

(D) $0.1\ \text{A}$

8. A galvanometer of resistance $100\ \Omega$ is converted into an ammeter of range $(0 - 1\ \text{A})$ using a resistance of $0.1\ \Omega$. The ammeter will show full scale deflection for a current of about

1

(A) $0.1\ \text{mA}$

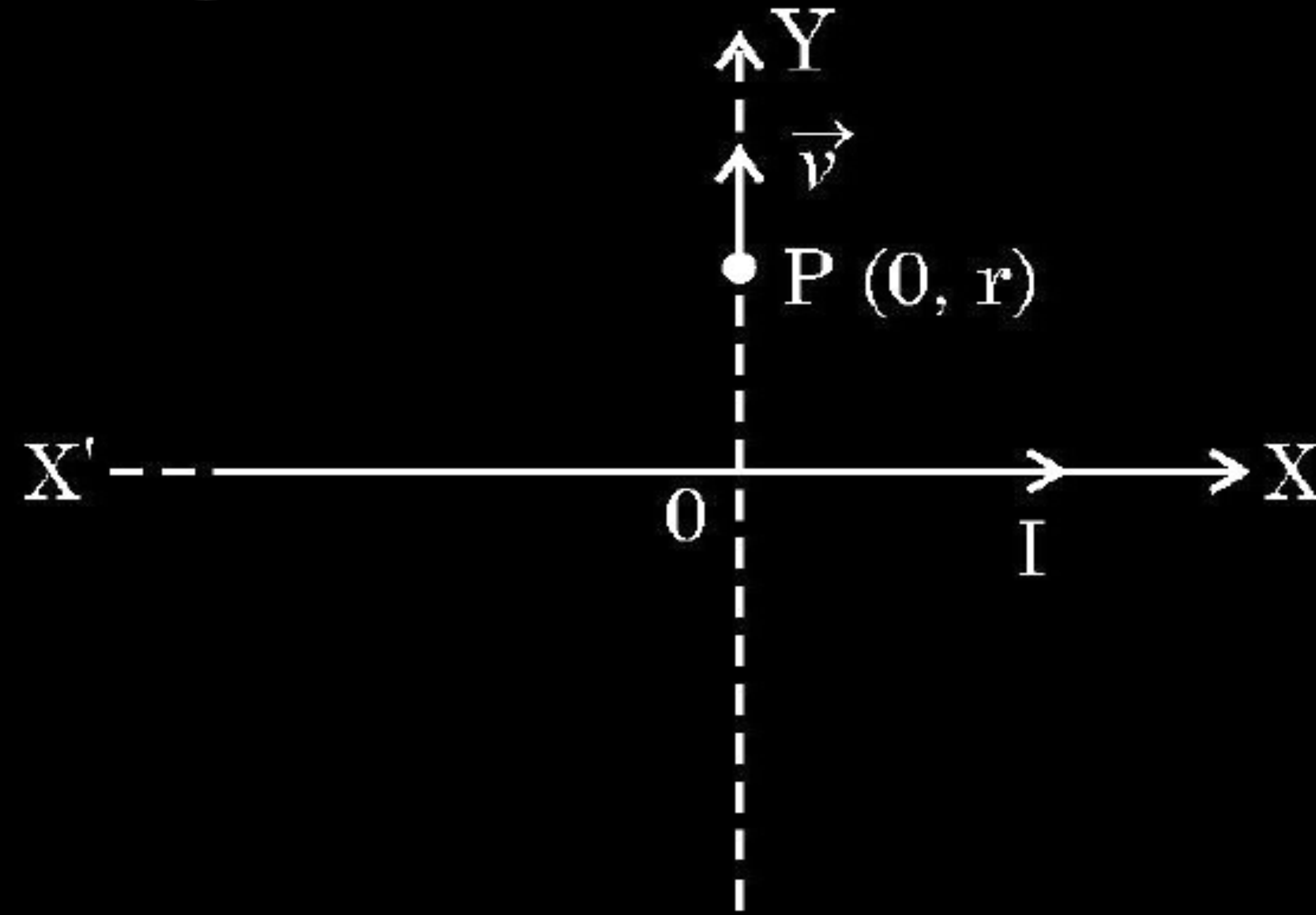
(B) $1\ \text{mA}$

(C) $10\ \text{mA}$

(D) $0.1\ \text{A}$

(B) 1mA

26. An infinite straight conductor is kept along $X'X$ axis and carries a current I . A charge q at point $P(0, r)$ starts moving with velocity $\vec{v} = v_0 \hat{j}$ as shown in figure. Find the direction and magnitude of force initially experienced by the charge.



Finding magnitude of force
Finding direction of force

2 1/2
1/2

Magnetic field at P due to infinite straight conductor carrying current

$$\vec{B} = \frac{\mu_0 I}{2\pi r} \hat{k}$$

1/2

Force on charge q in this magnetic field

$$\vec{F} = q(\vec{v} \times \vec{B})$$

1/2

$$\vec{F} = q \left[(v_0 \hat{j}) \times \left(\frac{\mu_0 I}{2\pi r} \right) \hat{k} \right]$$

1/2

$$\vec{F} = \frac{\mu_0 q v_0 I}{2\pi r} \hat{i}$$

1/2

The magnitude of force $F = \frac{\mu_0 q v_0 I}{2\pi r}$

1/2

The direction of force on charge is along +ve X-axis.

1/2

- (ii) An electron is revolving around a proton in an orbit of radius r with a speed v . Obtain expression for magnetic moment associated with the electron.

- (ii) An electron is revolving around a proton in an orbit of radius r with a speed v . Obtain expression for magnetic moment associated with the electron.

(ii) The equivalent current

$$I = \frac{q}{t} = \frac{e}{\frac{2\pi r}{v}} = \frac{ev}{2\pi r}$$

Magnetic moment of revolving electron

$$m = IA$$

$$= \frac{ev}{2\pi r} \times \pi r^2$$

$$= \frac{1}{2} evr$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

5. A particle of mass m and charge q is moving with velocity $\vec{v} = v_x \hat{i} + v_y \hat{j}$.

If it is subjected to a magnetic field $\vec{B} = B_0 \hat{i}$, it will move in a –

1

(A) straight line path

(B) circular path

(C) helical path

(D) parabolic path

5. A particle of mass m and charge q is moving with velocity $\vec{v} = v_x \hat{i} + v_y \hat{j}$.

If it is subjected to a magnetic field $\vec{B} = B_0 \hat{i}$, it will move in a –

1

(A) straight line path

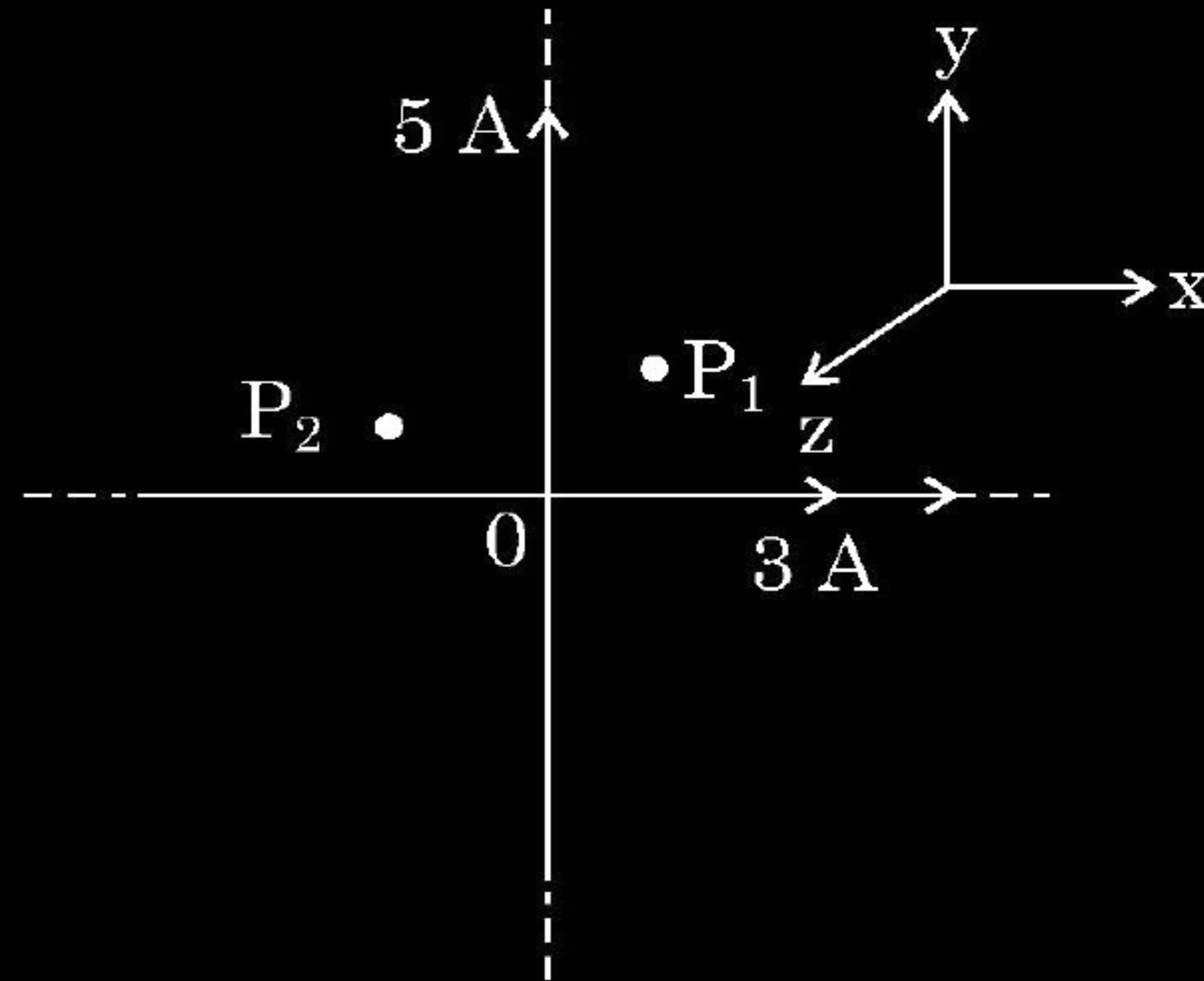
(B) circular path

(C) helical path

(D) parabolic path

(C) Helical path.

25. Two long insulated straight wires carrying currents of 3 A and 5 A are arranged in XY plane as shown in figure. Find the magnitude and direction of the net magnetic fields at points $P_1(2\text{m}, 2\text{m})$ and $P_2(-1\text{m}, 1\text{m})$. 3



Finding magnitude and direction of the net magnetic field at point P_1	1 ½
Finding magnitude and direction of the net magnetic field at point P_2	1 ½

Net magnetic field at point P₁

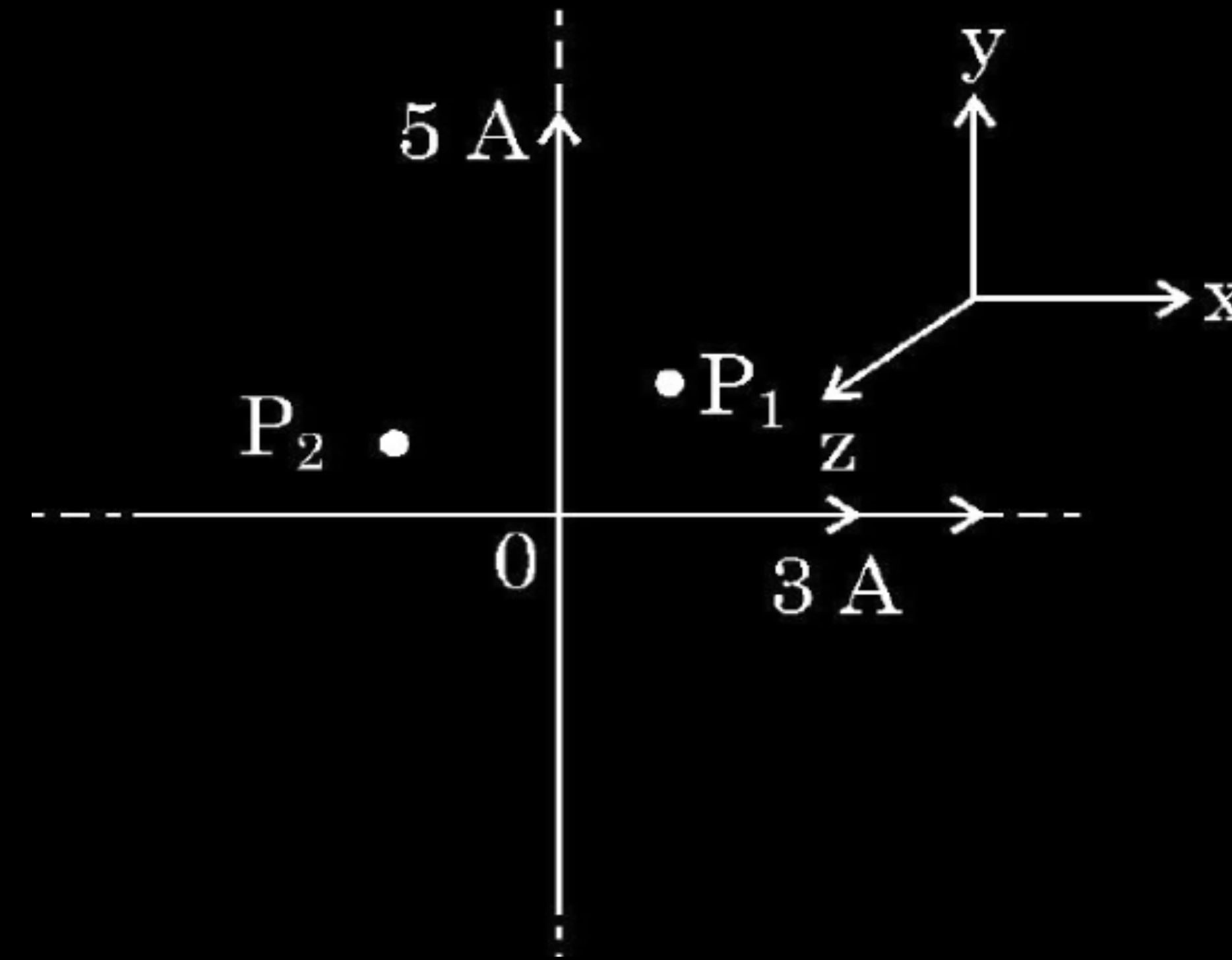
$$\begin{aligned}
 B &= B_{y(wire)} - B_{x(wire)} \\
 &= \frac{\mu_0 I_1}{2\pi r} - \frac{\mu_0 I_2}{2\pi r} \\
 &= \frac{\mu_0}{2\pi \times 2} (5 - 3) \\
 B &= 2 \times 10^{-7} \text{ T}
 \end{aligned}$$

The direction of net magnetic field is along –ve z-axis.

Net magnetic field at point P₂

$$\begin{aligned}
 B &= B_{y(wire)} + B_{x(wire)} \\
 &= \frac{\mu_0 I_1}{2\pi r} + \frac{\mu_0 I_2}{2\pi r} \\
 &= \frac{4\mu_0}{2\pi \times 1} (5 + 3) \\
 &= \frac{4\mu_0}{\pi} \\
 &= 16 \times 10^{-7} \text{ T}
 \end{aligned}$$

The direction of net magnetic field is along +ve z-axis.



1/2

1/2

1/2

1/2

1/2

1/2

5. A straight wire is kept horizontally along east-west direction. If a steady current flows in wire from east to west, the magnetic field at a point above the wire will point towards

(A) East
(C) North

(B) West
(D) South

5. A straight wire is kept horizontally along east-west direction. If a steady current flows in wire from east to west, the magnetic field at a point above the wire will point towards

1

(A) East

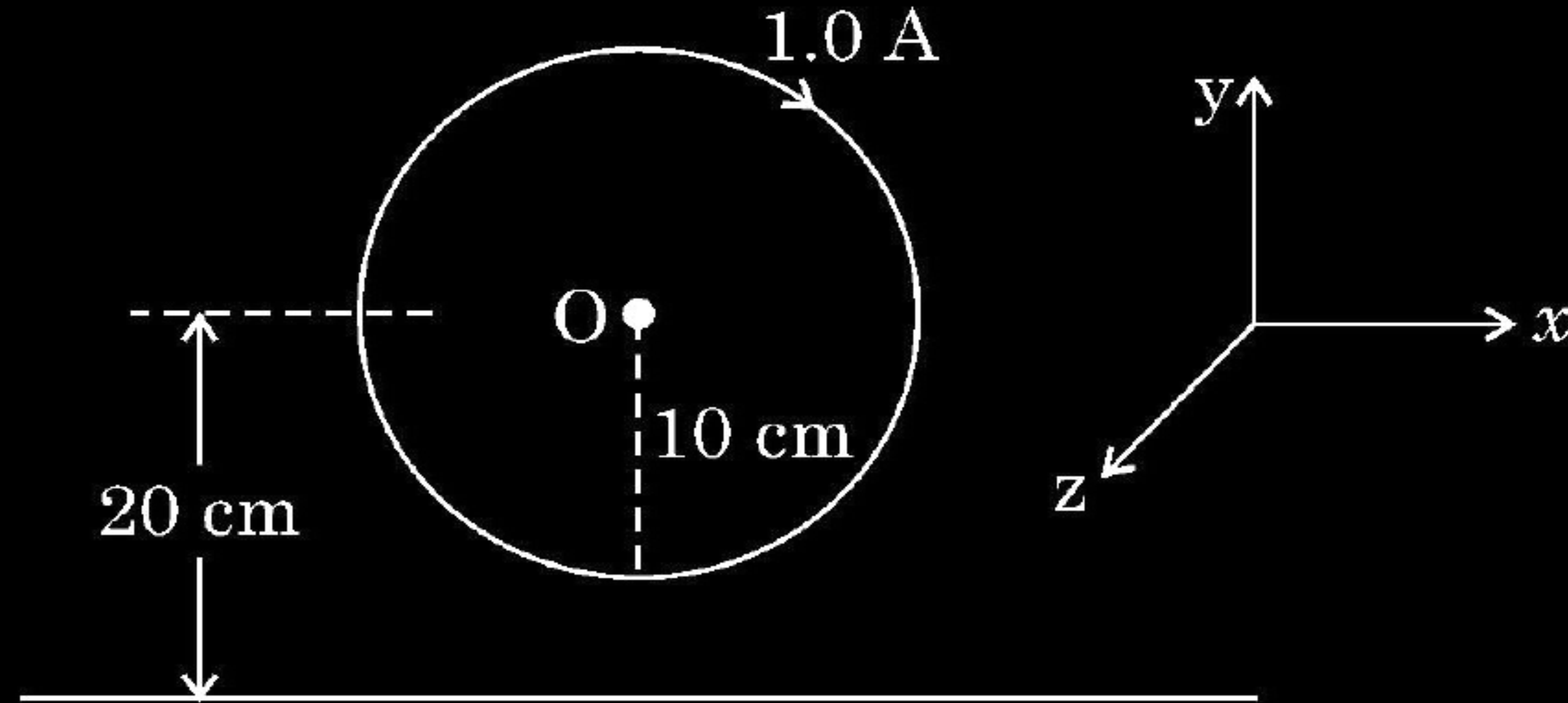
(B) West

(C) North

(D) South

(C) North

26. A circular loop of radius 10 cm carrying current of 1.0 A lies in x - y plane. A long straight wire lies in the same plane parallel to x -axis at a distance of 20 cm as shown in figure.



Find the direction and value of current that has to be maintained in the wire so that the net magnetic field at O is zero.

3

Finding value of current in a long straight wire

$2\frac{1}{2}$

Finding direction of current in a long straight wire

$\frac{1}{2}$

Magnetic field due to circular current loop at its centre O.

$$B_1 = \frac{\mu_0 I_1}{2r}$$

$$= \frac{\mu_0 \times 1}{2 \times 0.1}$$

$$= 5\mu_0 \text{ T}$$

The magnetic field B_1 is perpendicular to plane of loop and directed inwards.

Magnetic field due to long current carrying straight wire at O.

$$B_2 = \frac{\mu_0 I_2}{2\pi r}$$

$$B_2 = \frac{\mu_0 I_2}{2\pi \times 0.2} \text{ T}$$

For net magnetic field at O to be zero, B_1 should be equal and opposite to B_2 .

$$5\mu_0 = \frac{\mu_0 I_2}{0.4\pi}$$

$$I_2 = 2\pi \text{ A}$$

$$= 6.28 \text{ A}$$

Direction of current in the straight wire is along +ve x axis.

Alternatively

Net magnetic field at O is zero.

$$B_{\text{loop}} = B_{\text{wire}}$$

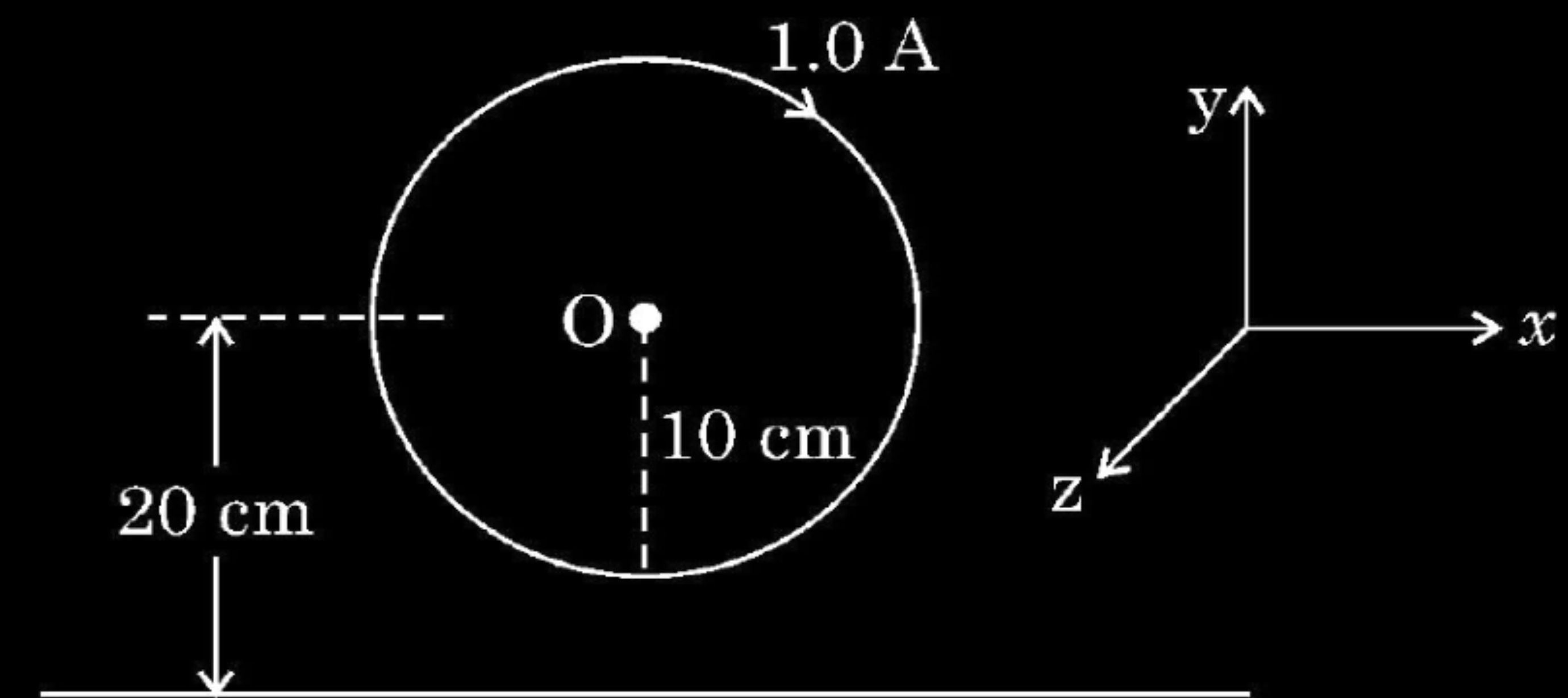
$$\frac{\mu_0 I_1}{2r_1} = \frac{\mu_0 I_2}{2\pi r_2}$$

$$\frac{\mu_0 \times 1}{2 \times 0.1} = \frac{\mu_0 I_2}{2\pi \times 0.2}$$

$$I_2 = 2\pi \text{ A}$$

$$= 6.28 \text{ A}$$

Direction of current is along + x-axis.



1/2

1/2

1/2

1/2

1/2

1/2

1/2

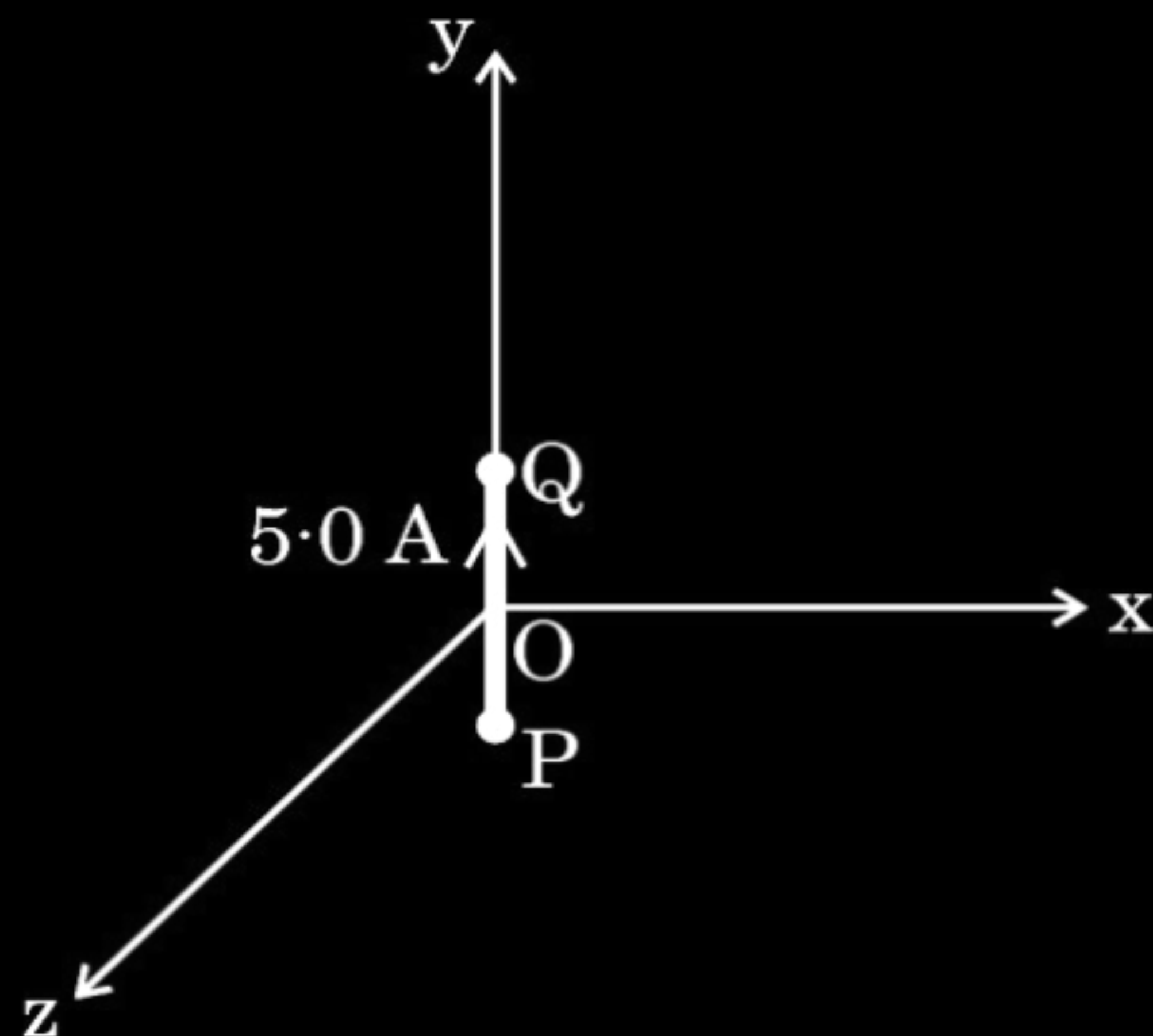
1

1/2

1/2

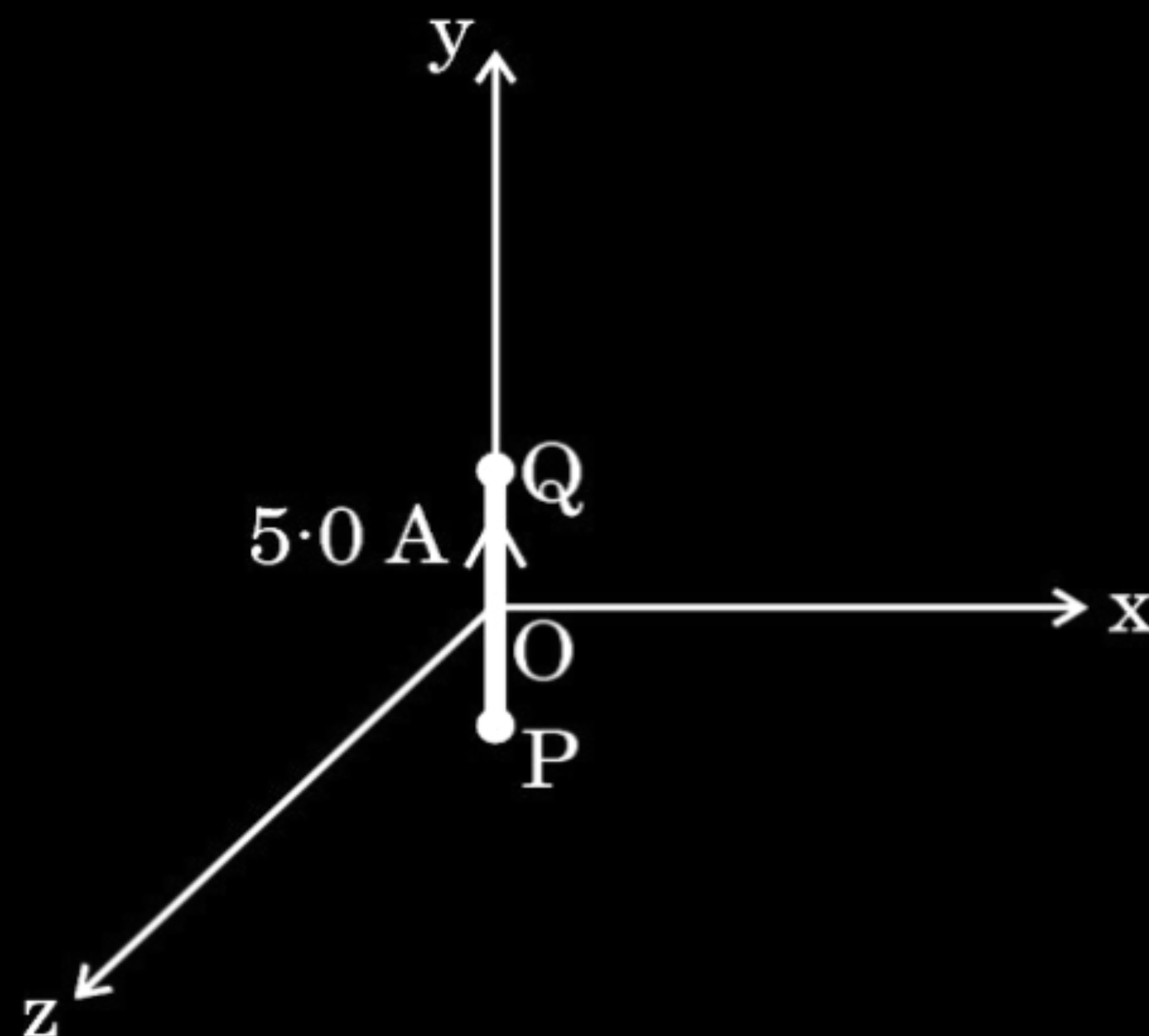
1/2

3. A 2.0 cm segment of wire, carrying 5.0 A current in positive y-direction lies along y-axis, as shown in the figure. The magnetic field at a point (3 m, 4 m, 0) due to this segment (part of a circuit) is :



- (A) $(0.12 \text{ nT}) \hat{j}$
(B) $-(0.10 \text{ nT}) \hat{j}$
(C) $-(0.24 \text{ nT}) \hat{k}$
(D) $(0.24 \text{ nT}) \hat{k}$

3. A 2.0 cm segment of wire, carrying 5.0 A current in positive y-direction lies along y-axis, as shown in the figure. The magnetic field at a point (3 m, 4 m, 0) due to this segment (part of a circuit) is :



- (A) $(0.12 \text{ nT}) \hat{j}$
(B) $-(0.10 \text{ nT}) \hat{j}$
(C) $-(0.24 \text{ nT}) \hat{k}$
(D) $(0.24 \text{ nT}) \hat{k}$

(C) $-(0.24 \text{ nT}) \hat{k}$

- (a) (i) (1) What is meant by current sensitivity of a galvanometer ?
Mention the factors on which it depends.
- (2) A galvanometer of resistance G is converted into a voltmeter of range $(0 - V)$ by using a resistance R . Find the resistance, in terms of R and G , required to convert it into a voltmeter of range $\left(0 - \frac{V}{2}\right)$.

(i)(1) Meaning of current sensitivity, mentioning factors	2
(2) Finding the required resistance	$1\frac{1}{2}$
(ii) Finding the induced current	$1\frac{1}{2}$

(i) (1) Current sensitivity of galvanometer is defined as the deflection per unit current.

Alternatively,

$$\frac{\phi}{I} = \frac{NBA}{K}$$

Factors

Number of turns in coil, Magnetic field intensity, Area of coil, Torsional Constant
(Any two)

$$(2) \quad R = \frac{V}{I} - G \quad \text{for (0-V) Range}$$

$$R_1 = \frac{V}{2I} - G \quad \text{for (0-}\frac{V}{2}\text{) Range}$$

$$\frac{V}{I} = R + G$$

$$R_1 = \left(\frac{R+G}{2} \right) - G$$

$$R_1 = \frac{R-G}{2}$$

1

$\frac{1}{2} + \frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

4. Two thin long parallel wires separated by a distance 'a' carry current 'I' in opposite directions. The wires will :

- (A) Repel each other with a force $\frac{\mu_0 I^2}{2\pi a^2}$, per unit length.
- (B) Attract each other with a force $\frac{\mu_0 I^2}{2\pi a^2}$, per unit length.
- (C) Attract each other with a force $\frac{\mu_0 I^2}{2\pi a}$, per unit length.
- (D) Repel each other with a force $\frac{\mu_0 I^2}{2\pi a}$, per unit length.

4. Two thin long parallel wires separated by a distance 'a' carry current 'I' in opposite directions. The wires will :

- (A) Repel each other with a force $\frac{\mu_0 I^2}{2\pi a^2}$, per unit length.
- (B) Attract each other with a force $\frac{\mu_0 I^2}{2\pi a^2}$, per unit length.
- (C) Attract each other with a force $\frac{\mu_0 I^2}{2\pi a}$, per unit length.
- (D) Repel each other with a force $\frac{\mu_0 I^2}{2\pi a}$, per unit length.

(D) Repel each other with a force $\frac{\mu_o I^2}{2\pi a}$, per unit length

15. *Assertion (A) :* A proton and an electron enter a uniform magnetic field $\vec{B}$ with the same momentum $\vec{p}$ such that $\vec{p}$ is perpendicular to $\vec{B}$. They describe circular paths of the same radius.

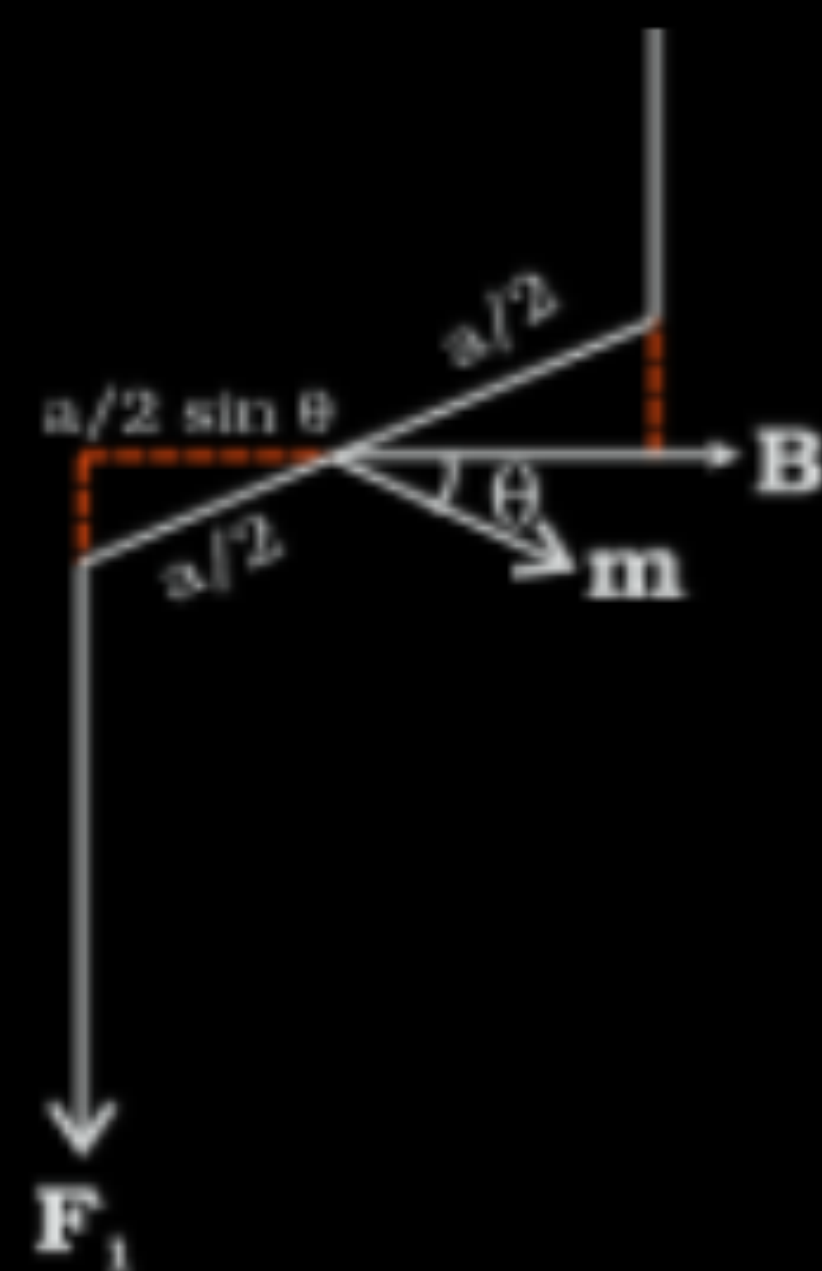
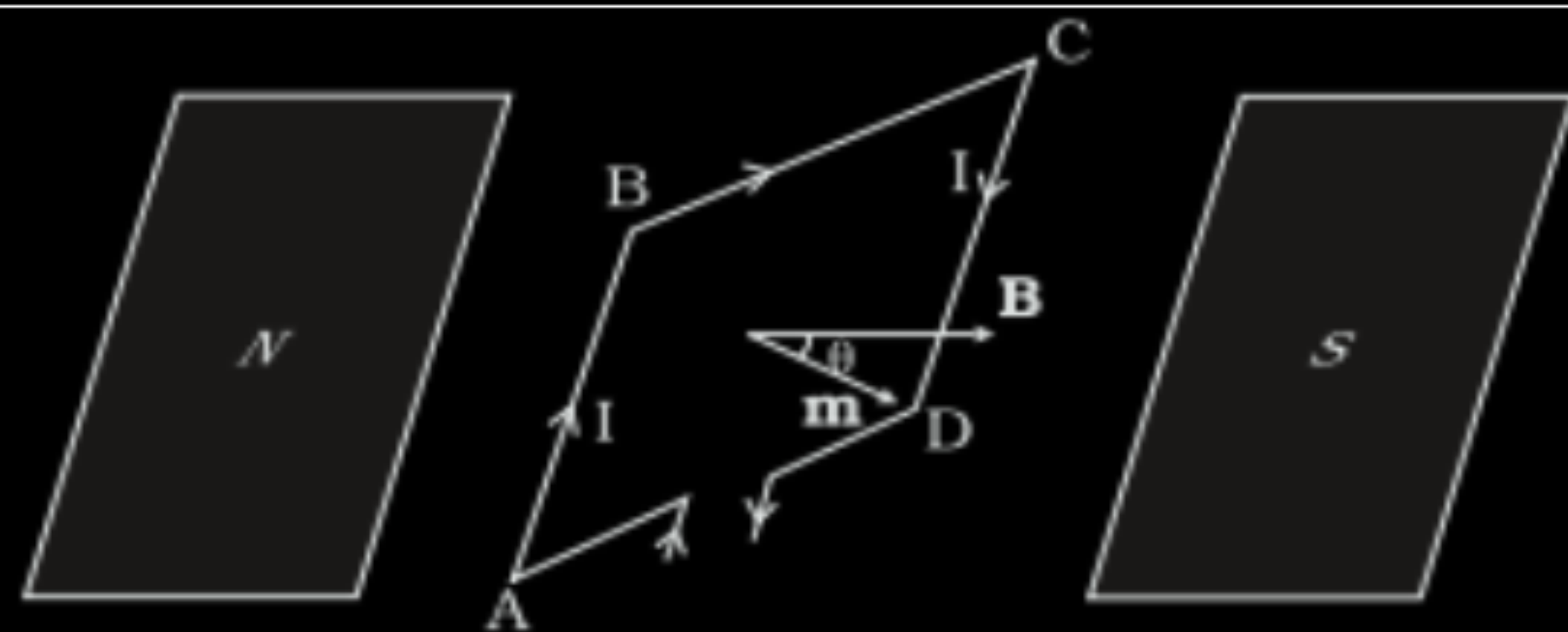
Reason (R) : In a magnetic field, orbital radius r is equal to $\frac{p}{qB}$.

15. *Assertion (A) :* A proton and an electron enter a uniform magnetic field $\vec{B}$ with the same momentum $\vec{p}$ such that $\vec{p}$ is perpendicular to $\vec{B}$. They describe circular paths of the same radius.

Reason (R) : In a magnetic field, orbital radius r is equal to $\frac{p}{qB}$.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion(A)

- 23.** A rectangular loop of area $\vec{A}$, carrying current I , is placed in a uniform magnetic field $\vec{B}$. With the help of a suitable diagram, derive an expression, in vector form, for the torque acting on the loop.



1

Forces on the arms BC and DA are, equal opposite and collinear. Hence they will cancel each other.

The forces on arms AB and CD are $\vec{F}_1$ and $\vec{F}_2$, equal but not collinear. The magnitude of the torque on the loop is

$$\tau = F_1 \frac{a}{2} \sin \theta + F_2 \frac{a}{2} \sin \theta$$

$$= IabB \sin \theta$$

$$= mB \sin \theta \quad (m = IA)$$

$$\vec{\tau} = \vec{m} \times \vec{B}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

4. A circular loop of wire, carrying a current 'I' is lying in xy-plane with its centre coinciding with the origin. It is subjected to a uniform magnetic field pointing along + z-axis. The loop will :
- | | |
|-----------------------|-------------------------|
| (A) move along x-axis | (B) move along – y-axis |
| (C) move along z-axis | (D) remain stationary |

4. A circular loop of wire, carrying a current 'I' is lying in xy-plane with its centre coinciding with the origin. It is subjected to a uniform magnetic field pointing along + z-axis. The loop will :
- | | |
|-----------------------|-------------------------|
| (A) move along x-axis | (B) move along – y-axis |
| (C) move along z-axis | (D) remain stationary |

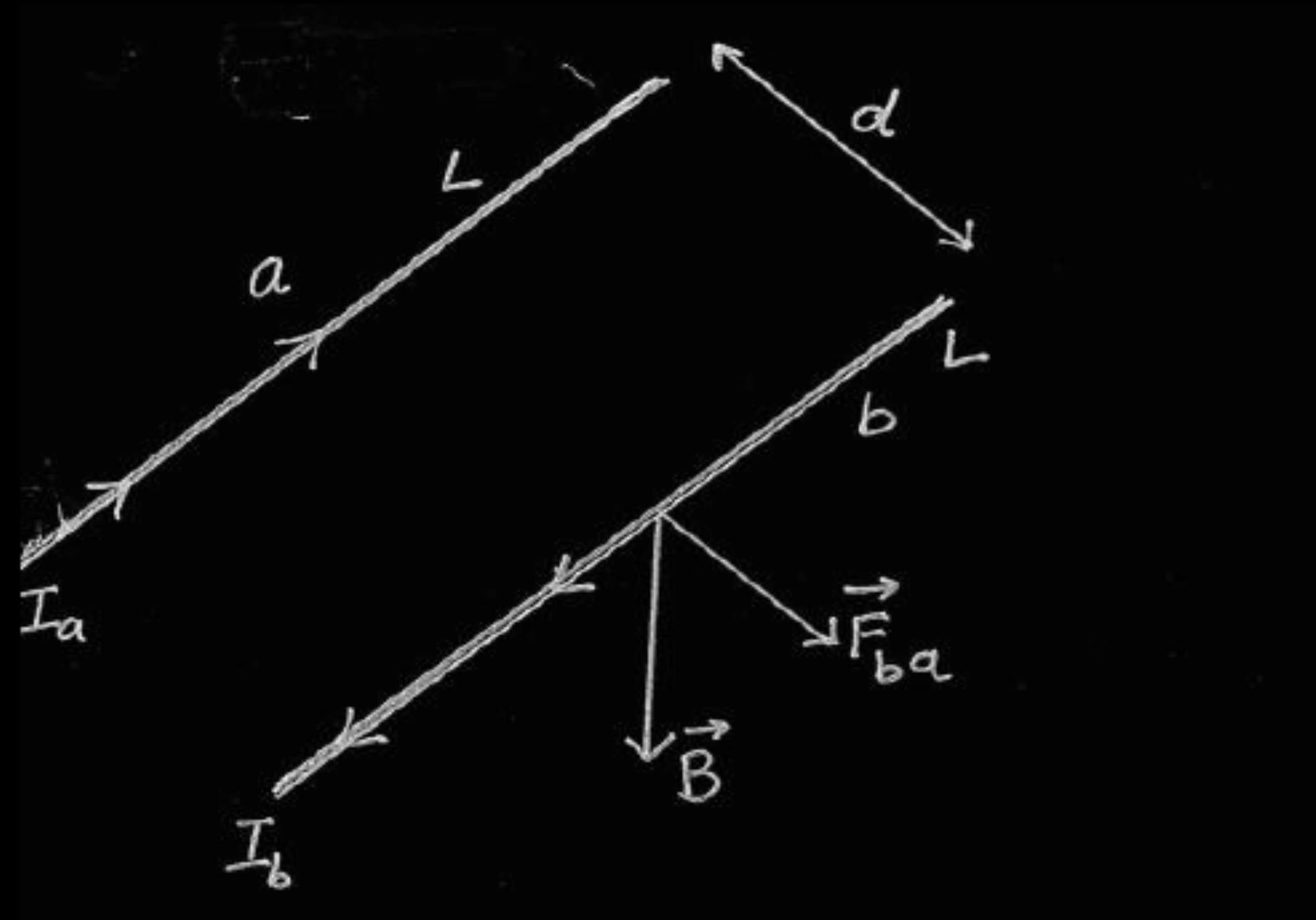
(D) remain stationary

23. Two long straight parallel conductors carrying currents, exert a force on each other. Why ? Derive an expression for the force per unit length between two long straight parallel conductors carrying currents in opposite directions. Explain the nature of the force between these conductors.

3

Reason for exerting force on straight parallel conductors	$\frac{1}{2}$
Derivation for force per unit length	2
Explanation of nature of Force	$\frac{1}{2}$

One conductor experiences a force due to magnetic field of the other conductor



Magnetic field produced by conductor 'a' at all points along the length of conductor 'b'

$$B_a = \frac{\mu_0 I_a}{2\pi d}$$

Force on conductor 'b' due to this magnetic field

$$F_{ba} = I_b L B_a$$

$$F_{ba} = \frac{\mu_0 I_a I_b L}{2\pi d}$$

$$f_{ba} = \frac{F_{ba}}{L} = \frac{\mu_0 I_a I_b}{2\pi d}$$

directed away from a

$$f_{ab} = \frac{F_{ab}}{L} = \frac{\mu_0 I_a I_b}{2\pi d}$$

directed away from b

Repulsive, the forces acting on them are away from each other.

1/2

1/2

1/2

1/2

1/2

1/2

- 22.** (a) Two long, straight, parallel conductors carry steady currents in opposite directions. Explain the nature of the force of interaction between them. Obtain an expression for the magnitude of the force between the two conductors. Hence define one ampere. 3

OR

- (b) Obtain an expression for the torque $\vec{\tau}$ acting on a current carrying loop in a uniform magnetic field $\vec{B}$. Draw the necessary diagram. 3

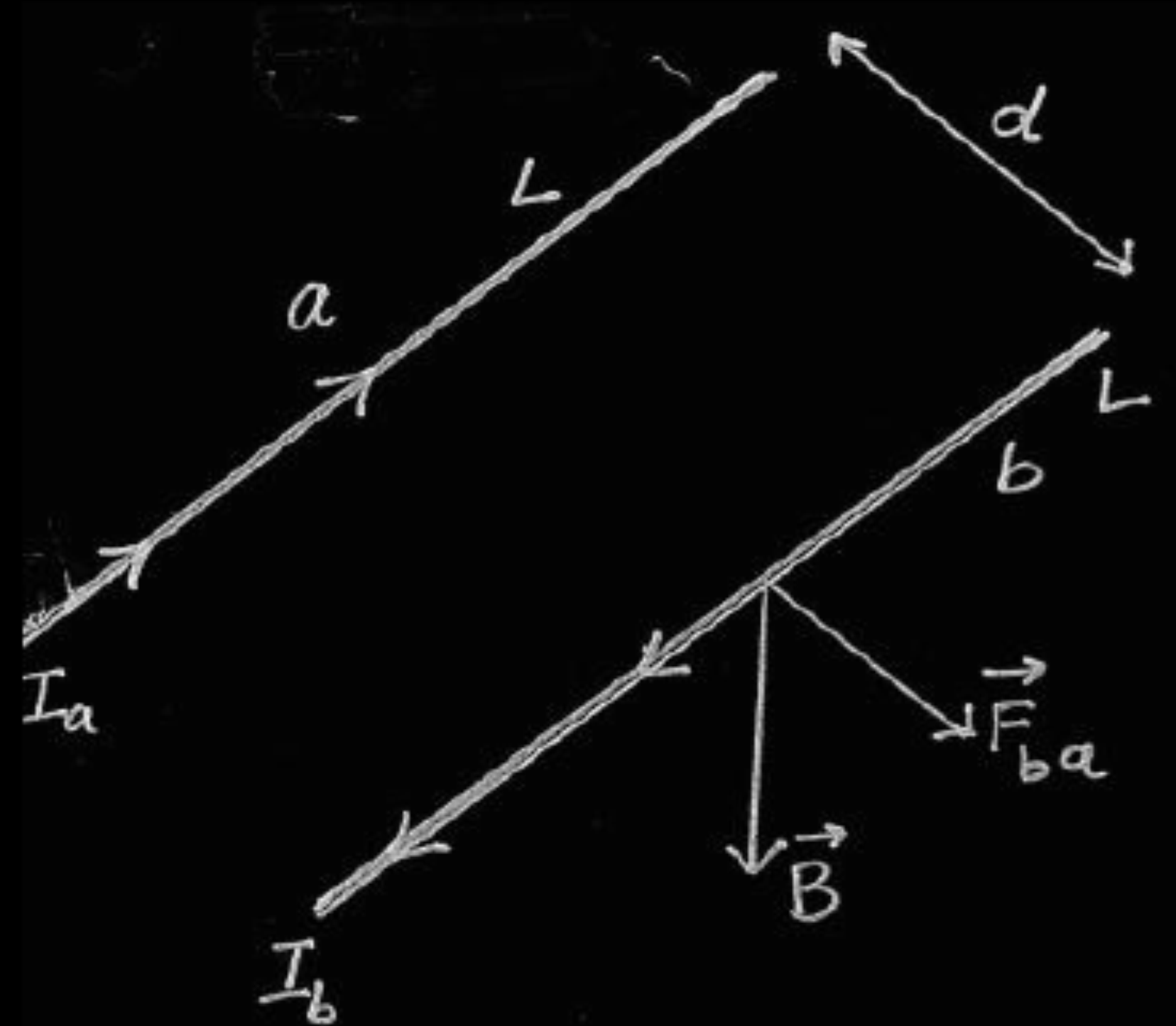
(a)

Explaining nature of force	$\frac{1}{2}$
Obtaining expression of force	$1\frac{1}{2}$
Defining one ampere	1

(b)

Obtaining expression of torque	2
Drawing diagram	1

Nature of force is repulsive.



Magnetic field due to current I_a at all points of conductor b

$$B_{ab} = \frac{\mu_0 I_a}{2\pi d} \quad \text{directed downwards}$$

Force experienced by conductor b on its segment of length l

$$F_{ab} = I_b l B_{ab}$$

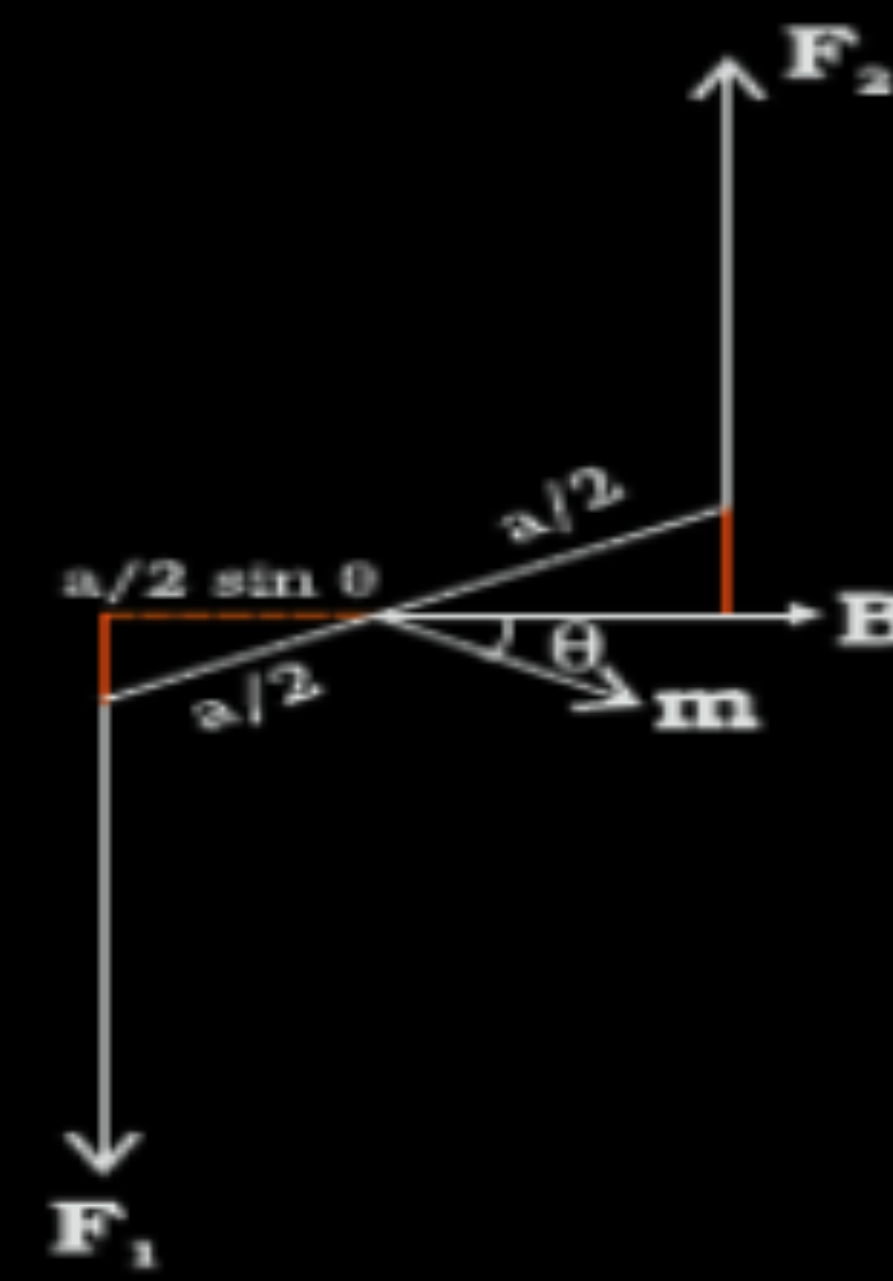
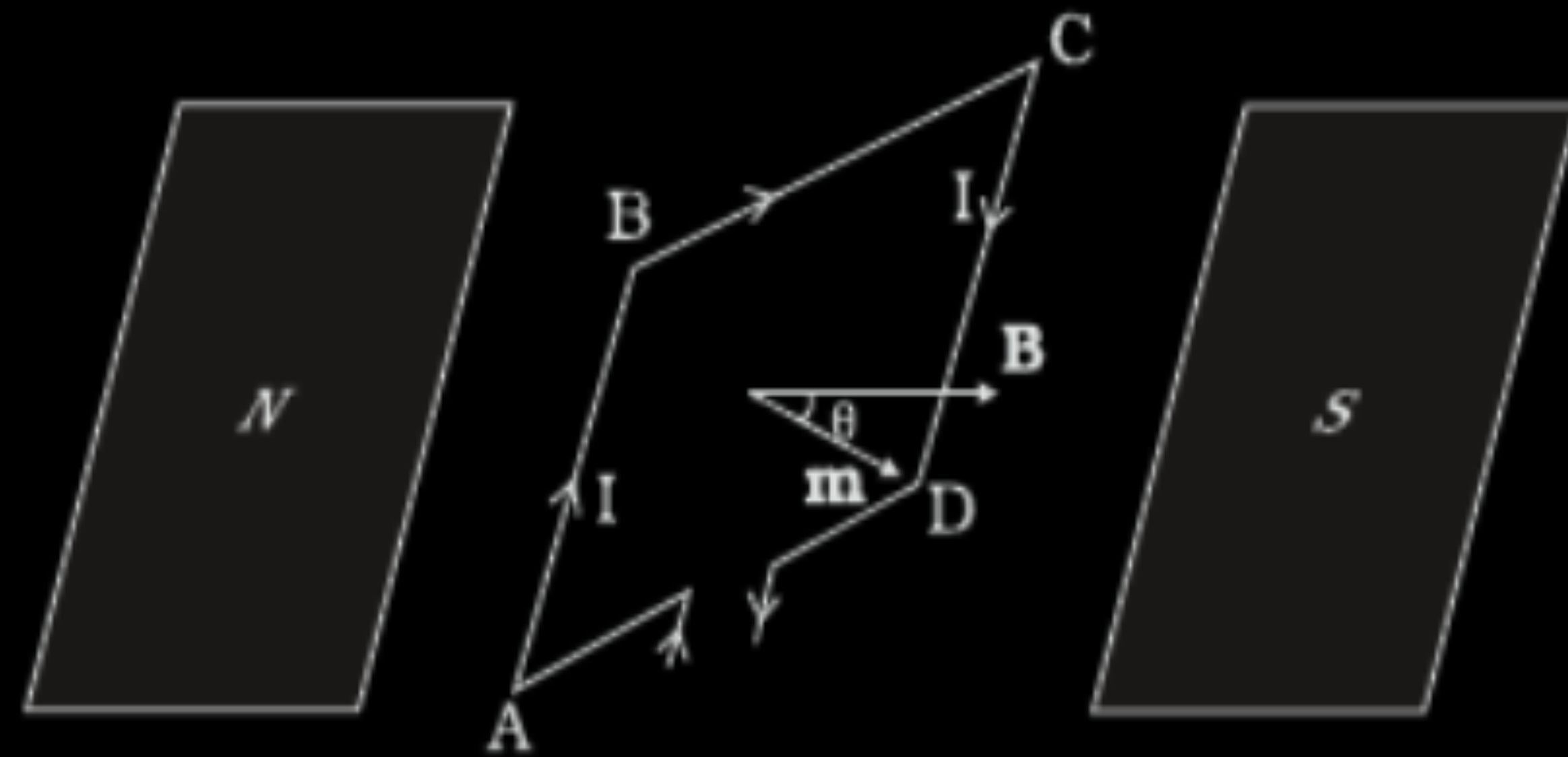
$$= \frac{\mu_0 I_a I_b}{2\pi d} l \quad \text{directed towards left}$$

Similarly

Force experienced by conductor a on its segment of length l

$$F_{ba} = \frac{\mu_0 I_a I_b}{2\pi d} l \quad \text{directed towards right}$$

One ampere is that steady current which when maintained in each of two very long straight parallel conductors of negligible cross-section, placed one metre apart in vacuum produces a force of 2×10^{-7} N/m on each conductor.



1

$\frac{1}{2}$

Forces on arm BC and DA are equal and opposite and act along the axis of the coil. Being collinear they cancel each other.

Forces on arms AB and CD are equal and opposite but not collinear. They form a couple.

$$F_1 = F_2 = I b B$$

$$\tau = F_1 \frac{a}{2} \sin \theta + F_2 \frac{a}{2} \sin \theta$$

$\frac{1}{2}$

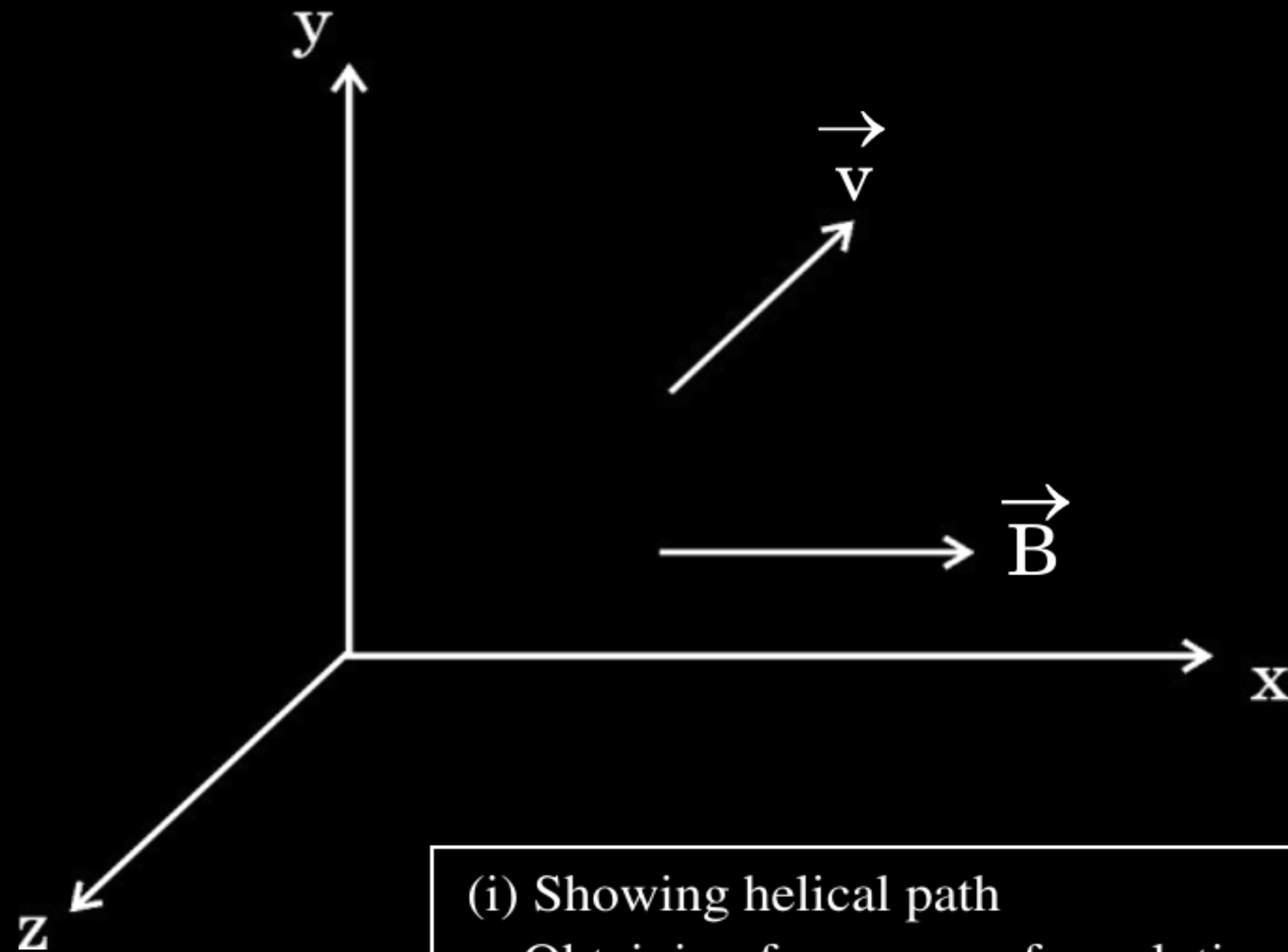
$$\tau = I a b B \sin \theta$$

$$\tau = I A B \sin \theta \quad (\text{where } A = ab \text{ \& } m = IA)$$

$$\vec{\tau} = \vec{m} \times \vec{B}$$

$\frac{1}{2}$

- 33.** (a) (i) A particle of mass m and charge q is moving with a velocity $\vec{v}$ in a magnetic field $\vec{B}$ as shown in the figure. Show that it follows a helical path. Hence, obtain its frequency of revolution.

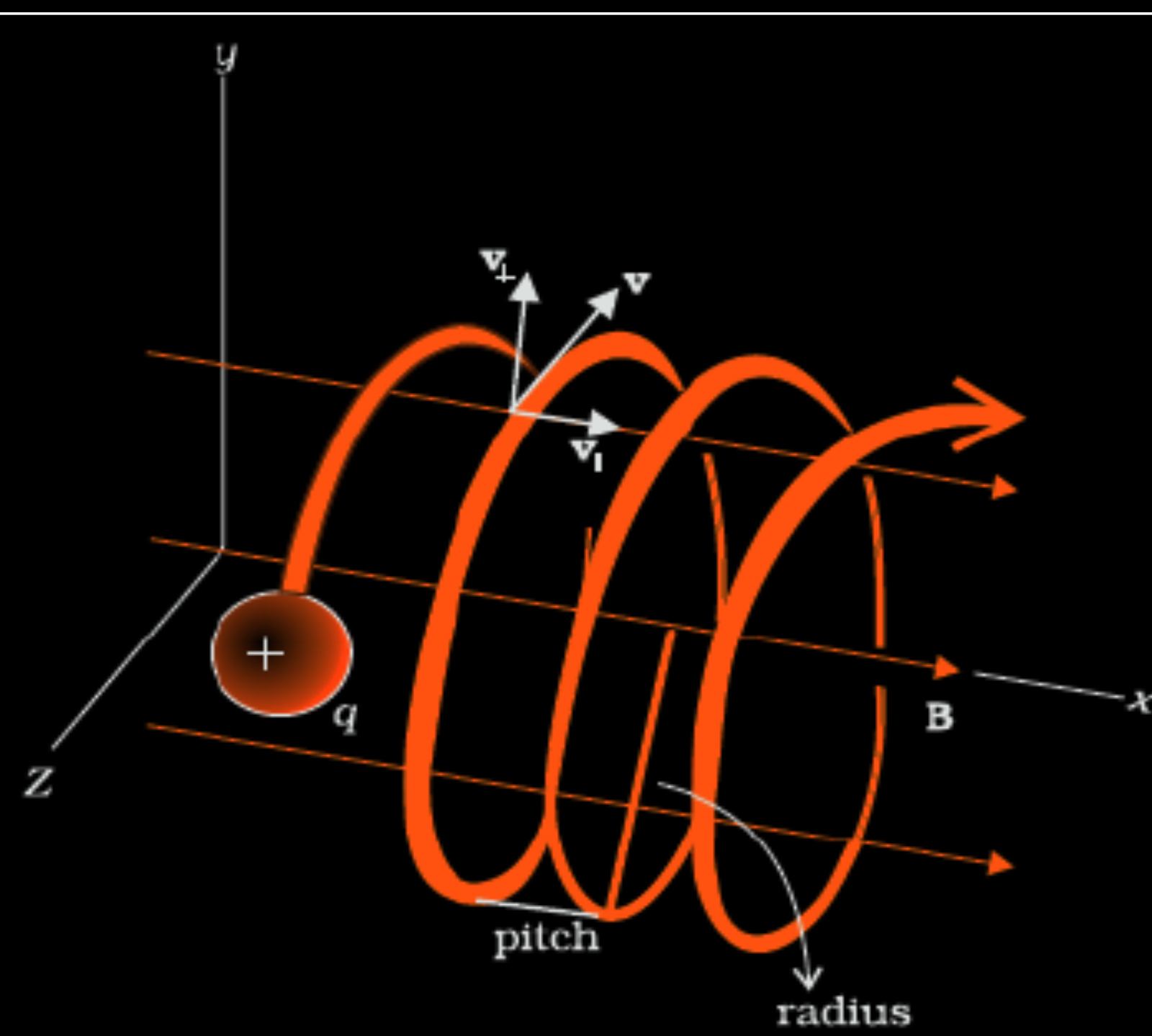


(i) Showing helical path

1 ½

Obtaining frequency of revolution

1 ½



1/2

$v_\perp = v \sin \theta$ is perpendicular to $\vec{B}$ and

$v_\parallel = v \cos \theta$ is parallel to $\vec{B}$

Due to $v_\perp$ the charge describes circular path and $v_\parallel$ pushes it in the direction of $\vec{B}$. Therefore under the combined effect of two components the charged particle describes helical path, as shown in the figure.

The centripetal force

$$\frac{mv_\perp^2}{r} = B q v_\perp$$

1/2

$$v_\perp = \frac{Bqr}{m} \quad (v_\perp = v \sin \theta)$$

1/2

$$\begin{aligned} \text{Time period} = T &= \frac{2\pi r}{v_\perp} \\ &= \frac{2\pi m}{Bq} \end{aligned}$$

$$\text{frequency } \nu = \frac{1}{T} = \frac{Bq}{2\pi m}$$

1/2

- (ii) In a hydrogen atom, the electron moves in an orbit of radius $2 \text{ \AA}$ making 8×10^{14} revolutions per second. Find the magnetic moment associated with the orbital motion of the electron.

- (ii) In a hydrogen atom, the electron moves in an orbit of radius $2\text{ \AA}$ making 8×10^{14} revolutions per second. Find the magnetic moment associated with the orbital motion of the electron.

(ii) Magnetic moment $m = I A$

$$I = \frac{e}{T} = ev$$

$$= 1.6 \times 10^{-19} \times 8 \times 10^{14}$$

$$= 1.28 \times 10^{-4} \text{ A}$$

$$M = 1.28 \times 10^{-4} \times 3.14 \times (2 \times 10^{-10})^2$$

$$= 5.12\pi \times 10^{-24} \text{ Am}^2 = 1.6 \times 10^{-23} \text{ Am}^2$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

(ii) Finding magnetic moment of electron

2

- (b) (i) What is current sensitivity of a galvanometer ? Show how the current sensitivity of a galvanometer may be increased. “Increasing the current sensitivity of a galvanometer may not necessarily increase its voltage sensitivity.” Explain.
- (ii) A moving coil galvanometer has a resistance $15\ \Omega$ and takes $20\ \text{mA}$ to produce full scale deflection. How can this galvanometer be converted into a voltmeter of range 0 to $100\ \text{V}$?

(i) Definition of current sensitivity	1
Showing dependence of current sensitivity & explanation	1+1
(ii) Calculation of resistance	2

(i) Deflection produced per unit current is called its current sensitivity.

$$I_s = \frac{\theta}{I} = \frac{NBA}{K}$$

Current sensitivity can be increased by

- (a) increasing number of turns in coil
 - (b) increasing area of coil in magnetic field
 - (c) decreasing K (Torsional Constant)
- (any one)**

$$V_s = \frac{\theta}{V} = \frac{NBA}{KR}$$

If current sensitivity is increased by increasing number of turns of the coil, the resistance of the galvanometer will also increase. Thus voltage sensitivity may not increase.

(ii) $V = I_G(R + G)$

$$R = \frac{V}{I_G} - G$$

$$= \frac{100}{20 \times 10^{-3}} - 15$$

$$= 5000 - 15$$

$$= 4985 \Omega$$

By connecting 4985Ω in series with galvanometer it is converted to voltmeter of range (0-100V)

1

1

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

14. *Assertion (A) :* An electron and a proton enter with the same momentum $\vec{p}$ in a magnetic field $\vec{B}$ such that $\vec{p} \perp \vec{B}$. Then both describe a circular path of the same radius.

Reason (R) : The radius of the circular path described by the charged particle (charge q , mass m) moving in the magnetic field $\vec{B}$ is given by $r = \frac{mv}{qB}$.

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Reason (R) : The radius of the circular path described by the charged particle (charge q , mass m) moving in the magnetic field $\vec{B}$ is given by $r = \frac{mv}{qB}$.

(A) Both Assertion (A) and Reason (R) are true and Reason(R) is the correct explanation of the Assertion (A)

3. A wire of length 4.4 m is bent round in the shape of a circular loop and carries a current of 1.0 A. The magnetic moment of the loop will be :

(A) 0.7 Am^2

(B) 1.54 Am^2

(C) 2.10 Am^2

(D) 3.5 Am^2

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(B) 1.54 Am^2

(C) 2.10 Am^2

(D) 3.5 Am^2

(B) 1.54 Am^2

4. Two charged particles, P and Q, each having charge q but of masses m_1 and m_2 , are accelerated through the same potential difference V . They enter a region of magnetic field $\vec{B} (\perp \vec{v})$ and describe the circular paths of radii a and b respectively. Then $\left(\frac{m_1}{m_2}\right)$ is equal to :

(A) $\frac{a}{b}$ (B) $\frac{b}{a}$ (C) $\left(\frac{a}{b}\right)^2$ (D) $\left(\frac{b}{a}\right)^2$

4. Two charged particles, P and Q, each having charge q but of masses m_1 and m_2 , are accelerated through the same potential difference V . They enter a region of magnetic field $\vec{B} (\perp \vec{v})$ and describe the circular paths of radii a and b respectively. Then $\left(\frac{m_1}{m_2}\right)$ is equal to :

(A) $\frac{a}{b}$ (B) $\frac{b}{a}$ (C) $\left(\frac{a}{b}\right)^2$ (D) $\left(\frac{b}{a}\right)^2$

(C) $\left(\frac{a}{b}\right)^2$

5. A galvanometer of resistance $G \Omega$ is converted into an ammeter of range 0 to I A. If the current through the galvanometer is 0.1% of I A, the resistance of the ammeter is :

(A) $\frac{G}{999} \Omega$ (B) $\frac{G}{1000} \Omega$ (C) $\frac{G}{1001} \Omega$ (D) $\frac{G}{100.1} \Omega$

5. A galvanometer of resistance $G \Omega$ is converted into an ammeter of range 0 to $I A$. If the current through the galvanometer is 0.1% of $I A$, the resistance of the ammeter is :

(A) $\frac{G}{999} \Omega$ (B) $\frac{G}{1000} \Omega$ (C) $\frac{G}{1001} \Omega$ (D) $\frac{G}{100.1} \Omega$

(B) $\frac{G}{1000} \Omega$

6. A 10 cm long wire lies along y -axis. It carries a current of 1.0 A in positive y -direction. A magnetic field $\vec{B} = (5 \text{ mT})\hat{j} - (8 \text{ mT})\hat{k}$ exists in the region. The force on the wire is :

(A) $(0.8 \text{ mN})\hat{i}$

(B) $-(0.8 \text{ mN})\hat{i}$

(C) $(80 \text{ mN})\hat{i}$

(D) $-(80 \text{ mN})\hat{i}$

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(A) $(0.8 \text{ mN})\hat{i}$

(B) $-(0.8 \text{ mN})\hat{i}$

(C) $(80 \text{ mN})\hat{i}$

(D) $-(80 \text{ mN})\hat{i}$

(B) $(-0.8 \text{ mN})\hat{i}$

13. *Assertion (A) :* Two long parallel wires, freely suspended and connected in series to a battery, move apart.

Reason (R) : Two wires carrying current in opposite directions repel each other.

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(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A)

- 24.** A circular coil with cross-sectional area 0.2 cm^2 carries a current of 4 A . It is kept in a uniform magnetic field of magnitude 0.5 T normal to the plane of the coil. Calculate :
- (a) the net force on the coil.
 - (b) the torque on the coil.
 - (c) the average force on each electron in the coil due to the magnetic field. The free electron density in the material of the coil is 10^{28} m^{-3} .

Calculating		
(a)	Net force	1
(b)	Torque	1
(c)	Average force	1

(a)

Net force = zero

By symmetry, force on each element of the coil is equal and opposite to the force on the diametrically opposite element of the coil. Hence, the net force is zero.

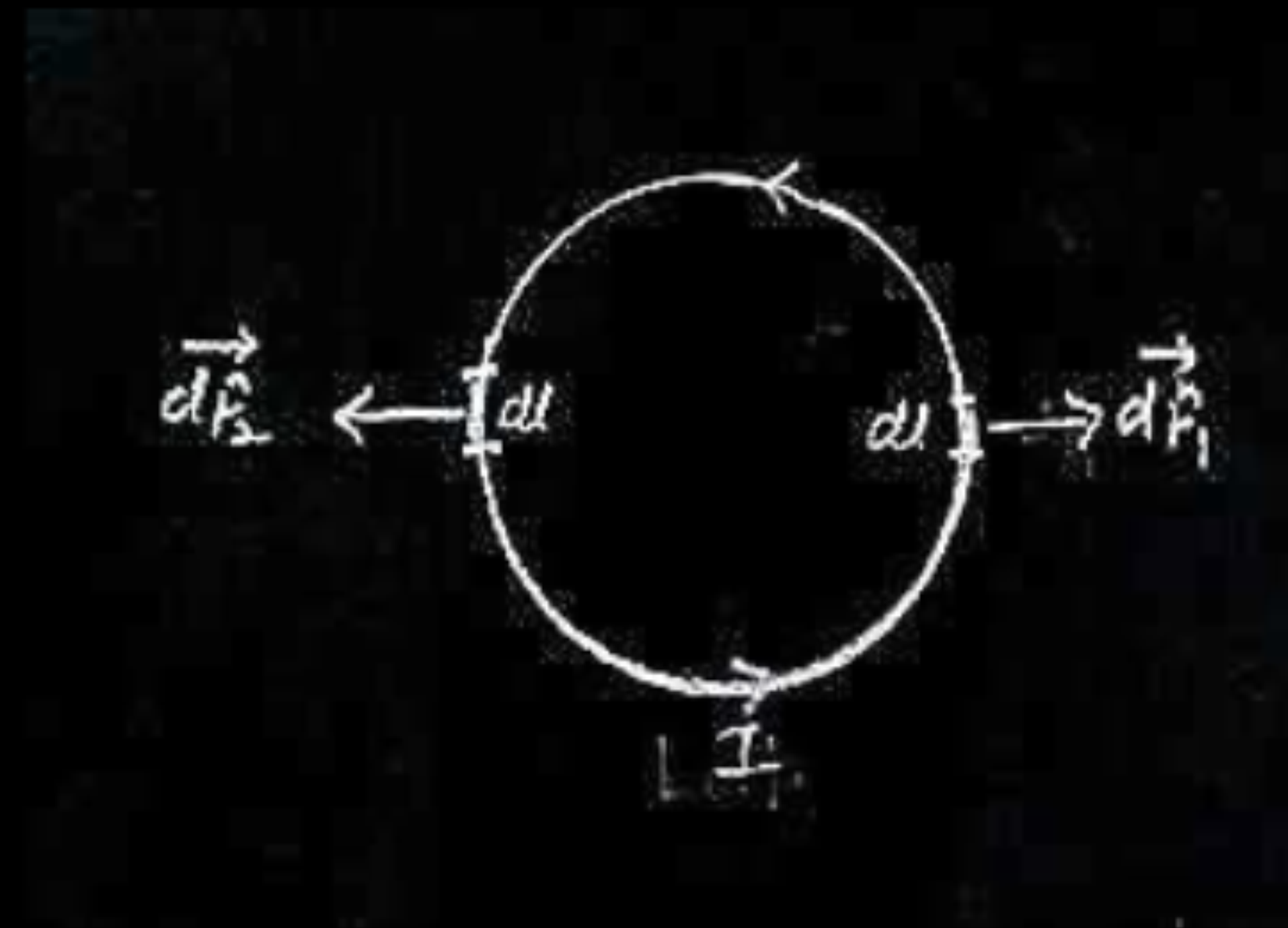
Alternatively :

$$d\vec{f}_1 = I d\vec{l} \times \vec{B}$$

$$d\vec{f}_2 = I d\vec{l} \times \vec{B}$$

$$\vec{F}_1 = -\vec{F}_2$$

On coil magnetic force are equal and opposite, so they will cancel out each other.



(b)

Torque on the coil.

$$\vec{\tau} = \vec{m} \times \vec{B}$$

$$\tau = m B \sin \theta \quad \left[\vec{m} \parallel \vec{B} \right]$$

$$\tau = m B \sin 0^\circ \quad \theta = 0^\circ$$

$$\tau = 0$$

1

1

(c)

$$f_{avg} = e(\vec{v}_d \times \vec{B})$$

$$= e v_d B \sin 90^\circ$$

$$= e v_d B$$

$$\therefore I = n e A v_d$$

$$v_d = \frac{I}{n e A}$$

$$f_{avg} = e \times \frac{I}{n e A} B$$

$$= \frac{IB}{nA}$$

$$= \frac{4 \times 0.5}{10^{28} \times 0.2 \times 10^{-4}}$$

$$= 10^{-23} \text{ N}$$

$\frac{1}{2}$

$\frac{1}{2}$

4. A piece of wire bent in the form of a circular loop A carries a current I . The wire is then bent into a circular loop B of two turns and carries the same current. The ratio of magnetic fields at the centre of loop A to that of loop B will be :

(A) $\frac{1}{16}$

(B) 16

(C) 4

(D) $\frac{1}{4}$

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(A) $\frac{1}{16}$

(B) 16

(C) 4

(D) $\frac{1}{4}$

(D) $\frac{1}{4}$

4. A loop carrying a current I clockwise is placed in $x - y$ plane, in a uniform magnetic field directed along z -axis. The tendency of the loop will be to :
- | | |
|--------------------------|--------------------------|
| (A) move along x -axis | (B) move along y -axis |
| (C) shrink | (D) expand |

4. A loop carrying a current I clockwise is placed in $x - y$ plane, in a uniform magnetic field directed along z -axis. The tendency of the loop will be to :
- (A) move along x -axis (B) move along y -axis
(C) shrink (D) expand

(C) Shrink

3. The magnetic field due to a current carrying circular coil at a point far off on its axis at a distance 'r' from the centre of the coil is proportional to :

(A) $\frac{1}{r}$

(B) $\frac{1}{r^2}$

(C) $\frac{1}{r^{3/2}}$

(D) $\frac{1}{r^3}$

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(A) $\frac{1}{r}$

(B) $\frac{1}{r^2}$

(C) $\frac{1}{r^{3/2}}$

(D) $\frac{1}{r^3}$

(D) $\frac{1}{r^3}$

4. The magnetic field B within the solenoid of length L having N turns and carrying current I will be :

(A) $\mu_0 N I L$ (B) $\frac{\mu_0 N I}{L}$

(C) $\frac{\mu_0 N I}{4\pi L}$ (D) $\mu_0 N I$

(B) $\frac{\mu_0 N I}{L}$

5. A circular loop of radius 'a' carries a current $3I$ through it. Another concentric circular loop of radius $2a$ carrying a current I is placed perpendicular to the first loop. The dipole moment of the equivalent magnetic dipole for the system is :

(A) $\pi I a^2$

(B) $5Ia$

(C) $5\pi I a^2$

(D) $7\pi I a^2$

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(C) $5\pi I a^2$

(D) $7\pi I a^2$

(C) $5\pi I a^2$

15. *Assertion (A) :* When current is passed through a spring, two adjacent turns move towards each other.

Reason (R) : There is a force between any two adjacent turns of a current carrying spring.

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Reason (R) : There is a force between any two adjacent turns of a current carrying spring.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

- 23.** Two parallel straight wires of lengths 15 m and 20 m carrying currents 6.0 A and 8.0 A respectively in opposite directions, are 10 cm apart. Calculate the force on a 10 cm long section of the second wire near its centre. Also mention the nature of the force between these wires.

3

Calculation of force on wire

2

Nature of the force

1

$$\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d}$$

$$= \frac{4\pi \times 10^{-7} \times 6 \times 8}{2\pi \times 10 \times 10^{-2}}$$

$$= 96 \times 10^{-6} \text{ N/m}$$

Force on 10 cm long section of II wire

$$F = 96 \times 10^{-6} \times 10 \times 10^{-2}$$

$$= 96 \times 10^{-7} \text{ N}$$

Nature of force is repulsive.

1

1/2

1/2

1

7. A current of 5 A is flowing in a wire of length 1.5 cm. A force of 7.5 mN acts on it when it is placed in a uniform magnetic field of 0.2 T. The angle between the magnetic field and the direction of current is :

(A) 30°

(B) 45°

(C) 60°

(D) 90°

7. A current of 5 A is flowing in a wire of length 1.5 cm. A force of 7.5 mN acts on it when it is placed in a uniform magnetic field of 0.2 T. The angle between the magnetic field and the direction of current is :

(A) 30°

(B) 45°

(C) 60°

(D) 90°

(A) 30°

16. *Assertion (A) :* The torque acting on a current carrying coil is maximum when it is suspended in a radial magnetic field.

Reason (R) : The torque tends to rotate the coil on its own axis.

16. *Assertion (A) :* The torque acting on a current carrying coil is maximum when it is suspended in a radial magnetic field.

Reason (R) : The torque tends to rotate the coil on its own axis.

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion(A).

18. An electron is passing through a region and experiences no force. Under what condition is it possible when the region has (a) only the magnetic field, and (b) both the electric and the magnetic fields ? Justify your answers.

2

Condition for experiencing no force on electron in

- | | |
|---|-----------------------------|
| (a) Only magnetic field and justification | $\frac{1}{2} + \frac{1}{2}$ |
| (b) In electric and magnetic field both and justification | $\frac{1}{2} + \frac{1}{2}$ |

18. An electron is passing through a region and experiences no force. Under what condition is it possible when the region has (a) only the magnetic field, and (b) both the electric and the magnetic fields ? Justify your answers.

2

(a) Force on an electron in magnetic field is zero when it moves parallel/ antiparallel to the direction of magnetic field.

$$F = qvB \sin\theta$$

Hence $F = 0$, when $\theta = 0^\circ, \theta = 180^\circ$

(b) Force is zero when electron moves in a crossed field of electric and magnetic fields with condition.

$$\vec{F}_E + \vec{F}_B = 0$$

$$\mathbf{v} = \frac{\mathbf{E}}{\mathbf{B}}$$

1/2

1/2

1/2

1/2

Condition for experiencing no force on electron in

- | | |
|---|-----------|
| (a) Only magnetic field and justification | 1/2 + 1/2 |
| (b) In electric and magnetic field both and justification | 1/2 + 1/2 |

(a) An electron is revolving with speed v around the proton in a hydrogen atom, in a circular orbit of radius r . The magnitude of magnetic dipole moment of the electron is : I

(A) 4 evr

(B) 2 evr

(C) $\frac{1}{2}\text{ evr}$

(D) $\frac{1}{4}\text{ evr}$

(a) An electron is revolving with speed v around the proton in a hydrogen atom, in a circular orbit of radius r . The magnitude of magnetic dipole moment of the electron is : I

(A) 4 evr

(B) 2 evr

(C) $\frac{1}{2}\text{ evr}$

(D) $\frac{1}{4}\text{ evr}$

(C) $\frac{1}{2}evr$

(b) A square loop of side 5.0 cm carries a current of 2.0 A. The magnitude of magnetic dipole moment associated with the loop is : l

(A) $1.0 \times 10^{-3} \text{ Am}^2$

(B) $5.0 \times 10^{-3} \text{ Am}^2$

(C) $1.0 \times 10^{-2} \text{ Am}^2$

(D) $5.0 \times 10^{-2} \text{ Am}^2$

(b) A square loop of side 5.0 cm carries a current of 2.0 A. The magnitude of magnetic dipole moment associated with the loop is : l

(A) $1.0 \times 10^{-3} \text{ Am}^2$

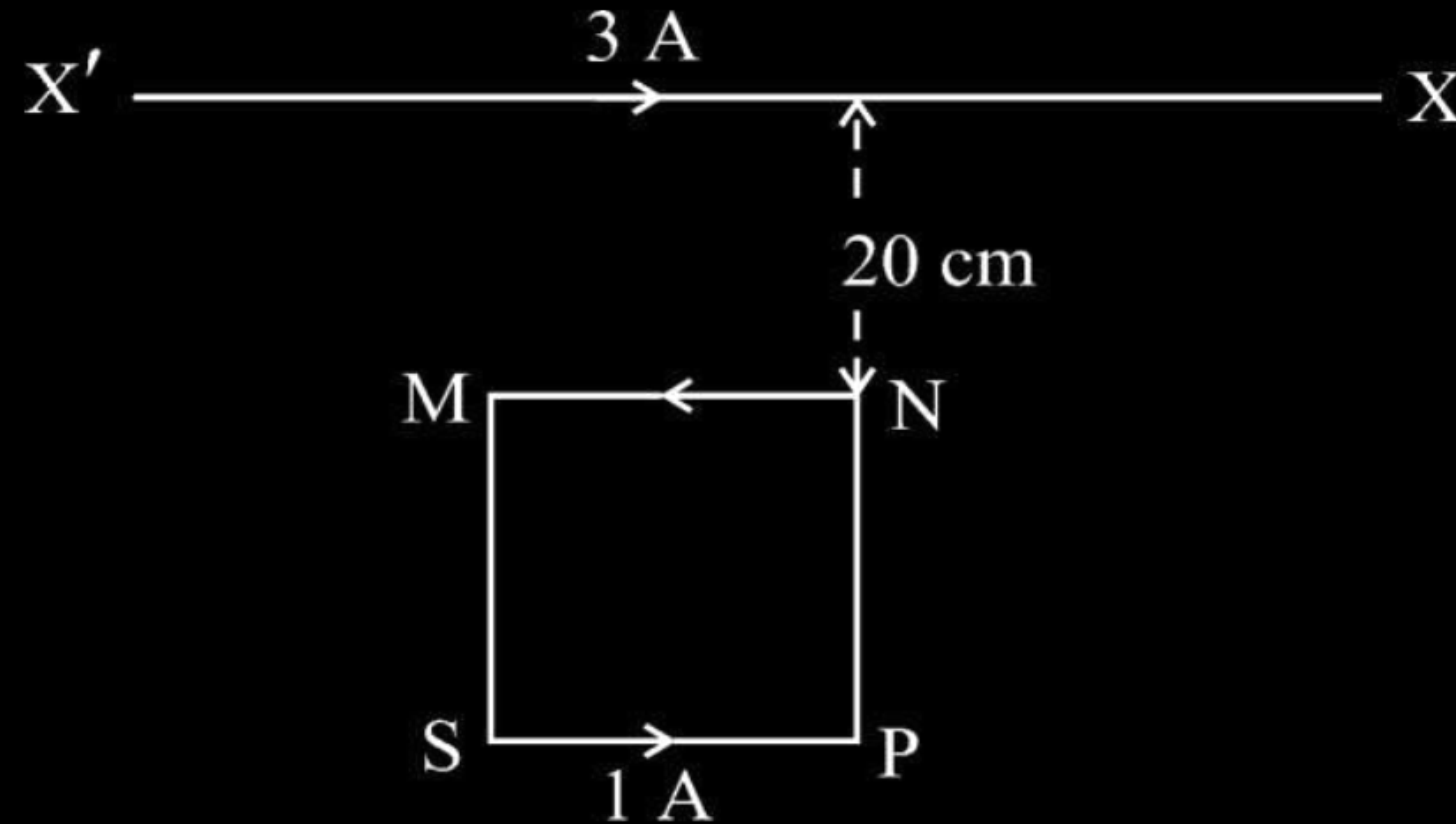
(B) $5.0 \times 10^{-3} \text{ Am}^2$

(C) $1.0 \times 10^{-2} \text{ Am}^2$

(D) $5.0 \times 10^{-2} \text{ Am}^2$

(B) $5.0 \times 10^{-3} \text{ Am}^2$

- (ii) A long, straight horizontal wire $X'X$ is held stationary and carries a current of 3.0 A . A square loop $MNPS$ of side 10 cm , carrying a current of 1.0 A is kept near the wire $X'X$ as shown in the figure. Find the magnitude and direction of the net magnetic force acting on the loop due to the wire.



2. Two circular loops of areas A and $4A$ carry currents $2I$ and I respectively. The magnetic fields at their centres will be in the ratio of :

(A) $3 : 1$

(B) $4 : 1$

(C) $1 : 1$

(D) $1 : 2$

2. Two circular loops of areas A and $4A$ carry currents $2I$ and I respectively. The magnetic fields at their centres will be in the ratio of :

(A) $3 : 1$

(B) $4 : 1$

(C) $1 : 1$

(D) $1 : 2$

(B) 4:1

12. The current sensitivity of a galvanometer does *not* depend on the :

- (A) magnetic field in which the coil is suspended.
- (B) current flowing in the coil.
- (C) torsional constant of the spring.
- (D) area of the coil.

12. The current sensitivity of a galvanometer does *not* depend on the :
- (A) magnetic field in which the coil is suspended.
 - (B) current flowing in the coil.
 - (C) torsional constant of the spring.
 - (D) area of the coil.

(B) Current flowing in the coil.

22. A particle of mass m and charge q is moving in a magnetic field $\vec{B}$ with a velocity $\vec{v}$. Discuss, giving reasons, the shape of its trajectory when the angle between $\vec{v}$ and $\vec{B}$ is :

3

- (a) 0°
- (b) 90°
- (c) 120°

Shape of trajectory with reasons when the angle between $\vec{V}$ & $\vec{B}$ is

- | | |
|-----------------|-----------------------------|
| (a) 0° | $\frac{1}{2} + \frac{1}{2}$ |
| (b) 90° | $\frac{1}{2} + \frac{1}{2}$ |
| (c) 120° | $\frac{1}{2} + \frac{1}{2}$ |

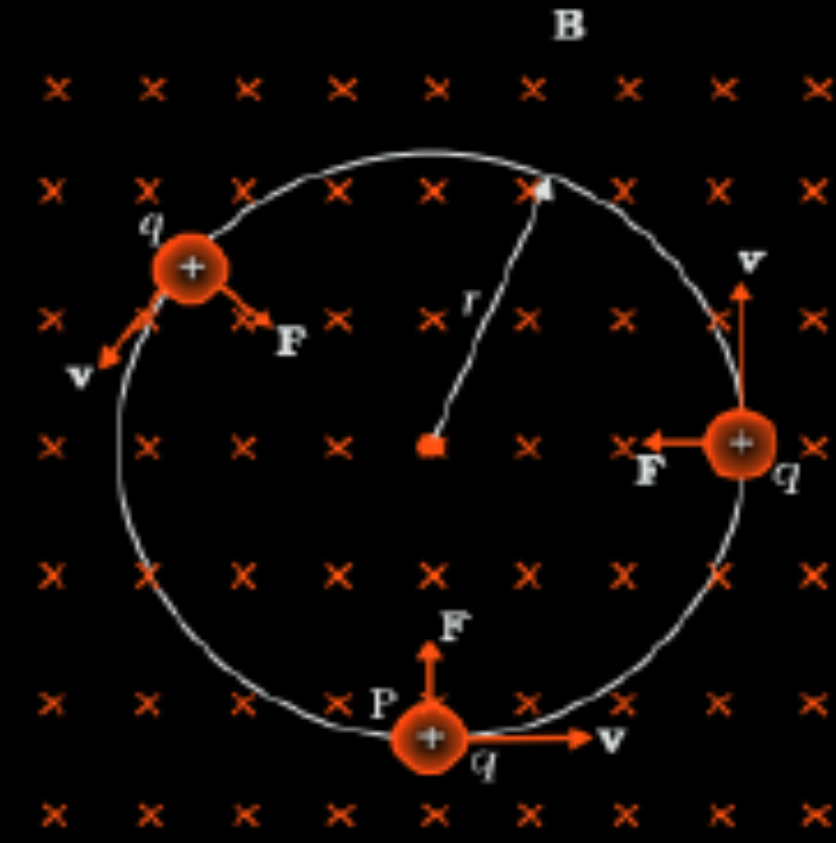
(a) We know $\vec{F} = q(\vec{v} \times \vec{B})$



Alternatively:- The particle moves in a straight line.

As the angle between $\vec{v}$ & $\vec{B}$ is 0° hence $\vec{F} = 0$

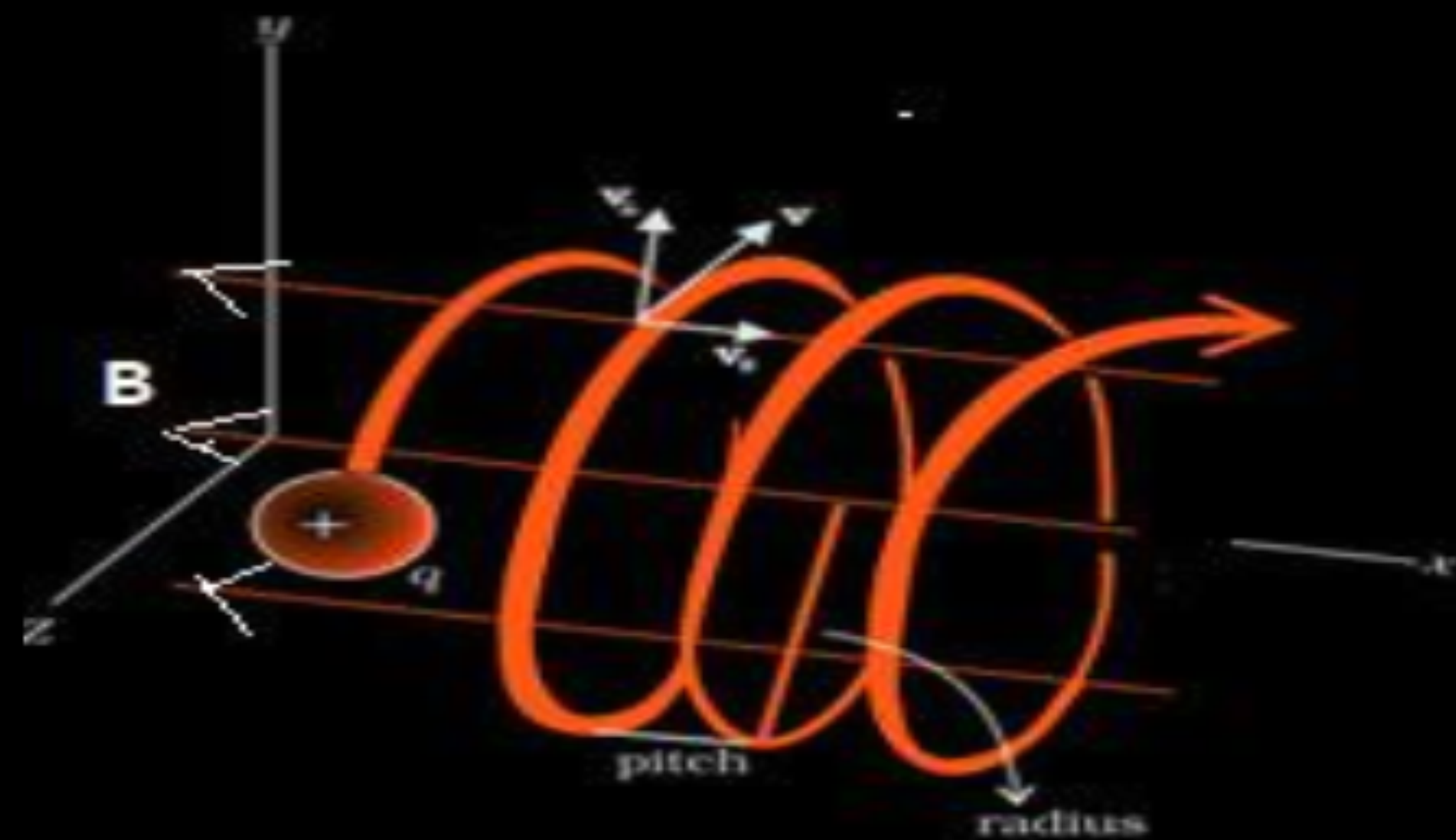
(b)



Alternatively- The particle moves in a circular path.

$\vec{F}$ is perpendicular to both $\vec{v}$ & $\vec{B}$

(c)



Alternatively The particle follows a helical path.

$$F = q v B \sin\theta$$

Component of velocity parallel to magnetic field tends to move the particle along linear path while the component perpendicular to magnetic field tends to move the particle in circular path. As a consequence the particle moves in a helical path.

27. Two circular coils of radius R each and having equal number of turns N , carry equal currents I in the same direction. They are placed coaxially at a distance $2\sqrt{3} R$. Find the magnitude and direction of the net magnetic field produced at the midpoint of the line joining their centres.

3

Finding the magnitude of net magnetic field	2
Direction of net magnetic field	1

Magnetic field due to a circular coil having N number of turns at point on the axis of the coil

$$B = \frac{N\mu_0 IR^2}{2(a^2 + R^2)^{3/2}}$$

Net field at mid point = $B_1 + B_2$

$$= \frac{N\mu_0 IR^2}{(a^2 + R^2)^{3/2}}$$

(As $a = \sqrt{3}R$)

$$\text{Net magnetic field } B = \frac{N\mu_0 IR^2}{(3R^2 + R^2)^{3/2}}$$

$$B = \frac{N\mu_0 I}{8R}$$

Direction of the net field will be towards left or right along the axis depending on direction of current in the two loops.

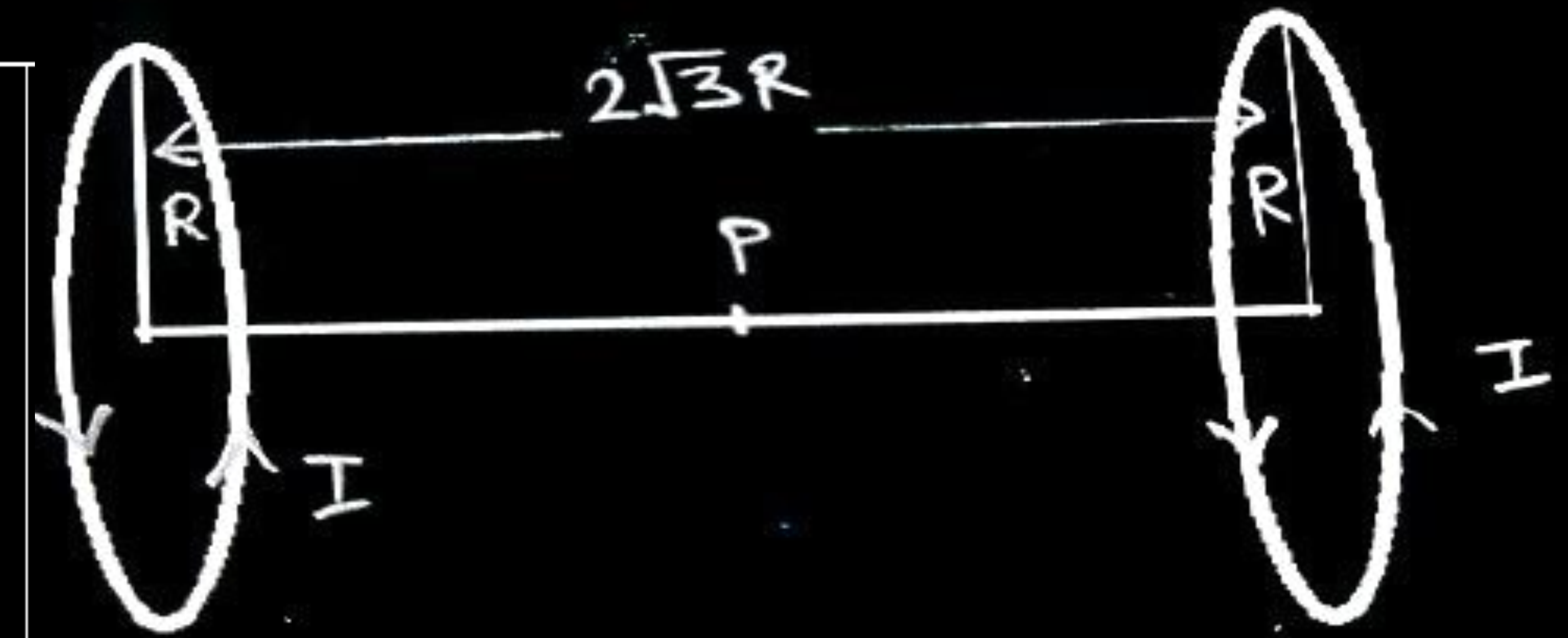
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1



7. A straight wire of length 1.0 m is placed along x-axis, in a region with magnetic field $\vec{B} = (3\hat{i} + 2\hat{j})$ T. A current of 2.0 A flows in the wire along +x direction. The magnetic force acting on the wire is :

(A) 2.0 N, along z-axis

(B) 2.0 N, along -z-axis

(C) 4.0 N, along z-axis

(D) 4.0 N, along -z-axis

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(A) 2.0 N, along z-axis

(B) 2.0 N, along -z-axis

(C) 4.0 N, along z-axis

(D) 4.0 N, along -z-axis

(C) 4.0 N, along z-axis

18. A long straight horizontal wire is carrying a current I . At an instant, an alpha particle at a distance r from it, is travelling with speed v parallel to the wire in a direction opposite to the current. Find the magnitude and direction of the force experienced by the particle at this instant.

2

- | | |
|--------------------------------------|----------------|
| • Finding the magnitude of force | $1\frac{1}{2}$ |
| • Identifying the direction of force | $\frac{1}{2}$ |

Magnetic field due to current carrying wire at distance r

$$B = \frac{\mu_o I}{2\pi r}$$

1/2

Charge on alpha particle, q = 2e

Force acting on a charged particle in motion in a uniform magnetic field.

$$\vec{F} = q(\vec{v} \times \vec{B})$$

1/2

$$\therefore F = 2ev \left(\frac{\mu_o I}{2\pi r} \right) \quad (\because \theta = 90^\circ)$$

$$F = \frac{\mu_o e v I}{\pi r}$$

1/2

Direction of the force on alpha particle is away from the wire.

1/2

5. Two parallel straight wires carrying currents I_1 and I_2 are separated by a distance x . The current in the first wire is increased from I_1 to $2I_1$ and their separation is reduced to $\frac{x}{2}$. The value of force between the wires will :
- (A) become four times
 - (B) become two times
 - (C) becomes one-fourth
 - (D) remain the same

5. Two parallel straight wires carrying currents I_1 and I_2 are separated by a distance x . The current in the first wire is increased from I_1 to $2I_1$ and their separation is reduced to $\frac{x}{2}$. The value of force between the wires will :
- (A) become four times
 - (B) become two times
 - (C) becomes one-fourth
 - (D) remain the same

(A) become four times

15. *Assertion (A) :* A magnetic field does not interact with a moving charge.

Reason (R) : A moving charge does not produce a magnetic field.

15. *Assertion (A) :* A magnetic field does not interact with a moving charge.

Reason (R) : A moving charge does not produce a magnetic field.

(D) Assertion (A) is false and Reason (R) is also false.

32. (a) (i) Discuss the motion of a charged particle of mass m and charge q in a uniform magnetic field $\vec{B}$ with initial velocity $\vec{v}$
- (1) at an angle θ with field direction,
 - (2) perpendicular to the field direction ($\theta = 90^\circ$), and
 - (3) parallel to the field direction ($\theta = 0^\circ$).
- (ii) An electron is moving along a circular path of radius 18.2×10^{-2} m in a uniform magnetic field 2×10^{-3} T. Calculate :
- (1) the speed of the electron, and
 - (2) the potential difference through which the electron must be accelerated to acquire this speed.

5

(i) Discussing the motion of charged particle when it is moving

- | | |
|---|---|
| (1) at an angle θ with field direction | 1 |
| (2) Perpendicular to the field direction | 1 |
| (3) Parallel to the field direction | 1 |

(ii) Calculating

- | | |
|---------------------------|---|
| (1) The speed of electron | 1 |
| (2) Potential difference | 1 |

(i) The force on a charge particle of mass m and charge q moving in a uniform magnetic field $\vec{B}$ velocity $\vec{v}$

$$\vec{F} = q(\vec{v} \times \vec{B})$$

(1) The particle follows a helical path. As the component of velocity parallel to magnetic field tends to move the particle along linear path while the component perpendicular to magnetic field tends to move the particle in circular path. Hence, the particle follows a helical path.

(2) The particle follows a circular path. As total magnetic force $(F = qvB)$

(3) The particle moves in a straight line.
As force on particle will be zero.

(ii)

(1) Speed of the electron $v=\frac{qBr}{m}$

$$v = \frac{1.6 \times 10^{-19} \times 2 \times 10^{-3} \times 18.2 \times 10^{-2}}{9.1 \times 10^{-31}}$$

$$v = 6.4 \times 10^7 \text{ m/s}$$

(2) Accelerating Potential Difference

$$V = \frac{mv^2}{2e}$$

$$V = \frac{9.1 \times 10^{-31} \times (6.4 \times 10^7)^2}{2 \times 1.6 \times 10^{-19}}$$

$$=11.65 \text{ kV}$$

1

1

1

1/2

1/2

1/2

1/2

- (b) (i) Deduce an expression for the torque experienced by a current carrying coil placed in an external uniform magnetic field.
- (ii) The maximum torque acting on a current carrying loop of area 0.08 m^2 is $8 \times 10^{-8} \text{ N-m}$. The current in the loop is $1.6 \mu\text{A}$. Calculate :
- (1) the magnetic field in which the loop is kept, and
 - (2) the magnetic dipole moment of the loop

5

(i) Deducing expression for Torque experienced by the coil	3
(ii) Calculating	
(1) The magnetic field	1
(2) The magnetic dipole moment	1

The field exerts no force on the two arms AD and BC of the coil.

The force on arm AB= $F_1 = I b B$

The force on arm CD= $F_2=I b B$

The torque on the coil due to the pair of forces F_1 & F_2 tends to rotate the coil.

Torque acting on the coil,

$$\tau = F_1 \frac{a}{2} + F_2 \frac{a}{2}$$

$$= I b B \frac{a}{2} + I b B \frac{a}{2}$$

$$\tau = I A B$$

(ii)

(1) Torque $\tau = I A B$

$$B = \frac{\tau}{I A}$$

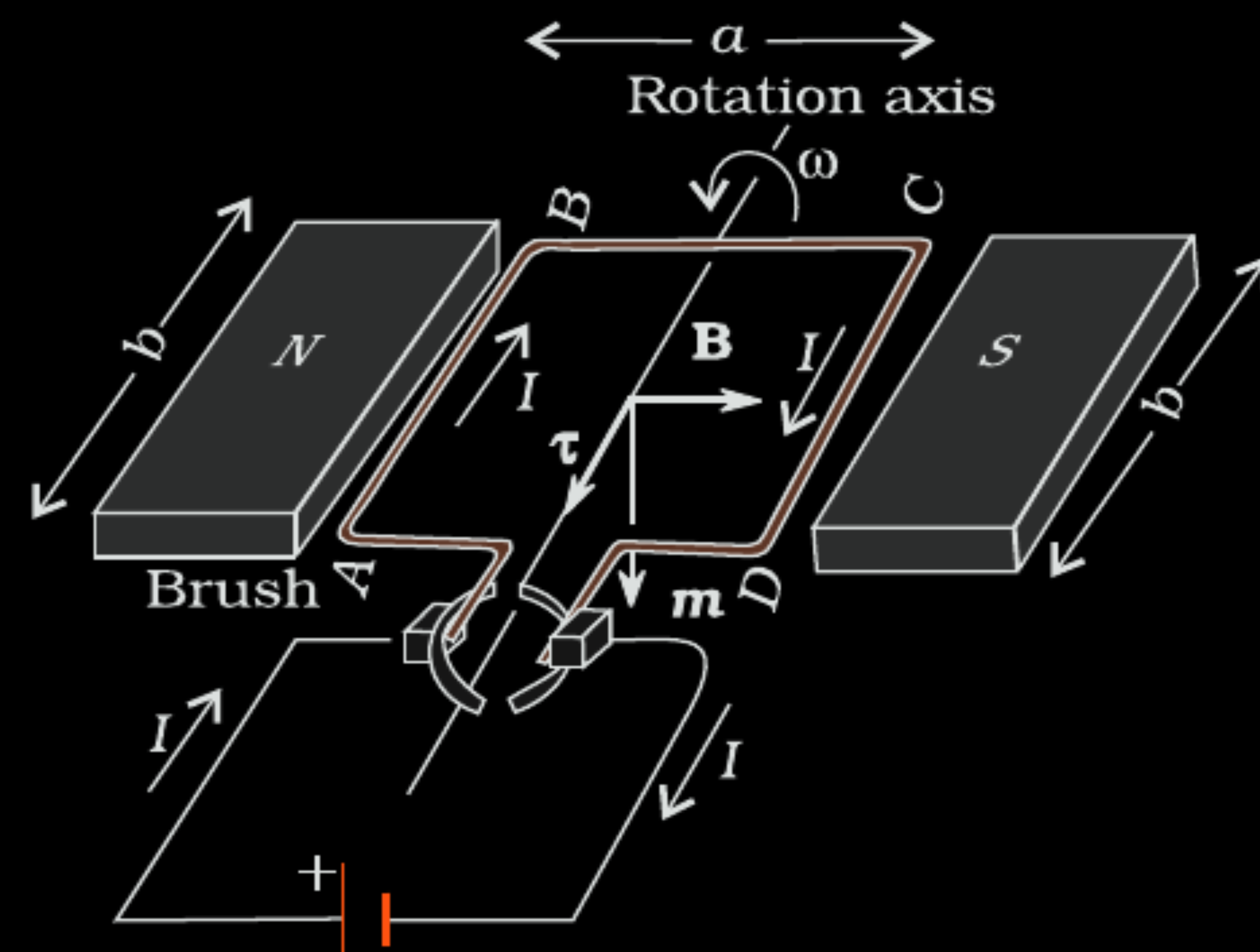
$$= \frac{8 \times 10^{-8}}{1.6 \times 10^{-6} \times 0.08}$$

$$B = 0.62 T$$

(2) Magnetic dipole moment (m)=IA

$$m = 1.6 \times 10^{-6} \times 0.08$$

$$m = 1.28 \times 10^{-7} \text{ Am}^2$$



14. *Assertion (A) :* A charged particle is moving with velocity v in x - y plane, making an angle θ ($0 < \theta < \frac{\pi}{2}$) with x -axis. If a uniform magnetic field $\vec{B}$ is applied in the region, along y -axis, the particle will move in a helical path with its axis parallel to x -axis.

Reason (R) : The direction of the magnetic force acting on a charged particle moving in a magnetic field is along the velocity of the particle.

14. *Assertion (A) :* A charged particle is moving with velocity v in x - y plane, making an angle θ ($0 < \theta < \frac{\pi}{2}$) with x -axis. If a uniform magnetic field $\vec{B}$ is applied in the region, along y -axis, the particle will move in a helical path with its axis parallel to x -axis.

Reason (R) : The direction of the magnetic force acting on a charged particle moving in a magnetic field is along the velocity of the particle.

(D) Both Assertion (A) and Reason (R) are false.

- 25.** (a) (i) Write Biot-Savart's law in vector form.
- (ii) Two identical circular coils A and B, each of radius R, carrying currents I and $\sqrt{3}I$ respectively, are placed concentrically in XY and YZ planes respectively. Find the magnitude and direction of the net magnetic field at their common centre.

3

(i) Writing Biot-Savart's Law in vector form	1
(ii) Finding magnitude & direction of net magnetic field at centre of two current carrying coils	2

$$(i) \vec{dB} = \frac{\mu_0}{4\pi} \frac{I(\vec{dl} \times \vec{r})}{r^3}$$

$$(ii) B_1 = \frac{\mu_0 I}{2R}$$

$$B_2 = \frac{\mu_0 \sqrt{3} I}{2R}$$

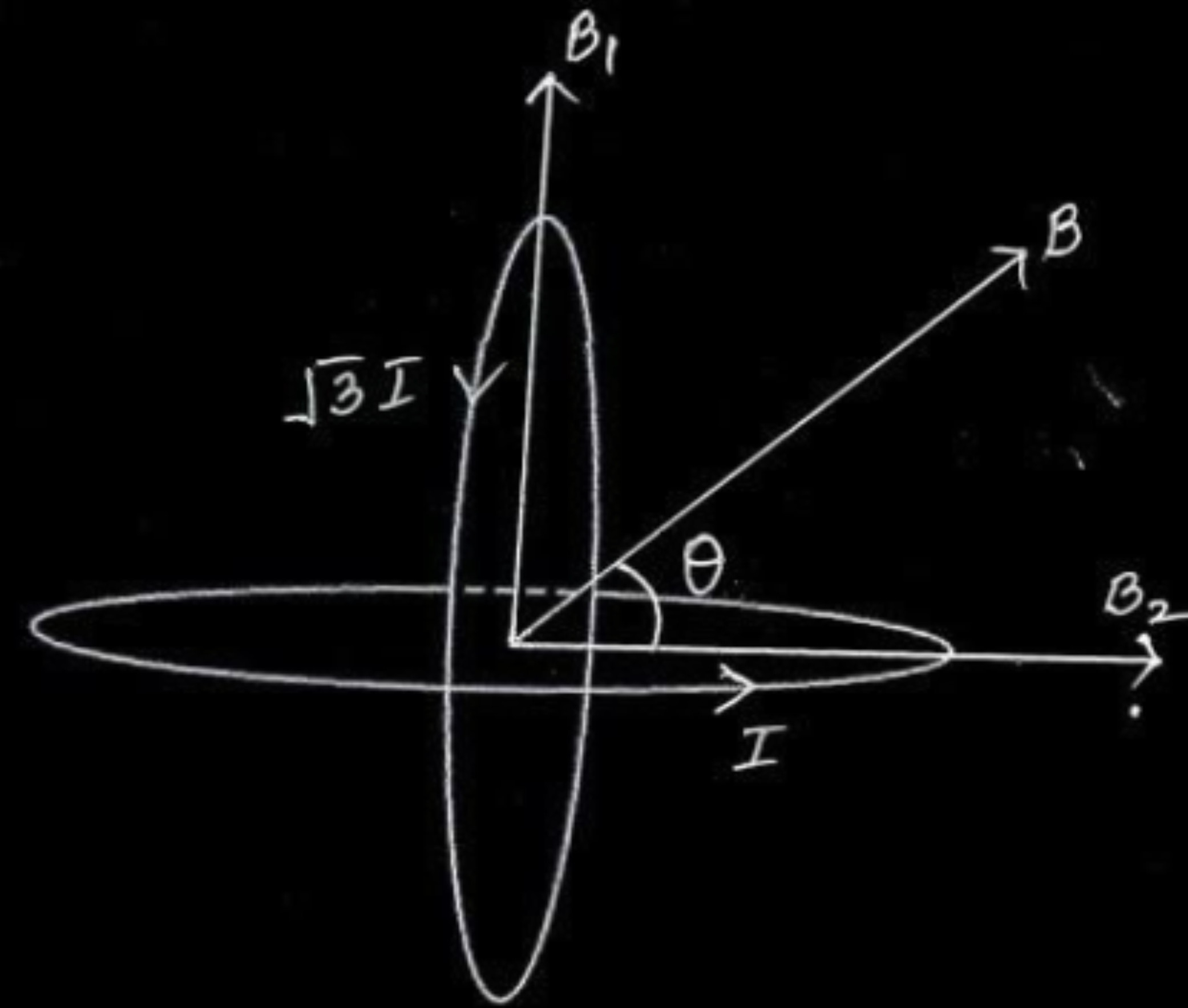
$$B = \sqrt{B_1^2 + B_2^2}$$

$$\therefore B = \frac{\mu_0 I}{2R} \sqrt{1+3}$$

$$B = \frac{\mu_0 I}{R}$$

$$\tan \theta = \frac{B_1}{B_2} = \frac{1}{\sqrt{3}}$$

Direction of net magnetic field is 30° with direction of B_2 / 60° with the direction of B_1 .



1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- (b) (i) A rectangular loop of sides l and b carries a current I clockwise. Write the magnetic moment $\vec{m}$ of the loop and show its direction in a diagram.
- (ii) The loop is placed in a uniform magnetic field $\vec{B}$ and is free to rotate about an axis which is perpendicular to $\vec{B}$. Prove that the loop experiences no net force, but a torque $\vec{\tau} = \vec{m} \times \vec{B}$.

Writing the expression for magnetic moment &
showing its direction

1

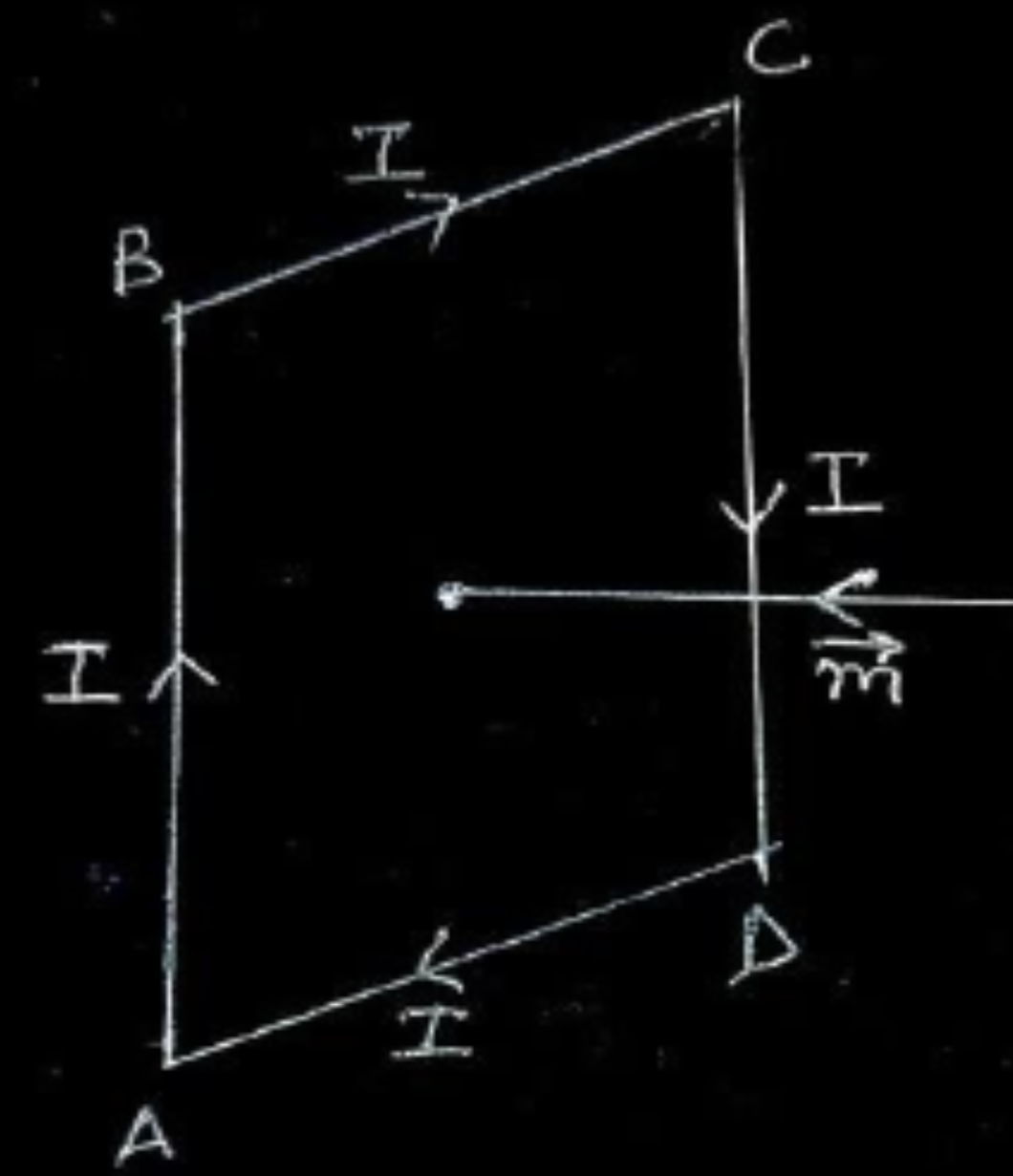
Proving no net force

1

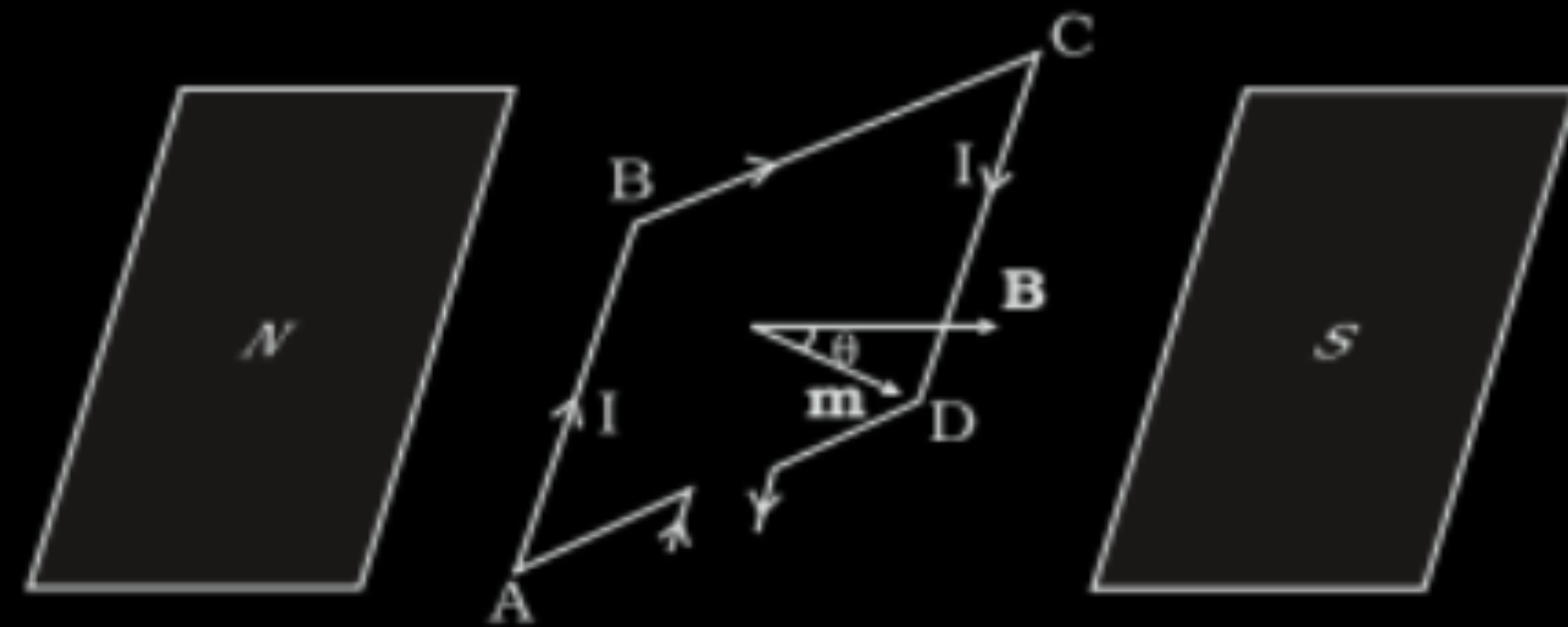
Torque $(\vec{\tau}) = \vec{m} \times \vec{B}$

1

(i) $\vec{m} = I\vec{A}$

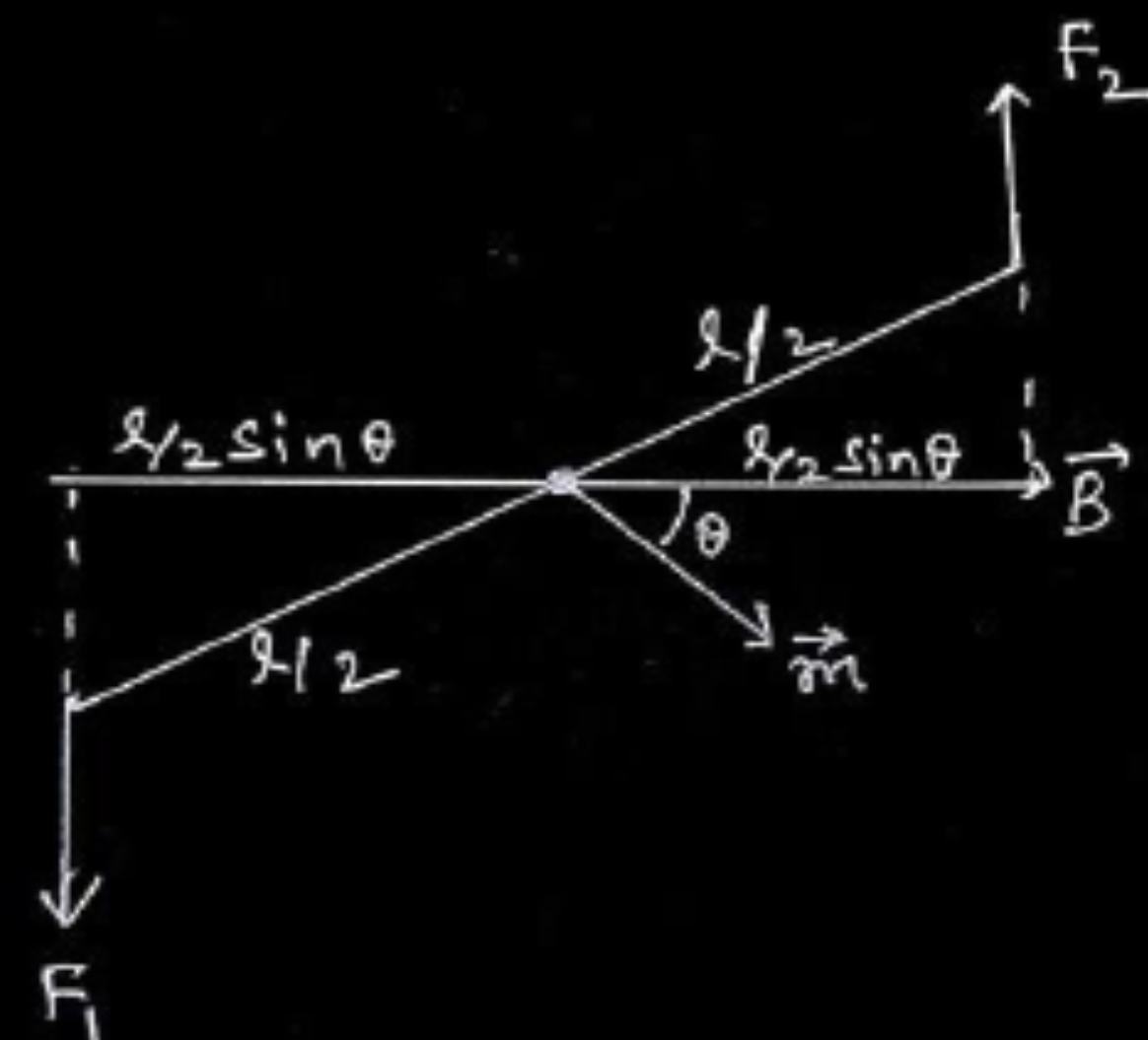


(ii) $F_1 = F_2 = I b B$ $F_1 = \text{Force on AB into the plane}$
 $F_2 = \text{Force on CD out of the plane}$



Since forces are equal & opposite so net force = 0
 Both forces form a couple, magnitude of torque acting on the coil is

$$\begin{aligned}\therefore \tau &= F_1 \frac{l}{2} \sin \theta + F_2 \frac{l}{2} \sin \theta \\ &= I b B l \sin \theta \\ &= I A B \sin \theta \\ &= m B \sin \theta\end{aligned}$$



$$\vec{\tau} = \vec{m} \times \vec{B}$$

Alternatively:

$$\frac{1}{2}$$

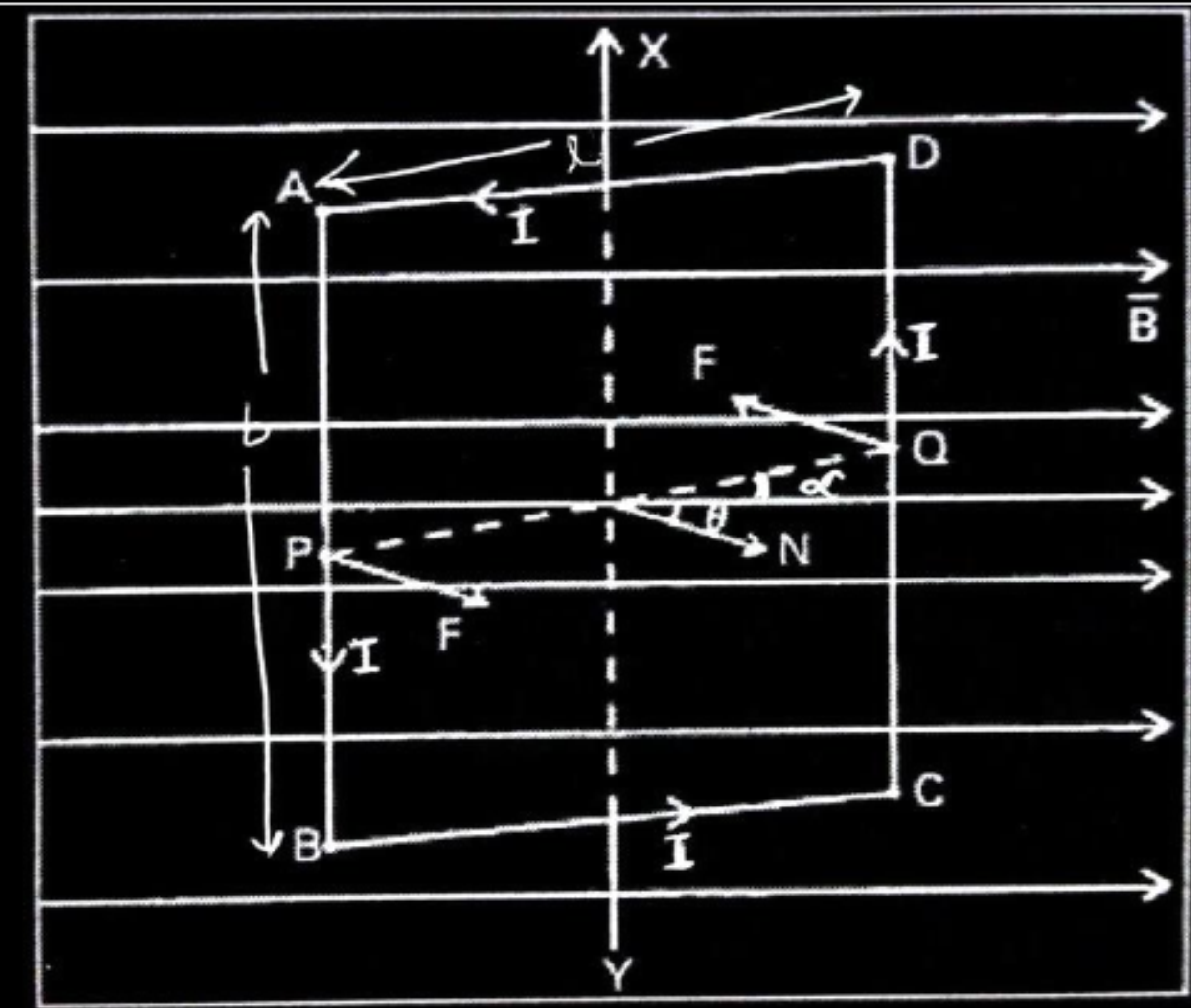
$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$



If the plane of the current carrying coil makes an angle α with the magnetic field

$$\vec{F}_{DA} = -\vec{F}_{BC} \text{ (cancel each other)}$$

Force on the arm DC is into the plane of the paper

$$|F_{DC}| = I b B$$

Force on the arm AB is out of the plane of the paper.

$$|F_{AB}| = I b B$$

Since forces are equal & opposite so net force = 0

Both of them form a couple and magnitude of torque acting on the coil is

$\tau = \text{either force} \times \text{perpendicular distance between the two forces.}$

$$\begin{aligned}\tau &= I b B \times l \sin \theta \\ &= I A B \sin \theta\end{aligned}$$

$$\vec{\tau} = I\vec{A} \times \vec{B}$$

$$\vec{\tau} = \vec{m} \times \vec{B}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

- 31.** (a) (i) With the help of a labelled diagram, explain the principle of working of a moving coil galvanometer. Write the purpose of using (i) radial magnetic field, and (ii) soft iron core, in it.
- (ii) Define current sensitivity of a galvanometer. “Increasing the current sensitivity may not necessarily increase the voltage sensitivity.” Give reason.

5

(i) Labelled diagram	1
Working principle of moving coil galvanometer	1
Use of (i) Radial magnetic field	$\frac{1}{2}$
(ii) Soft iron core	$\frac{1}{2}$
(ii) Defining current sensitivity	1
Reason	1

Principle:A current carrying coil placed in uniform magnetic field experiences a torque.

(i) Radial magnetic field makes the scale linear
Alternatively: Radial magnetic field provides maximum Torque.

(ii) Use of soft iron core is to increase the strength of magnetic field/
increase sensitivity of the galvanometer.

(ii) **Current sensitivity** is defined as deflection per unit current.
Alternatively:

$$I_s = \frac{\Phi}{I} = \frac{NAB}{k}$$

Voltage sensitivity $V_s = \frac{\Phi}{V} = \left(\frac{NAB}{k}\right) \frac{I}{V} = \left(\frac{NAB}{k}\right) \frac{1}{R}$

Increase in number of turns, increases the current sensitivity and resistance of the galvanometer in the same proportion of current sensitivity therefore Voltage sensitivity remains unchanged.

1

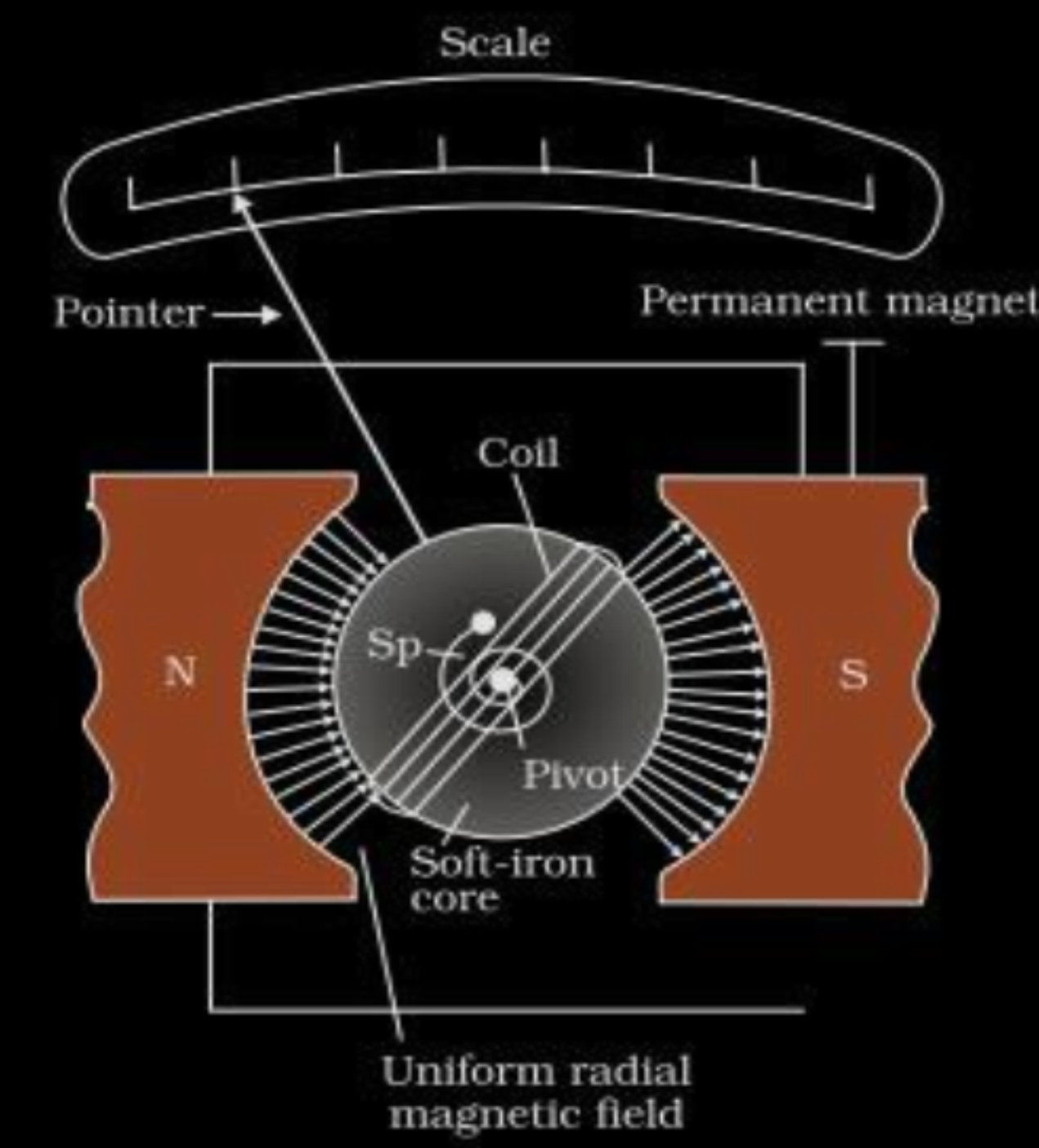
1/2

1/2

1

1/2

1/2

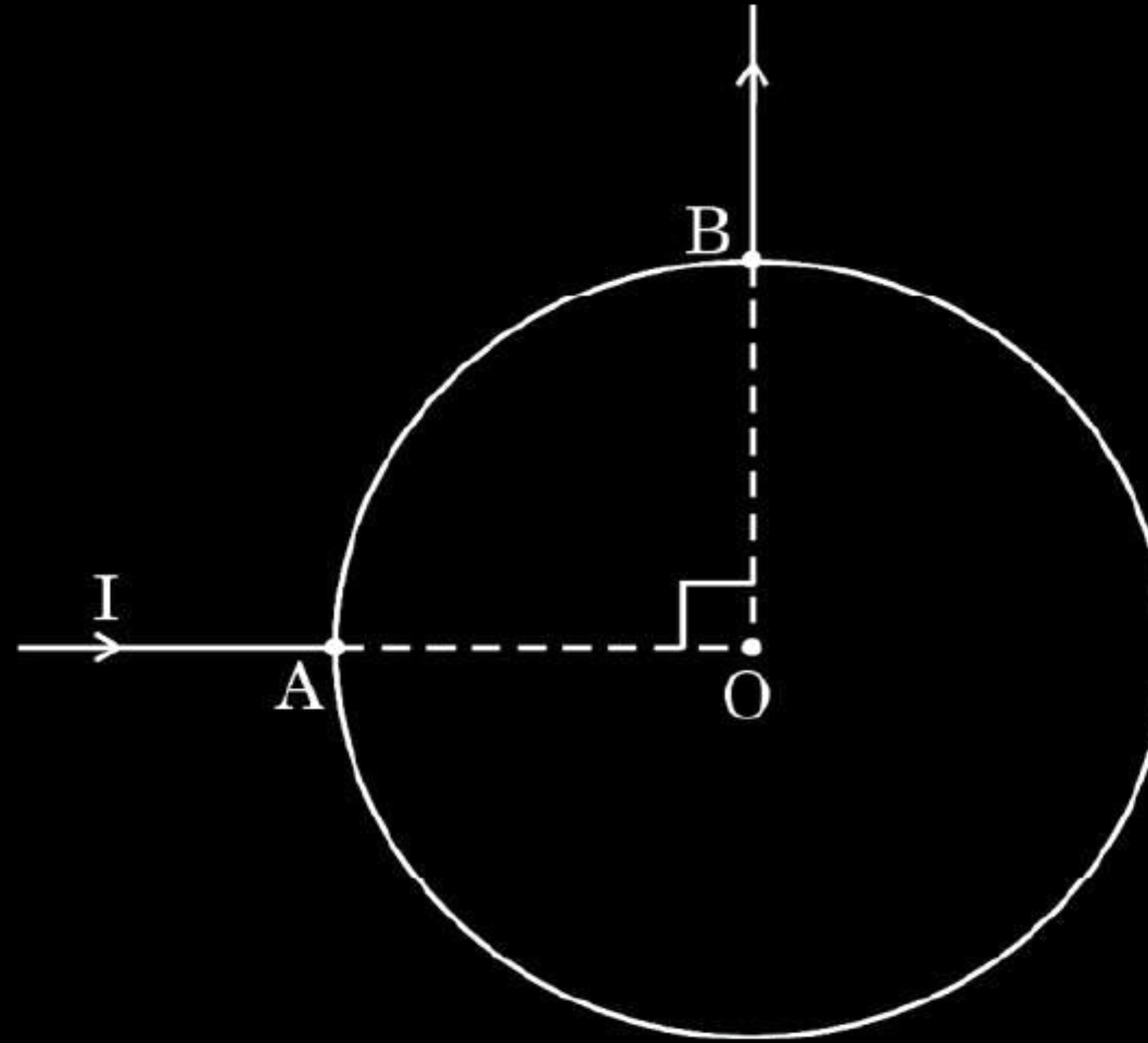


- (b) (i) (I) Write Ampere's circuital law in mathematical form and explain the terms used.
- (II) As the current carrying solenoid is made longer, the magnetic field produced outside it approaches zero. Why ?
- (III) A flexible loop of irregular shape carrying current when located in an external magnetic field, changes to a circular shape. Give reason.
- (ii) A galvanometer of resistance G is converted into a voltmeter to measure up to V volts, by connecting a resistance R_1 in series with the coil. If R_1 is replaced by R_2 , then it can only measure up to $\frac{V}{2}$ volt. Find the value of the resistance R_3 (in terms of R_1 and R_2) needed to convert it into a voltmeter that can read up to $2V$.

(i) (I) Writing Ampere circuital law & explaining the terms.	1
(II) Reason for magnetic field outside long solenoid approaching zero	1
(III) Reason for irregular shaped loop changing to circular loop in uniform magnetic field	1
(ii) Finding the value of Resistance R_3	2

(i) (I) $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_e$ I_e = Total current through the surface B = Magnetic field dl = length of small element	1/2
(II) As length of solenoid increases, it appears like a long cylindrical metal sheet so field outside approaches zero.	1/2
(III) For a given perimeter, a circle encloses greater area than any other shape, which maximizes the flux.	1
(ii) $R_1 = \frac{V}{I_g} - G \Rightarrow \frac{V}{I_g} = R_1 + G$ -----(1)	1/2
$R_2 = \frac{V}{2I_g} - G \Rightarrow \frac{V}{2I_g} = R_2 + G$ -----(2)	1/2
Solving (1) & (2) $G = R_1 - 2R_2$	1/2
$R_3 = \frac{2V}{I_g} - G$ -----(3)	
Solving using eq (1) & (3) $R_3 = 3R_1 - 2R_2$	1/2

3. In a circular loop of radius R , current I enters at point A and exits at point B, as shown in the figure. The value of the magnetic field at the centre O of the loop is :



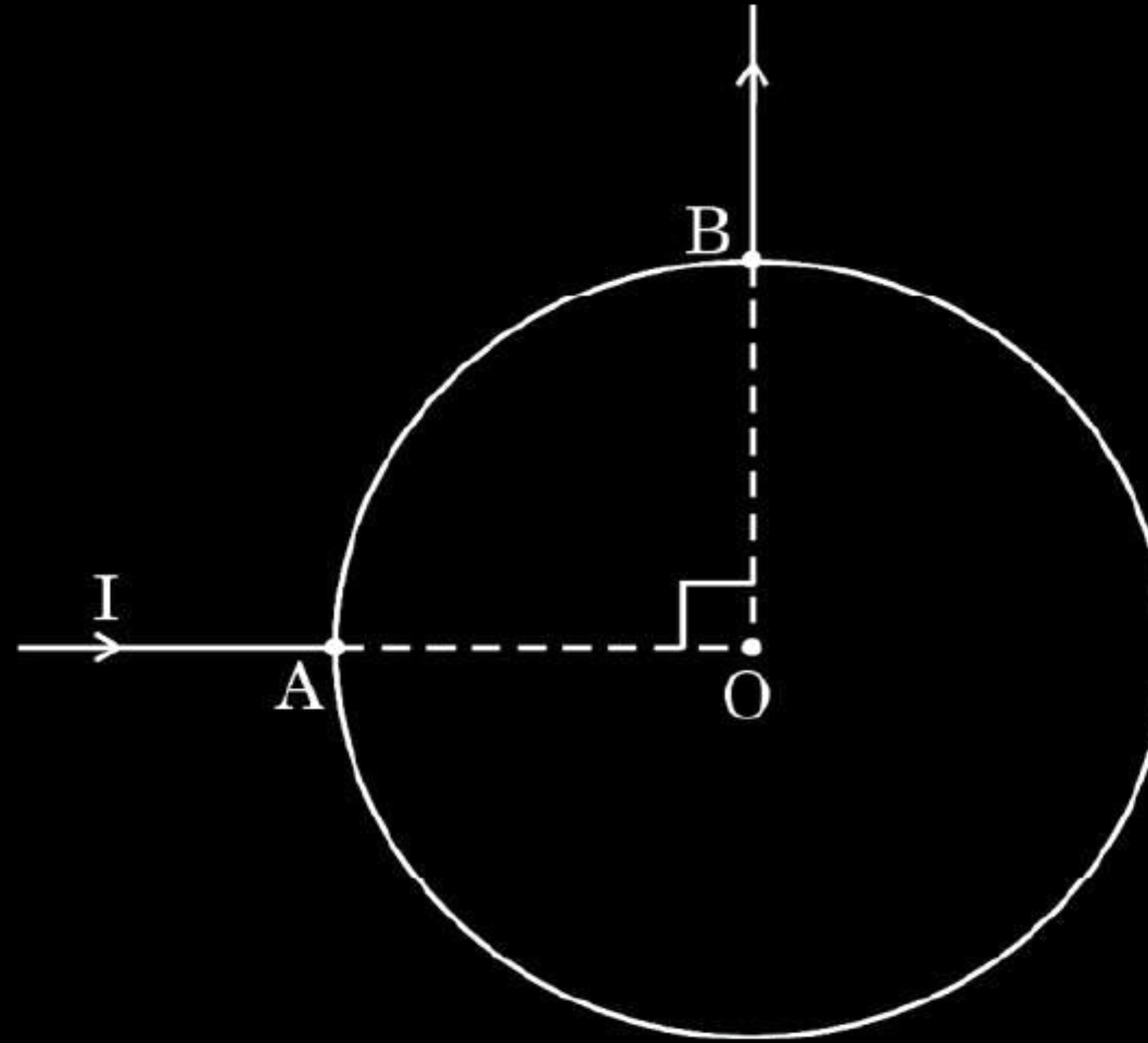
(A) $\frac{\mu_0 I}{R}$

(B) zero

(C) $\frac{\mu_0 I}{2R}$

(D) $\frac{\mu_0 I}{4R}$

3. In a circular loop of radius R , current I enters at point A and exits at point B, as shown in the figure. The value of the magnetic field at the centre O of the loop is :



(A) $\frac{\mu_0 I}{R}$

(B) zero

(C) $\frac{\mu_0 I}{2R}$

(D) $\frac{\mu_0 I}{4R}$

(B) zero

7. A proton and an α -particle enter with the same velocity $\vec{v}$ in a uniform magnetic field $\vec{B}$ such that $\vec{v} \perp \vec{B}$. The ratio of the radii of their paths is :

(A) 2

(B) $\frac{1}{2}$

(C) $\frac{1}{4}$

(D) 4

7. A proton and an α -particle enter with the same velocity $\vec{v}$ in a uniform magnetic field $\vec{B}$ such that $\vec{v} \perp \vec{B}$. The ratio of the radii of their paths is :

(A) 2

(B) $\frac{1}{2}$

(C) $\frac{1}{4}$

(D) 4

(B) $\frac{1}{2}$

11. A galvanometer can be converted into an ammeter of desired range by connecting a :

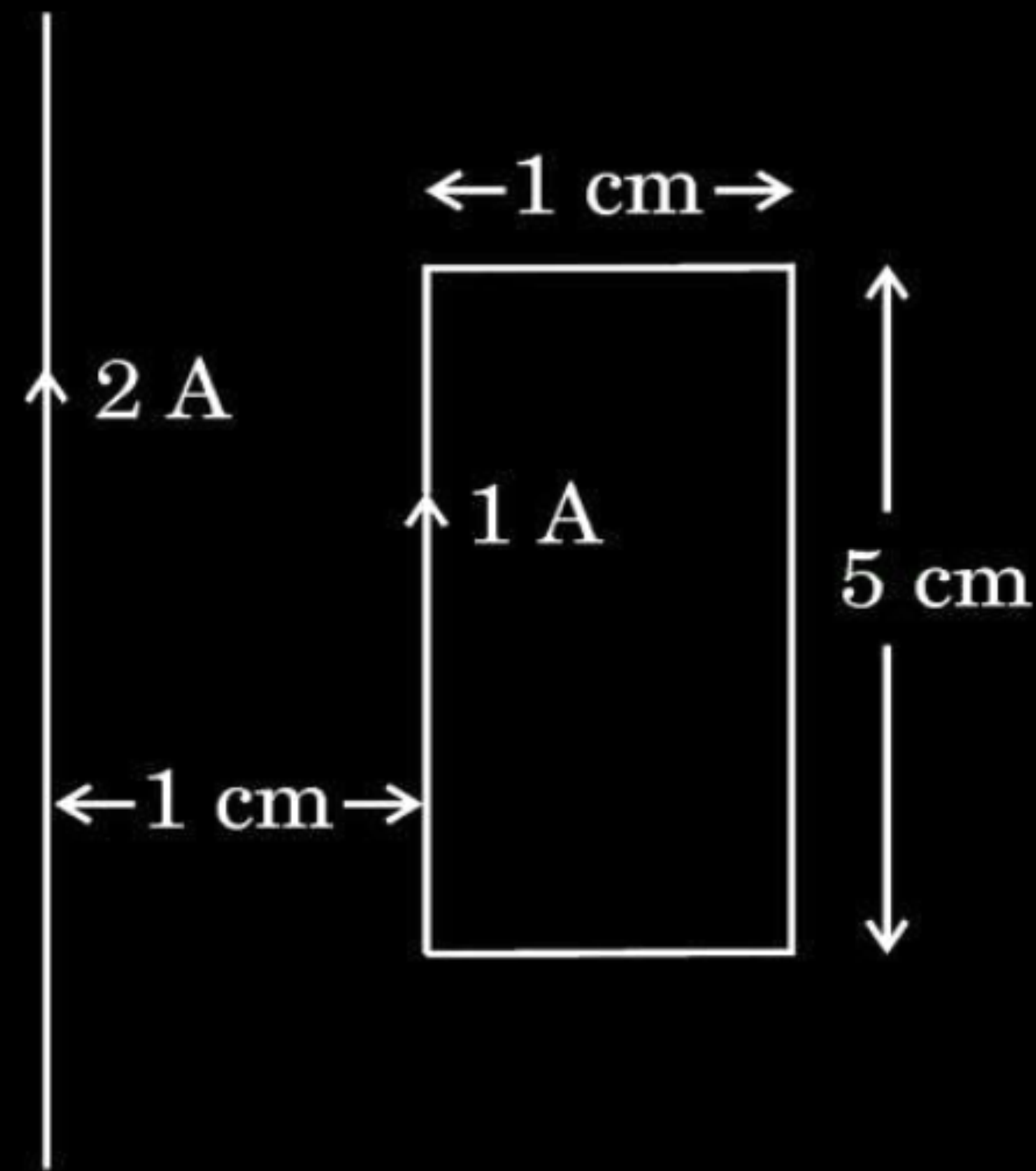
- | | |
|----------------------------------|----------------------------------|
| (A) small resistance in series | (B) large resistance in series |
| (C) small resistance in parallel | (D) large resistance in parallel |

11. A galvanometer can be converted into an ammeter of desired range by connecting a :

- | | |
|----------------------------------|----------------------------------|
| (A) small resistance in series | (B) large resistance in series |
| (C) small resistance in parallel | (D) large resistance in parallel |

(C) small resistance in parallel

24. A rectangular loop carries a current of 1 A. A straight long wire carrying 2 A current is kept near the loop in the same plane as shown in the figure.



Find :

- (i) the torque acting on the loop, and
- (ii) the magnitude and direction of the net force on the loop.

Finding-

- (i) The torque acting on the loop
- (ii) The magnitude and direction of net force

1

2

(i) $\tau = mB \sin\theta$	$\frac{1}{2}$
As $\vec{m}$ and $\vec{B}$ are in same direction, $\theta=0^\circ$	
$\tau=0$	$\frac{1}{2}$
(ii) $F = \frac{\mu_0 I_1 I_2 l}{2\pi r}$	$\frac{1}{2}$
$F_{net} = \frac{\mu_0 I_1 I_2 l}{2\pi} \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$ $= \frac{4\pi \times 10^{-7} \times 2 \times 1 \times 5 \times 10^{-2}}{2\pi \times 10^{-2}} \left(1 - \frac{1}{2} \right)$	$\frac{1}{2}$
$F_{net} = 1 \times 10^{-6} \text{ N}$	$\frac{1}{2}$
Net force on the loop is towards the long straight wire.	$\frac{1}{2}$

3. A straight conductor is carrying a current of 2 A in +x direction along it. A uniform magnetic field $\vec{B} = (0.6\hat{j} + 0.8\hat{k})$ T is switched on, in the region. The force acting on 10 cm length of the conductor is :

(A) $(0.12\hat{j} - 0.16\hat{k})$ N

(B) $(-0.16\hat{j} + 0.12\hat{k})$ N

(C) $(-0.12\hat{j} + 0.16\hat{k})$ N

(D) $(0.16\hat{j} - 0.12\hat{k})$ N

3. A straight conductor is carrying a current of 2 A in +x direction along it. A uniform magnetic field $\vec{B} = (0.6\hat{j} + 0.8\hat{k})$ T is switched on, in the region. The force acting on 10 cm length of the conductor is :

(A) $(0.12\hat{j} - 0.16\hat{k})$ N

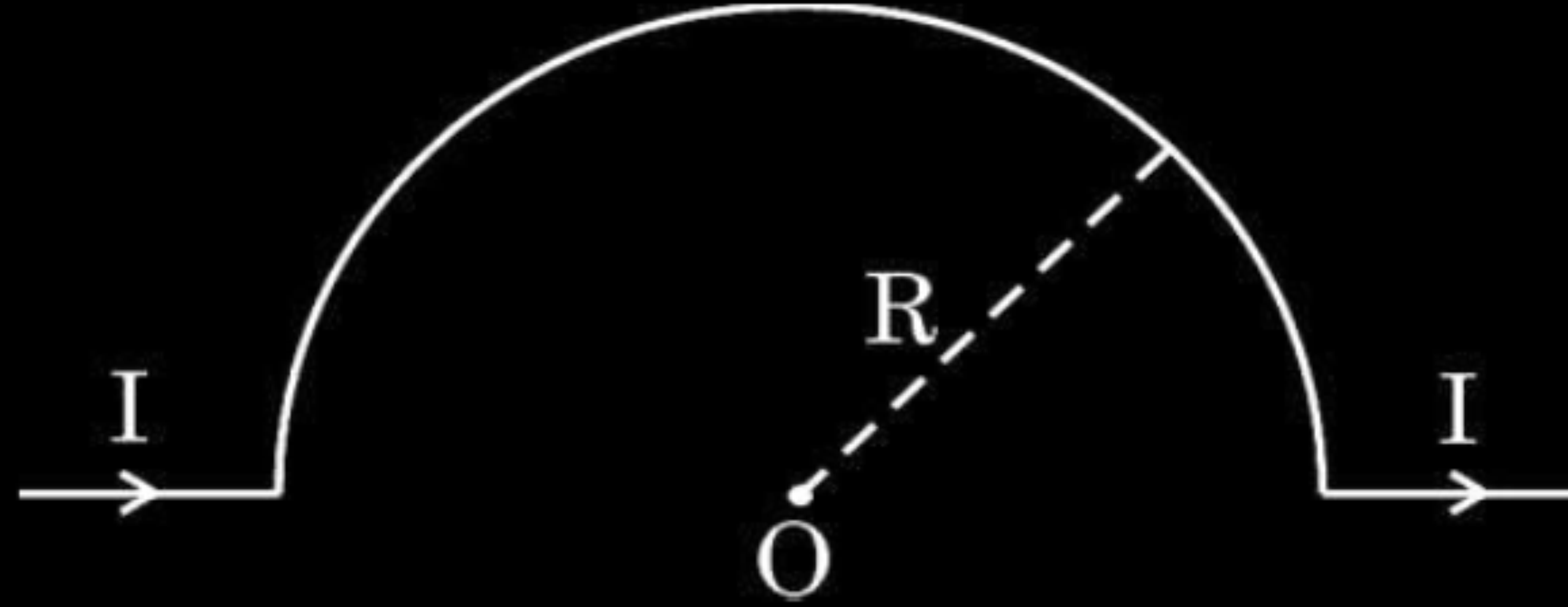
(B) $(-0.16\hat{j} + 0.12\hat{k})$ N

(C) $(-0.12\hat{j} + 0.16\hat{k})$ N

(D) $(0.16\hat{j} - 0.12\hat{k})$ N

(B) $(-0.16\hat{j} + 0.12\hat{k})$ N

3. The value of magnetic field at point O in the given figure is :



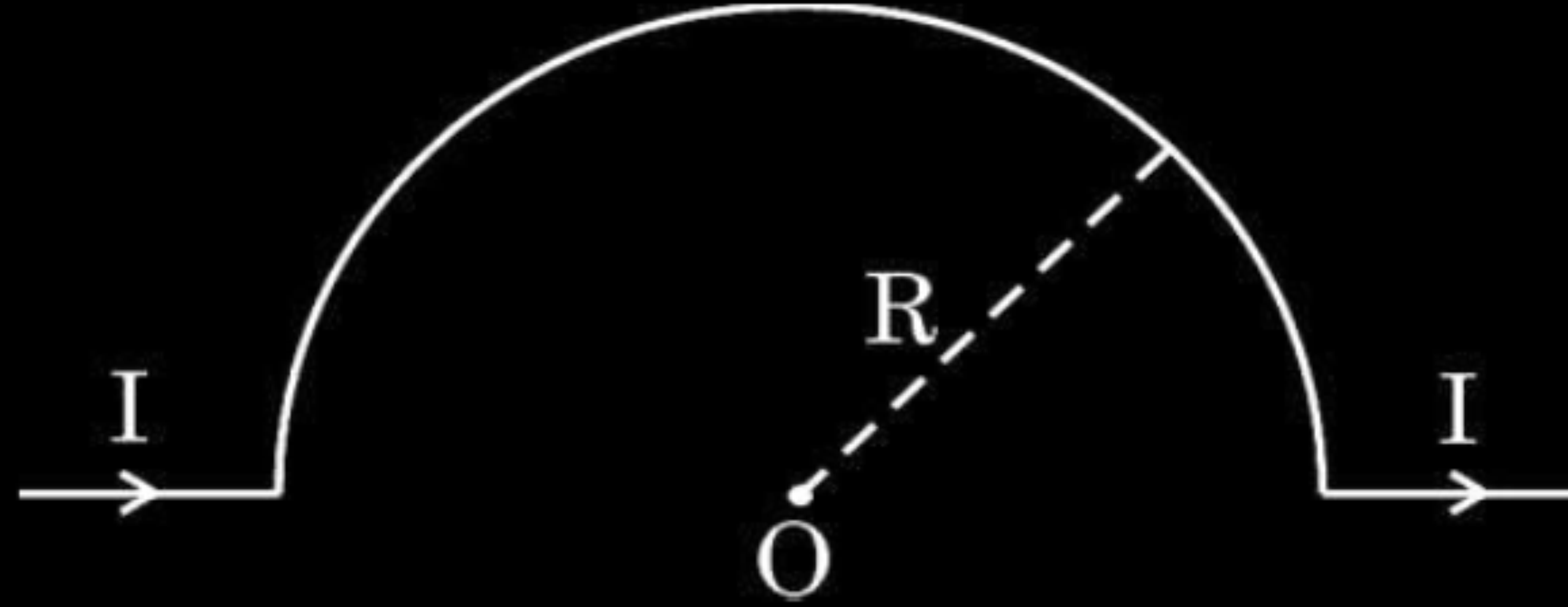
(A) $\frac{\mu_0 I}{2\pi R}$

(B) $\frac{\mu_0 I}{\pi R}$

(C) $\frac{\mu_0 I}{4R}$

(D) $\frac{\mu_0 I}{R}$

3. The value of magnetic field at point O in the given figure is :



(A) $\frac{\mu_0 I}{2\pi R}$

(B) $\frac{\mu_0 I}{\pi R}$

(C) $\frac{\mu_0 I}{4R}$

(D) $\frac{\mu_0 I}{R}$

(C) $\frac{\mu_0 I}{4R}$

3. A particle of mass m and charge q moving with a velocity $\vec{V}$ at right angle to a magnetic field $\vec{B}$, experiences a force $\vec{F}$. Another particle of mass $2m$ and charge $2q$ enters the same field $\vec{B}$ with a velocity $\frac{\vec{V}}{2}$, in the same direction. This particle will experience a force

(A) $4\vec{F}$

(B) $2\vec{F}$

(C) $\vec{F}$

(D) $\frac{\vec{F}}{2}$

3. A particle of mass m and charge q moving with a velocity $\vec{V}$ at right angle to a magnetic field $\vec{B}$, experiences a force $\vec{F}$. Another particle of mass $2m$ and charge $2q$ enters the same field $\vec{B}$ with a velocity $\frac{\vec{V}}{2}$, in the same direction. This particle will experience a force

(A) $4\vec{F}$

(B) $2\vec{F}$

(C) $\vec{F}$

(D) $\frac{\vec{F}}{2}$

1

(C) F

4. A 5.0 cm long wire carrying 1.0 A current is kept in a region in which a uniform magnetic field of 0.3 T is applied perpendicular to the length of the wire. The magnitude of the force acting on the wire is –

1

(A) zero

(B) 0.45 N

(C) 0.03 N

(D) 0.015 N

4. A 5.0 cm long wire carrying 1.0 A current is kept in a region in which a uniform magnetic field of 0.3 T is applied perpendicular to the length of the wire. The magnitude of the force acting on the wire is –

1

(A) zero

(B) 0.45 N

(C) 0.03 N

(D) 0.015 N

(D) 0.015 N

23. (a) Write one point of similarity and one point of difference between magnetic field and the electrostatic field. **3**
- (b) A wire of length l is first bent into a circular loop and then into a circular coil of two turns. For the same current passing through them, find the ratio of the magnetic fields at their centres.

- | | |
|--|---------------|
| (a) Writing one | |
| • Similarity | $\frac{1}{2}$ |
| • Difference | $\frac{1}{2}$ |
| (b) Finding ratio of magnetic field in two cases | 2 |

32. (a) (A) When a current passes through a conductor it produces a magnetic field.
- (i) State the right-hand thumb rule that gives the direction of the magnetic field.
 - (ii) How does the magnetic field vary with the distance from the conductor ?
 - (iii) Name the factors (other than the distance) which affect the magnitude of the magnetic field.
- (B) Explain how a magnetic dipole in a uniform magnetic field attains (i) stable equilibrium (ii) unstable equilibrium.

5

(A) (i) Stating the right-hand thumb rule	1
(ii) Variation of magnetic field with distance	1
(iii) Naming the factors affecting magnitude of magnetic field	1
(B) Explanation of magnetic dipole to attain	
(i) Stable equilibrium	1
(ii) Unstable equilibrium	1

(a) (i) Right hand thumb rule When we hold a conductor in right hand in such a way that a thumb stretches in the direction of flow of current than the curled fingers shows the direction of magnetic field.	1
(ii) Magnetic field due to straight conductor at a distance r is given by Since $B \propto \frac{1}{r}$ As r increases , B decreases	1
(iii) B depends on current ,nature of medium and angle θ between current element and displacement r.(any two factors)	$\frac{1}{2} + \frac{1}{2}$
(B) Explanation (i) Stable equilibrium When the magnetic dipole aligned along the direction of external magnetic field. (ii) Unstable equilibrium When the magnetic dipole align in opposite direction of external magnetic field .	1

- (b) (A) A particle of mass m and charge q is moving with velocity v in a circular path under the influence of a uniform magnetic field $\vec{B}$. Explain how will the path followed be affected when
- the strength of magnetic field is decreased.
 - the velocity of the particle is decreased.
 - the velocity of charged particle is such that one of its component is perpendicular to B and another component is parallel to $\vec{B}$.
 - an electric field is applied in such a way that electrostatic force balances the magnetic force on the charged particle.
 - the magnetic field is suddenly removed.
- In each of the above cases, other factors remain the same.

(A) Explaining, the reason for changing the path of a charge particle, when

- | | |
|---|-----------------|
| (i) the strength of magnetic field is decreased | $\frac{1}{2}$ |
| (ii) the velocity of charge particle is decreased | $\frac{1}{2}$ |
| (iii) the component of velocity of charged particle is along magnetic field and perpendicular to magnetic field | $\frac{1}{2}$ |
| (iv) electrostatic force balance, the magnetic force | $\frac{1}{2}$ |
| (v) the magnetic field is switched off | $\frac{1}{2}$ |
| (B) Deriving the expression for torque | $2 \frac{1}{2}$ |

(i) Radius of circular path in magnetic field is

$$r = \frac{mv}{qB}$$

$$r \propto \frac{1}{B}$$

If B decreases , r increases.

(ii) $r \propto v$

If v decreases , r decreases.

(iii) Under the influence of these two components the charge particle follow helical path.

(iv) The charge particle will move in the field undeviated.

(v) The charge particle will fly off tangentially along the direction of velocity.

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

(B) Derive an expression for torque acting as a rectangular loop carrying a steady current when placed in a uniform magnetic field.

(B) Derivation of torque

The magnetic field exerts no force on two arms of length 'a' parallel to magnetic field.

 $\frac{1}{2}$

The field exerts a force on two arm of length b perpendicular to magnetic field is,

$$F_1 = F_2 = IbB$$

 $\frac{1}{2}$

F_1 is directed into the plane of the loop and F_2 is directed out of the plane of paper. Thus the net force on the loop is zero .

 $\frac{1}{2}$

There is a torque on loop due to pair of force $\vec{F}_1$ and $\vec{F}_2$. The torque on the loop tends to rotate anticlockwise.

The magnitude of torque -

$$\tau = F_1 \frac{a}{2} + F_2 \frac{a}{2}$$

 $\frac{1}{2}$

$$= IbB \frac{a}{2} + IbB \frac{a}{2} = I(ab)B$$

 $\frac{1}{2}$

$$\tau = IAB$$

Where $A=ab$ is the area of rectangle.

6. A charged particle gains a speed of 10^6 ms^{-1} , when accelerated from rest through a potential difference 10 kV. It enters a region of magnetic field of 0.4 T such that $\vec{v} \perp \vec{B}$. The radius of circular path described by it is :
- (A) 2.5 cm (B) 5 cm
(C) 8 cm (D) 10 cm

6. A charged particle gains a speed of 10^6 ms^{-1} , when accelerated from rest through a potential difference 10 kV. It enters a region of magnetic field of 0.4 T such that $\vec{v} \perp \vec{B}$. The radius of circular path described by it is :
- (A) 2.5 cm (B) 5 cm
(C) 8 cm (D) 10 cm

(B) 5 cm

7. A current of $\left(\frac{10}{\pi}\right)$ A is maintained in a circular loop of radius 14 cm. The value of dipole moment associated with the loop is :

(A) 0.019 Am^2

(B) 0.14 Am^2

(C) 0.196 Am^2

(D) 0.615 Am^2

7. A current of $\left(\frac{10}{\pi}\right)$ A is maintained in a circular loop of radius 14 cm. The value of dipole moment associated with the loop is :

(A) 0.019 Am^2

(B) 0.14 Am^2

(C) 0.196 Am^2

(D) 0.615 Am^2

(C) 0.196 Am^2

18. A current of 5 A is passing along +X direction through a wire lying along X-axis. Find the magnetic field $\vec{B}$ at a point $\vec{r} = (3\hat{i} + 4\hat{j})$ m due to 1 cm element of the wire, centered at origin.

18. A current of 5 A is passing along +X direction through a wire lying along X-axis. Find the magnetic field $\vec{B}$ at a point $\vec{r} = (3\hat{i} + 4\hat{j})$ m due to 1 cm element of the wire, centered at origin.

2

Finding magnetic field

2

Using Biot – Savart's law

$$\begin{aligned}\vec{B} &= \frac{\mu_0}{4\pi} \frac{I \vec{dl} \times \vec{r}}{r^3} \\ &= \frac{(10^{-7})[5(10^{-2})\hat{i} \times (3\hat{i} + 4\hat{j})]}{5^3} \\ &= 1.6 \times 10^{-10} \hat{k} T\end{aligned}$$

$\frac{1}{2}$

1

$\frac{1}{2}$

- 29.** A galvanometer is an instrument used to show the direction and strength of the current passing through it. In a galvanometer, a coil placed in a magnetic field experiences a torque and hence gets deflected when a current passes through it. The name is derived from the surname of Italian scientist L. Galvani, who in 1791 discovered that electric current makes a dead frog's leg jerk. A spring attached with the coil provides a counter torque.

In equilibrium, the deflecting torque is balanced by the restoring torque of the spring and we have :

$$NBAI = k\phi$$

where N is the total number of turns in the coil

A is the area of cross-section of each turn

B is the radial magnetic field

k is the torsional constant of the spring

ϕ is the angular deflection of the coil

As the current (I_g) which produces full scale deflection in the galvanometer is very small, the galvanometer cannot as such be used to measure current in electric circuits. A small resistance, called shunt, of a suitable value is connected with the galvanometer to convert it into an ammeter of desired range. By using a higher resistance, a galvanometer can also be converted into a voltmeter.

- (i) (a) A galvanometer is converted into a voltmeter of range $(0 - V)$ by connecting with it, a resistance R_1 . If R_1 is replaced by R_2 , the range becomes $(0 - 2 V)$. The resistance of the galvanometer is :

(A) $(R_2 - 2R_1)$

(B) $(R_2 - R_1)$

(C) $(R_1 + R_2)$

(D) $(R_1 - 2R_2)$

- (i) (a) A galvanometer is converted into a voltmeter of range $(0 - V)$ by connecting with it, a resistance R_1 . If R_1 is replaced by R_2 , the range becomes $(0 - 2 V)$. The resistance of the galvanometer is :

1

- (A) $(R_2 - 2R_1)$ (B) $(R_2 - R_1)$
(C) $(R_1 + R_2)$ (D) $(R_1 - 2R_2)$

(A) $(R_2 - 2R_1)$

- (b) A current of 5 mA flows through a galvanometer. Its coil has 100 turns, each of area of cross-section 18 cm^2 and is suspended in a magnetic field 0.20 T. The deflecting torque acting on the coil will be :

1

- (A) $3.6 \times 10^{-3} \text{ Nm}$ (B) $1.8 \times 10^{-4} \text{ Nm}$
(C) $2.4 \times 10^{-3} \text{ Nm}$ (D) $1.2 \times 10^{-4} \text{ Nm}$

(B) $1.8 \times 10^{-4} \text{ Nm}$

- (ii) The value of resistance of the ammeter in case (ii) will be : *1*
- (A) $0.20\ \Omega$ (B) $0.24\ \Omega$
- (C) $6.0\ \Omega$ (D) $6.25\ \Omega$

Award 1 mark for this question to all students .

(iii) A galvanometer of resistance $6\ \Omega$ shows full scale deflection for a current of 0.2 A . The value of shunt to be used with this galvanometer to convert it into an ammeter of range $(0 - 5\text{ A})$ is : *1*

(A) $0.25\ \Omega$

(B) $0.30\ \Omega$

(C) $0.50\ \Omega$

(D) $6.0\ \Omega$

(iii) A galvanometer of resistance $6\ \Omega$ shows full scale deflection for a current of 0.2 A . The value of shunt to be used with this galvanometer to convert it into an ammeter of range $(0 - 5\text{ A})$ is : *1*

(A) $0.25\ \Omega$

(B) $0.30\ \Omega$

(C) $0.50\ \Omega$

(D) $6.0\ \Omega$

(A) $0.25\ \Omega$

(iv) The value of the current sensitivity of a galvanometer is given by : *1*

(A) $\frac{k}{NBA}$

(B) $\frac{NBA}{k}$

(C) $\frac{kBA}{N}$

(D) $\frac{kNB}{A}$

(iv) The value of the current sensitivity of a galvanometer is given by : 1

(A) $\frac{k}{NBA}$

(B) $\frac{NBA}{k}$

(C) $\frac{kBA}{N}$

(D) $\frac{kNB}{A}$

(B) $\frac{NBA}{K}$

- 19.** A circular coil of wire having 200 turns, each of radius 4.0 cm is placed in a horizontal plane. It carries a current of 0.40 A in clockwise direction. Find the magnitude and direction of the magnetic field at the centre of the coil.

- 19.** A circular coil of wire having 200 turns, each of radius 4.0 cm is placed in a horizontal plane. It carries a current of 0.40 A in clockwise direction. Find the magnitude and direction of the magnetic field at the centre of the coil.

2

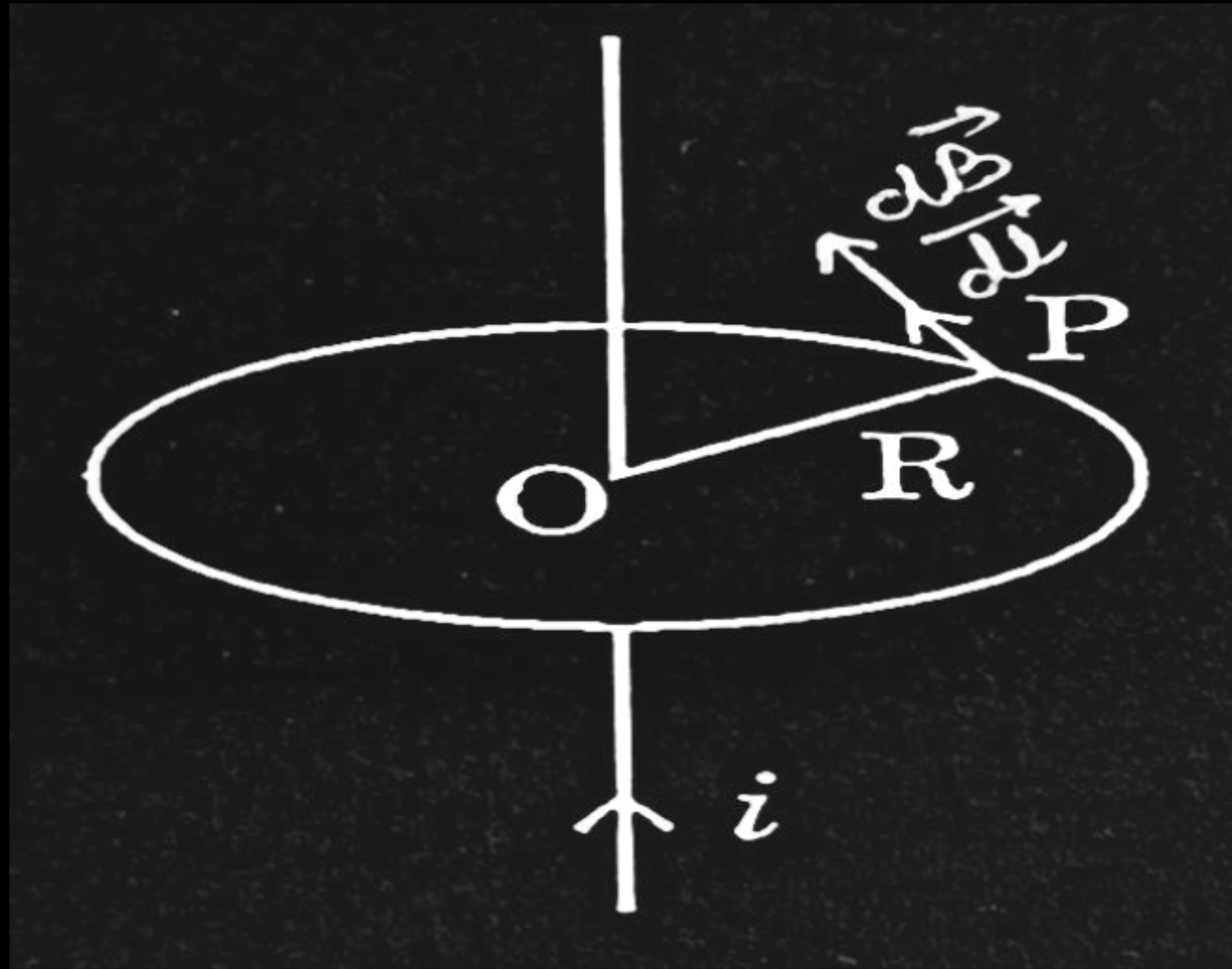
$B = \frac{\mu_0 NI}{2r}$	$\frac{1}{2}$
$= \frac{4\pi \times 10^{-7} \times 200 \times 0.40}{2 \times 4 \times 10^{-2}}$	$\frac{1}{2}$
$= 1.256 \times 10^{-3} \text{ T}$	$\frac{1}{2}$
Direction is perpendicularly inward into the horizontal plane.	$\frac{1}{2}$

- 27.** (a) Use Ampere's law to derive the expression for the magnetic field due to a long straight current carrying wire of infinite length.
- (b) Why is Ampere's law used for the derivation in (a) above and not Biot-Savart's law ? Explain.

3

(a) Deriving expression of magnetic field for long straight wire	2
(b) Explanation	1

(a)



$\frac{1}{2}$

Consider an amperian loop of radius R around the wire carrying current I .

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 i$$

$\frac{1}{2}$

$$B \times 2\pi r = \mu_0 i$$

$\frac{1}{2}$

$$B = \frac{\mu_0 i}{2\pi r}$$

$\frac{1}{2}$

(b) The expression for magnetic field due to a long straight current carrying infinite wire can be obtained easily using Ampere's circuital law whereas Biot- Savart's law requires lengthy calculation.

1

19. A voltmeter of resistance $1000\ \Omega$ can measure up to 25 V. How will you convert it so that it can read up to 250 V ?

19. A voltmeter of resistance $1000\ \Omega$ can measure up to $25\ \text{V}$. How will you convert it so that it can read up to $250\ \text{V}$?

2

Conversion of voltmeter to read upto 250V 2	
$I = \frac{V}{R}$ $= \frac{25}{1000}$ $= 25 \times 10^{-3}\ \text{A}$	$\frac{1}{2}$
Resistance to be connected to voltmeter $R' = \frac{V'}{I} - R$ $= \frac{250}{25 \times 10^{-3}} - 1000$ $= 9000\ \Omega$	$\frac{1}{2}$ $\frac{1}{2}$
This $9000\ \Omega$ is in series with voltmeter.	$\frac{1}{2}$

26. An electron of mass m and charge $-e$ is revolving anticlockwise around the nucleus of an atom.

(a) Obtain the expression for the magnetic dipole moment (μ) of the atom.

(b) If $\vec{L}$ is the angular momentum of electron, show that

$$\vec{\mu} = - \left(\frac{e}{2m} \right) \vec{L} .$$

3

a) Obtaining expression for magnetic dipole moment	1½
b) To Show $\vec{\mu} = - \left(\frac{e}{2m} \right) \vec{L}$	1½

a) $\mu = IA$ $= \frac{e}{T} \times A$	$\frac{1}{2}$
$= \frac{e}{\frac{2\pi r}{v}} \times \pi r^2$	$\frac{1}{2}$
$= \frac{1}{2} evr$	$\frac{1}{2}$
b) $L = mvr$ $\mu = \frac{evr \times m}{2 \times m}$	$\frac{1}{2}$
$= \left(\frac{e}{2m} \right) L$	$\frac{1}{2}$
Direction of $\vec{\mu}$ is opposite to that of $\vec{L}$ $\vec{\mu} = - \left(\frac{e}{2m} \right) \vec{L}$	$\frac{1}{2}$

3. A particle of mass m and charge q moves along y -axis in a region in which a uniform magnetic field $\vec{B}$ is pointing along x -axis. The Lorentz force acting on the charge will point along :

(A) x -axis

(B) y -axis

(C) z -axis

(D) negative z -axis

3. A particle of mass m and charge q moves along y -axis in a region in which a uniform magnetic field $\vec{B}$ is pointing along x -axis. The Lorentz force acting on the charge will point along :

(A) x -axis

(B) y -axis

(C) z -axis

(D) negative z -axis

(D) Negative Z axis

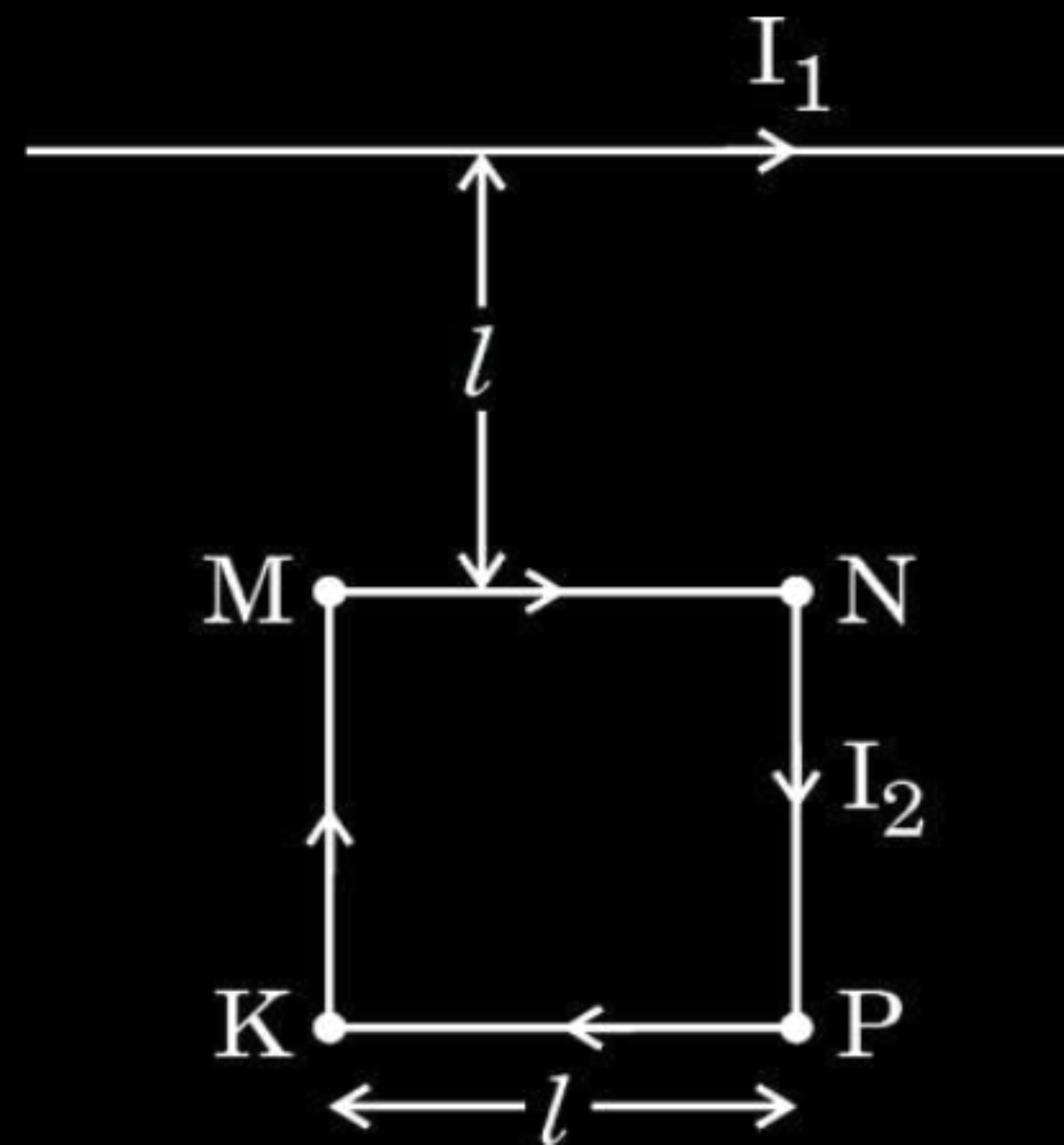
8. A long straight wire is held vertically and carries a steady current in upward direction. The shape of magnetic field lines produced by the current-carrying wire are :
- (A) horizontal straight lines directed radially out from the wire.
 - (B) straight lines parallel to the current-carrying wire.
 - (C) concentric horizontal circles around the wire.
 - (D) coaxial helixes around the wire.

8. A long straight wire is held vertically and carries a steady current in upward direction. The shape of magnetic field lines produced by the current-carrying wire are :
- (A) horizontal straight lines directed radially out from the wire.
 - (B) straight lines parallel to the current-carrying wire.
 - (C) concentric horizontal circles around the wire.
 - (D) coaxial helixes around the wire.

(C) Concentric horizontal circles around the wire

- 23.** (a) In which cases does a charged particle not experience a force in a magnetic field ?
- (b) A square loop MNPK of side ' l ' carrying a current ' I_2 ' is kept close to a long straight wire in the same plane and the wire carries a steady current I_1 as shown in the figure. Obtain the magnitude of magnetic force exerted by the wire on the loop.

3



- | | |
|--|---|
| (a) Conditions for, no force experienced by charged particle in magnetic field | 1 |
| (b) Obtaining the magnitude of magnetic force exerted by wire on the loop | 2 |

3. A particle with charge q moving with velocity $\vec{v} = v_0 \hat{i}$ enters a region with magnetic field $\vec{B} = B_1 \hat{j} + B_2 \hat{k}$. The magnitude of force experienced by the particle is :

(A) $qv_0 (B_1 + B_2)$

(B) $q\sqrt{v_0 (B_1 + B_2)}$

(C) $q v_0 \sqrt{(B_1^2 + B_2^2)}$

(D) $q \sqrt{v_0 (B_1^2 + B_2^2)}$

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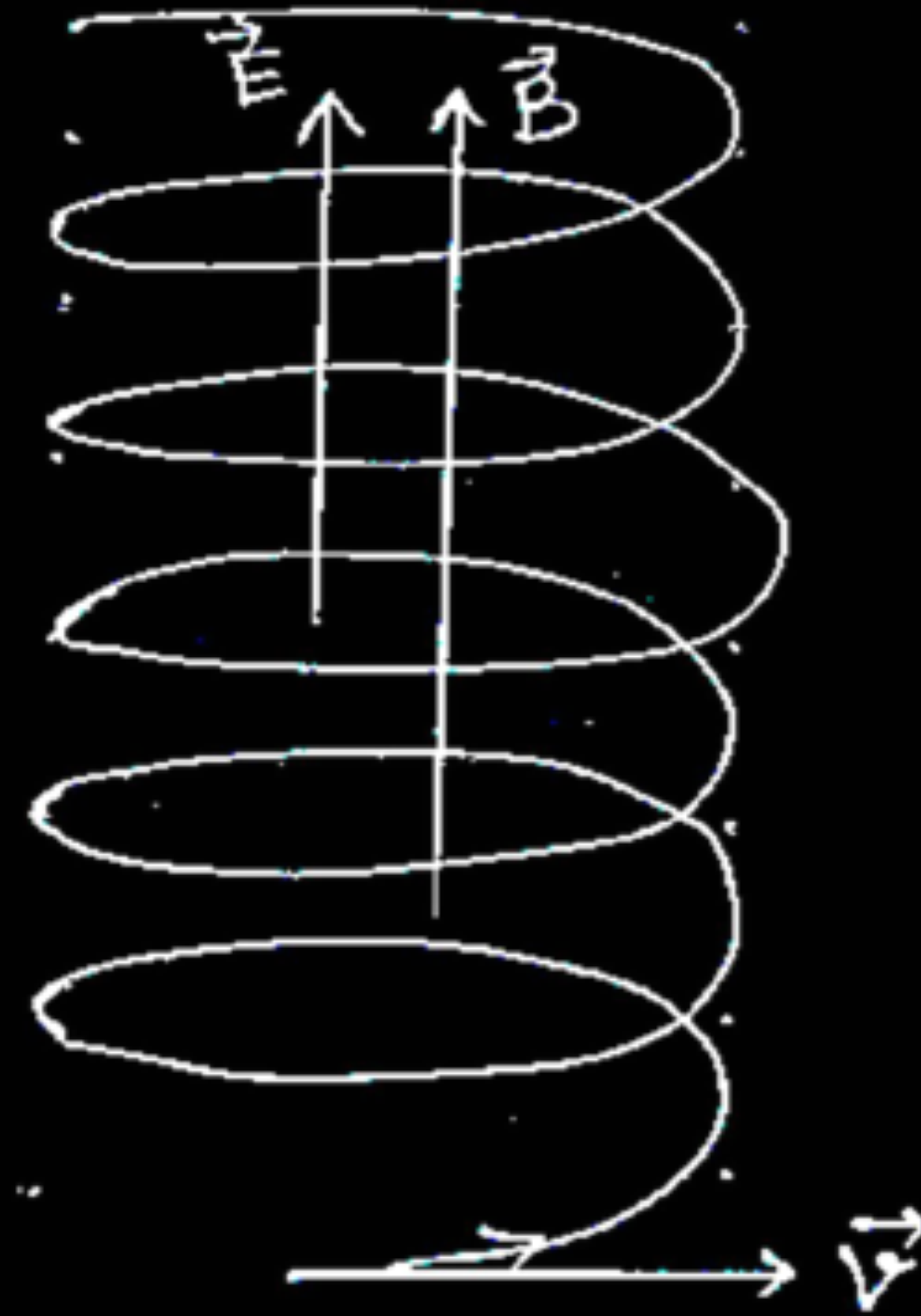
(C) $qv_0 \sqrt{B_1^2 + B_2^2}$

- 23.** (a) A charged particle q moving with a velocity $\vec{v}$ is subjected to a uniform magnetic field $\vec{B}$ acting perpendicular to $\vec{v}$. If a uniform electric field $\vec{E}$ is also set up in the region along the direction of $\vec{B}$, describe the path followed by the particle and draw its shape.
- (b) How will the magnetic field inside a long solenoid be affected when :
- the radius of the turns of the solenoid is increased,
 - the length of solenoid as well as the total number of its turns are doubled ?

3

(a)		
• Explanation of a path followed by the particle		1
• Shape of path	1	
(b) Effect on magnetic field when		
(i) Radius of turns of solenoid is increased		$\frac{1}{2}$
(ii) Length and number of turns are doubled		$\frac{1}{2}$

(a) Due to magnetic field particle will follow circular path and due to electric field, particle will accelerate along the electric field.
As a result particle will follow a helical path with constant radius but increasing pitch.



(b) (i) $B = \frac{\mu_0 NI}{l}$

No change

(ii) $\frac{N}{l} = \frac{2N}{2l}$

No Change

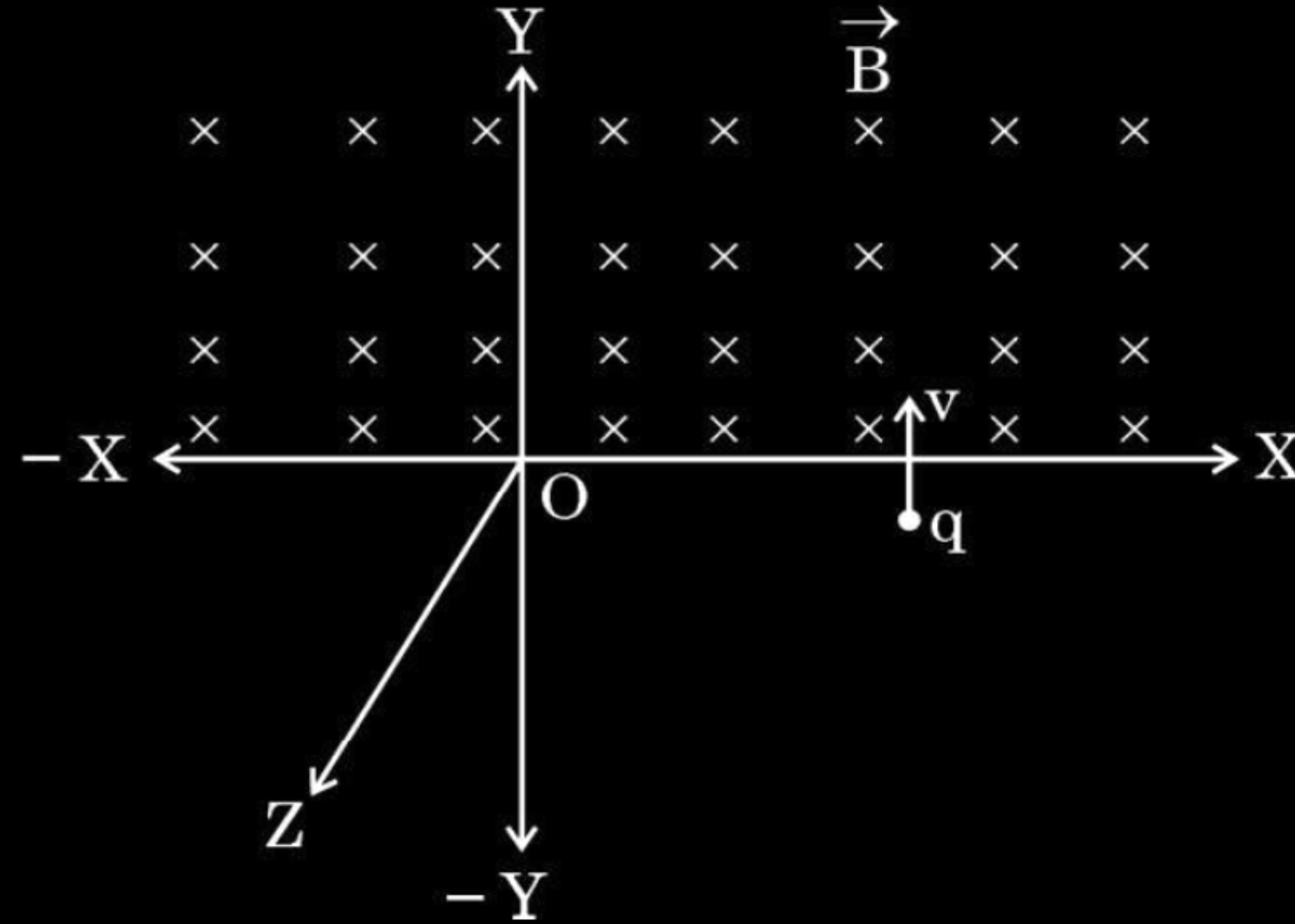
1

1

$\frac{1}{2}$

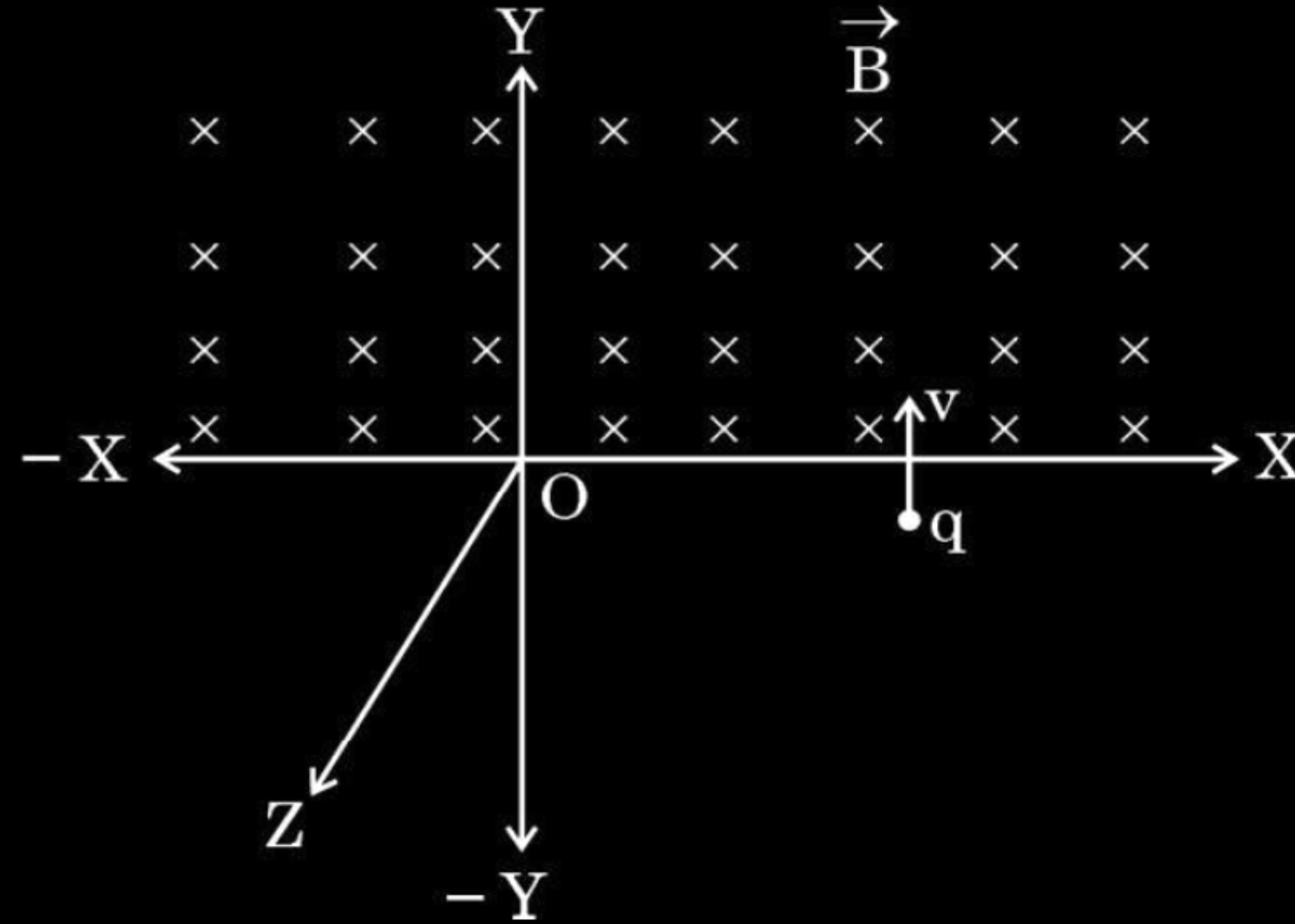
$\frac{1}{2}$

3. A particle having charge $+q$ enters a uniform magnetic field $\vec{B}$ as shown in the figure. The particle will describe :



- (A) a circular path in XZ plane
- (B) a semicircular path in XY plane
- (C) a helical path with its axis parallel to Y-axis
- (D) a semicircular path in YZ plane

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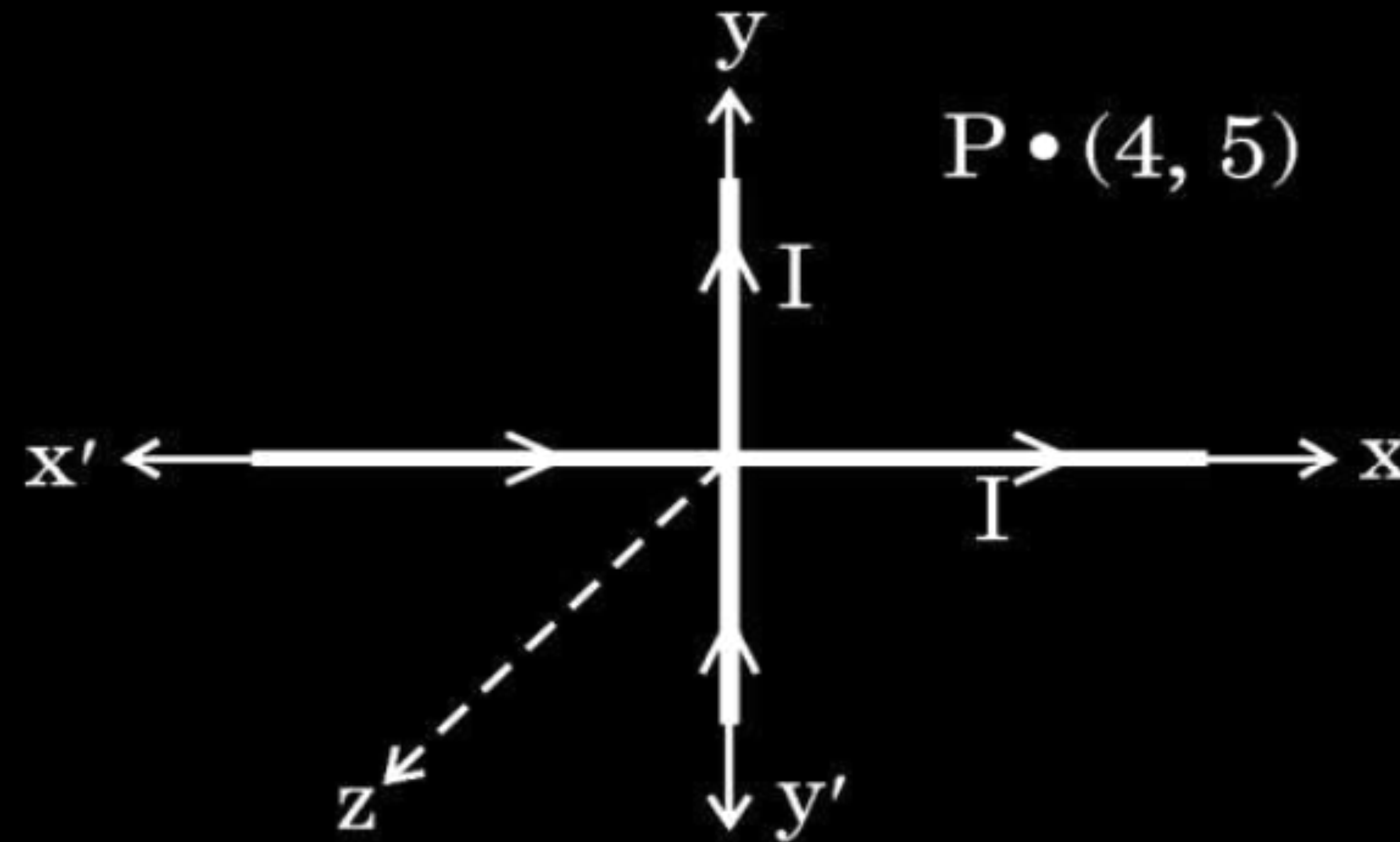


- (A) a circular path in XZ plane
- (B) a semicircular path in XY plane
- (C) a helical path with its axis parallel to Y-axis
- (D) a semicircular path in YZ plane

(B) a semicircular path in XY plane

- 23.** (a) Write vector form of Biot-Savart law.
- (b) Two insulated long straight wires, each carrying 2.0 A current are kept along xx' and yy' axis as shown in the figure. Find the magnitude and direction of resultant magnetic field at point P (4m, 5m).

3



(a) Vector form of Biot – Savart Law	1
(b) Finding ,	
• Magnitude of resultant magnetic field	$1\frac{1}{2}$
• Direction of resultant magnetic field	$\frac{1}{2}$

(a) $d\vec{B} = \frac{\mu_o}{4\pi} \frac{I(\vec{dl} \times \vec{r})}{r^3}$	1
(b) As $B = \frac{\mu_o}{4\pi} \frac{2I}{a}$	
$\therefore B_1 = \text{Magnetic field due to current carrying wire along } XX'$	
$\vec{B}_1 = \frac{\mu_o}{4\pi} \frac{2I}{a} = \frac{2 \times 2 \times 10^{-7}}{5} = (8 \times 10^{-8} \text{ T})(\hat{k})$	$\frac{1}{2}$
$B_2 = \text{Magnetic field due to current carrying wire along } YY'$	
$\vec{B}_2 = \frac{\mu_o}{4\pi} \frac{2I}{a} = \frac{2 \times 2 \times 10^{-7}}{4} = 10 \times 10^{-8} \text{ T } (-\hat{k})$	$\frac{1}{2}$
$\vec{B}_{\text{net}} = \vec{B}_2 + \vec{B}_1$	$\frac{1}{2}$
$\quad = 2 \times 10^{-8} \text{ T } (-\hat{k})$	
• Direction of resultant magnetic field along $-Z$ axis.	$\frac{1}{2}$

6. A 1 cm straight segment of a conductor carrying 1 A current in x direction lies symmetrically at origin of Cartesian coordinate system. The magnetic field due to this segment at point (1m, 1m, 0) is

1

(A) $1.0 \times 10^{-9} \hat{k} \text{ T}$

(B) $-1.0 \times 10^{-9} \hat{k} \text{ T}$

(C) $\frac{5.0}{\sqrt{2}} \times 10^{-10} \hat{k} \text{ T}$

(D) $-\frac{5.0}{\sqrt{2}} \times 10^{-10} \hat{k} \text{ T}$

6. A 1 cm straight segment of a conductor carrying 1 A current in x direction lies symmetrically at origin of Cartesian coordinate system. The magnetic field due to this segment at point (1m, 1m, 0) is

1

(A) $1.0 \times 10^{-9} \hat{k} \text{ T}$

(B) $-1.0 \times 10^{-9} \hat{k} \text{ T}$

(C) $\frac{5.0}{\sqrt{2}} \times 10^{-10} \hat{k} \text{ T}$

(D) $-\frac{5.0}{\sqrt{2}} \times 10^{-10} \hat{k} \text{ T}$

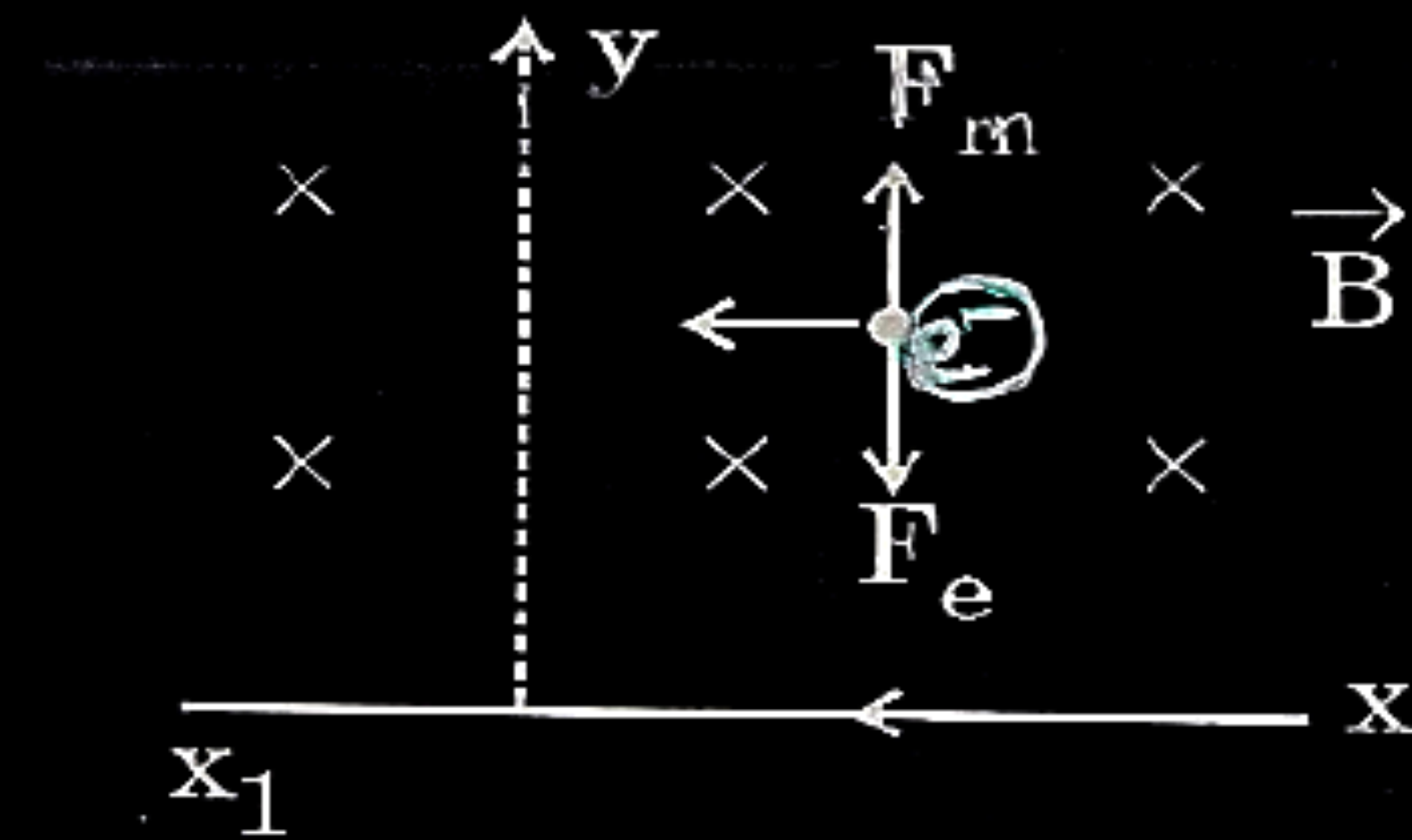
(C) $\frac{5.0}{\sqrt{2}} \times 10^{-10} \hat{k} \text{ T}$

27. (a) In a region of a uniform electric field $\vec{E}$, a negatively charged particle is moving with a constant velocity $\vec{v} = -v_0 \hat{i}$ near a long straight conductor coinciding with XX' axis and carrying current I towards $-X$ axis. The particle remains at a distance d from the conductor. **3**

- (i) Draw diagram showing direction of electric and magnetic fields.
- (ii) What are the various forces acting on the charged particle ?
- (iii) Find the value of v_0 in terms of E , d and I .

(i) Diagram showing direction of electric and magnetic fields	1
(ii) Naming forces acting on the charged particle	1
(iii) Finding the value of v_0	1

(i)



1

(ii) Electric force
Magnetic force

 $\frac{1}{2}$
 $\frac{1}{2}$

Alternatively: -

$$F_E = eE$$

$$F_B = evB$$

(iii) $ev_o B = eE$

 $\frac{1}{2}$

$$v_o \times \left[\frac{\mu_o I}{2\pi d} \right] = E$$

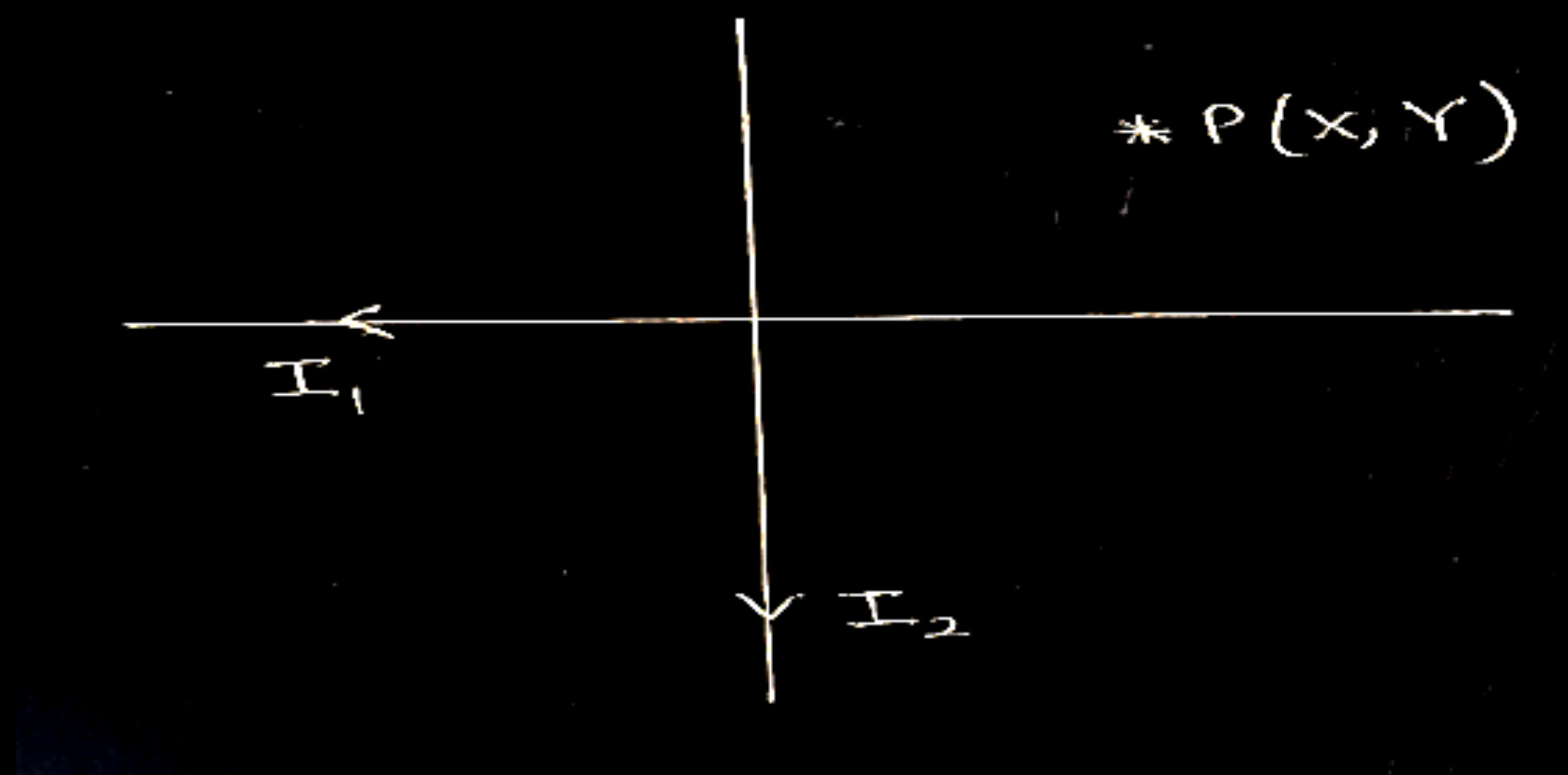
$$v_o = \frac{(2\pi d)E}{\mu_o I}$$

 $\frac{1}{2}$

- (b) Two infinitely long conductors kept along XX' and YY' axes are carrying current I_1 and I_2 along $-X$ axis and $-Y$ axis respectively. Find the magnitude and direction of the net magnetic field produced at point $P(X, Y)$.

Finding the magnitude and direction of the net magnetic field

2+1



Magnetic field due to conductor carrying current I_1 $(\vec{B}_1) = \frac{\mu_0 I_1}{2\pi Y} (-\hat{k})$

$\frac{1}{2}$

Magnetic field due to conductor Carrying current I_2 $(\vec{B}_2) = \frac{\mu_0 I_2}{2\pi X} (\hat{k})$

$\frac{1}{2}$

$$\vec{B}_p = \vec{B}_1 + \vec{B}_2$$

$$\vec{B}_p = \frac{\mu_0}{2\pi} \left[\frac{I_2}{X} - \frac{I_1}{Y} \right] \hat{k}$$

1

1

Direction will be along the Z-axis.

31. (a) (i) What is the source of force acting on a current-carrying conductor placed in a magnetic field ? Obtain the expression for force acting between two long straight parallel conductors carrying steady currents and hence define 'ampere'.
- (ii) A point charge q is moving with velocity $\vec{v}$ in a uniform magnetic field $\vec{B}$. Find the work done by the magnetic force on the charge.
- (iii) Explain the necessary conditions in which the trajectory of a charged particle is helical in a uniform magnetic field.

5

(i) Source of force	$\frac{1}{2}$
Obtaining expression for force	$1\frac{1}{2}$
Definition of 'ampere'	1
(ii) Finding work done by the magnetic force	1
(iii) Necessary conditions	1

(i) The source of force is the interaction between the field produced by the current carrying conductor and the external field in which it is placed.	1/2
Two long parallel conductors a & b, separated by a distance d, carrying currents I_a and I_b , respectively.	
The magnetic field due to a,	1/2
$B_a = \frac{\mu_0 I_a}{2\pi d}$	
The force F_{ba} , is the force on a segment L of 'b' due to 'a'.	
$F_{ba} = I_b L B_a$	
$= \frac{\mu_0 I_a I_b}{2\pi d} L$	1
Definition –	
The 'ampere' is that value of steady current which, when maintained in each of the two very long, straight, parallel conductors of negligible cross-section, and placed one metre apart in vacuum, would produce on each of these conductors a force equal to 2×10^{-7} newton per metre of length.	1
(ii) Work done by the magnetic force on the charge is zero as force is perpendicular to $\vec{v}$.	1
(iii) The velocity ($\vec{v}$) is at an arbitrary angle θ w.r.t the magnetic field ($\vec{B}$).	1

- (b) (i) A current carrying loop can be considered as a magnetic dipole placed along its axis. Explain.
- (ii) Obtain the relation for magnetic dipole moment $\vec{M}$ of current carrying coil. Give the direction of $\vec{M}$.
- (iii) A current carrying coil is placed in an external uniform magnetic field. The coil is free to turn in the magnetic field. What is the net force acting on the coil ? Obtain the orientation of the coil in stable equilibrium. Show that in this orientation the flux of the total field (field produced by the loop + external field) through the coil is maximum.

(i) Explanation	1
(ii) Obtaining relation for $\vec{M}$, and direction of $\vec{M}$.	1+1
(iii) Net force on coil	1
Obtaining orientation	$\frac{1}{2}$
Showing flux is maximum	$\frac{1}{2}$

(i) The two faces of a current carrying loop behave like two poles of a magnet therefore can be considered as a magnetic dipole placed along its axis.

1

(ii) Magnetic moment (M) $\propto$ Current (I)

$\propto$ Area (A)

$$\therefore \vec{M} = I\vec{A}$$

1

Direction is same as the area vector.

1

Alternatively: -

Magnetic moment is perpendicular to the plane of the coil.

(iii) Net force acting on the coil is zero.

1

The potential energy (U_B) of a current carrying loop in an external magnetic

$$\text{field} = -\vec{M} \cdot \vec{B}$$

For the coil to be in stable equilibrium U_B should be minimum so $\theta = 0^\circ$.

$\frac{1}{2}$

Therefore, magnetic flux (ϕ) due to the total field $= (B_{\text{coil}} + B_{\text{ext}})A$, which is its maximum value.

$\frac{1}{2}$

Alternatively: -

Orientation of stable equilibrium is one where the area vector A of the loop is in the direction of external magnetic field. In this orientation, the magnetic field produced by the loop is in the same direction as external field, both normal to the plane of the loop, thus giving rise to maximum flux of the total field.

6. A 1 cm straight segment of a conductor carrying 1 A current in x direction lies symmetrically at origin of Cartesian coordinate system. The magnetic field due to this segment at point (1m, 1m, 0) is

1

(A) $1.0 \times 10^{-9} \hat{k} \text{ T}$

(B) $-1.0 \times 10^{-9} \hat{k} \text{ T}$

(C) $\frac{5.0}{\sqrt{2}} \times 10^{-10} \hat{k} \text{ T}$

(D) $-\frac{5.0}{\sqrt{2}} \times 10^{-10} \hat{k} \text{ T}$

6. A 1 cm straight segment of a conductor carrying 1 A current in x direction lies symmetrically at origin of Cartesian coordinate system. The magnetic field due to this segment at point (1m, 1m, 0) is

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(D) $-\frac{5.0}{\sqrt{2}} \times 10^{-10} \hat{k} \text{ T}$

(C) $\frac{5.0}{\sqrt{2}} \times 10^{-10} \hat{k} \text{ T}$

3. A 1 cm segment of a wire lying along x-axis carries current of 0.5 A along +x direction. A magnetic field $\vec{B} = (0.4 \text{ mT}) \hat{j} + (0.6 \text{ mT}) \hat{k}$ is switched on, in the region. The force acting on the segment is

(A) $(2\hat{j} + 3\hat{k}) \text{ mN}$

(B) $(-3\hat{j} + 2\hat{k}) \mu\text{N}$

(C) $(6\hat{j} + 4\hat{k}) \text{ mN}$

(D) $(-4\hat{j} + 6\hat{k}) \mu\text{N}$

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1

(A) $(2\hat{j} + 3\hat{k}) \text{ mN}$

(B) $(-3\hat{j} + 2\hat{k}) \mu\text{N}$

(C) $(6\hat{j} + 4\hat{k}) \text{ mN}$

(D) $(-4\hat{j} + 6\hat{k}) \mu\text{N}$

(B) $(-3\hat{j} + 2\hat{k}) \mu\text{N}$

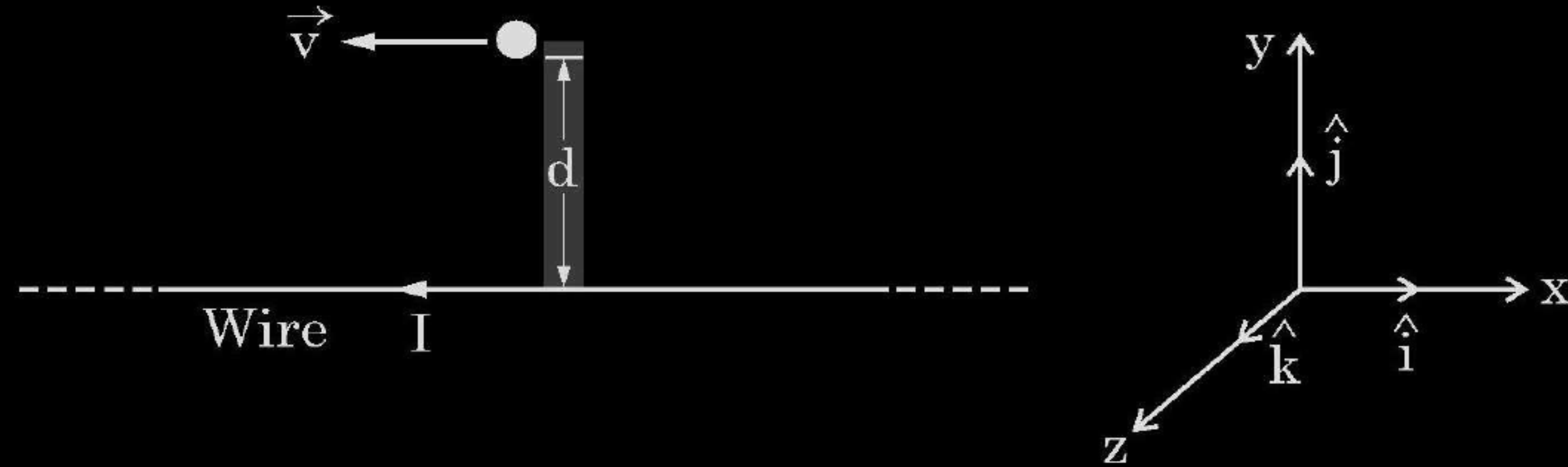
16. **Assertion (A)** : The deflection in a galvanometer is directly proportional to the current passing through it. **1**
- Reason (R)** : The coil of a galvanometer is suspended in a uniform radial magnetic field.

16. **Assertion (A)** : The deflection in a galvanometer is directly proportional to the current passing through it. **1**
- Reason (R)** : The coil of a galvanometer is suspended in a uniform radial magnetic field.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion(A).

23. A particle of charge q is moving with a velocity $\vec{v}$ at a distance 'd' from a long straight wire carrying a current 'I' as shown in figure. At this instant, it is subjected to a uniform electric field $\vec{E}$ such that the particle keeps moving undeviated. In terms of unit vectors $\hat{i}$, $\hat{j}$ and $\hat{k}$, find –

3



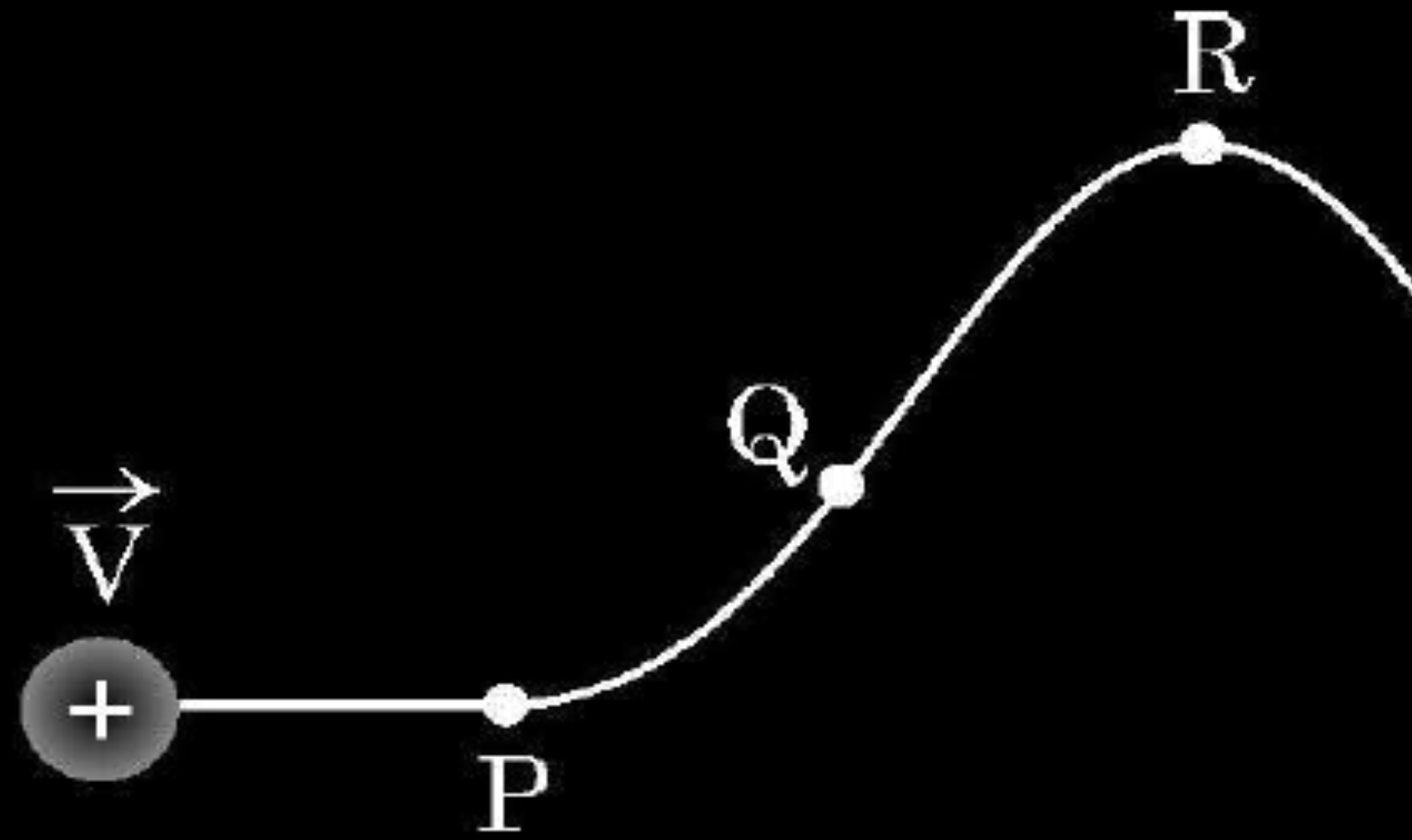
- the magnetic field $\vec{B}$,
- the magnetic force $\vec{F}_m$, and
- the electric field $\vec{E}$, acting on the charge.

Finding

- | | |
|-----------------------------------|---|
| a) The magnetic field $\vec{B}$ | 1 |
| b) The magnetic force $\vec{F}_m$ | 1 |
| c) The electric field $\vec{E}$ | 1 |

33. (a) (i) A proton moving with velocity $\vec{V}$ in a non-uniform magnetic field traces a path as shown in the figure.

5



The path followed by the proton is always in the plane of the paper. What is the direction of the magnetic field in the region near points P, Q and R ? What can you say about relative magnitude of magnetic fields at these points ?

- (ii) A current carrying circular loop of area A produces a magnetic field B at its centre. Show that the magnetic moment of the loop

is $\frac{2 BA}{\mu_0} \sqrt{\frac{A}{\pi}}$.

- | | |
|---|---|
| i) Finding the direction of magnetic field near points P, Q and R | $\frac{1}{2} + \frac{1}{2} + \frac{1}{2}$ |
| Conclusion about the relative magnitude of magnetic field. | $\frac{1}{2} + \frac{1}{2} + \frac{1}{2}$ |
| ii) Showing the given expression of magnetic moment. | 2 |

i) Near point P

Magnetic field is acting into the plane of the paper as Force is acting upwards.

Near point Q

Magnetic field is into the plane of paper as force is acting upwards.

Near point R

Magnetic field is acting out of the plane of the paper as $\vec{F}$ is acting downwards.

Relative Magnitude of the Magnetic field.

$$\text{As } B \propto \frac{1}{r}$$

Therefore,

Near point P, magnitude of B is small.

Near point Q, B is relatively smaller than point P.

Near point R, B is relatively larger than point P.

$$(B_Q < B_P < B_R)$$

ii) Let r be the radius of the circular coil and I is the current in the coil then

$$B = \frac{\mu_0 I}{2r} \quad \text{or} \quad I = \frac{2Br}{\mu_0}$$

$$A = \pi r^2 \quad r = \sqrt{\frac{A}{\pi}}$$

$$M = IA$$

$$= \frac{2Br}{\mu_0} A$$

$$= \frac{2BA}{\mu_0} \sqrt{\frac{A}{\pi}}$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

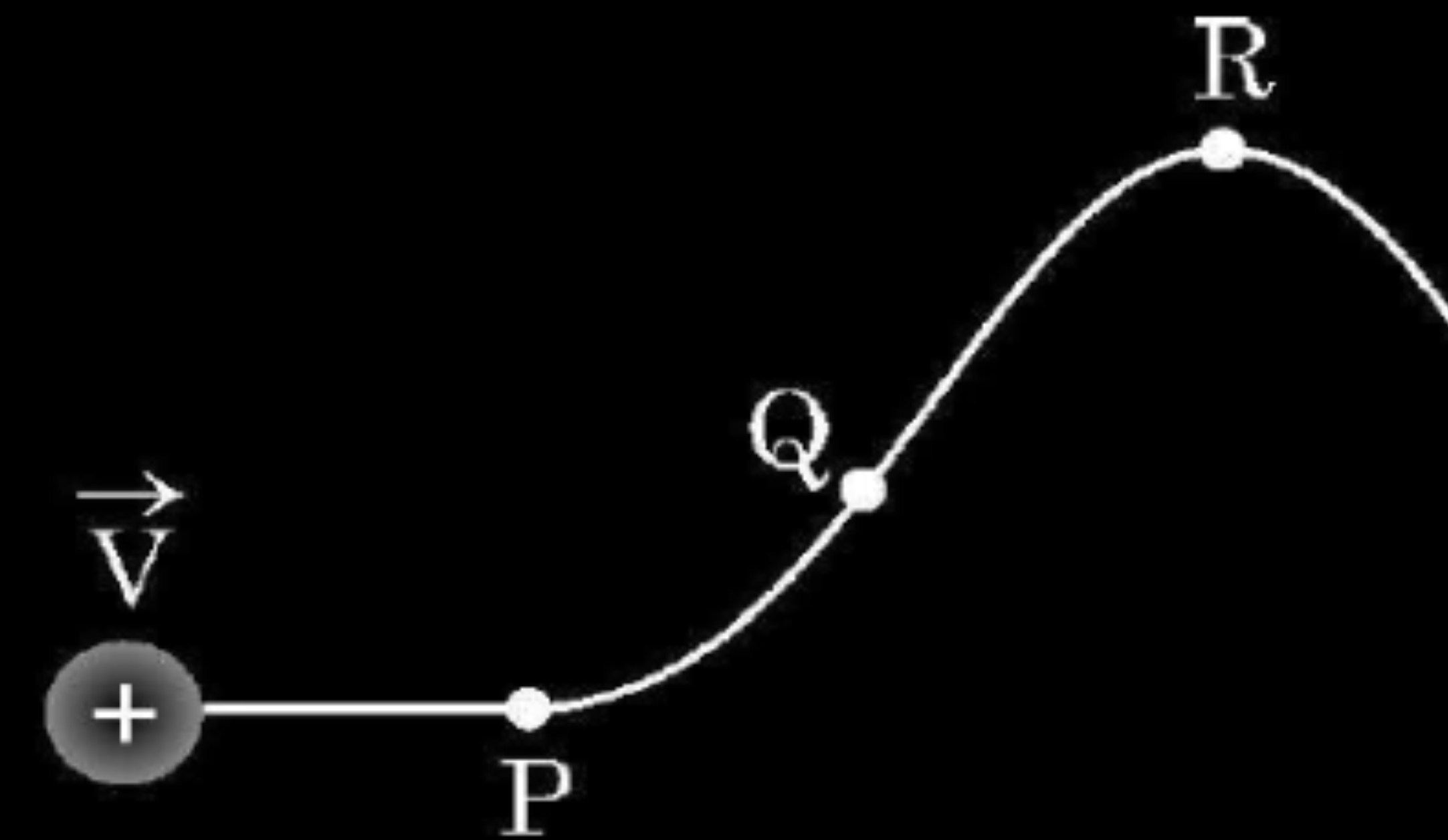
$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

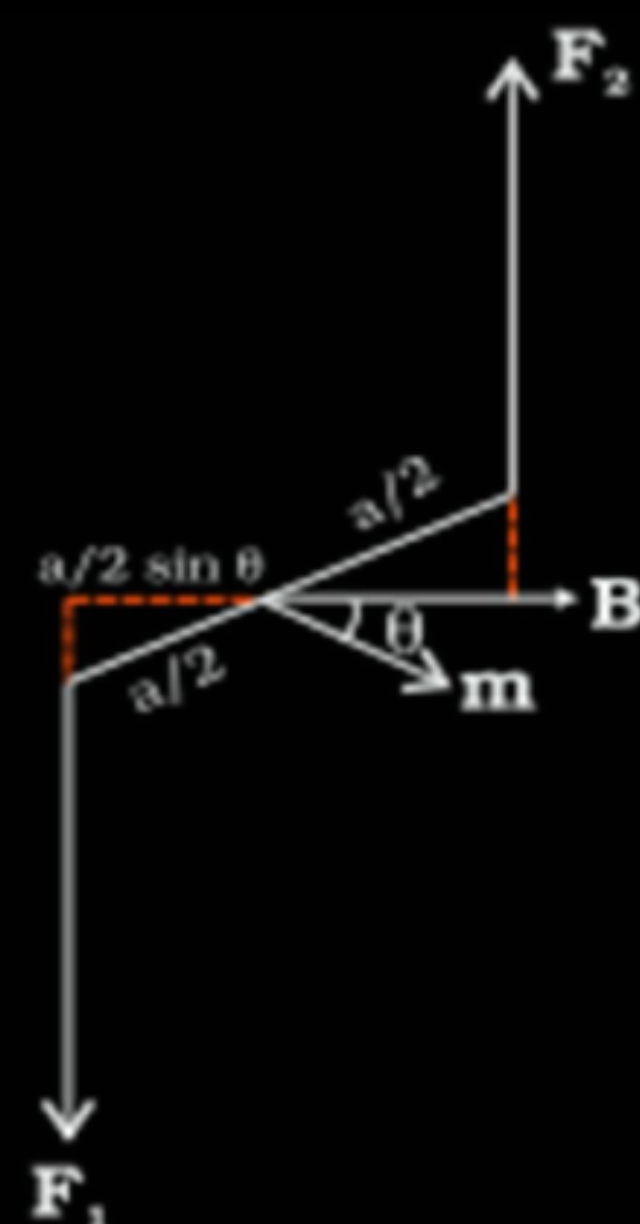
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- (b) (i) Derive an expression for the torque acting on a rectangular current loop suspended in a uniform magnetic field.
- (ii) A charged particle is moving in a circular path with velocity $\vec{V}$ in a uniform magnetic field $\vec{B}$. It is made to pass through a sheet of lead and as a consequence, it loses one half of its kinetic energy without change in its direction. How will (1) the radius of its path (2) its time period of revolution change ?

i) Deriving the expression for the torque.	3
ii) 1) Finding the change in radius.	1
2) Finding the change in time period of Revolution.	1

i)



$\vec{F}_1$ and $\vec{F}_2$ are the forces acting on two arms of the rectangular coil having sides a and b.

$$|\vec{F}_1| = |\vec{F}_2| = I b B \quad (b = \text{length of the arm})$$

Forces constitute a couple. The magnitude of Torque on the loop is –

$$\begin{aligned} \tau &= F_1 \frac{a}{2} \sin \theta + F_2 \frac{a}{2} \sin \theta \\ &= I a b B \sin \theta \\ &= I A B \sin \theta \end{aligned}$$

$$\vec{\tau} = I \vec{A} \times \vec{B}$$

ii) 1) $r = \frac{mv}{qB} = \frac{\sqrt{2mK}}{qB}$

$$\frac{r'}{r} = \frac{\alpha \sqrt{K}}{\sqrt{K/2}} = \frac{1}{\sqrt{2}}$$

$$r' = \frac{r}{\sqrt{2}}$$

2) $T = \frac{2\pi m}{qB}$

Time period does not depend on Kinetic Energy
 $\therefore$ Time period will not change.

1/2

1

1/2

1/2

1/2

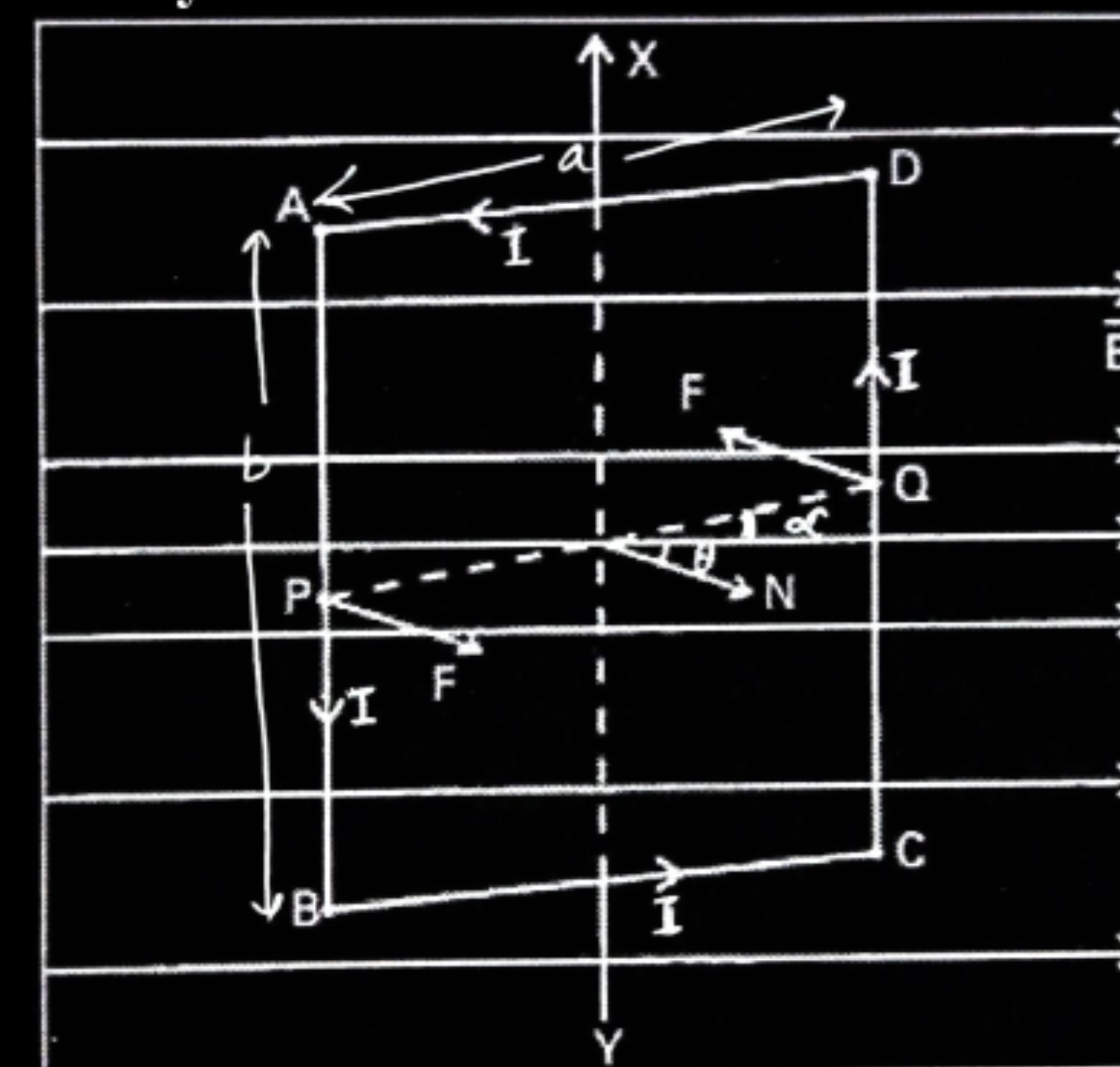
1/2

1/2

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1/2

Alternatively



If the plane of the current carrying coil makes an angle α with the magnetic field

$$\vec{F}_{DA} = -\vec{F}_{BC} \text{ (cancel each other) .}$$

Force on the arm DC is into the plane of the paper

$$|F_{DC}| = I b B$$

Force on the arm AB is out of the plane of the paper.

$$|F_{AB}| = I b B$$

Both of them form a couple and Torque acting on the coil is

$\tau = \text{either force} \times \text{perpendicular distance between the two forces.}$

$$\tau = I b B \times a \cos \alpha$$

$$= I a b B \cos \alpha$$

$$\tau = I A B \cos \alpha$$

Let $\hat{n}$ = outward drawn normal to the plane of the coil.

$$\theta + \alpha = 90^\circ$$

$$\alpha = 90^\circ - \theta$$

$$\tau = I A B \cos(90 - \theta)$$

$$= I A B \sin \theta$$

$$\vec{\tau} = I \vec{A} \times \vec{B}$$

1/2

1/2

1/2

1/2

1/2

1/2

23. (a) Define magnetic moment of a current-carrying coil. Write its SI unit.
- (b) A coil of 60 turns and area $1.5 \times 10^{-3} \text{ m}^2$ carrying 2A current lies in a vertical plane. It experiences a torque of 0.12 Nm when placed in a uniform horizontal magnetic field. The torque acting on the coil changes to 0.05 Nm after the coil is rotated about its diameter by 90° , in the magnetic field. Find the magnitude of the magnetic field.

3

a) Defining magnetic moment	1
SI unit of magnetic moment	$\frac{1}{2}$
b) Finding the magnitude of magnetic field	$1\frac{1}{2}$

a) Magnetic moment of a current carrying coil is defined as the product of current flowing through the coil and Area of the coil.

Alternatively

$$\vec{M} = I\vec{A}$$

S.I. unit is Am^2

b) $\tau = NIAB\sin\theta$

$$\tau_1 = 0.12 = 60 \times 2 \times 1.5 \times 10^{-3} \times B \sin \theta$$

$$B \sin \theta = \frac{2}{3}$$

$$\tau_2 = 0.05 = 60 \times 2 \times 1.5 \times 10^{-3} \times B \cos \theta$$

$$B \cos \theta = \frac{5}{18}$$

$$B = \sqrt{B^2 \sin^2 \theta + B^2 \cos^2 \theta}$$

$$= \sqrt{\left(\frac{2}{3}\right)^2 + \left(\frac{5}{18}\right)^2} = \frac{13}{18} \text{ T}$$

23. Using Biot-Savart law, derive expression for the magnetic field ($\vec{B}$) due to a circular current carrying loop at a point on its axis and hence at its centre.

3

Derivation for

Magnetic field on the axis

$2 \frac{1}{2}$

Magnetic field at the centre

$\frac{1}{2}$

From Biot Savart's Law

$$|d\vec{B}| = \frac{\mu_0}{4\pi} \frac{I |d\vec{l} \times \vec{r}|}{r^3}$$

Now $r^2 = x^2 + R^2$

Because $|d\vec{l} \times \vec{r}| = r dl$

$$\therefore dB = \frac{\mu_0}{4\pi} \frac{idl}{(x^2 + R^2)}$$

$d\vec{B}$ has two components.

All the components perpendicular to x- axis are summed over and we obtain a null result.

Only x- components contribute. The net contribution along x- direction

$$dB_x = dB \cos \theta$$

$$\cos \theta = \frac{R}{(R^2 + x^2)^{\frac{1}{2}}}$$

Thus :

$$dB_x = \frac{\mu_0 i}{4\pi} dl \frac{R}{(R^2 + x^2)^{\frac{3}{2}}}$$

Summing dB_x over the entire loop

$$\oint dl = 2\pi R$$

$$\vec{B} = B_x \hat{i} = \frac{\mu_0 i R^2}{2(x^2 + R^2)^{\frac{3}{2}}} \hat{i}$$

Magnetic field at the centre of the loop-

Here $x=0$

$$\therefore \vec{B} = \frac{\mu_0 i}{2R} \hat{i}$$

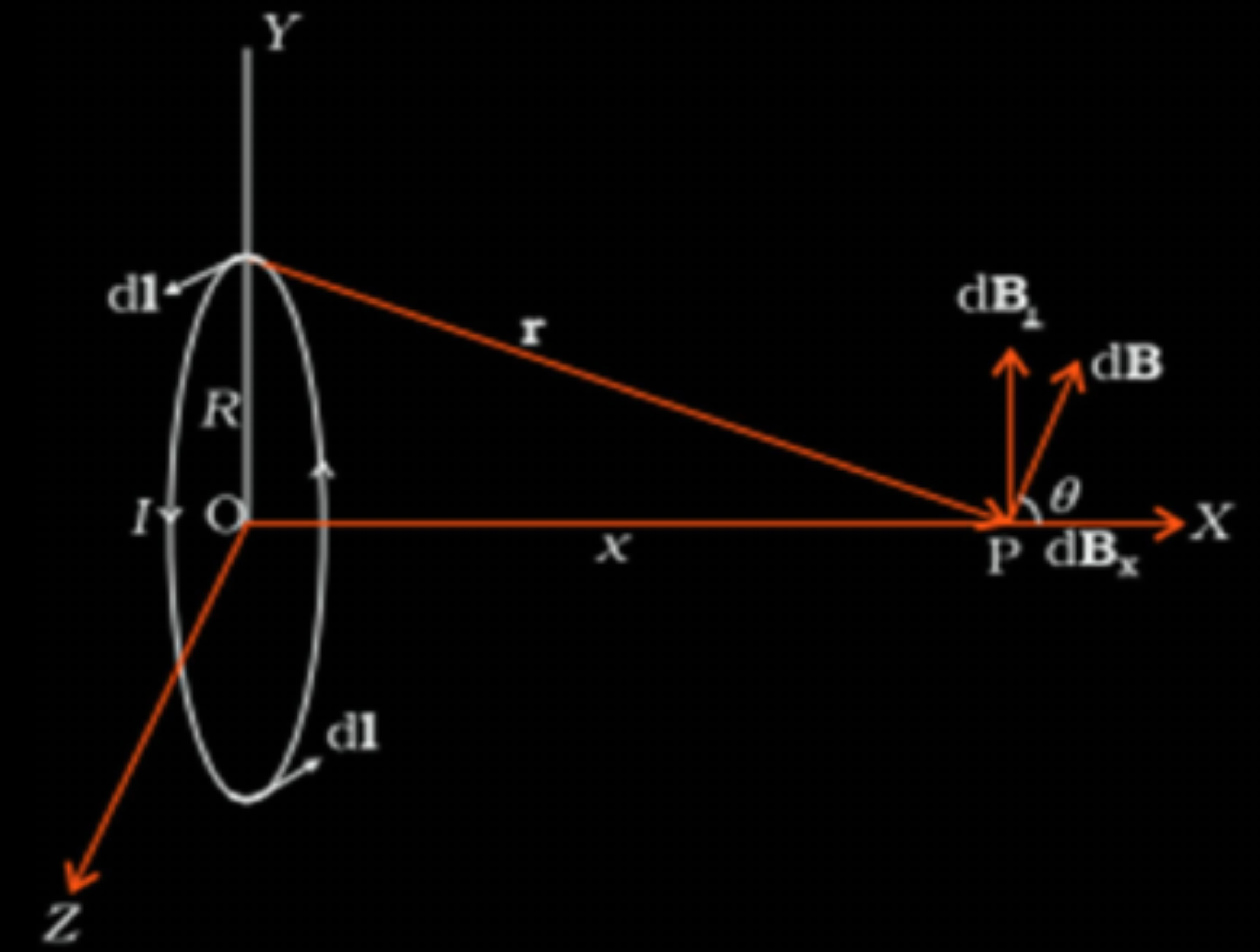
$\frac{1}{2}$

$\frac{1}{2}$

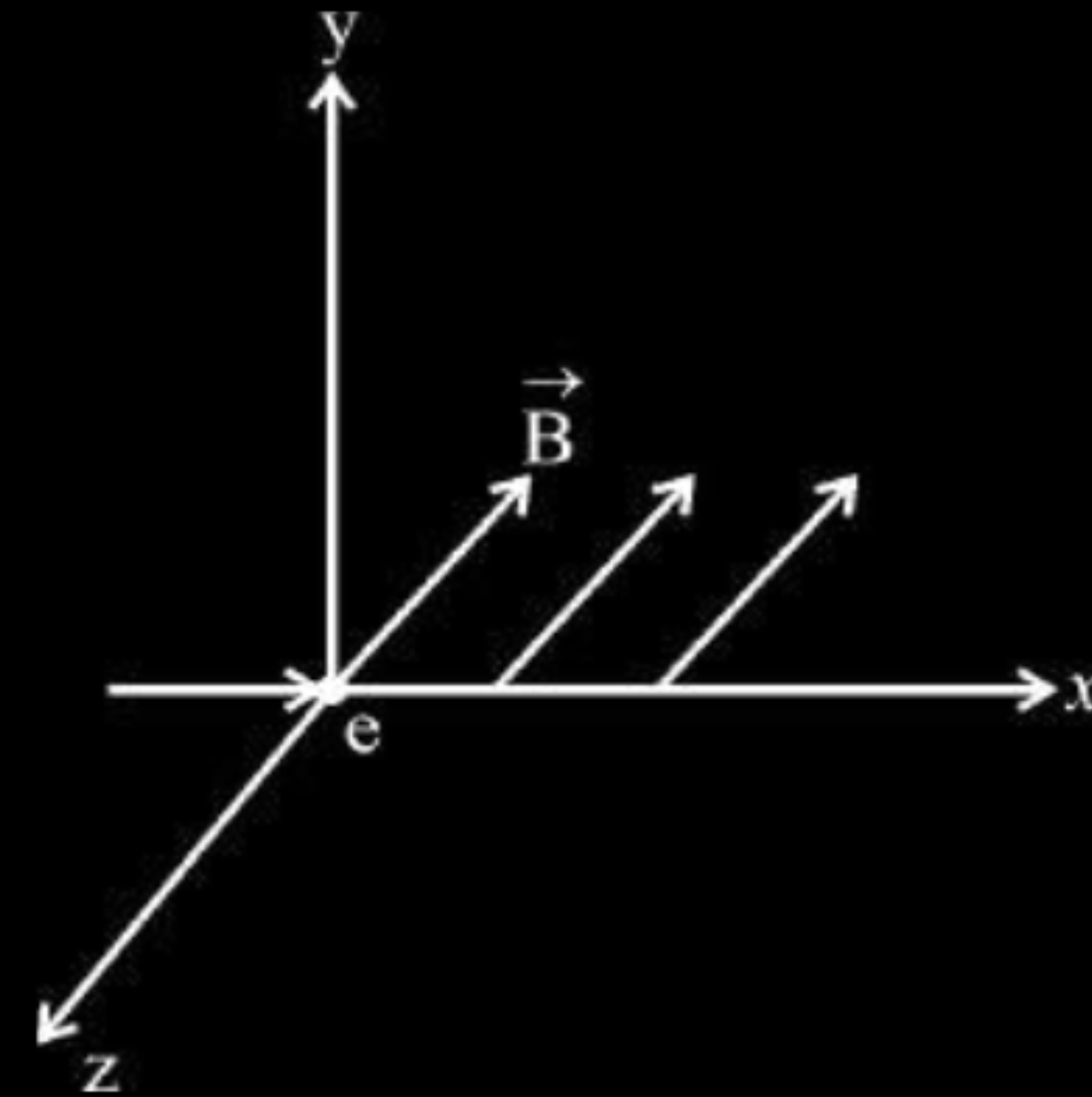
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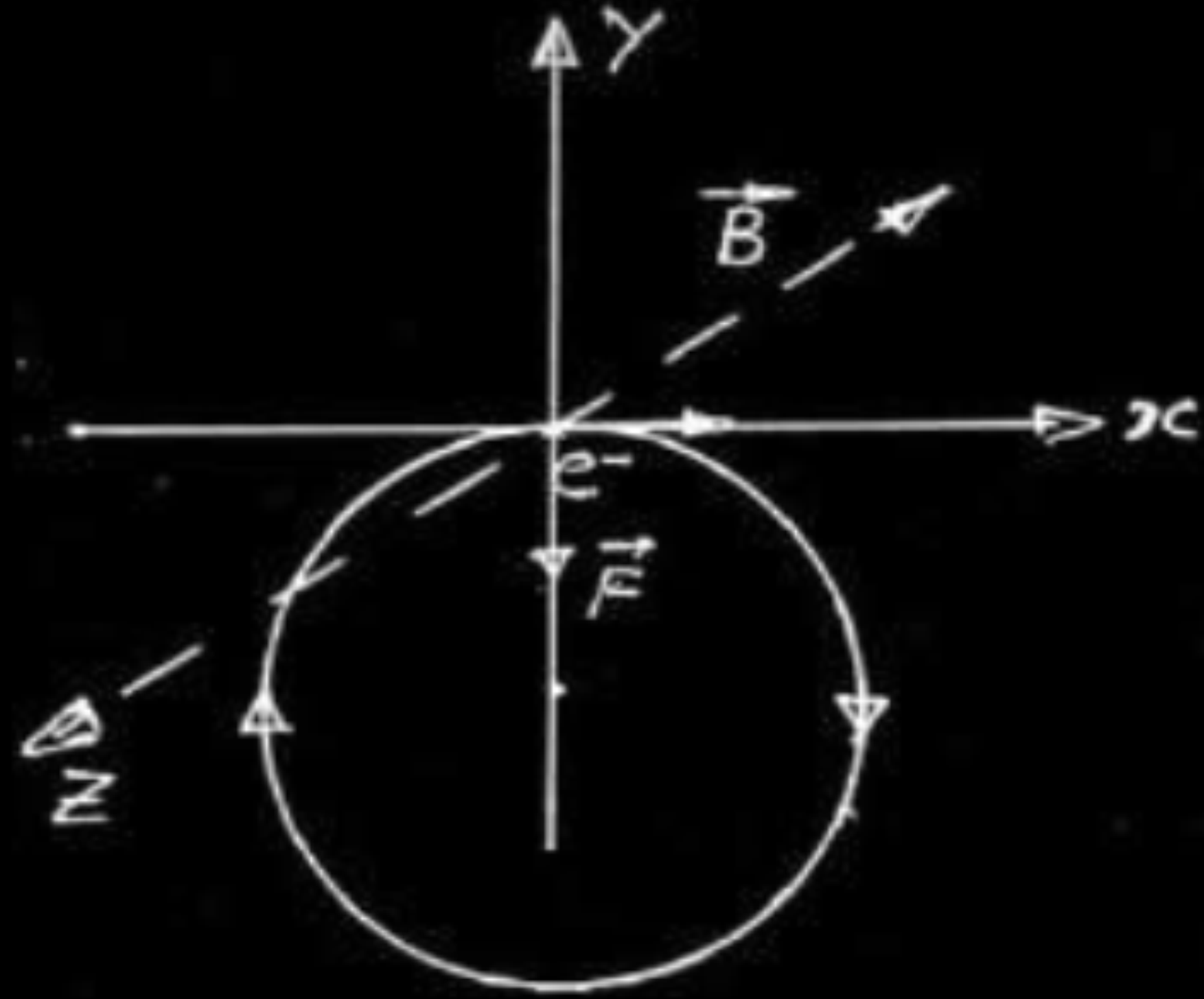
$\frac{1}{2}$

$\frac{1}{2}$



20. An electron moves along $+x$ direction. It enters into a region of uniform magnetic field $\vec{B}$ directed along $-z$ direction as shown in fig. Draw the shape of trajectory followed by the electron after entering the field.





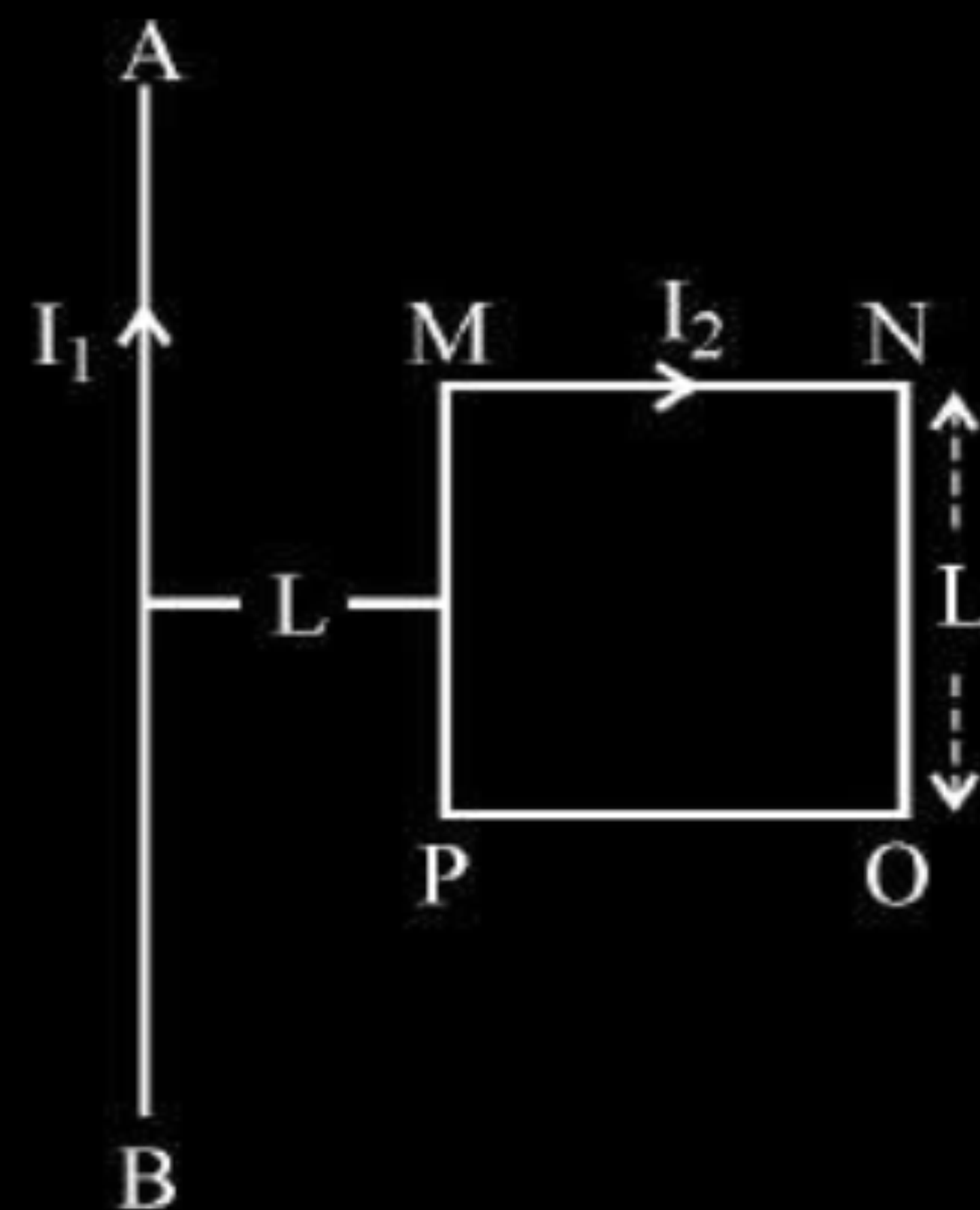
Alternatively: Circular path in the X-Y plane in clockwise sense.

[**Note:** If the student just writes, force on the electron will be along negative Y axis, i.e. $F = -e(v \hat{i}) \times (B(-\hat{k}) = evB(-\hat{j})$ award $\frac{1}{2}$ mark only)

1

$\frac{1}{2} + \frac{1}{2}$

A square shaped current carrying loop MNOP is placed near a straight long current carrying wire AB as shown in the fig. The wire and the loop lie in the same plane. If the loop experiences a net force F towards the wire, find the magnitude of the force on the side 'NO' of the loop.



Magnitude of force on side NO is $= F$

Alternatively

Let force on side MP be $= F_1$

Force on side $NO = \frac{F_1}{2}$

Magnitude of net force $= F_1 - \frac{F_1}{2} = \frac{F_1}{2} = F$

Therefore force on side NO $= \frac{F_1}{2} = F$

Give full credit if a student calculates the force as shown below.

$$F = \frac{\mu_0}{2\pi} I_1 I_2$$

$\frac{1}{2}$

$\frac{1}{2}$

3. A current I flows through a long straight conductor which is bent into a circular loop of radius R in the middle as shown in the figure.



The magnitude of the net magnetic field at point O will be

- (a) Zero (b) $\frac{\mu_0 I}{2R} (1 + \pi)$ (c) $\frac{\mu_0 I}{4\pi R}$ (d) $\frac{\mu_0 I}{2R} \left(1 - \frac{1}{\pi}\right)$ **1**

3. A current I flows through a long straight conductor which is bent into a circular loop of radius R in the middle as shown in the figure.



The magnitude of the net magnetic field at point O will be

- (a) Zero (b) $\frac{\mu_0 I}{2R} (1 + \pi)$ (c) $\frac{\mu_0 I}{4\pi R}$ (d) $\frac{\mu_0 I}{2R} \left(1 - \frac{1}{\pi}\right)$ **1**

(d) $\frac{\mu_0 I}{2R} \times \left(1 - \frac{1}{\pi}\right)$

4. A circular loop of radius r , carrying a current I lies in y - z plane with its centre at the origin. The net magnetic flux through the loop is :
- (a) directly proportional to r (b) zero
(c) inversely proportional to r (d) directly proportional to I

4. A circular loop of radius r , carrying a current I lies in y - z plane with its centre at the origin. The net magnetic flux through the loop is :
- (a) directly proportional to r (b) zero
(c) inversely proportional to r (d) directly proportional to I

1

(b) Zero

7. A current of 10A is flowing from east to west in a long straight wire kept on a horizontal table. The magnetic field developed at a distance of 10 cm due north on the table is :

- (a) 2×10^{-5} T, acting downwards (b) 2×10^{-5} T, acting upwards
(c) 4×10^{-5} T, acting downwards (d) 4×10^{-5} T, acting upwards

7. A current of 10A is flowing from east to west in a long straight wire kept on a horizontal table. The magnetic field developed at a distance of 10 cm due north on the table is :

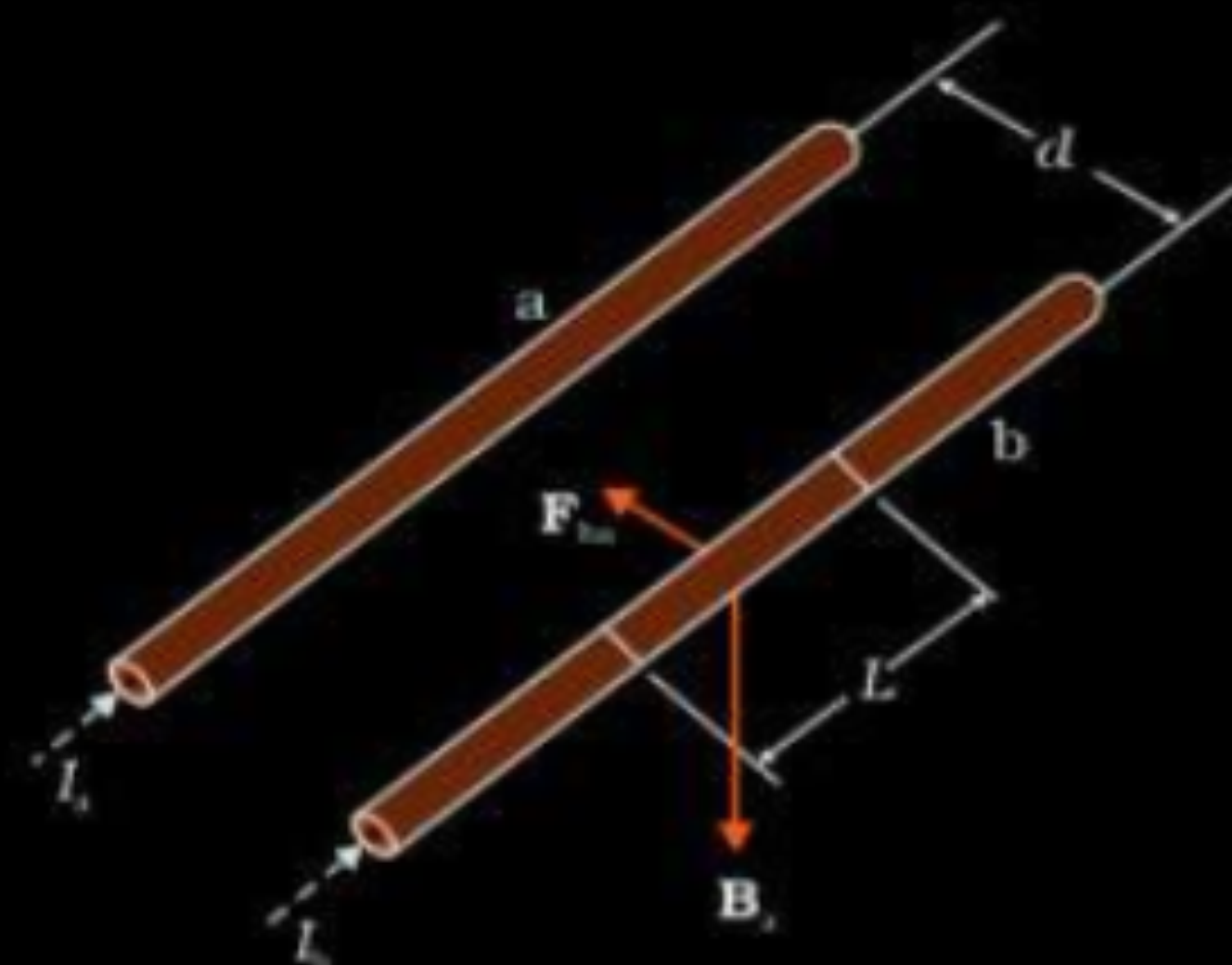
- (a) 2×10^{-5} T, acting downwards (b) 2×10^{-5} T, acting upwards
(c) 4×10^{-5} T, acting downwards (d) 4×10^{-5} T, acting upwards

1

(a) 2×10^{-5} T acting downwards

- (a) Derive the expression for the force acting between two long parallel current carrying conductors. Hence, define 1 A current.
- (b) A steady current of 2A flows through a circular coil having 5 turns of radius 7 cm. The coil lies in X-Y plane with its centre at the origin. Find the magnitude and direction of the magnetic dipole moment of the coil.

(a)



Magnetic field due to the current I_a flowing in conductor 'a' at any point on conductor 'b'

$$B_a = \frac{\mu_0 I_a}{2\pi d}$$

(Acting perpendicular downward)

Therefore force on conductor 'b' due to field B_a

$$\vec{F} = I_b (\vec{l_b} \times \vec{B_a})$$

$$|\vec{F_{ba}}| = I_b l_b \times \frac{\mu_0 I_a}{2\pi d}$$

$$= \frac{\mu_0 I_a I_b l_b}{2\pi d}$$

$$\frac{|\vec{F_{ba}}|}{l_b} = \frac{\mu_0 I_a I_b}{2\pi d}$$

Definition of 1 A :

Two straight infinitely long parallel conductors are said to carry 1 A current each when they interact each other with a force of $2 \times 10^{-7} Nm^{-1}$, when kept 1m apart in vacuum

$$(b) M = NI\pi a^2$$

$$= 5 \times 2 \times \frac{22}{7} \times 49 \times 10^{-4}$$

$$= 154 \times 10^{-3}$$

$$= 0.154 Am^2$$

$\vec{M}$ will be perpendicular to $x - y$ plane or parallel to Z axis

1/2

1/2

1/2

1/2

1

1/2

1/2

1/2

1/2

9. A region has a uniform magnetic field in it. A proton enters into the region with velocity making an angle of 45° with the direction of the magnetic field. In this region the proton will move on a path having the shape of a

- (A) straight line
- (B) circle
- (C) spiral
- (D) helix

9. A region has a uniform magnetic field in it. A proton enters into the region with velocity making an angle of 45° with the direction of the magnetic field. In this region the proton will move on a path having the shape of a

1

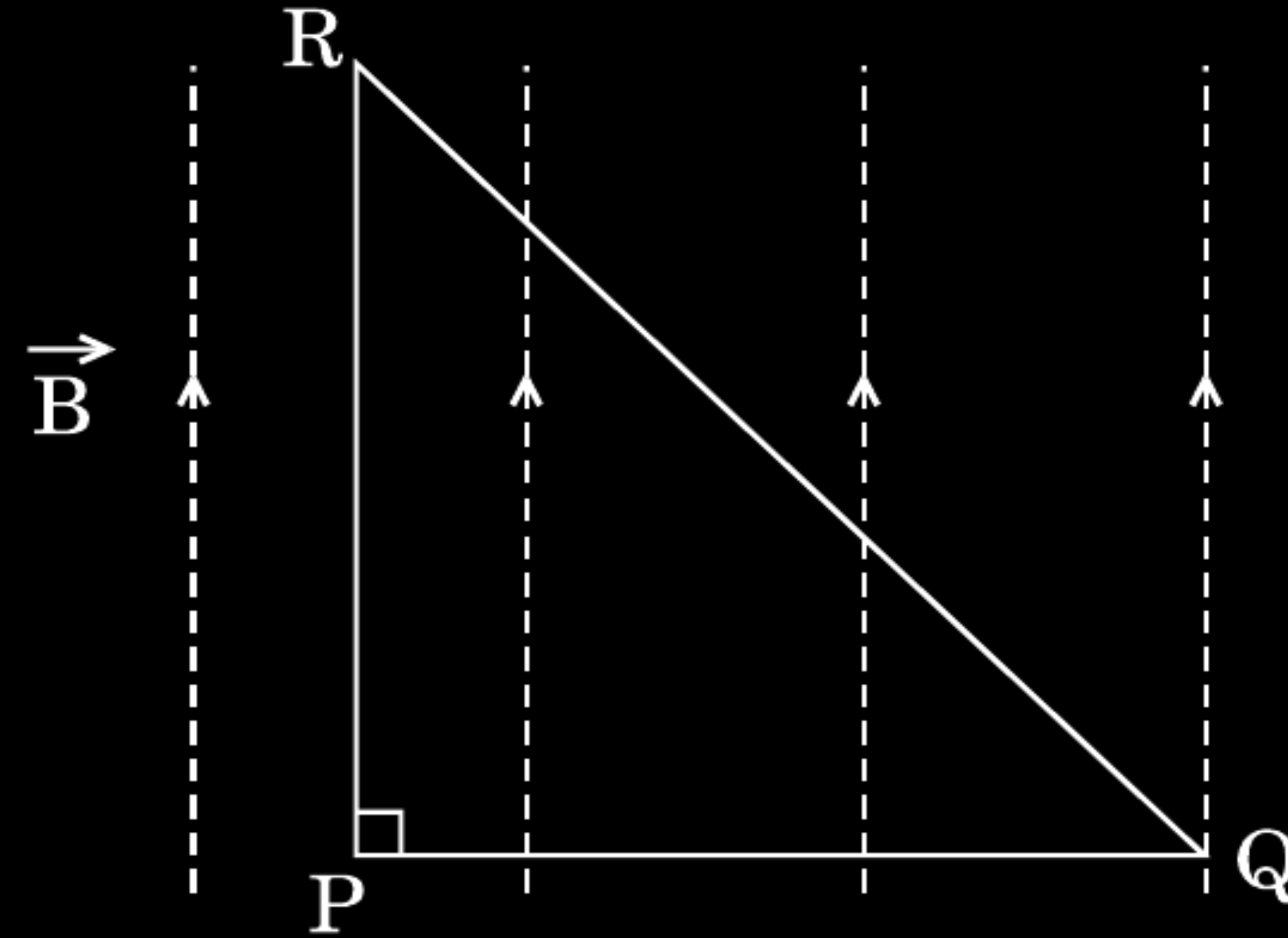
- (A) straight line
- (B) circle
- (C) spiral
- (D) helix

(D)

Helix

10. An isosceles right angled current carrying loop PQR is placed in a uniform magnetic field $\vec{B}$ pointing along PR. If the magnetic force acting on the arm PQ is F, then the magnetic force which acts on the arm QR will be

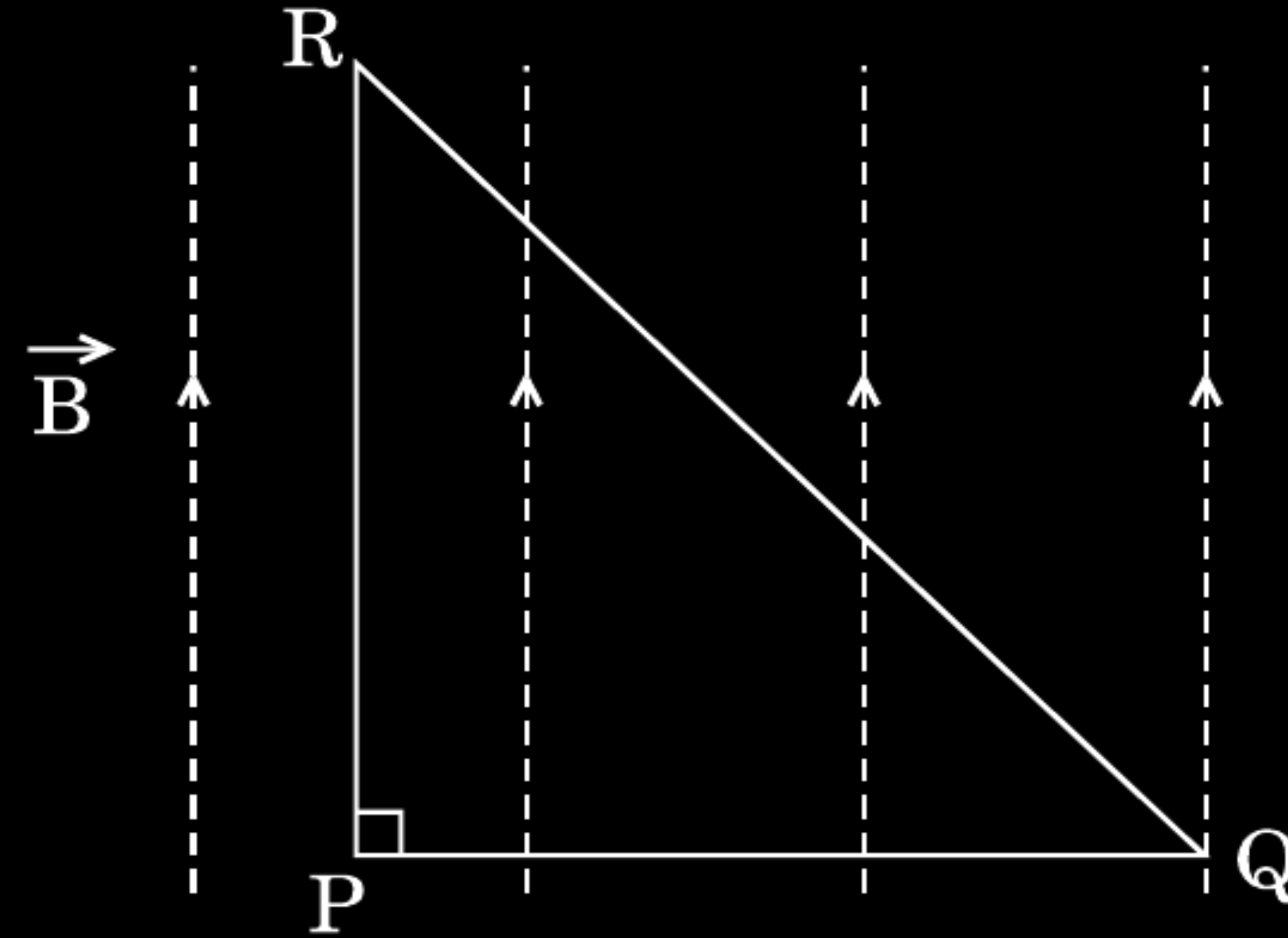
1



- (A) F
- (B) $\frac{F}{\sqrt{2}}$
- (C) $\sqrt{2} F$
- (D) $-F$

10. An isosceles right angled current carrying loop PQR is placed in a uniform magnetic field $\vec{B}$ pointing along PR. If the magnetic force acting on the arm PQ is F, then the magnetic force which acts on the arm QR will be

1



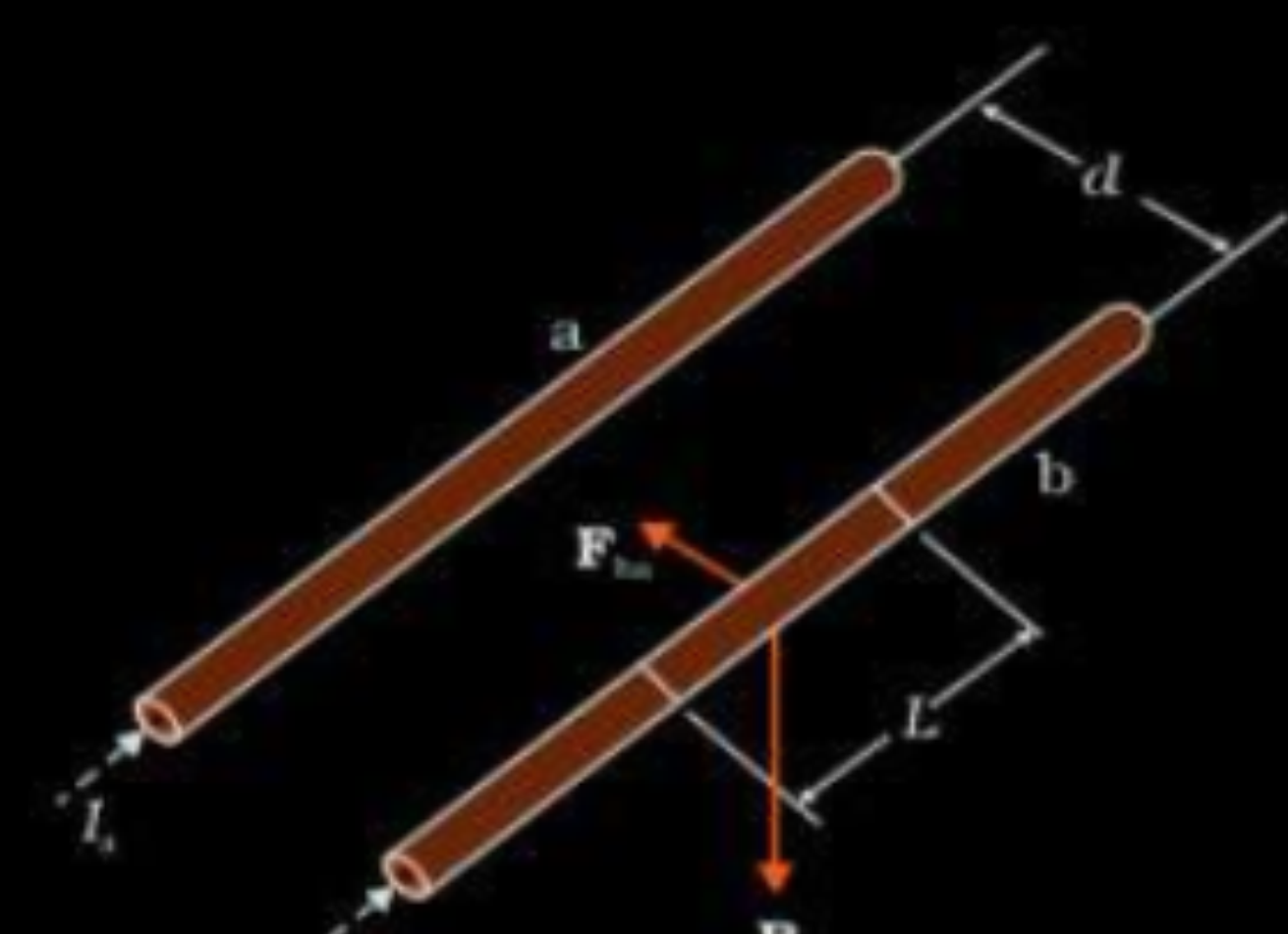
- (A) F
- (B) $\frac{F}{\sqrt{2}}$
- (C) $\sqrt{2} F$
- (D) $-F$

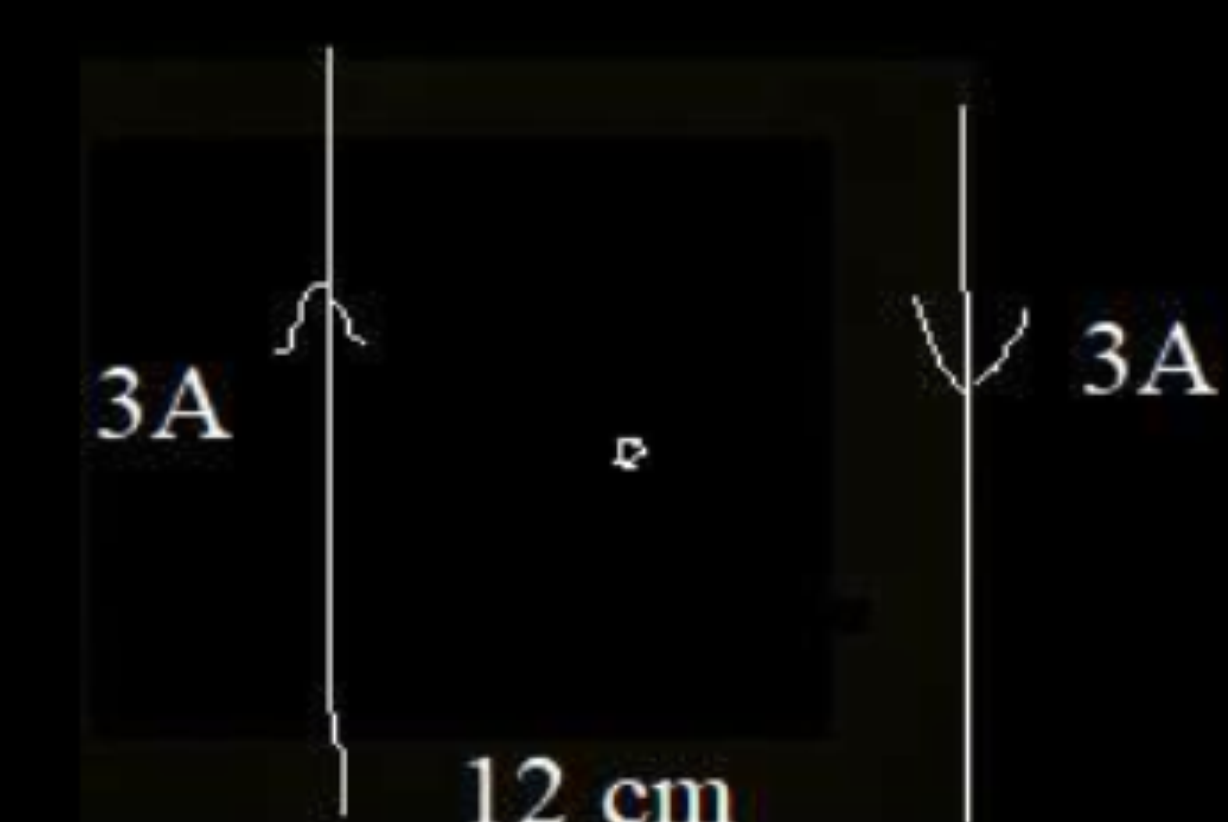
(D)

-F

- 36.** (a) Derive the expression for the force acting per unit length between two long straight parallel current carrying conductors. Hence define one ampere.
- (b) Two long parallel straight conductors are placed 12 cm apart in air. They carry equal currents of 3 A each. Find the magnitude and direction of the magnetic field at a point midway between them (drawing a figure) when the currents in them flow in opposite directions.

a) Diagram showing direction	1/2 mark
Derivation of expression	1 1/2 marks
Definition of 1 ampere	1 mark
b) Magnetic fields of the wires	1/2 mark
Net magnetic field and its direction	1 + 1/2 mark

(a) 	
Magnetic field due to the current I_a flowing in conductor 'a' at any point on conductor 'b' $B_a = \frac{\mu_0 I_a}{2\pi d}$	1/2
(Acting perpendicular downward) Therefore force on conductor 'b' due to field B_a	
$\vec{F} = I_b (\vec{l}_b \times \vec{B}_a)$ $ \vec{F}_{ba} = I_b l_b \times \frac{\mu_0 I_a}{2\pi d}$ $= \frac{\mu_0 I_a I_b l_b}{2\pi d}$ $\frac{ \vec{F}_{ba} }{l_b} = \frac{\mu_0 I_a I_b}{2\pi d}$	1/2
Definition of 1 A : Two straight infinitely long parallel conductors are said to carry 1 A current each when they interact each other with a force of $2 \times 10^{-7} \text{ Nm}^{-1}$, when kept 1m apart in vacuum	1

	
$\vec{B} = \vec{B}_1 + \vec{B}_2$	1/2
$B = \frac{\mu_0 I_1}{2\pi r_1} + \frac{\mu_0 I_2}{2\pi r_2}$	
$= \frac{\mu_0}{2\pi} \left(\frac{3}{6 \times 10^{-2}} + \frac{3}{6 \times 10^{-2}} \right)$	
$= \frac{4\pi \times 10^{-7} \times 3}{\pi \times 6 \times 10^{-2}}$	
$= 2 \times 10^{-5} \text{ tesla}$	1
Direction of $\vec{B}$ at midpoint is perpendicular to the plane containing the two conductors and pointing downwards. (Note: give full credit of this direction if student takes direction opposite to the shown in fig and answer accordingly)	1/2

- (b) To convert a given galvanometer into a voltmeter of ranges 2 V , V and $\frac{\text{V}}{2}$ volt, resistances R_1 , R_2 and R_3 ohm respectively, are required to be connected in series with the galvanometer. Obtain the relationship between R_1 , R_2 and R_3 .

b)

$$R = \frac{V}{i_g} - G$$

1/2

$$R_1 = \frac{2V}{i_g} - G = R_o - G$$

$$R_1 + G = 2R_o$$

1/2

$$\left[\text{Where } R_o = \frac{V}{i_g} \right]$$

Similarly

$$R_2 + G = R_o$$

$$R_3 + G = R_o/2$$

1/2

From the above equations,

$$R_1 - R_2 = 2(R_2 - R_3)$$

$$R_1 - 3R_2 + 2R_3 = 0$$

1/2

- 3.** An electron is released from rest in a region of uniform electric and magnetic fields acting parallel to each other. The electron will *1*
- (A) move in a straight line.
 - (B) move in a circle.
 - (C) remain stationary.
 - (D) move in a helical path.

3. An electron is released from rest in a region of uniform electric and magnetic fields acting parallel to each other. The electron will 1
- (A) move in a straight line.
 - (B) move in a circle.
 - (C) remain stationary.
 - (D) move in a helical path.

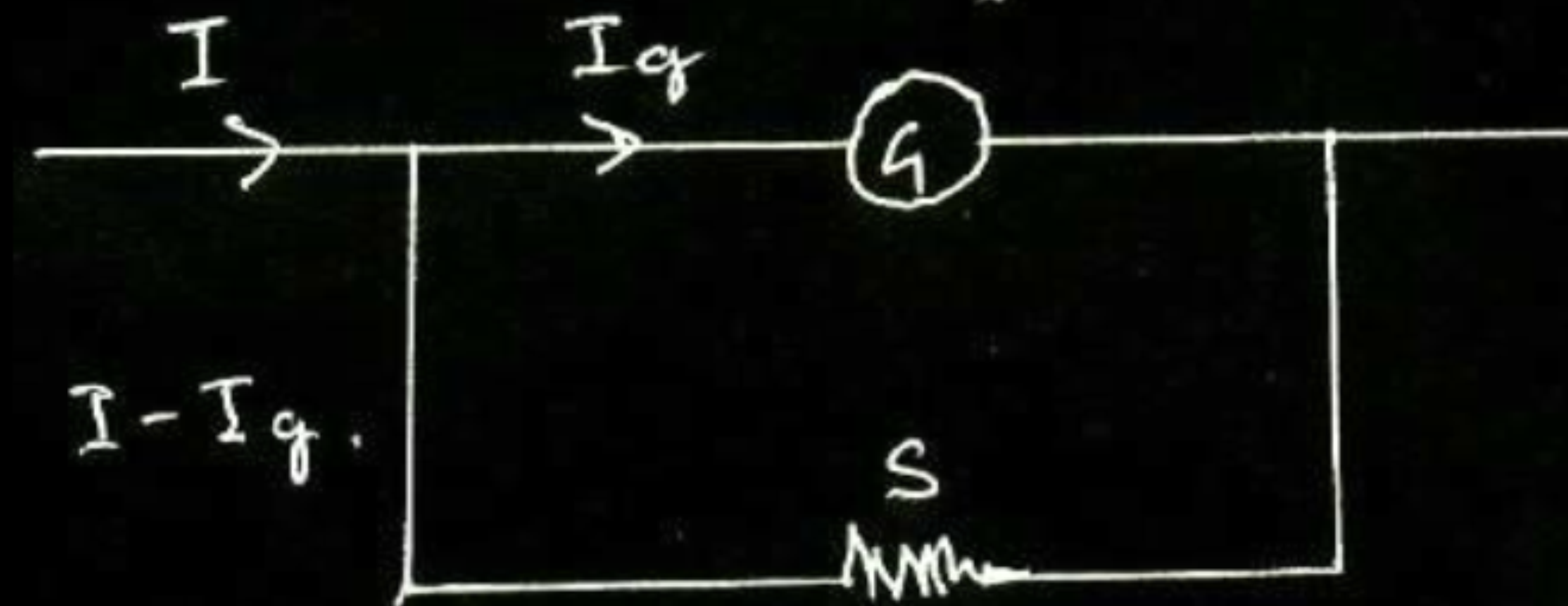
(A) move in a straight line.

- 22.** An ammeter of resistance $0.8\ \Omega$ can measure a current up to 1.0 A . Find the value of shunt resistance required to convert this ammeter to measure a current up to 5.0 A .

22. An ammeter of resistance 0.8Ω can measure a current up to 1.0 A . Find the value of shunt resistance required to convert this ammeter to measure a current up to 5.0 A .

2

Diagram	$\frac{1}{2}$
Formula	$\frac{1}{2}$
Calculation of value of shunt	1



$$\begin{aligned}
 &I_g = 1 \text{ A} \\
 &\text{Resistance of ammeter, } R_A = 0.8 \Omega \\
 &I_g R_A = (I - I_g) S \\
 &\Rightarrow 1 \times 0.8 = (5 - 1) S \\
 &\Rightarrow S = 0.2 \Omega
 \end{aligned}$$

$\frac{1}{2}$

1

$\frac{1}{2}$

- 36.** (a) A circular loop of radius R carries a current I . Obtain an expression for the magnetic field at a point on its axis at a distance x from its centre.
- (b) A conducting rod of length 2 m is placed on a horizontal table in north-south direction. It carries a current of 5 A from south to north. Find the direction and magnitude of the magnetic force acting on the rod. Given that the Earth's magnetic field at the place is 0.6×10^{-4} T and angle of dip is $\frac{\pi}{6}$.

5

(a) Magnetic field at a point on the axis of the current loop	3
(b) Magnitude and direction of the magnetic force	2

$$dB = \frac{\mu_0 Idl}{4\pi(x^2 + R^2)}$$

dB has two components dB_x and $dB_{\perp}$, perpendicular components from diametrically opposite elements dl cancel out, thus only dB_x components remain effective

$$dB_x = dB \cos\theta$$

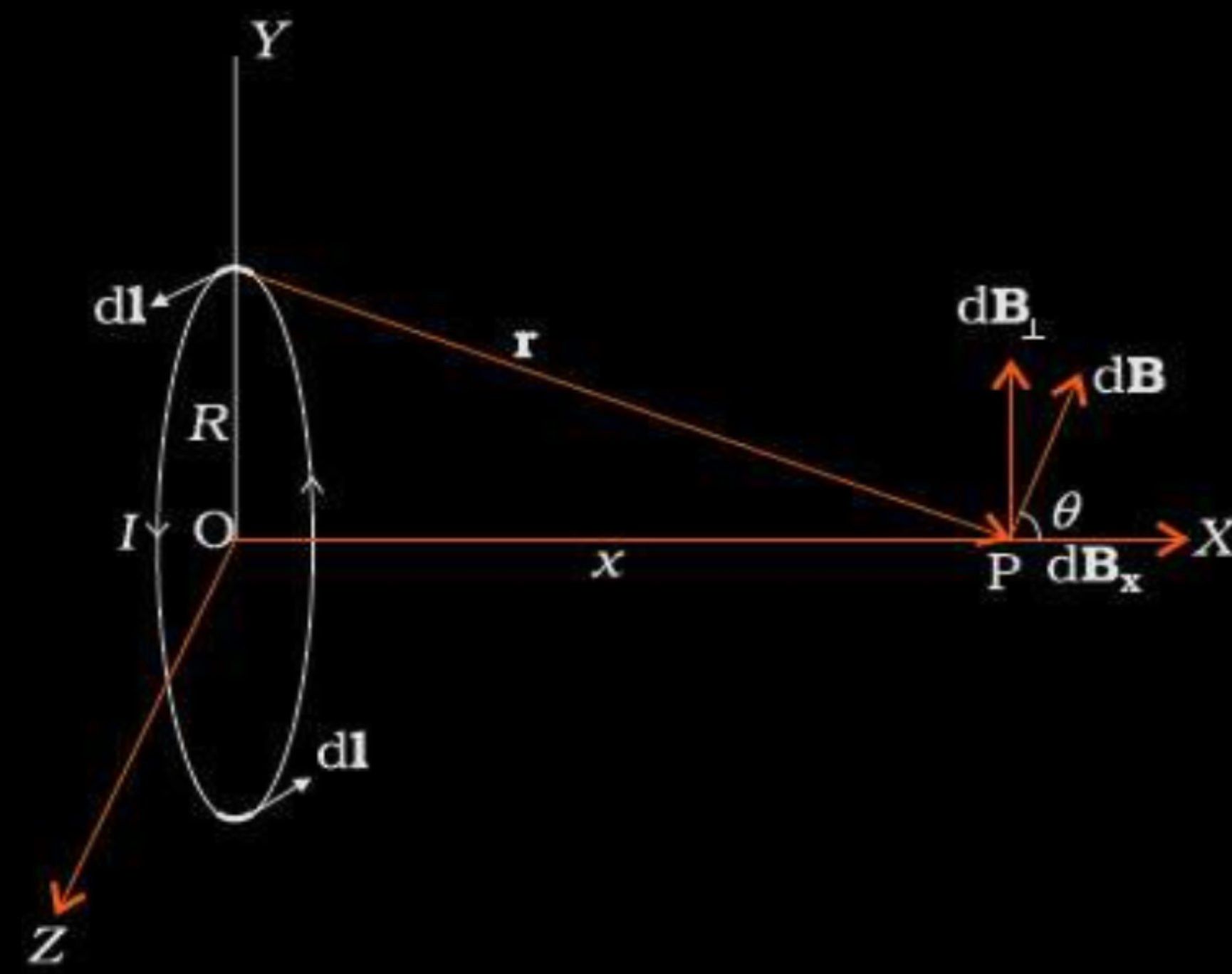
$$\text{and } \cos\theta = \frac{R}{(x^2 + R^2)^{\frac{1}{2}}}$$

$$\therefore B = \int dB_x$$

$$= \int_0^{2\pi R} \frac{\mu_0 IdlR}{4\pi(x^2 + R^2)^{\frac{3}{2}}}$$

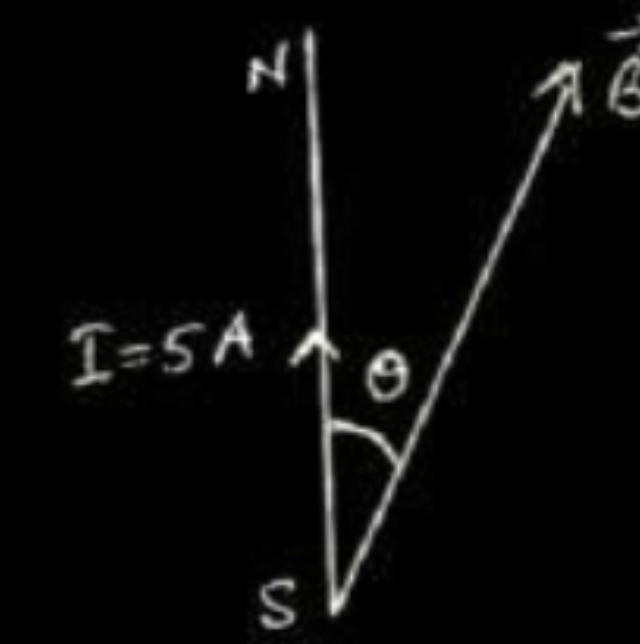
$$= \frac{\mu_0 IR^2}{2(x^2 + R^2)^{\frac{3}{2}}}$$

along the axis of the loop



(b)

(i)



$$\vec{F} = I[\vec{l} \times \vec{B}]$$

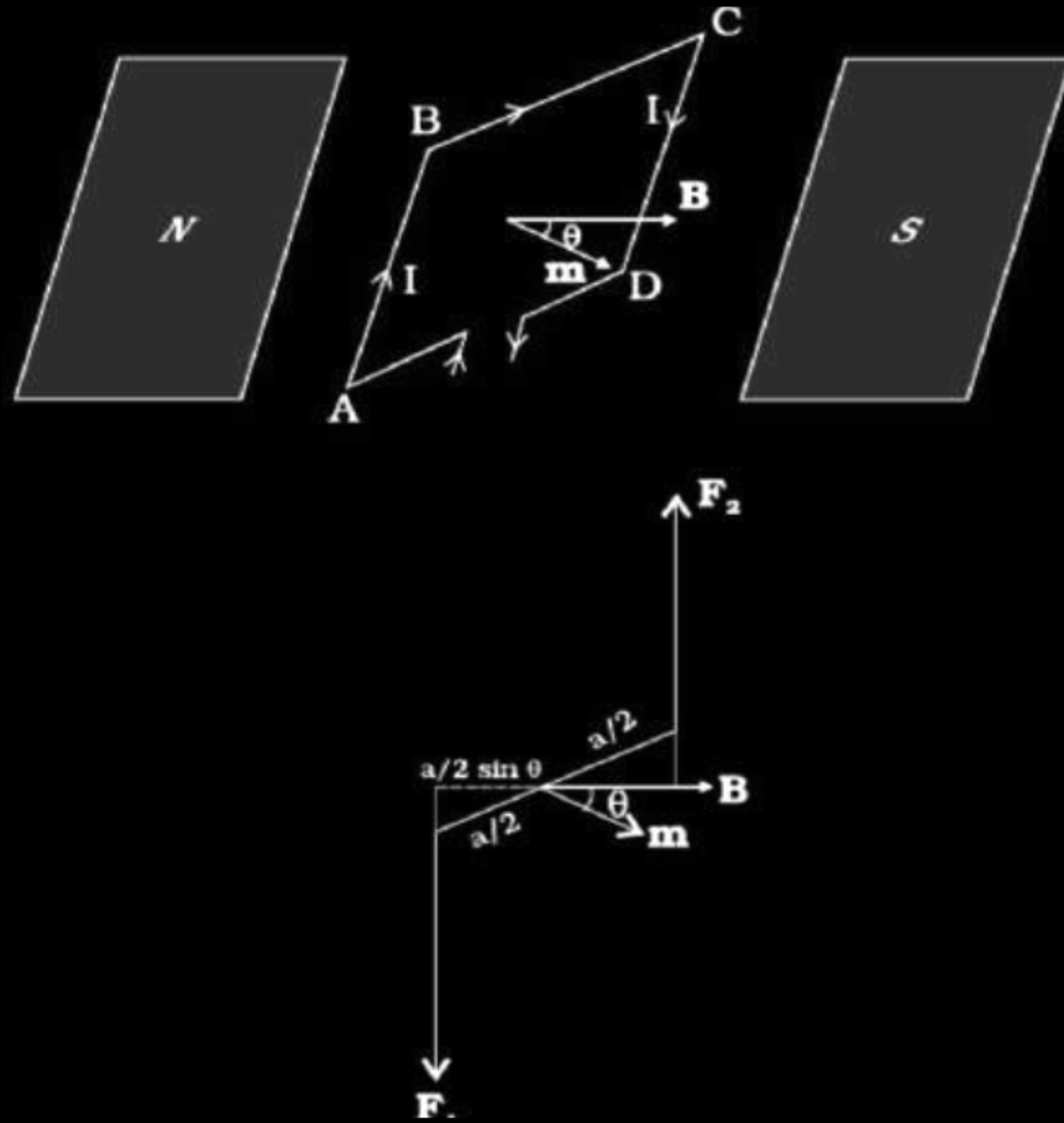
$$F = IlB \sin\theta$$

$$= 5 \times 2 \times 0.6 \times 10^{-4} \times 0.5 = 3 \times 10^{-4} \text{ N}$$

(ii) Towards east

- (a) Obtain the expression for the deflecting torque acting on the current carrying rectangular coil of a galvanometer in a uniform magnetic field. Why is a radial magnetic field employed in the moving coil galvanometer ?

(a)

 $\frac{1}{2}$

Arms AD and BC experience no net force whereas arm AB and CD experience forces which constitute torque
 $F_1 = F_2 = I b B$

Therefore magnitude of torque

$$\tau = \left(F_1 \frac{a}{2} + F_2 \frac{a}{2} \right) \sin \theta$$

$$\tau = I(ab) \sin \theta$$

where $ab = A$ (area of the loop)

$$\Rightarrow \tau = IAB \sin \theta$$

for N number of turns

$$\tau = NIAB \sin \theta$$

$$\tau = \vec{M} \times \vec{B}$$

Where magnetic moment $M = NIA$

Galvanometer has Radial magnetic field to increase the field strength and to make torque independent of orientation θ / it maximise the torque

 $\frac{1}{2}$ $\frac{1}{2}$ $\frac{1}{2}$

1

5. The magnetic dipole moment of a current carrying coil does *not* depend upon

1

- (A) number of turns of the coil.
- (B) cross-sectional area of the coil.
- (C) current flowing in the coil.
- (D) material of the turns of the coil.

5. The magnetic dipole moment of a current carrying coil does *not* depend upon

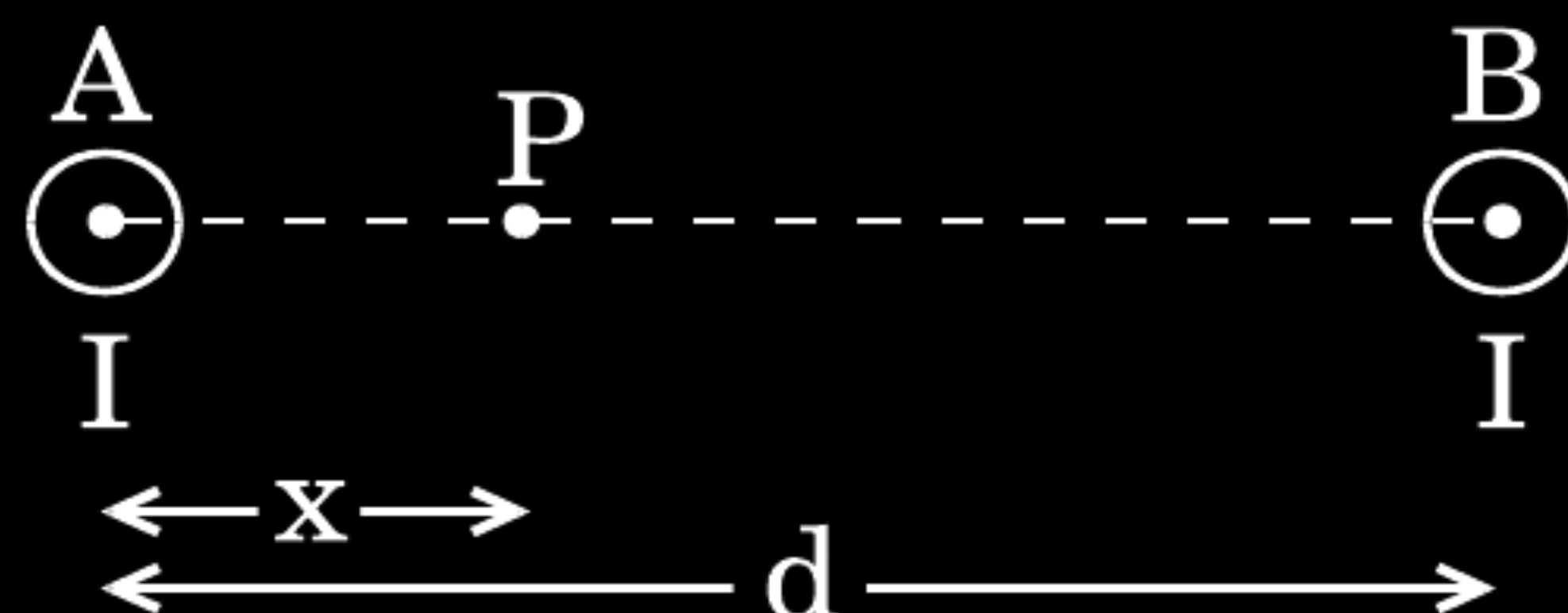
1

- (A) number of turns of the coil.
- (B) cross-sectional area of the coil.
- (C) current flowing in the coil.
- (D) material of the turns of the coil.

(D)

material of the turns of the coil

25. Two long straight parallel wires A and B separated by a distance d , carry equal current I flowing in same direction as shown in the figure.



- (a) Find the magnetic field at a point P situated between them at a distance x from one wire.
- (b) Show graphically the variation of the magnetic field with distance x for $0 < x < d$.

2

Magnetic field at point P

1 ½ mark

Curve

½ mark

a)

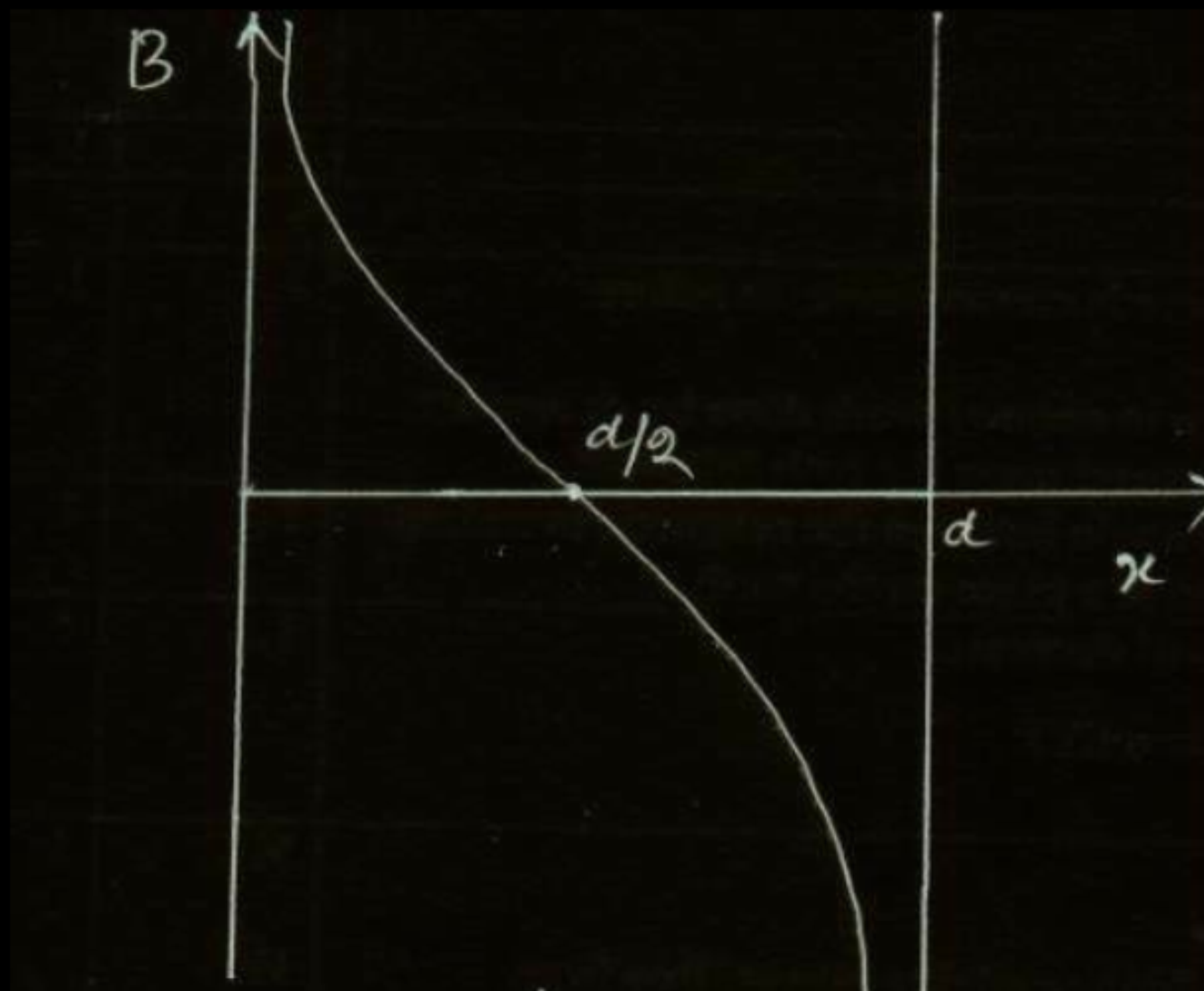
$$B = \frac{\mu_0 I}{2\pi x}$$

$$B_P = B_1 - B_2 = \frac{\mu_0 I}{2\pi x} - \frac{\mu_0 I}{2\pi(d-x)} = \frac{\mu_0 I(d-2x)}{2\pi(d-x)x}$$

$\frac{1}{2}$

1

b)



$\frac{1}{2}$

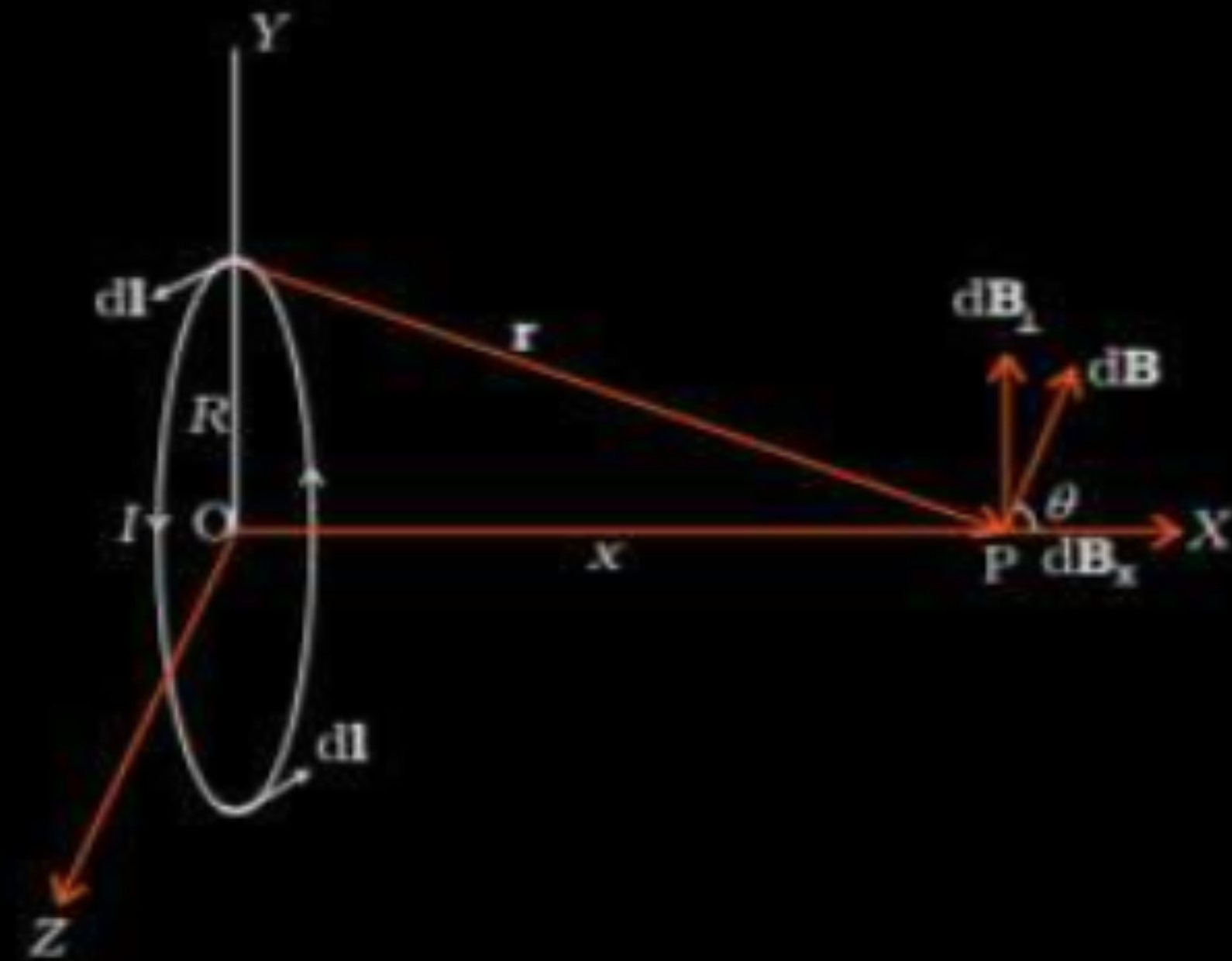
- 29.** (a) Write an expression of magnetic moment associated with a current (I) carrying circular coil of radius r having N turns.
- (b) Consider the above mentioned coil placed in YZ plane with its centre at the origin. Derive expression for the value of magnetic field due to it at point $(x, 0, 0)$.

3

a) Writing expression for magnetic moment	$\frac{1}{2}$ mark
b) Figure	$\frac{1}{2}$ mark
Magnetic field and calculation	2 mark

(a) magnetic moment = $M = NIA$
 $M = NI\pi r^2$

1/2



1/2

According to Biot-sevart law

$$\overrightarrow{dB} = \frac{\mu_0 I}{4\pi} \frac{|\overrightarrow{dl} \times \vec{r}|}{r^3}$$

1/2

$$dB = \frac{\mu_0 I}{4\pi} \frac{dl}{r^2}$$

$dB_{\perp}$ components due to diametrically opposite components cancel out. Only dB_x components remain

$$dB_x = \frac{\mu_0 I dl}{4\pi r^2} \cdot \cos\theta$$

1/2

$$B = \int dB_x$$

$$B = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}} \text{ (along } x \text{ axis)}$$

1

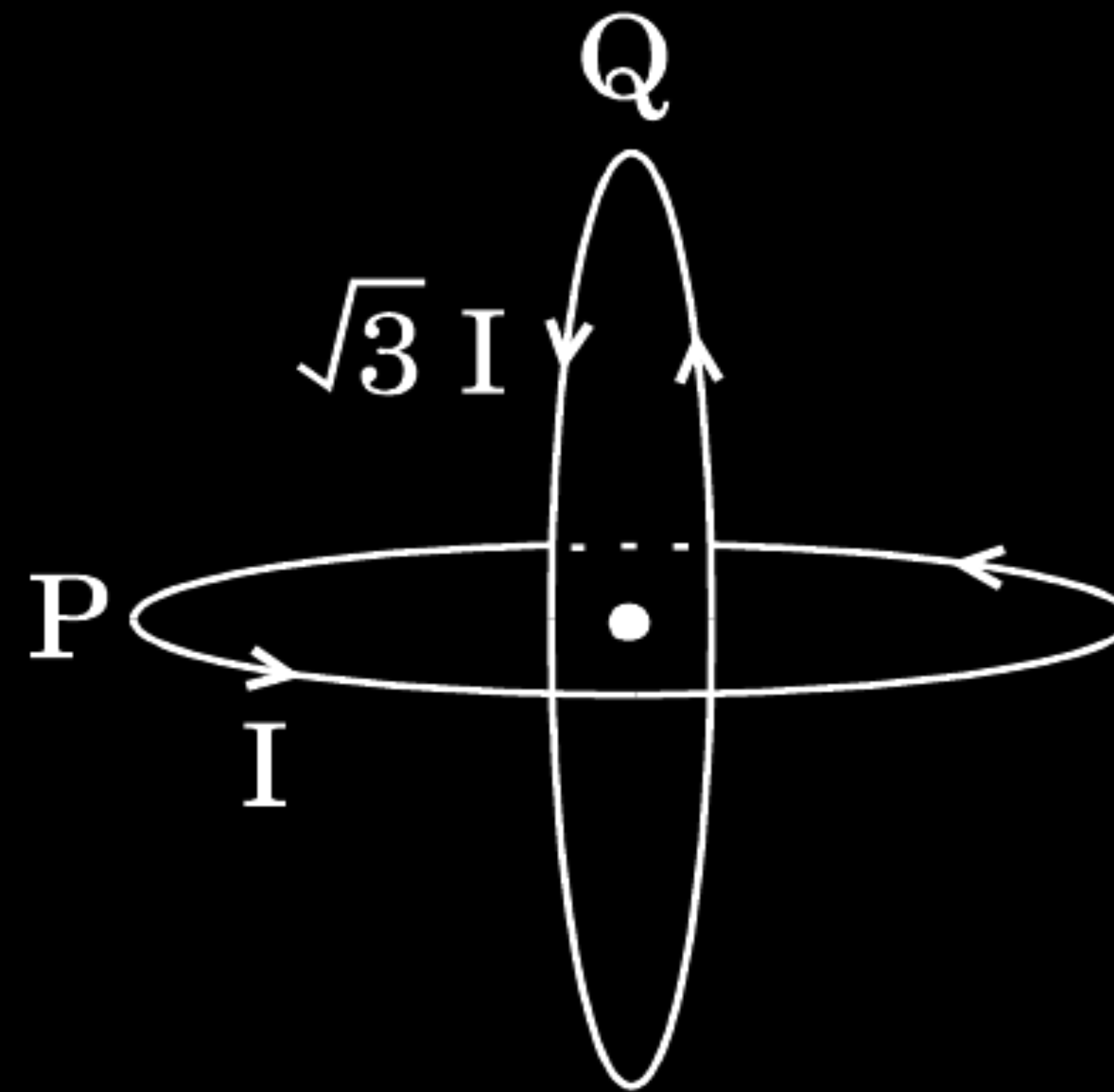
- (a) Define current sensitivity of a galvanometer. Write its expression.
- (b) A galvanometer has resistance G and shows full scale deflection for current I_g .
- (i) How can it be converted into an ammeter to measure current up to I_0 ($I_0 > I_g$) ?
- (ii) What is the effective resistance of this ammeter ?

3

a) Definition and expression	1 mark
b) Conversion of Galvanometer	
(i) into ammeter	1 mark
(ii) Effective resistance	1 mark

a) Deflection per unit current	$\frac{1}{2}$
$I_s = \frac{\theta}{I} = \frac{BNA}{K}$	$\frac{1}{2}$
b) (i) By connecting a low resistance (R_s) in parallel to galvanometer such that	$\frac{1}{2}$
$(I_0 - I_g)R_s = I_g G$	$\frac{1}{2}$
(ii) effective resistance	
$\frac{1}{R_A} = \frac{1}{R_s} + \frac{1}{G} = \frac{G + R_s}{R_s G}$	
$\therefore R_A = \frac{R_s G}{G + R_s}$	1

Two identical coils P and Q each of radius R are lying in perpendicular planes such that they have a common centre. Find the magnitude and direction of the magnetic field at the common centre when they carry currents equal to I and $\sqrt{3}$ I respectively.



Expression for resultant magnetic field -

$1\frac{1}{2}$

Direction of resultant field -

$\frac{1}{2}$

$$B_1 = \frac{\mu_0 I}{2R}$$

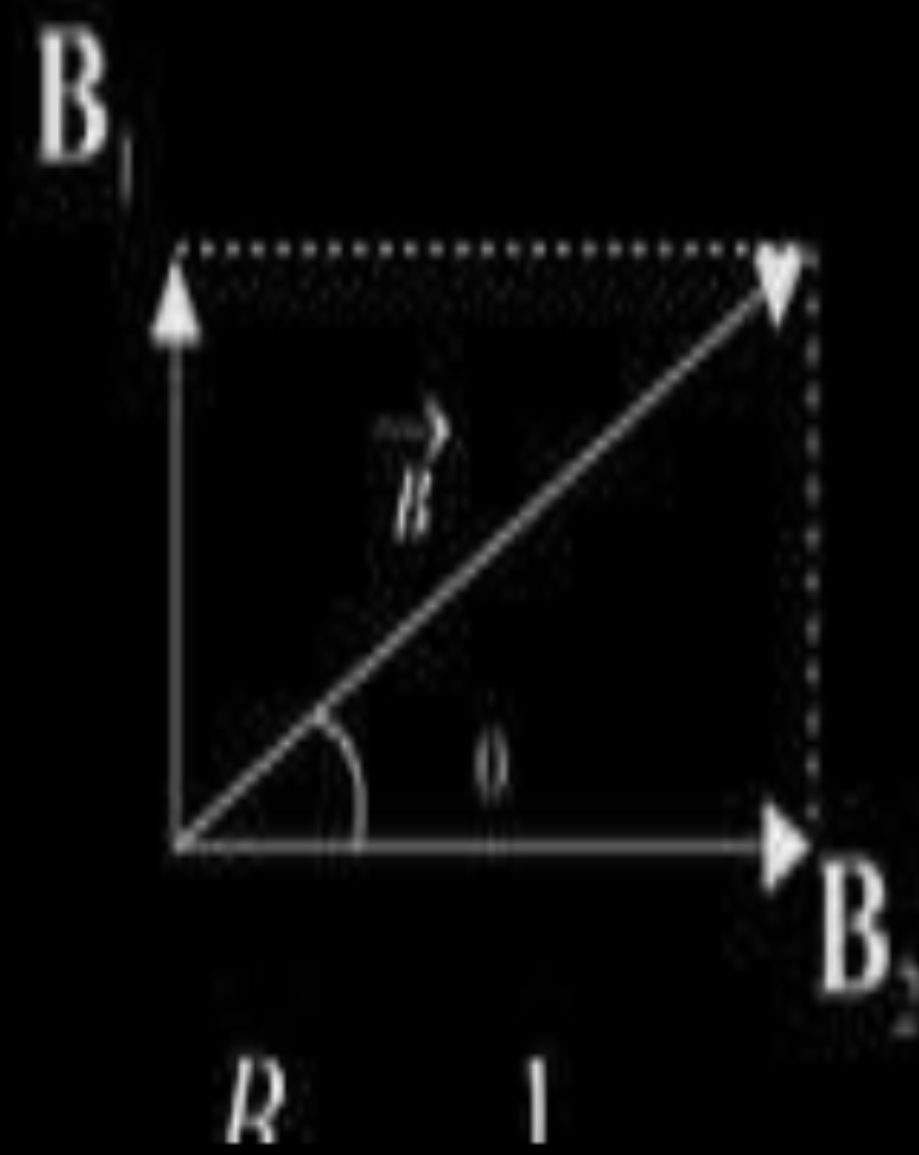
$$B_2 = \frac{\mu_0 \sqrt{3} I}{2R}$$

$$B = \sqrt{B_1^2 + B_2^2}$$

$$= \sqrt{\left(\frac{\mu_0 I}{2R}\right)^2 + \left(\frac{\mu_0 \sqrt{3} I}{2R}\right)^2}$$

$$B = \frac{\mu_0 I}{R}$$

For Direction



$$\tan \theta = \frac{B_1}{B_2} \Rightarrow \theta = 30^\circ$$

- (a) Obtain the conditions under which an electron does not suffer any deflection while passing through a magnetic field.
- (b) Two protons P and Q moving with the same speed pass through the magnetic fields $\vec{B}_1$ and $\vec{B}_2$ respectively, at right angles to the field directions. If $|\vec{B}_2| > |\vec{B}_1|$, which of the two protons will describe the circular path of smaller radius ? Explain.

a) Condition for no deflection -	1
b) Conclusion for greater radius -	1

(a) No deflection if electron moves parallel or anti parallel to the magnetic field	1
(b)	
$r = \frac{mv}{Bq}$	1/2
$\frac{r_1}{r_2} = \frac{B_2}{B_1}$	
As $B_1 < B_2$ $r_2 < r_1$	1/2
Alternatively	
$[r = \frac{l}{B}]$	1/2
r_2 is smaller because B	1/2

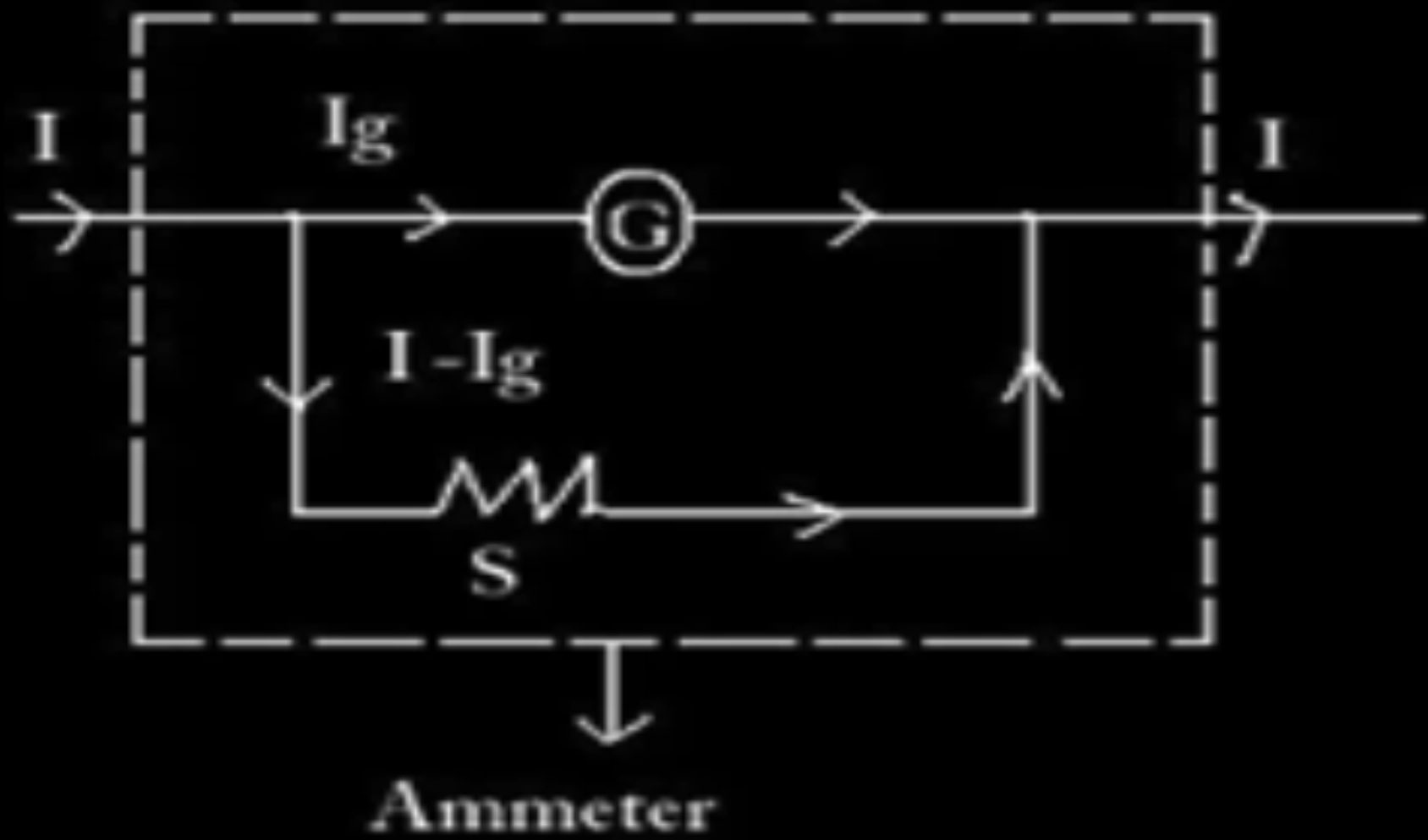
State the principle of a moving coil galvanometer. Explain its working and obtain the expression for the deflection produced due to the current passed through the coil. Define current sensitivity.

OR

Explain how a galvanometer can be converted into an ammeter of a given range. Derive an expression for shunt resistance and current for full scale deflection. Find the effective resistance of the ammeter.

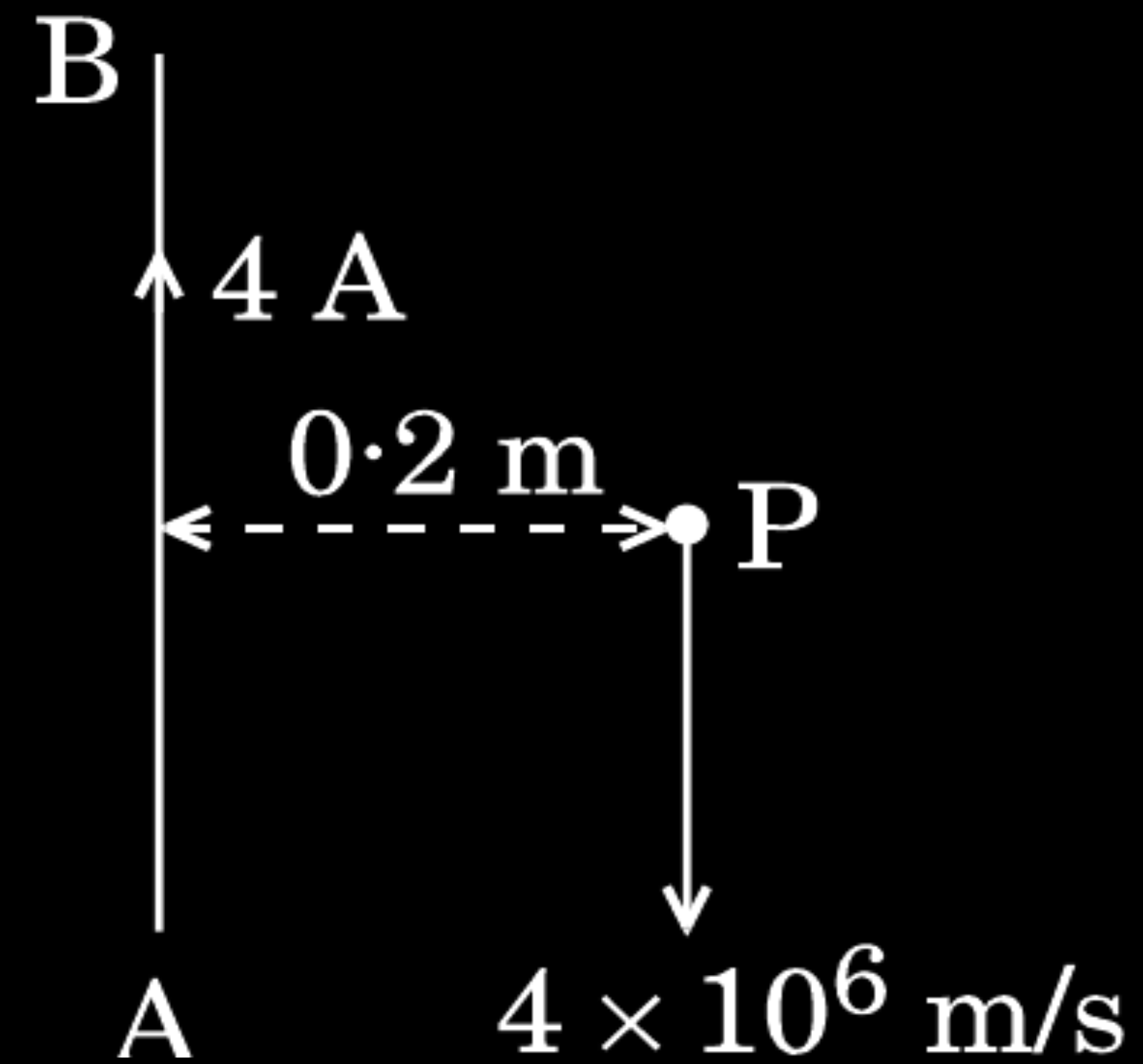
Principle -	1
Working & expression for deflection -	1
Current Sensitivity -	1

Conversion of galvanometer into ammeter -	1
Expression for shunt resistance -	1
Effective resistance of ammeter -	1

Principle - A current carrying coil experiences a torque in a magnetic field.	1
Working - When current is passed through the coil torque produced is $\tau = NIAB \sin \theta = NIAB \quad (\theta=90^\circ)$ Restoring torque $\tau' = k\phi$ At equilibrium $\tau = \tau'$ $NIAB = k\phi$ $\phi = \left(\frac{NAB}{k} \right) I$ Current sensitivity: $\frac{\phi}{I} = \frac{NAB}{k}$ [Alternatively: It is deflection per unit current] A galvanometer is converted into an ammeter when a suitable shunt resistance is connected in parallel with the galvanometer. 	1/2
[Award full one mark even if no figure drawn or only the figure is drawn]	1/2

$G I_g = (I - I_g) S$	1/2
$S = \left(\frac{I_g}{I - I_g} \right) G$	1/2
$R_{eq} = \frac{GS}{G + S}$	1

A long straight wire AB carries a current of 4 A. A proton P travels at $4 \times 10^6 \text{ ms}^{-1}$ parallel to the wire 0.2 m from it and in a direction opposite to the current as shown in the figure. Calculate the force which the magnetic field due to the current carrying wire exerts on the proton. Also specify its direction.



- | | |
|-------------------------------|---------------|
| i) Formula for Magnetic field | $\frac{1}{2}$ |
| ii) Formula for force | $\frac{1}{2}$ |
| iii) Calculation & Results | $\frac{1}{2}$ |
| iv) Direction of force | $\frac{1}{2}$ |

Magnetic field at Point P due current carrying straight wire AB

$$B = \frac{\mu_0}{2\pi} \frac{I}{r}$$

1/2

Force acting on the moving proton in the magnetic field

$$F = Bqv \sin \theta$$

1/2

$$\text{Therefore } F = \frac{\mu_0}{2\pi} \frac{I}{r} \times qv \sin \theta$$

$$= \frac{2 \times 10^{-7} \times 4 \times 1.6 \times 10^{-19} \times 4 \times 10^6 \sin 90}{0.2}$$

$$= 2.56 \times 10^{-18} \text{ N}$$

1/2

Direction of force at point P is towards right. (away from AB)

1/2

A proton, a deuteron and an alpha particle, are accelerated through the same potential difference and then subjected to a uniform magnetic field $\vec{B}$, perpendicular to the direction of their motions. Compare (i) their kinetic energies, and (ii) if the radius of the circular path described by proton is 5 cm, determine the radii of the paths described by deuteron and alpha particle.

i) Relation between K.E. and accelerating potential	$\frac{1}{2}$
ii) Comparison of Kinetic Energies	1
iii) Determination of radii of path of deuteron and alpha particle	$1\frac{1}{2}$

i)

Since $qV = \frac{1}{2}mv^2$

for proton $\frac{1}{2}m_p v_1^2 = qV$

For deuteron $\frac{1}{2}m_d v_2^2 = qV$

For alpha particle $\frac{1}{2}m_\alpha v_3^2 = 2qV$

$(K.E.)_p : (K.E.)_d : (K.E.)_\alpha : 1 : 1 : 2$

ii)

$$Bqv = \frac{mv^2}{r}$$

$$\text{so } r = \frac{mv}{Bq} = 5 \text{ cm;}$$

$$r_1 : r_2 : r_3 = 5 : 5\sqrt{2} : 5\sqrt{2} \text{ OR } 1 : \sqrt{2} : \sqrt{2}$$

$\frac{1}{2}$

1

$\frac{1}{2}$

1

- (a) Briefly explain how a galvanometer is converted into an ammeter.
- (b) A galvanometer coil has a resistance of $15\ \Omega$ and it shows full scale deflection for a current of $4\ \text{mA}$. Convert it into an ammeter of range 0 to $6\ \text{A}$.

i) Conversion of a galvanometer into an ammeter	$1\frac{1}{2}$
ii) Formula for shunt	$\frac{1}{2}$
iii) Calculation and Result	1

a)	By connecting a small resistance called shunt (S) in parallel to coil of the galvanometer The value of S is related to the maximum current (I) to be measured as $S = I_g G / I - I_g$. (Note : If the student just draws the diagram, full marks may be awarded).	1 1/2
b)	$G = 15 \Omega$ $I_g = 4 \times 10^{-3} \text{ A}$ $I = 6 \text{ A}$ $I_g G = (I - I_g) S$ $S = \frac{I_g G}{I - I_g} = \frac{4 \times 10^{-3} \times 15}{6 - 4 \times 10^{-3}}$ $= 0.01 \Omega$	 1/2 1/2 1/2

- (a) Briefly explain how a galvanometer is converted into a voltmeter.
- (b) A voltmeter of a certain range is constructed by connecting a resistance of $980\ \Omega$ in series with a galvanometer. When the resistance of $470\ \Omega$ is connected in series, the range gets halved. Find the resistance of the galvanometer.

i) Conversion of a galvanometer into a voltmeter	$1\frac{1}{2}$
ii) Formula	$\frac{1}{2}$
iii) Calculation and results	1

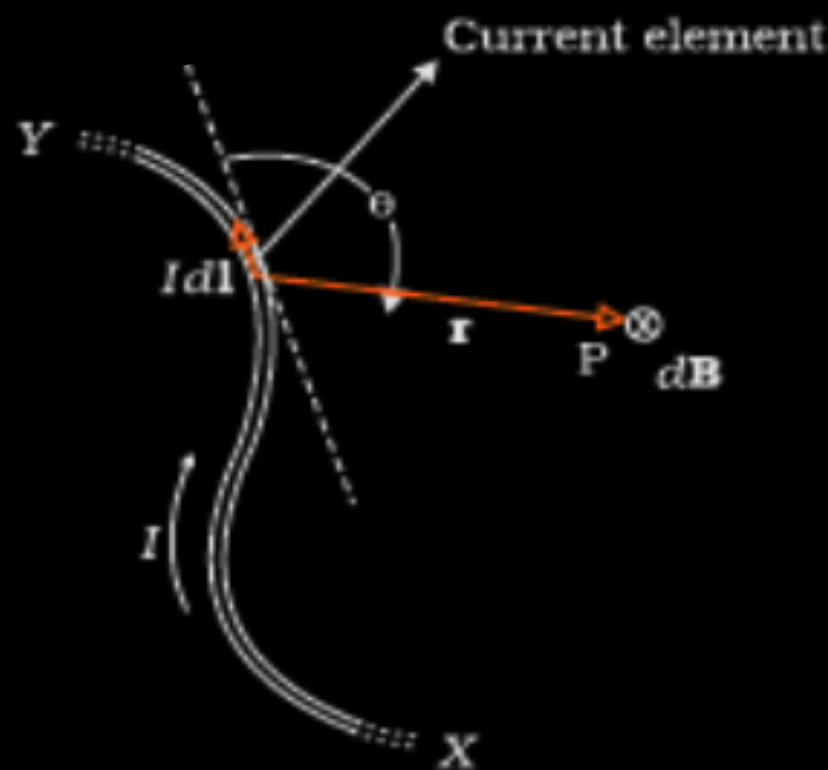
a) A galvanometer may be converted into voltmeter by connecting a high value resistance R in series with coil of the galvanometer. The value of (R) is related to the maximum voltage (V) to be measured as $R = \frac{V}{I_g} - G$ Note - Award full marks if a student just draws the labelled diagram.	1½
b) $I_g = \frac{V}{R_g + R}$ $\frac{V}{R_g + 980} = \frac{V}{2(R_g + 470)}$ $2R_g + 940 = R_g + 980$ $R_g = 40\Omega$	½ ½ ½

- (a) State and explain the law used to determine magnetic field at a point due to a current element. Derive the expression for the magnetic field due to a circular current carrying loop of radius r at its centre.
- (b) A long wire with a small current element of length 1 cm is placed at the origin and carries a current of 10 A along the X-axis. Find out the magnitude and direction of the magnetic field due to the element on the Y-axis at a distance 0.5 m from it.

- | |
|--|
| <p>a. Statement of the law and expression for the magnetic field - 1+2</p> <p>b. Finding the magnitude and direction of magnetic field - $1\frac{1}{2} + \frac{1}{2}$</p> |
|--|

a. According to Biot Savart law, the magnetic field due to a current element is given by

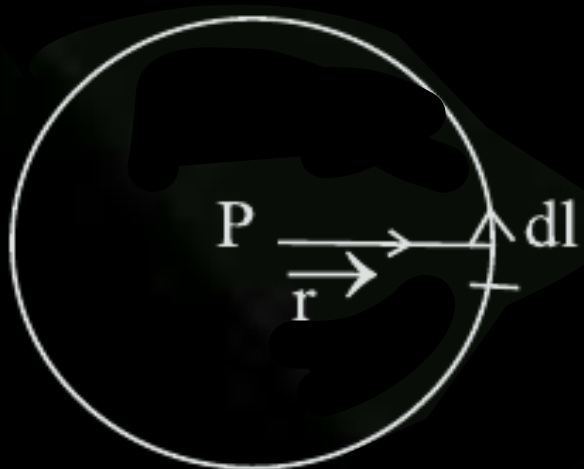
$$\vec{dB} = \frac{\mu_o}{4\pi} I \frac{\vec{dl} \times \vec{r}}{[\vec{r}]^3}$$



1

Alternatively award this 1 mark if a student makes statement of Biot Savart law.

Derivation of magnetic field



Magnetic field due to current element dl

$$\vec{dB} = \frac{\mu_o}{4\pi} I \frac{\vec{dl} \times \hat{r}}{[\vec{r}]^2} \quad \text{where } \hat{r} \text{ is a unit vector along } \vec{r}$$

$$\vec{r} \perp \vec{dl}$$

Direction of $\vec{dB}$ is perpendicular, pointing outward.

1/2

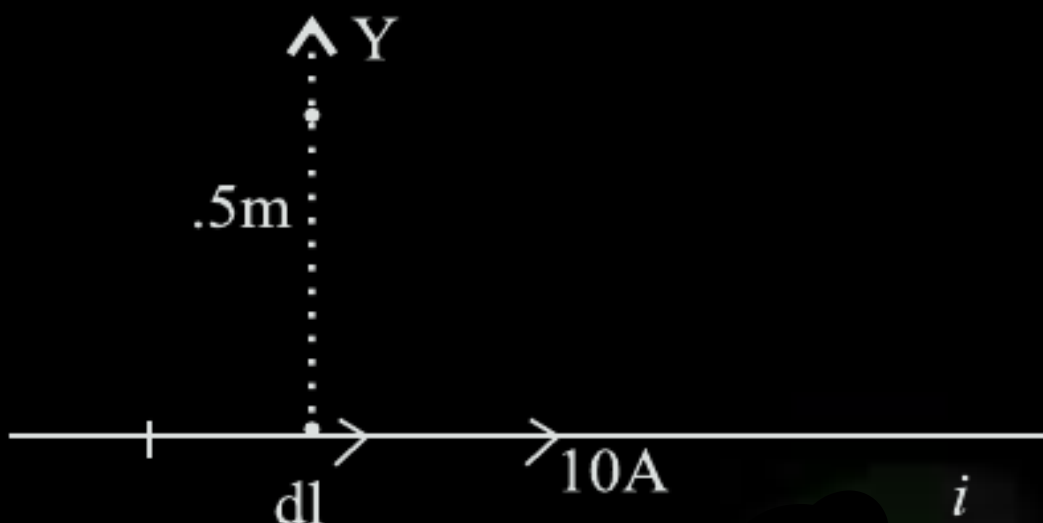
∴ Field due to the whole loop

$$\left| \vec{B} \right| = \int dB = \frac{\mu_o I}{4\pi r^2} \int dl = \frac{\mu_o I}{4\pi r^2} \times 2\pi r$$

$$\left| \vec{B} \right| = \frac{\mu_o I}{2r}$$

b.

$$\begin{aligned} \left| \vec{dB} \right| &= \frac{\mu_o I}{4\pi} \frac{dl \sin \theta}{[\vec{r}]^2} \\ &= \frac{4\pi \times 10^{-7}}{4\pi} \times \frac{10 \times (1 \times 10^{-2}) \times \sin 90^\circ}{(0.5)^2} \\ &= 4 \times 10^{-8} T \end{aligned}$$



1/2

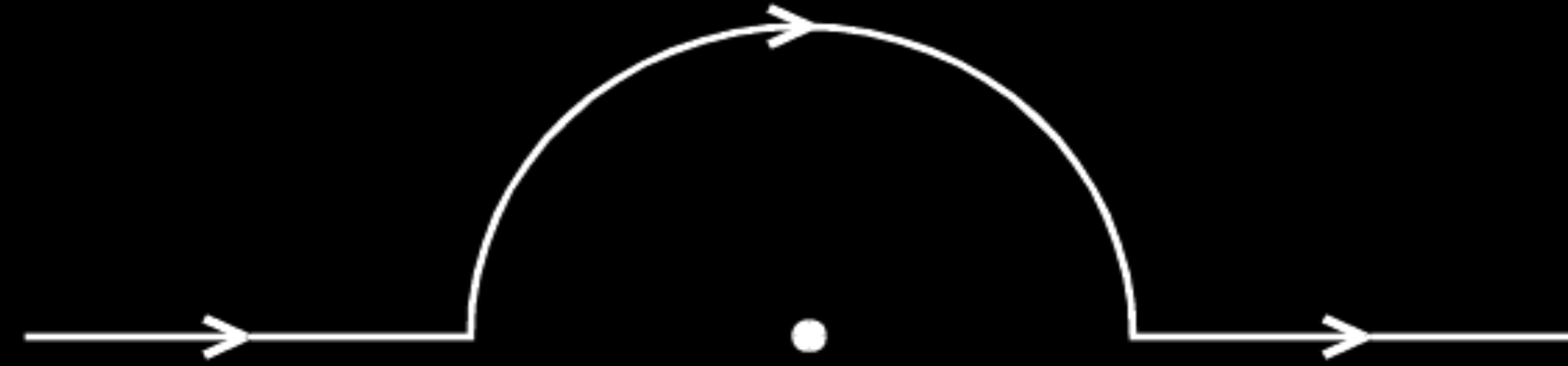
1/2

1/2

1

1/2

- (a) Derive the expression for the magnetic field due to a current carrying coil of radius r at a distance x from the centre along the X-axis.
- (b) A straight wire carrying a current of 5 A is bent into a semicircular arc of radius 2 cm as shown in the figure. Find the magnitude and direction of the magnetic field at the centre of the arc.



- | | | |
|----|--|----------------|
| a) | Derivation of expression with diagram | 3 |
| b) | Calculation of magnitude of magnetic field at the center of the arc. | $1\frac{1}{2}$ |
| | Direction of field | $\frac{1}{2}$ |

According to Biot Savart law

$$|\vec{dB}| = \frac{\mu_0}{4\pi} \frac{I dl \sin 90^\circ}{|\vec{r}_1|^2}$$

$$\text{Where } r_1 = \sqrt{x^2 + r^2}$$

$$|\vec{dB}| = \frac{\mu_0}{4\pi} \frac{I dl}{(x^2 + r^2)^{\frac{3}{2}}}$$

Direction of $\vec{dB}$ is perpendicular to $d\vec{l}$ and $\vec{r}_1$.

It has components dB_x and $dB_\perp$. The components $dB_\perp$ due to the whole coil cancel out in pairs.

$$\text{Net field } B = \int dB_x = \int dB \cos\theta$$

$$\vec{B} = \frac{\mu_0 I r^2}{2 (r^2 + x^2)^{\frac{3}{2}}} \hat{i}$$

$$\begin{aligned} \text{b) } B &= \frac{\mu_0 I}{4r} \\ &= \frac{4\pi \times 10^{-7} \times 5}{4 \times 2 \times 10^{-2}} \\ &= 7.85 \times 10^{-5} \text{ T} \end{aligned}$$

The field is directed inwards perpendicular to the plane of the page.

1/2

1/2

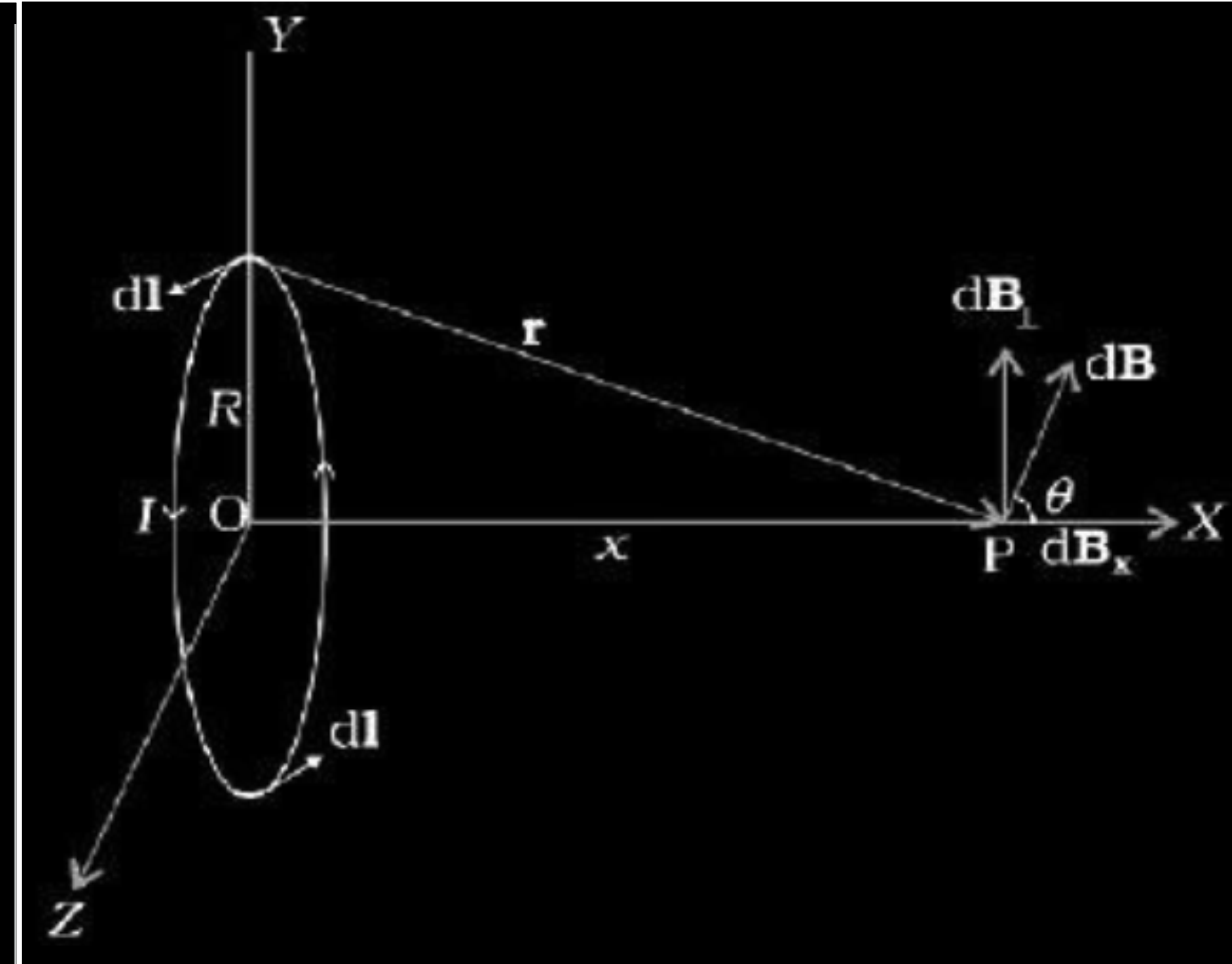
1

1/2

1/2

1/2

1/2



A proton is accelerated through a potential difference V , subjected to a uniform magnetic field acting normal to the velocity of the proton. If the potential difference is doubled, how will the radius of the circular path described by the proton in the magnetic field change ?

- | | |
|--|-----|
| • For writing the expression for radius | 1/2 |
| • To find the change in the radius of the circular orbit | 1/2 |

$$r = \frac{1}{B} \sqrt{\frac{2mV}{q}}$$

1/2

$$r' = \sqrt{2}r$$

1/2

Alternatively $r \propto \sqrt{V}$

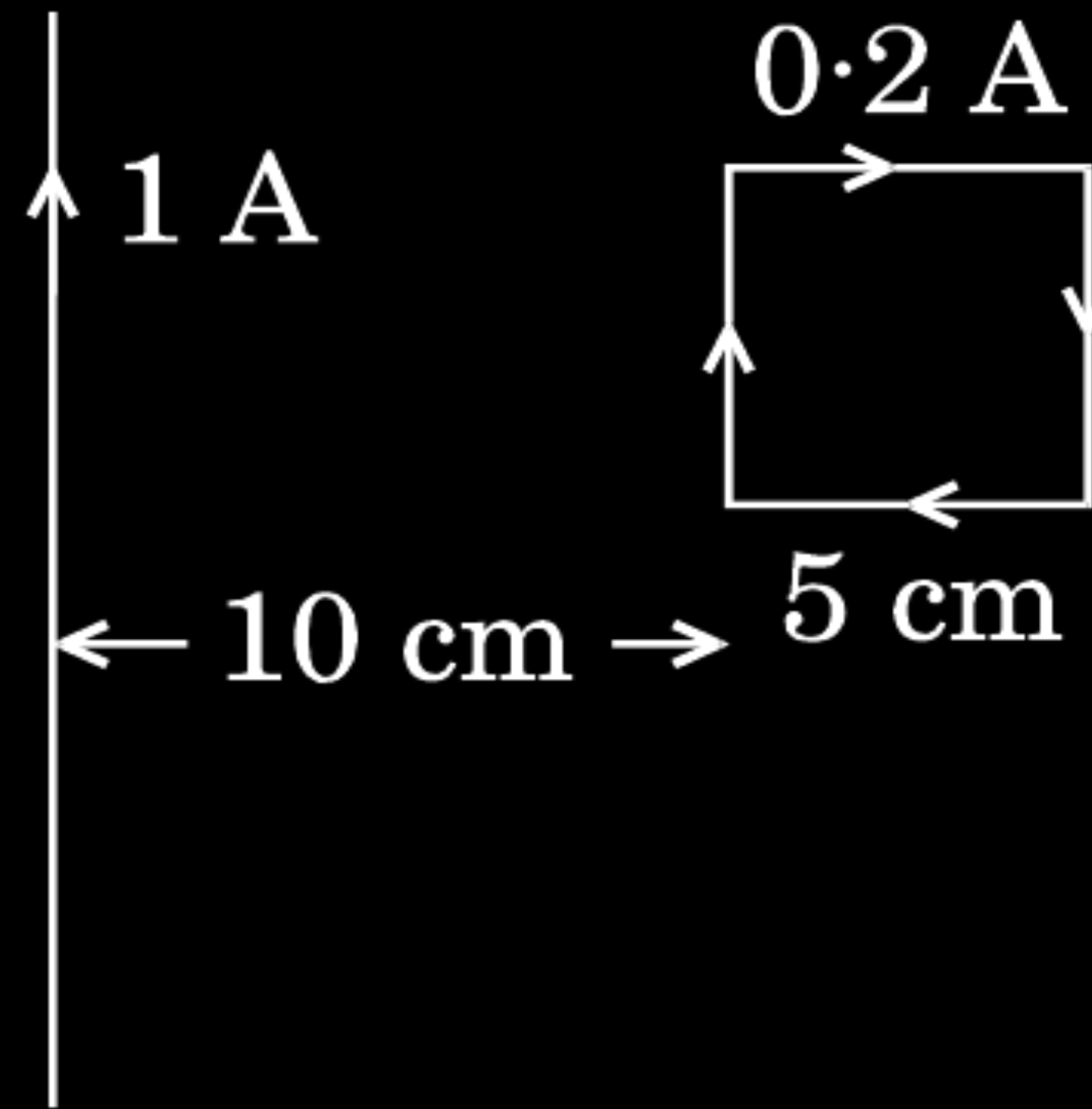
$$\frac{r'}{r} = \sqrt{2}$$

1/2

1/2

Even if student writes $Bqv = \frac{mv^2}{r}$ award one mark

- (b) A square loop of sides 5 cm carrying a current of 0.2 A in the clockwise direction is placed at a distance of 10 cm from an infinitely long wire carrying a current of 1 A as shown. Calculate (i) the resultant magnetic force, and (ii) the torque, if any, acting on the loop.



Force of attraction experienced by the length SP of the loop per unit length

$$F_1 = \frac{2\mu_0 I_1 I_2}{4\pi r_1}$$

$$f_1 = \frac{2 \times 10^{-7} \times 1 \times 0.2}{10 \times 10^{-2}} = 4 \times 10^{-7} \text{ Nm}^{-1}$$

Force is attractive

$$f_2 = \frac{2\mu_0 I_1 I_2}{4\pi r_2}$$

$$f_2 = \frac{2 \times 10^{-7} \times 1 \times 0.2}{15 \times 10^{-2}} = 2.6 \times 10^{-7} \text{ Nm}^{-1}$$

Force is repulsive

So the net force experienced by the loop is (per unit length)

$$f = (f_1 - f_2)$$

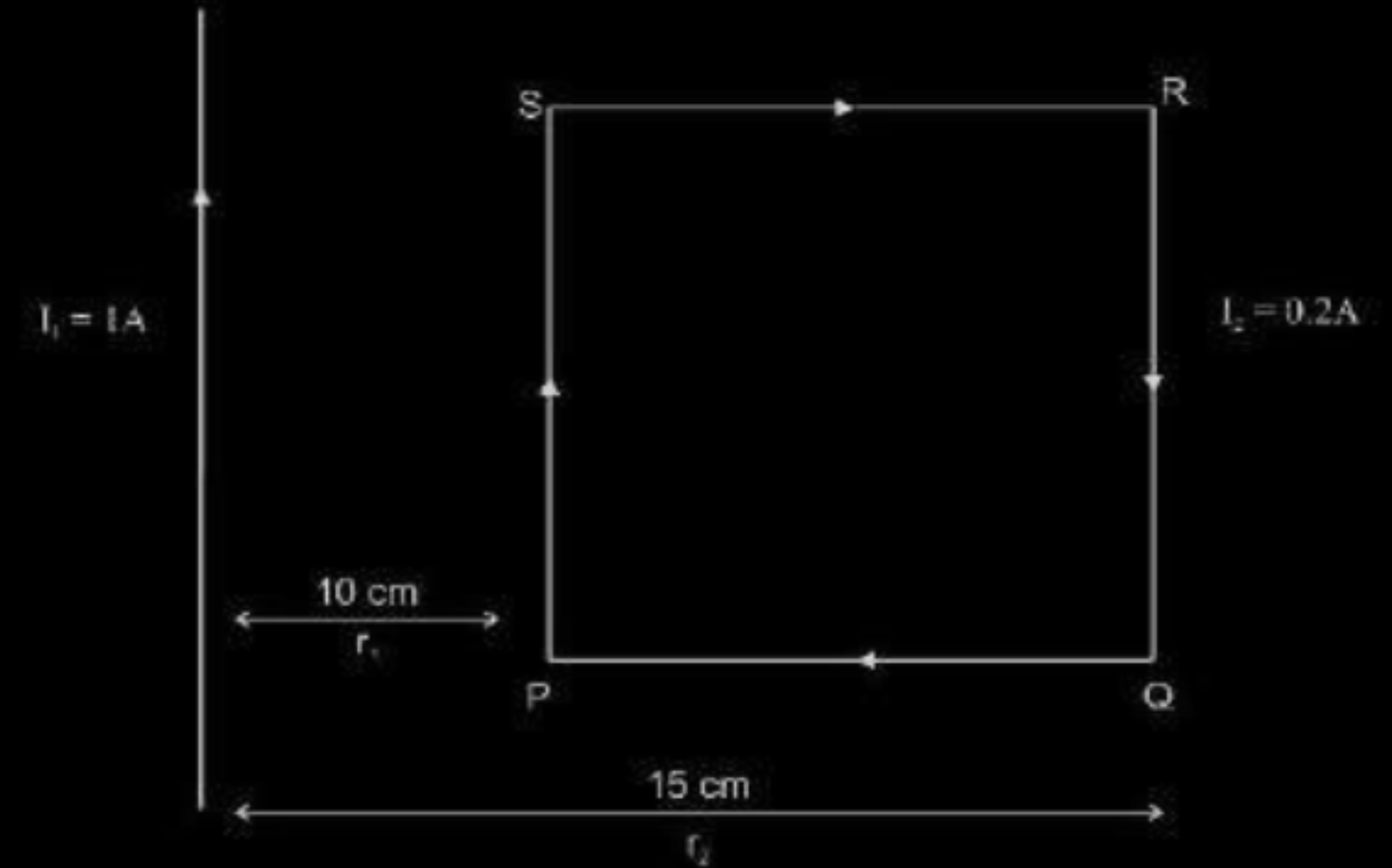
Total force experienced by the loop is:

$$F = (f_1 - f_2)l = (1.4 \times 10^{-7}) \times 5 \times 10^{-2}$$

$$F = 7 \times 10^{-7} \text{ N}$$

Net force is attractive in nature

As the lines of action of forces coincide torque is zero.



An α -particle and a proton of the same kinetic energy are in turn allowed to pass through a magnetic field $\vec{B}$, acting normal to the direction of motion of the particles. Calculate the ratio of radii of the circular paths described by them.

Formula	$\frac{1}{2}$
Calculation of ratio of radii	$1\frac{1}{2}$

$$\text{radius } r = \frac{mv}{qB} = \frac{\sqrt{2mk}}{qB}$$

$$K_{\alpha} = K_{\text{proton}}$$

$$M_{\alpha} = 4 m_p$$

$$q_{\alpha} = 2q_p$$

$$\frac{r_{\alpha}}{r_p} = \frac{\frac{\sqrt{2m_{\alpha} K}}{q_{\alpha} B}}{\frac{\sqrt{2m_p K}}{q_p B}}$$

$$= \sqrt{\frac{m_{\alpha}}{m_p}} \times \sqrt{\frac{q_p}{q_{\alpha}}}$$

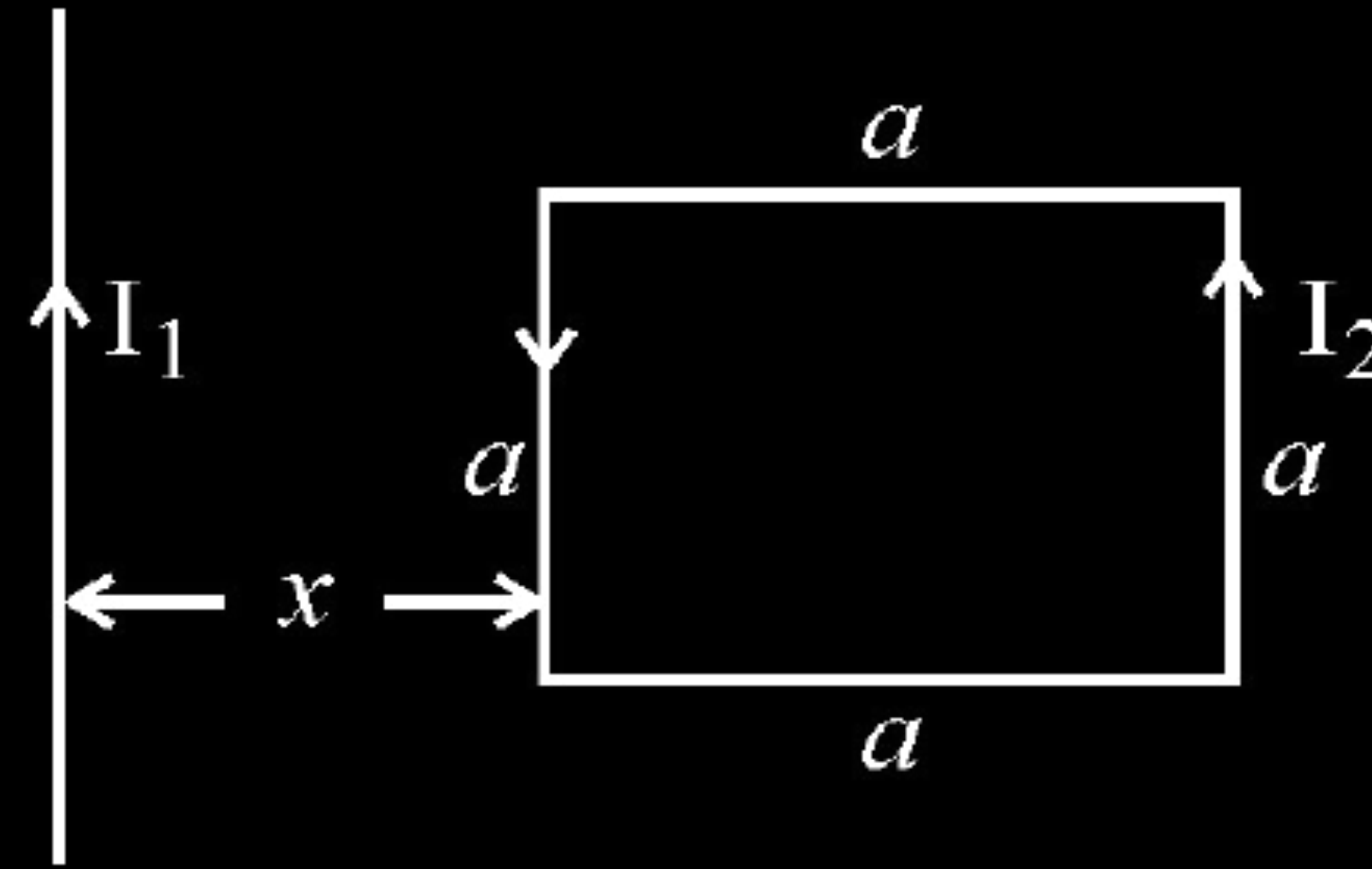
$$= \sqrt{4} \times \frac{1}{2} = 1$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

$$\frac{1}{2}$$

- (b) A square loop of side 'a' carrying a current I_2 is kept at distance x from an infinitely long straight wire carrying a current I_1 as shown in the figure. Obtain the expression for the resultant force acting on the loop.



(b) Force per unit length between two parallel straight conductors

$$F = \frac{\mu_0}{4\pi} \frac{2I_1 I_2}{d}$$

Force on the part of the loop which is parallel to infinite straight wire and at a distance x from it.

$$F_1 = \frac{\mu_0}{2\pi} \frac{I_1 I_2 a}{x} \quad (\text{away from the infinite straight wire})$$

Force on the part of the loop which is at a distance $(x + a)$ from it

$$F_2 = \frac{\mu_0}{2\pi} \frac{I_1 I_2 a}{(x + a)} \quad (\text{towards the infinite straight wire})$$

Net force $F = F_1 - F_2$

$$F = \frac{\mu_0}{2\pi} \frac{I_1 I_2 a}{x} \left[\frac{1}{x} - \frac{1}{x + a} \right]$$

$$F = \frac{\mu_0}{2\pi} \frac{I_1 I_2 a^2}{x(x + a)} \quad (\text{away from the infinite straight wire})$$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

- (a) Derive the expression for the torque acting on a current carrying loop placed in a magnetic field.
- (b) Explain the significance of a radial magnetic field when a current carrying coil is kept in it.

(a) Derivation of the expression for torque	2
(b) Significance of radial magnetic field	1

Force on arm AB = $F_1 = I b B$ (directed into the plane of the loop)
 Force on arm CD = $F_2 = I b B$ (directed out of the plane of the loop)

Therefore the magnitude of the torque on the loop due to these pair of forces

$$\begin{aligned}\tau &= F_1 \frac{a}{2} + F_2 \frac{a}{2} \\ &= I (ab) B \\ &= IAB = mB \\ & \text{(} A = ab = \text{area of the loop)}\end{aligned}$$

Alternatively

Also accept if the student does calculations for the general case and obtains the result

(b) When a current carrying coil is kept in a radial magnetic field the corresponding moving coil galvanometer would have a linear scale

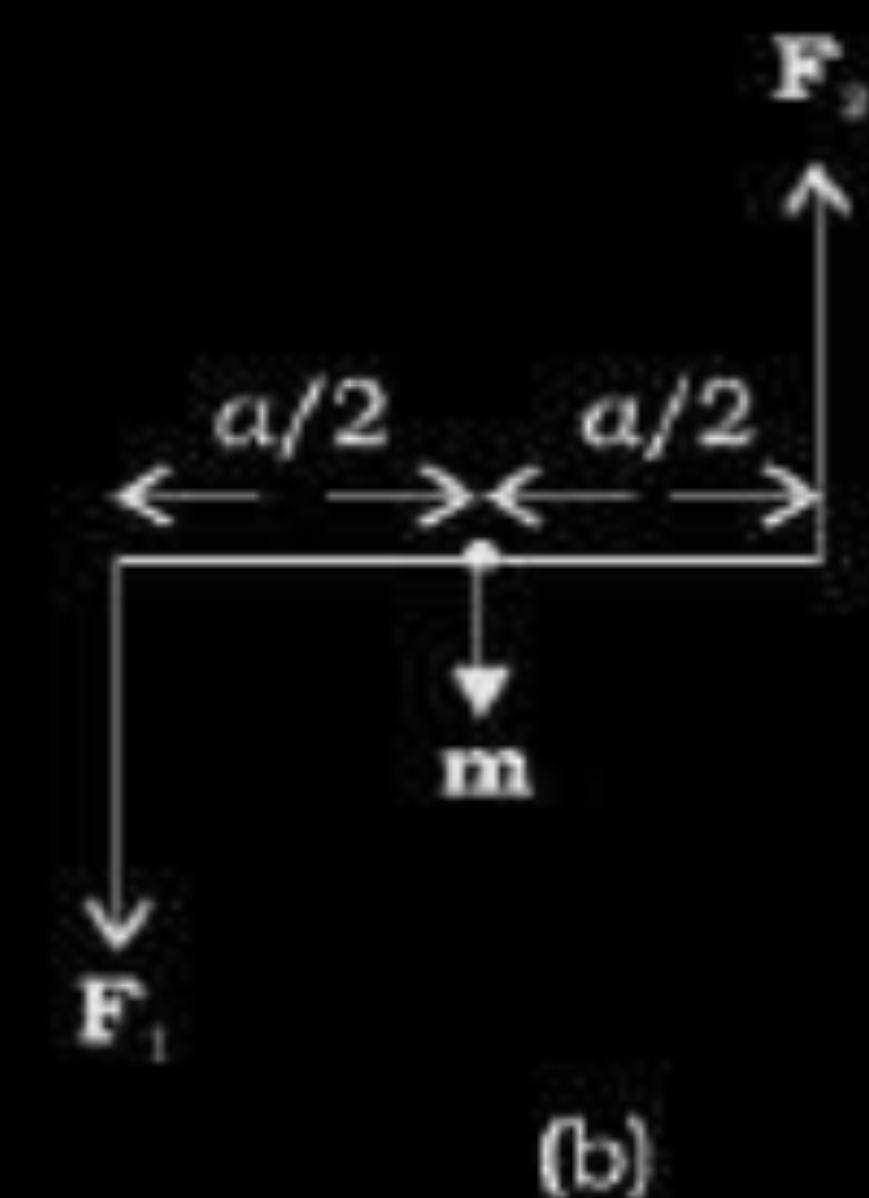
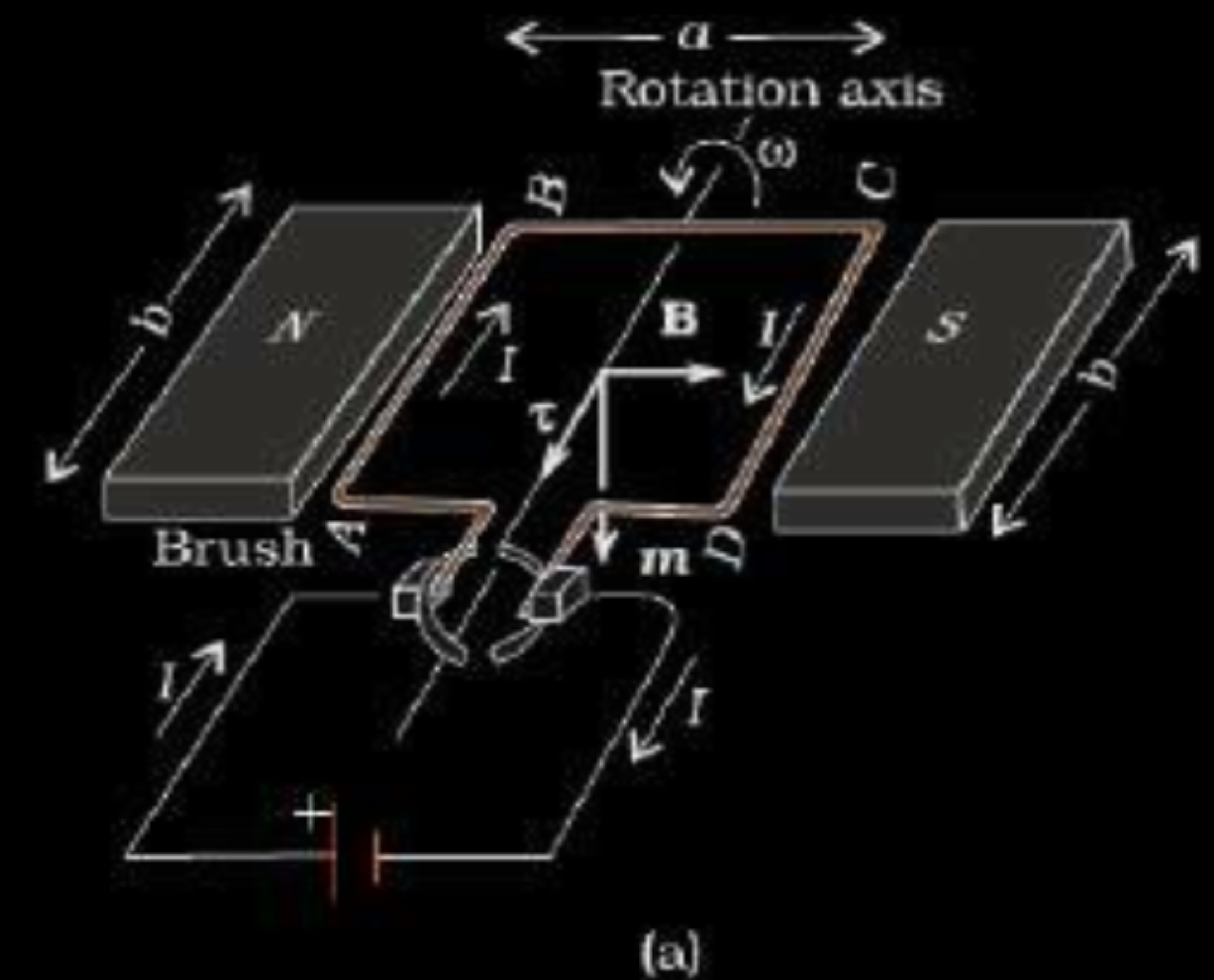
Alternatively " In a radial magnetic field two sides of the rectangular coil remain parallel to the magnetic field lines while its other two sides remain perpendicular to the magnetic field lines. This holds for all positions of the coil."

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

1



Two protons of equal kinetic energies enter a region of uniform magnetic field. The first proton enters normal to the field direction while the second enters at 30° to the field direction. Name the trajectories followed by them.

- (a) Define SI unit of current in terms of the force between two parallel current carrying conductors.
- (b) Two long straight parallel conductors carrying steady currents I_a and I_b along the same direction are separated by a distance d . How does one explain the force of attraction between them ? If a third conductor carrying a current I_c in the opposite direction is placed just in the middle of these conductors, find the resultant force acting on the third conductor.

- (a) State Biot – Savart law and express it in the vector form.
- (b) Using Biot – Savart law, obtain the expression for the magnetic field due to a circular coil of radius r , carrying a current I at a point on its axis distant x from the centre of the coil.

Find the condition under which the charged particles moving with different speeds in the presence of electric and magnetic field vectors can be used to select charged particles of a particular speed.

Condition	
i.	For directions of $\vec{E}, \vec{B}, \vec{v}$ 1
ii.	For magnitudes of $\vec{E}, \vec{B}, \vec{v}$ 1

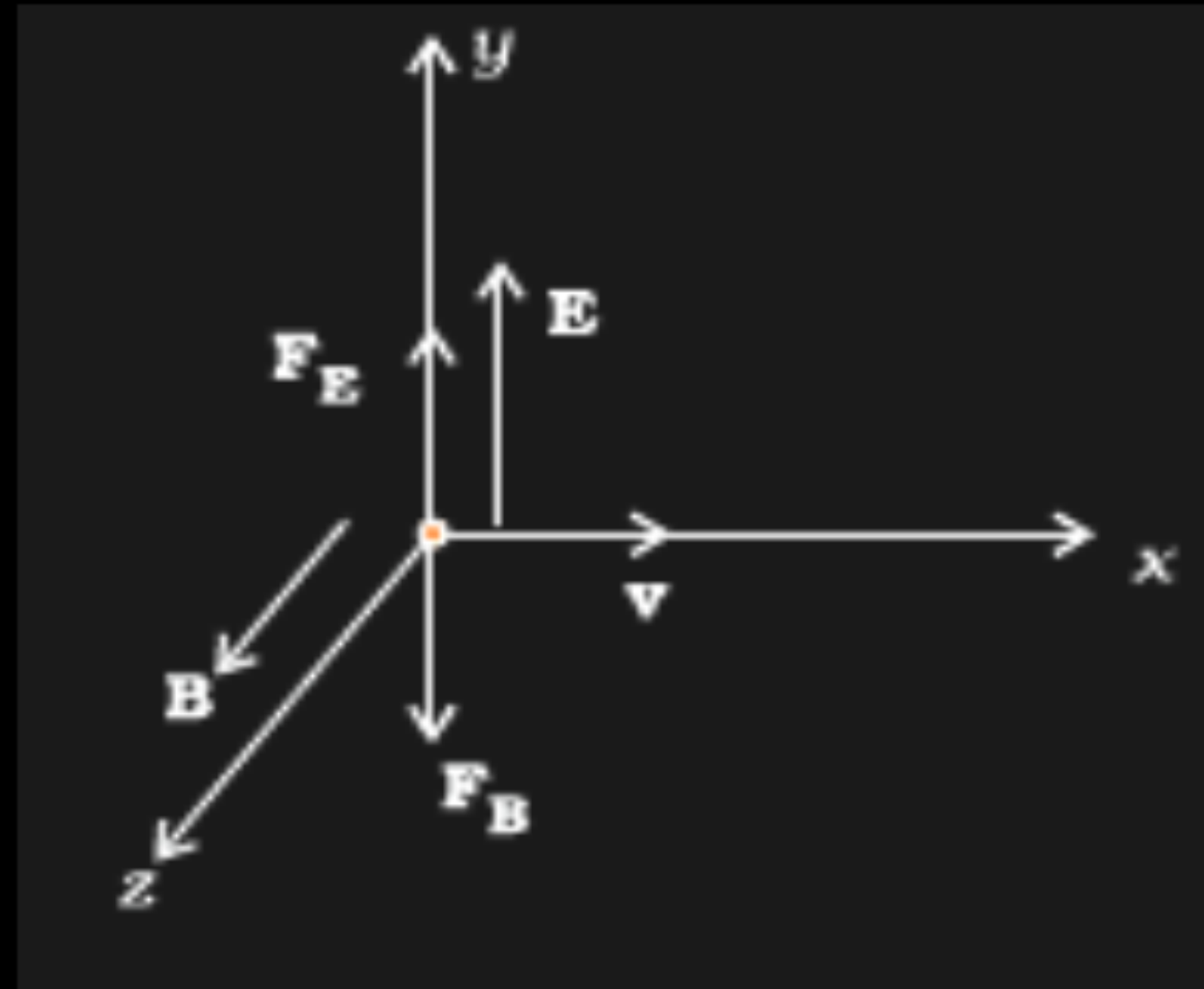
(i) The velocity $\vec{v}$, of the charged particles, and the $\vec{E}$ and $\vec{B}$ vectors, should be mutually perpendicular.

1/2

Also the forces on q , due to $\vec{E}$ and $\vec{B}$, must be oppositely directed.

1/2

(Also accept if the student draws a diagram to show the directions.)



(ii) $qE = qvB$
 $or\ v = \frac{E}{B}$

1/2

1/2

[Alternatively, The student may write:

Force due to electric field = $q\vec{E}$

1/2

Force due to magnetic field = $q(\vec{v} \times \vec{B})$

1/2

The required condition is

$$q\vec{E} = -q(\vec{v} \times \vec{B})$$

1/2

$$[or\ \vec{E} = -(\vec{v} \times \vec{B}) = (\vec{B} \times \vec{v})]$$

1/2

(Note: Award 1 mark only if the student just writes:

“The forces, on the charged particle, due to the electric and magnetic fields, must be equal and opposite to each other”)]

- (a) State Biot – Savart law and express this law in the vector form.
- (b) Two identical circular coils, P and Q each of radius R, carrying currents 1 A and $\sqrt{3}$ A respectively, are placed concentrically and perpendicular to each other lying in the XY and YZ planes. Find the magnitude and direction of the net magnetic field at the centre of the coils.

(a) Statement of Biot Savart law	1
Expression in vector form	$\frac{1}{2}$
(b) Magnitude of magnetic field at centre	1
Direction of magnetic field	$\frac{1}{2}$

- (a) It states that magnetic field strength, $d\vec{B}$, due to a current element, $I d\vec{l}$, at a point, having a position vector $\mathbf{r}$ relative to the current element, is found to depend (i) directly on the current element, (ii) inversely on the square of the distance $|\mathbf{r}|$, (iii) directly on the sine of angle between the current element and the position vector $\mathbf{r}$.

In vector notation,

$$\overrightarrow{d\mathbf{B}} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \vec{r}}{|\vec{r}|^3}$$

Alternatively,

$$\left(d\vec{B} = \frac{\mu_0}{4\pi} \frac{I d\vec{l} \times \hat{r}}{|\vec{r}|^2} \right)$$

(b) $B_p = \frac{\mu_0 \times 1}{2R} = \frac{\mu_0}{2R}$ (along z – direction)

$$B_Q = \frac{\mu_0 \times \sqrt{3}}{2R} = \frac{\mu_0 \sqrt{3}}{2R} \text{ (along x – direction)}$$

$$\therefore B = \sqrt{B_p^2 + B_Q^2} = \frac{\mu_0}{R}$$

This net magnetic field $\mathbf{B}$, is inclined to the field $\mathbf{B}_p$, at an angle Θ , where

$$\tan \theta = \sqrt{3}$$

$$(\therefore \theta = \tan^{-1} \sqrt{3} = 60^\circ)$$

(in XZ plane)

1

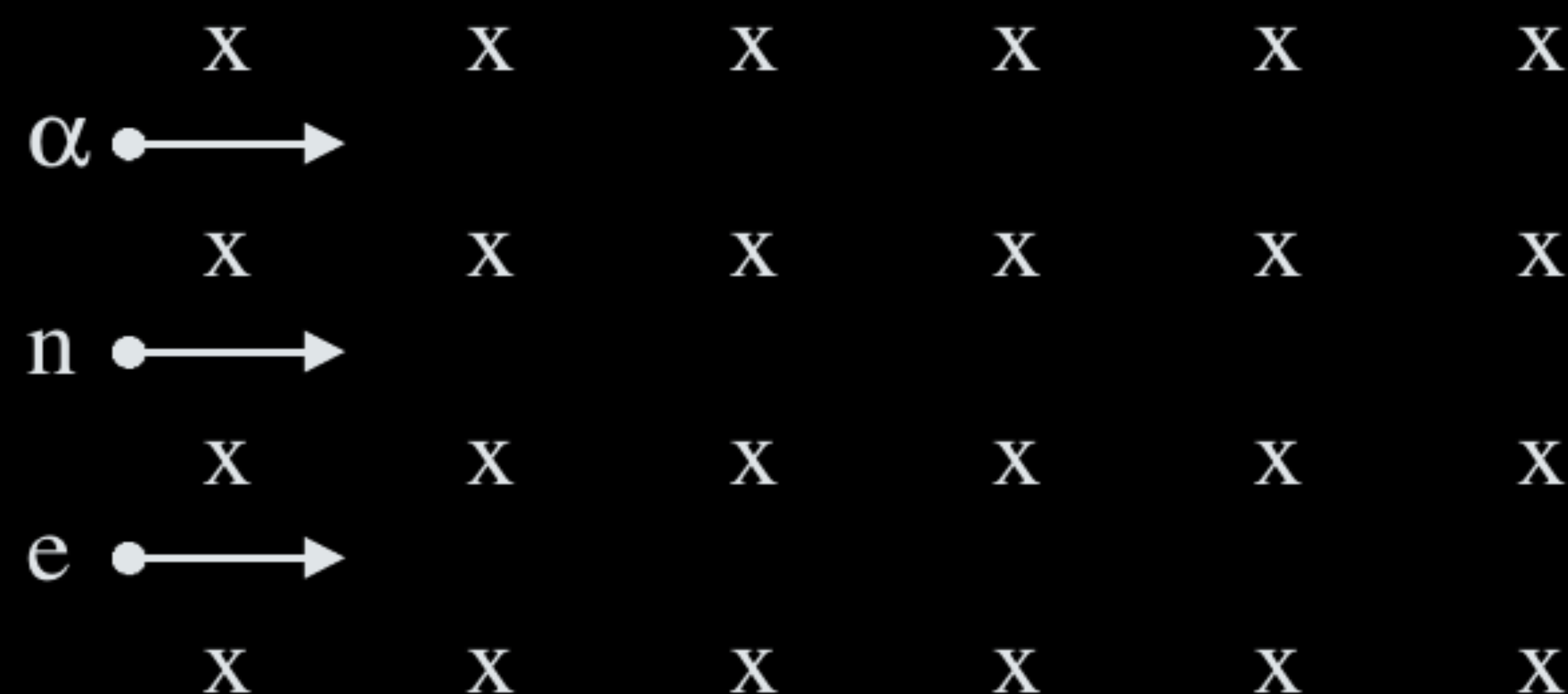
1/2

1/2

1/2

1/2

- (a) Write the expression for the magnetic force acting on a charged particle moving with velocity $\mathbf{v}$ in the presence of magnetic field $\mathbf{B}$.
- (b) A neutron, an electron and an alpha particle moving with equal velocities, enter a uniform magnetic field going into the plane of the paper as shown. Trace their paths in the field and justify your answer.



- a) Expression for the magnetic force
- b) Trace of paths
Justification

$$\frac{1}{2} + \frac{1}{2} + \frac{1}{2}$$

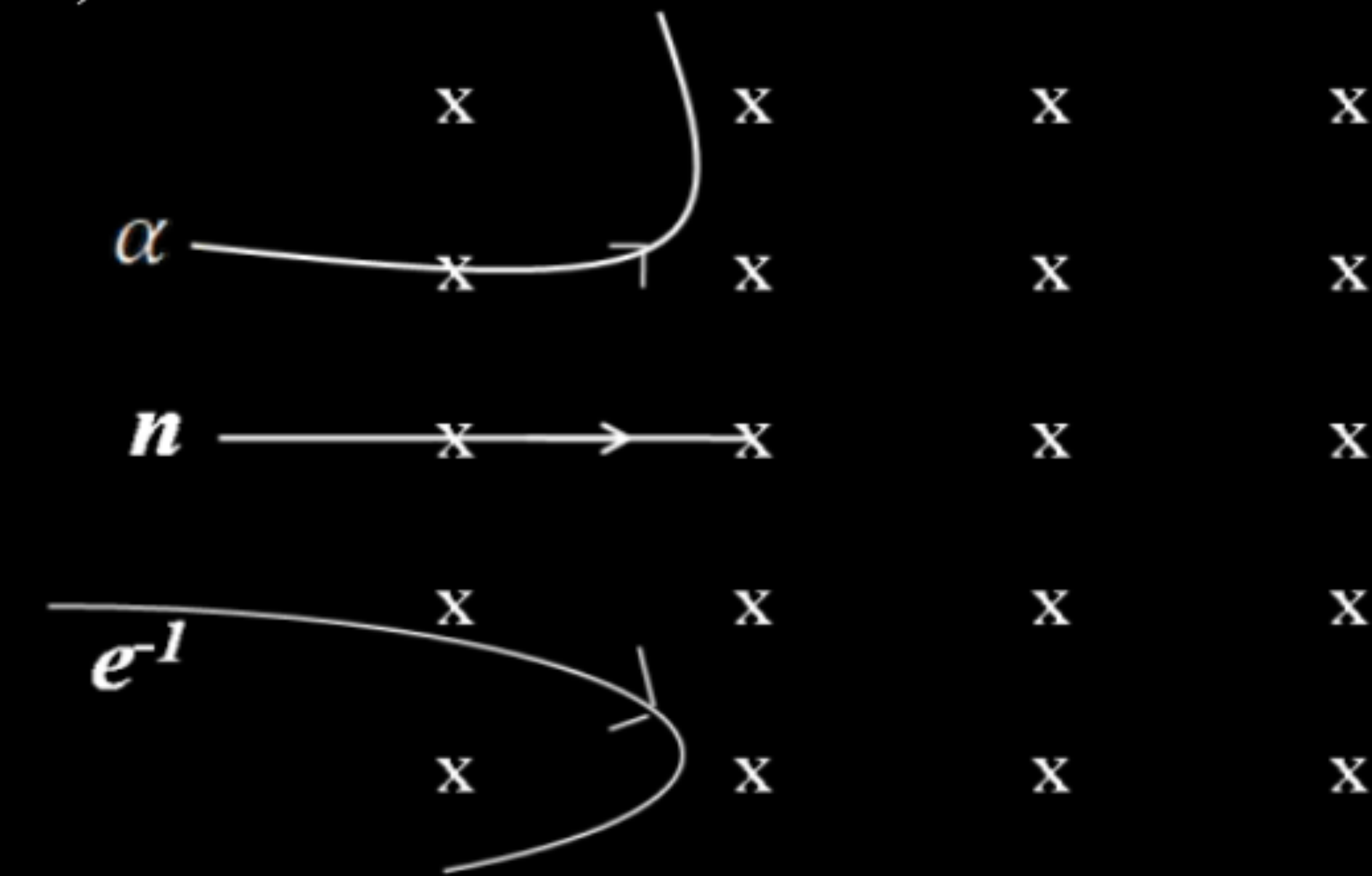
$$\vec{F} = q (\vec{v} \times \vec{B})$$

(Give Full credit of this part even if a student writes:

$$F = qvB \sin\theta \text{ and}$$

Force (F) acts perpendicular to the plane containing $\vec{v}$ and $\vec{B}$)

b)



Justification: Direction of force experienced by the particle will be according to the Fleming's Left hand rule / (any other alternative correct rule.)

1

$\frac{1}{2} +$
 $\frac{1}{2} +$
 $\frac{1}{2}$

$\frac{1}{2}$

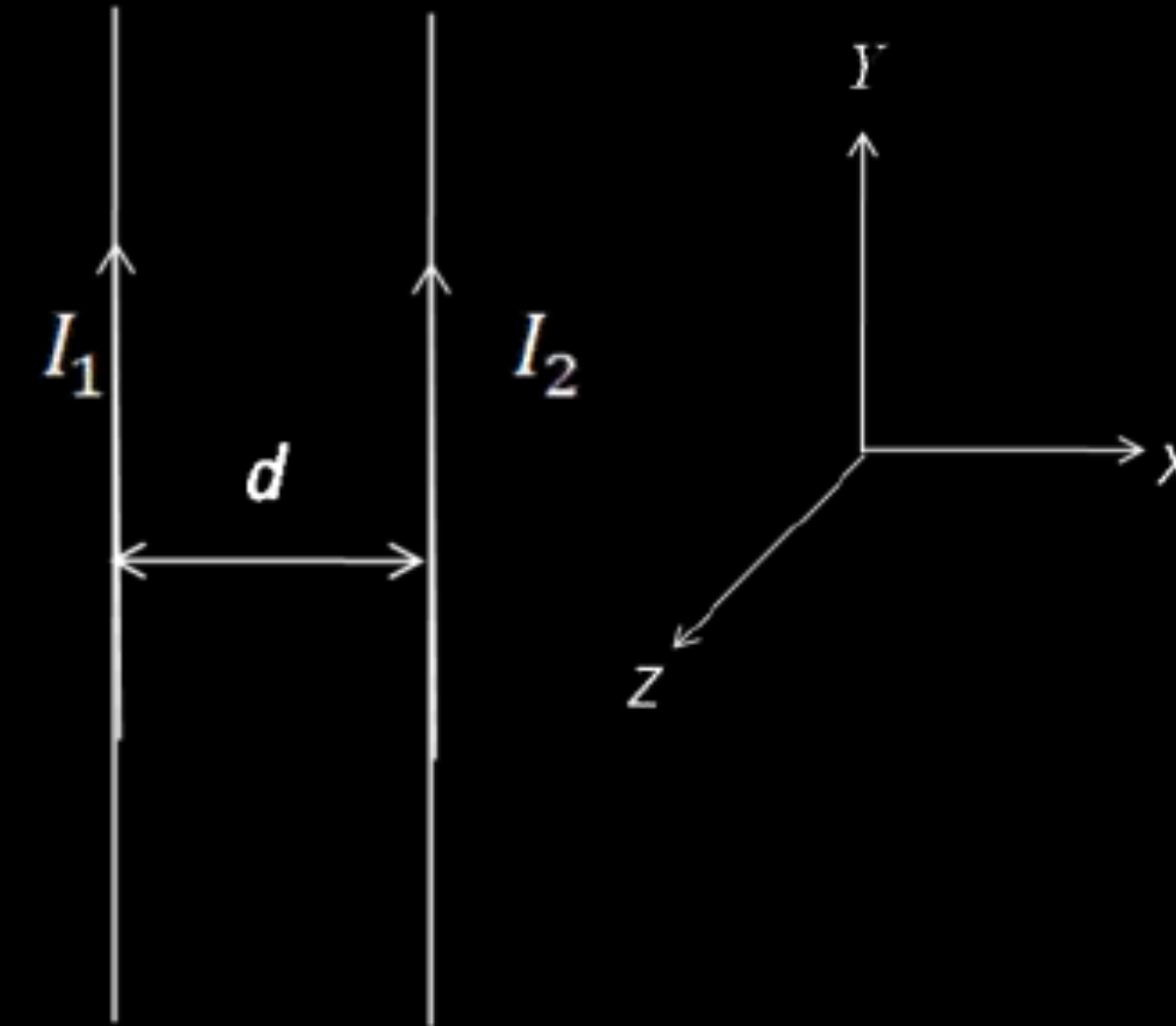
Two long straight parallel conductors carry steady current I_1 and I_2 separated by a distance d . If the currents are flowing in the same direction, show how the magnetic field set up in one produces an attractive force on the other. Obtain the expression for this force. Hence define one ampere.

Diagram showing attractive force on other wire.	1
Obtaining an expression for force.	1
Definition of one ampere.	1

As shown in Figure, the direction of force on conductor b is attractive
[Alternatively:

$\vec{B}$ at a point on wire 2, is along $-\hat{k}$

$\therefore \vec{F}$, on wire 2, due to the $\vec{B}$, is along $-\hat{i}$, i.e. towards wire1. Hence the force is attractive.



Magnetic field, due to current in conductor a,

$$B_1 = \frac{\mu_0 I_1}{2\pi d}$$

The magnitude of force on a length L of conductor b,

$$F_2 = I_2 L B_1$$

$$F_2 = \frac{\mu_0 I_1 I_2 L}{2\pi d}$$

One ampere is that steady current which, when maintained in each of the two very long, straight, parallel conductors, placed one meter apart in vacuum, would produce on each of these conductors a force equal to 2×10^{-7} newton per meter of their length.

1/2

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1

- (i) Express Biot-Savart law in the vector form.
- (ii) Use it to obtain the expression for the magnetic field at an axial point, distance d from the centre of a circular coil of radius R carrying current I .
- (iii) Also, find the ratio of the magnitudes of the magnetic field of this coil at the centre and at an axial point for which $x = R\sqrt{3}$.

(i) Biot-Savart law in vector form	1
(ii) Deriving an expression for the magnetic field at a point on the axial line of current carrying coil	3
(iii) Ratio of magnetic field at the centre and given outside point	1

$$(i) \quad \vec{dB} = \frac{\mu_0 I d\vec{\ell} \times \hat{r}}{4\pi r^2} = \frac{\mu_0 I d\vec{\ell} \times \vec{r}}{4\pi r^3}$$

$$(ii) \quad dB = \frac{\mu_0 I dl \sin \theta}{4\pi r^2} \text{ here } \theta = 90; \quad dB = \frac{\mu_0 I dl}{4\pi r^2}$$

$$= dB \sin \phi = \frac{\mu_0 I d\ell}{4\pi r^2} \sin \phi$$

$$B = \int_0^R \frac{\mu_0 I dl}{4\pi r^2} \sin \phi = \frac{\mu_0 I (2\pi R^2)}{4\pi r^3}$$

$$B = \frac{\mu_0 NI(R^2)}{2r^3} = \frac{\mu_0 NIR^2}{2(R^2 + d^2)^{\frac{3}{2}}}$$

$$(i) \quad \text{Magnetic field at the centre of the coil } B_1 = \frac{\mu_0 NI}{2R}$$

$$\text{Magnetic field at the outside point } B_2 = \frac{\mu_0 NIR^2}{2[R^2 + 3R^2]^{\frac{3}{2}}} = \frac{\mu_0 NIR^2}{2[4R^2]^{\frac{3}{2}}} = \frac{\mu_0 NI}{2 \cdot 8R}$$

$$\frac{B_1}{B_2} = 8$$

[Note : If the student takes $r = \sqrt{3} R$, the ratio of B centre to B axial would be $3\sqrt{3} : 1$. Award 1 mark in this case also.]

1

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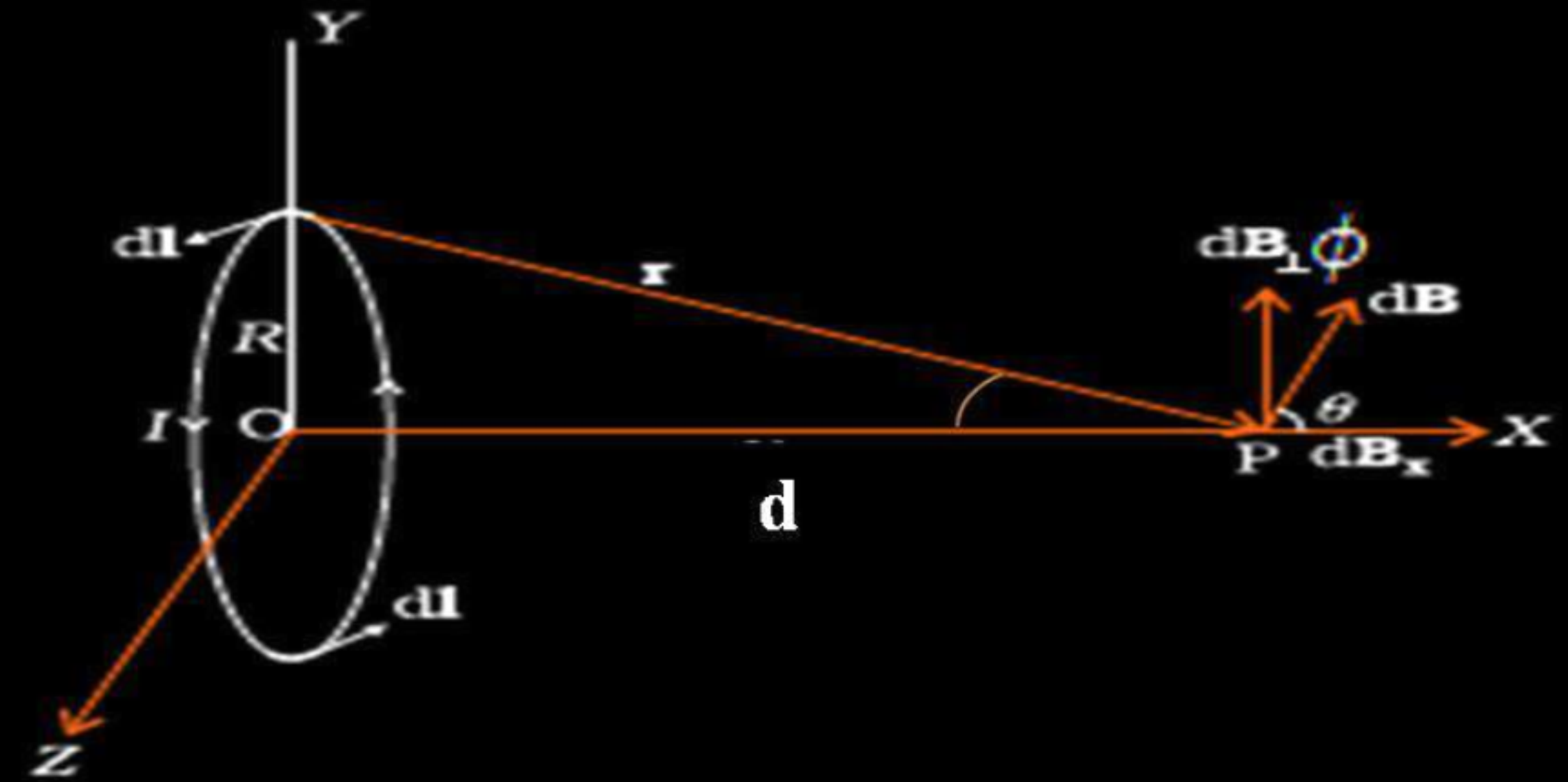
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2



- (a) Consider a beam of charged particles moving with varying speeds. Show how crossed electric and magnetic fields can be used to select charged particles of a particular velocity ?
- (b) Name another device/machine which uses crossed electric and magnetic fields. What does this machine do and what are the functions of magnetic and electric fields in this machine ? Where do these field exist in this machine ? Write about their natures.

a) Velocity selection condition	1
b) Name of device	$\frac{1}{2}$
What does the machine do	$\frac{1}{2}$
Use of two fields	$\frac{1}{2} + \frac{1}{2}$
Regions of existence of field	$\frac{1}{2} + \frac{1}{2}$
Nature of fields	$\frac{1}{2} + \frac{1}{2}$

a) $qE = Bqv$
 $v = E/B$

1

(b) Name of the device: Cyclotron

It accelerates charged particles/ions

$\frac{1}{2}$

Electric field accelerates the charged particles.

$\frac{1}{2}$

Magnetic field makes particles to move in circle.

1

Electric field exists between the Dees.

Magnetic field exists both inside and outside the dees.

1

Magnetic field is uniform / constant.

Electric field is oscillating/ alternating in nature.

1

What can be the cause of helical motion of a charged particle ?

What can be the cause of helical motion of a charged particle ?

Charged particle moves inclined to the magnetic field
(angle between $\vec{v}$ and $\vec{B}$ is neither $\pi/2$ nor 0)
(component of $\vec{v}$, parallel to $\vec{B}$, is not zero.)

State Ampere's circuital law. Use this law to find magnetic field due to straight infinite current carrying wire. How are the magnetic field lines different from the electrostatic field lines ?

Statement of Ampere's Circuital law	1
Finding Magnetic Field	1 ½
Differences between the two types of field lines	½

According to Ampers's circuital law, "The line integral of the magnetic field, around a closed loop, equals μ_o times the total current passing through the surface enclosed by that loop."

Alternatively,

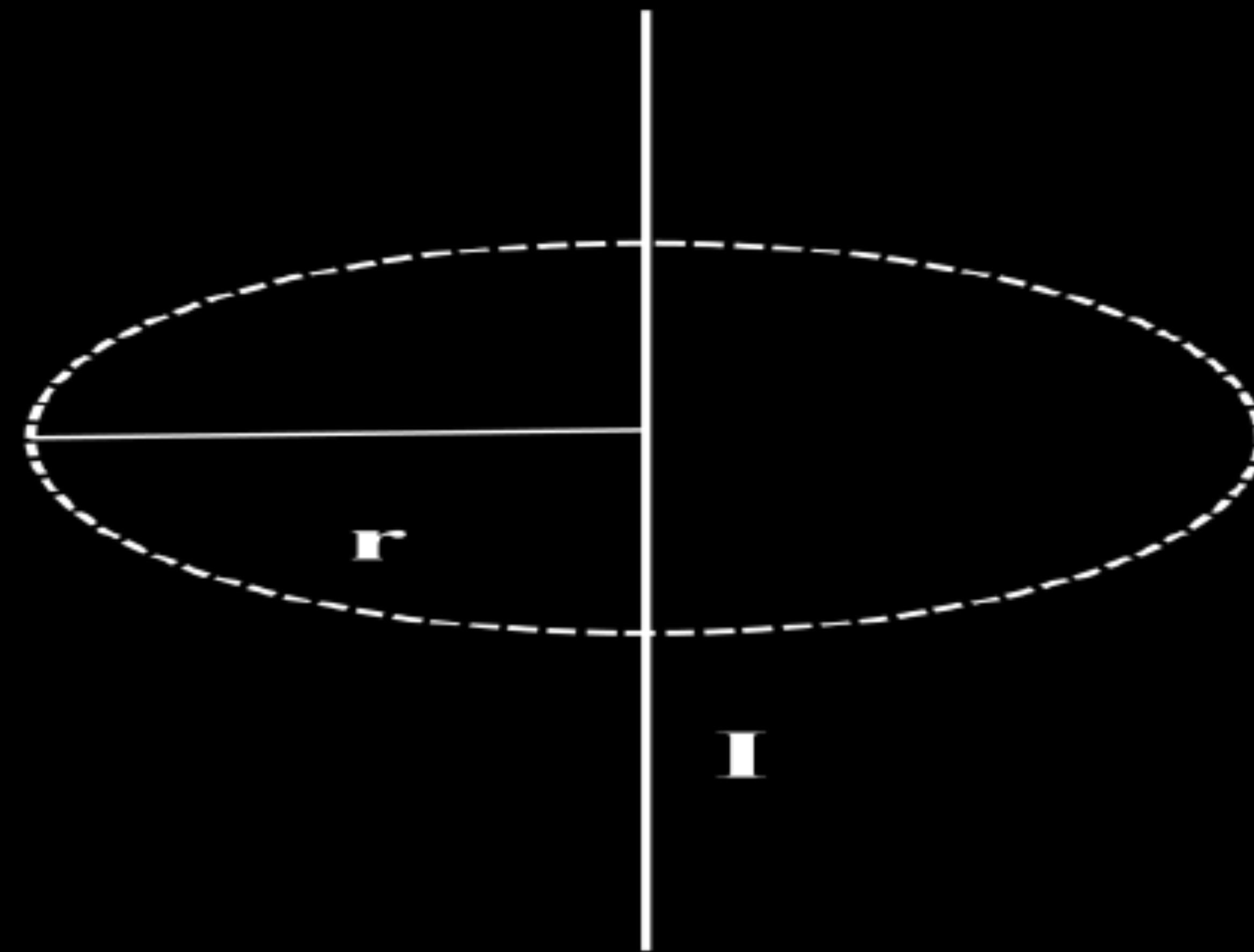
$$\oint \vec{B} \cdot d\vec{l} = \mu_o I$$

For the infinite current carrying wire, we get

$$\oint B \cdot dl = \mu_o I$$

$$\text{or } B 2\pi r = \mu_o I$$

$$\text{or } B = \frac{\mu_o I}{2\pi r}$$



The magnetic field lines form closed loops while the electric field lines originate from positive charges and end at negative charges.

1

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

$\frac{1}{2}$

How is a galvanometer converted into a voltmeter and an ammeter ? Draw the relevant diagrams and find the resistance of the arrangement in each case. Take resistance of galvanometer as G .

How galvanometer is converted in to a voltmeter and an Ammeter	$\frac{1}{2} + \frac{1}{2}$
Diagram for conversion of galvanometer into a voltmeter and an Ammeter.	$\frac{1}{2} + \frac{1}{2}$
Resistance of each arrangement	$\frac{1}{2} + \frac{1}{2}$

A galvanometer is converted into a voltmeter by connecting a high resistance 'R' in series with it.

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A galvanometer is converted into an ammeter by connecting a small resistance (called shunt) in parallel with it.

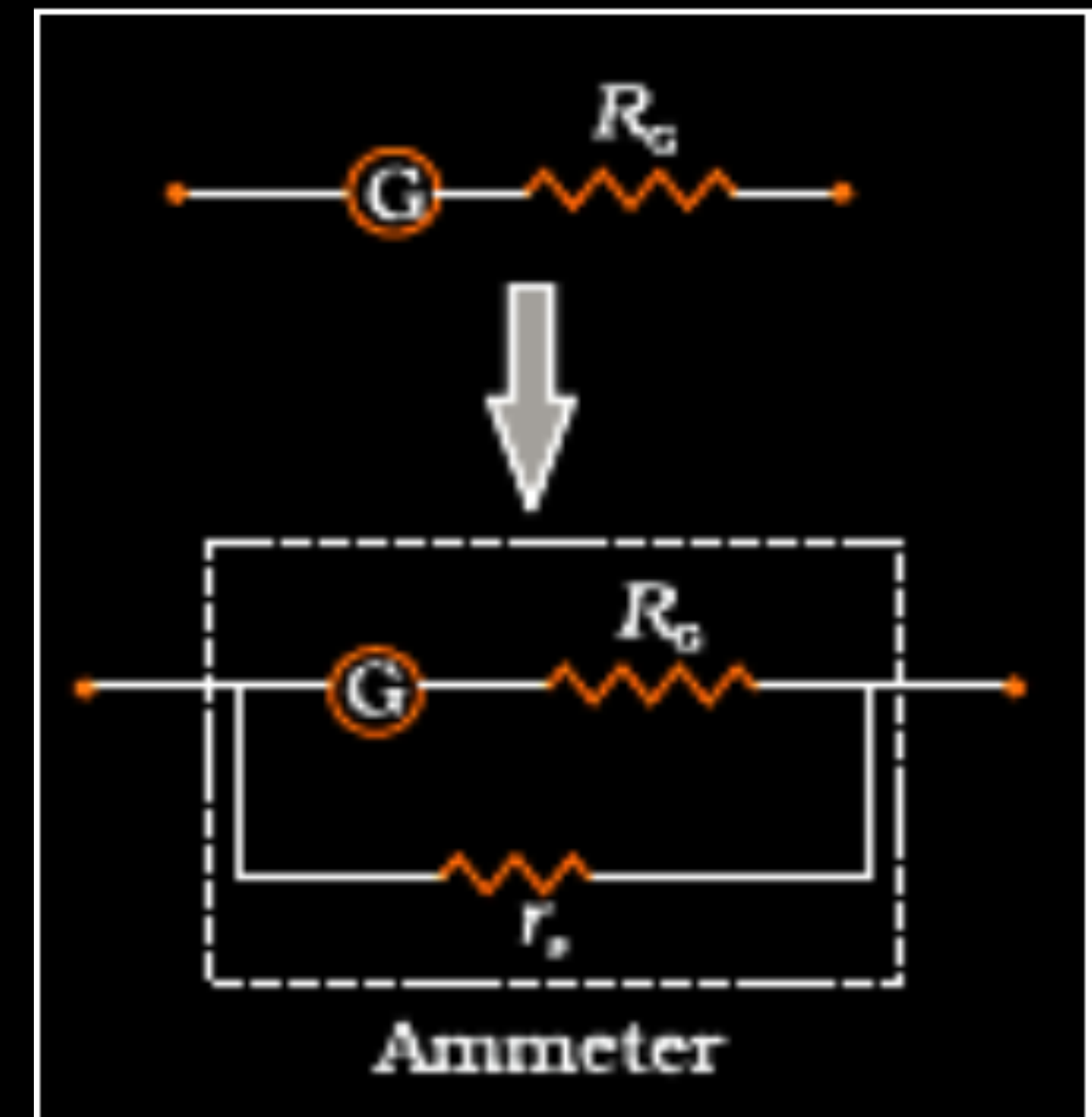
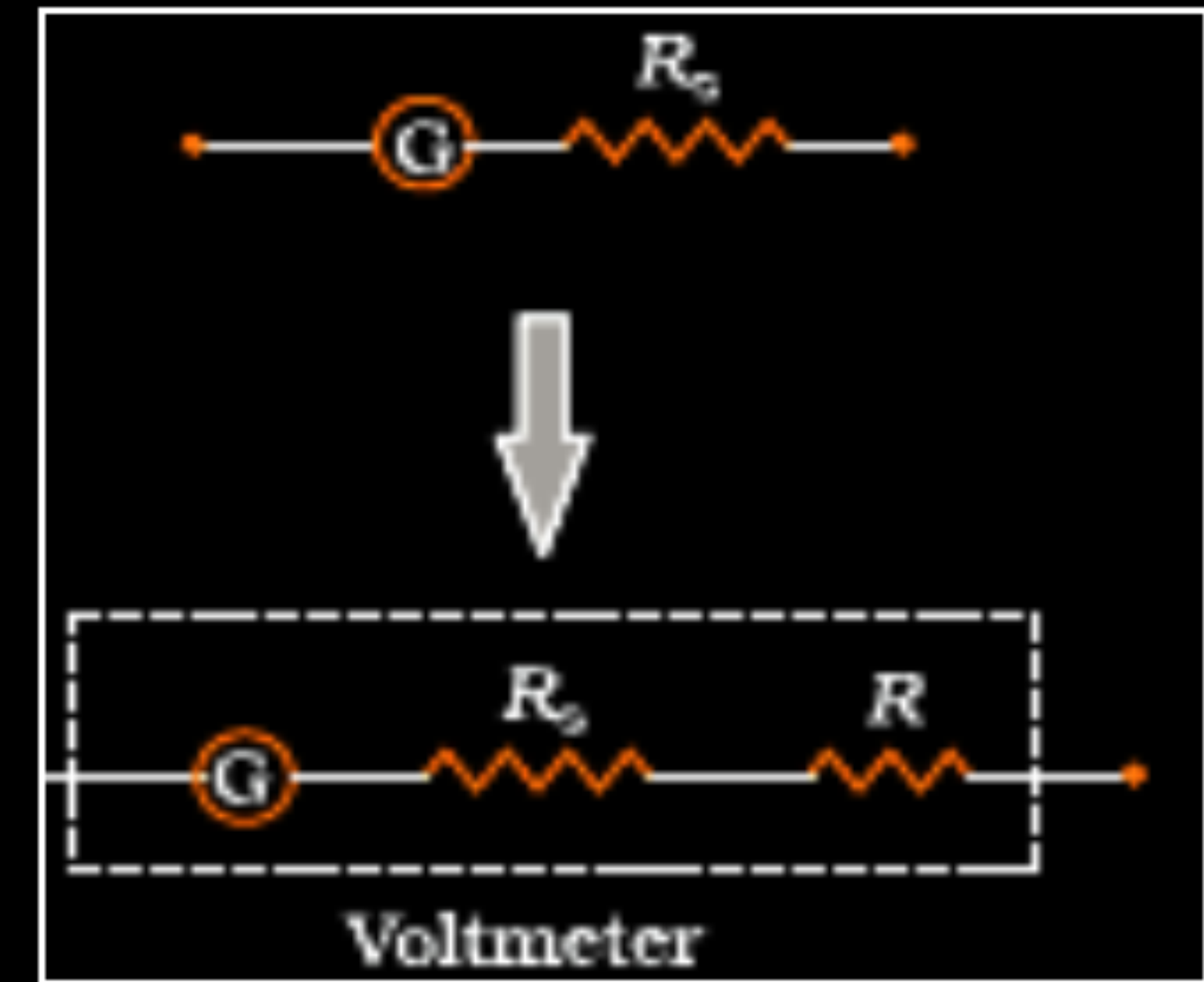
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Resistance of voltmeter, $R_V = G + R$

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Resistance for Ammeter, $R_A = \frac{G r_s}{G + r_s}$

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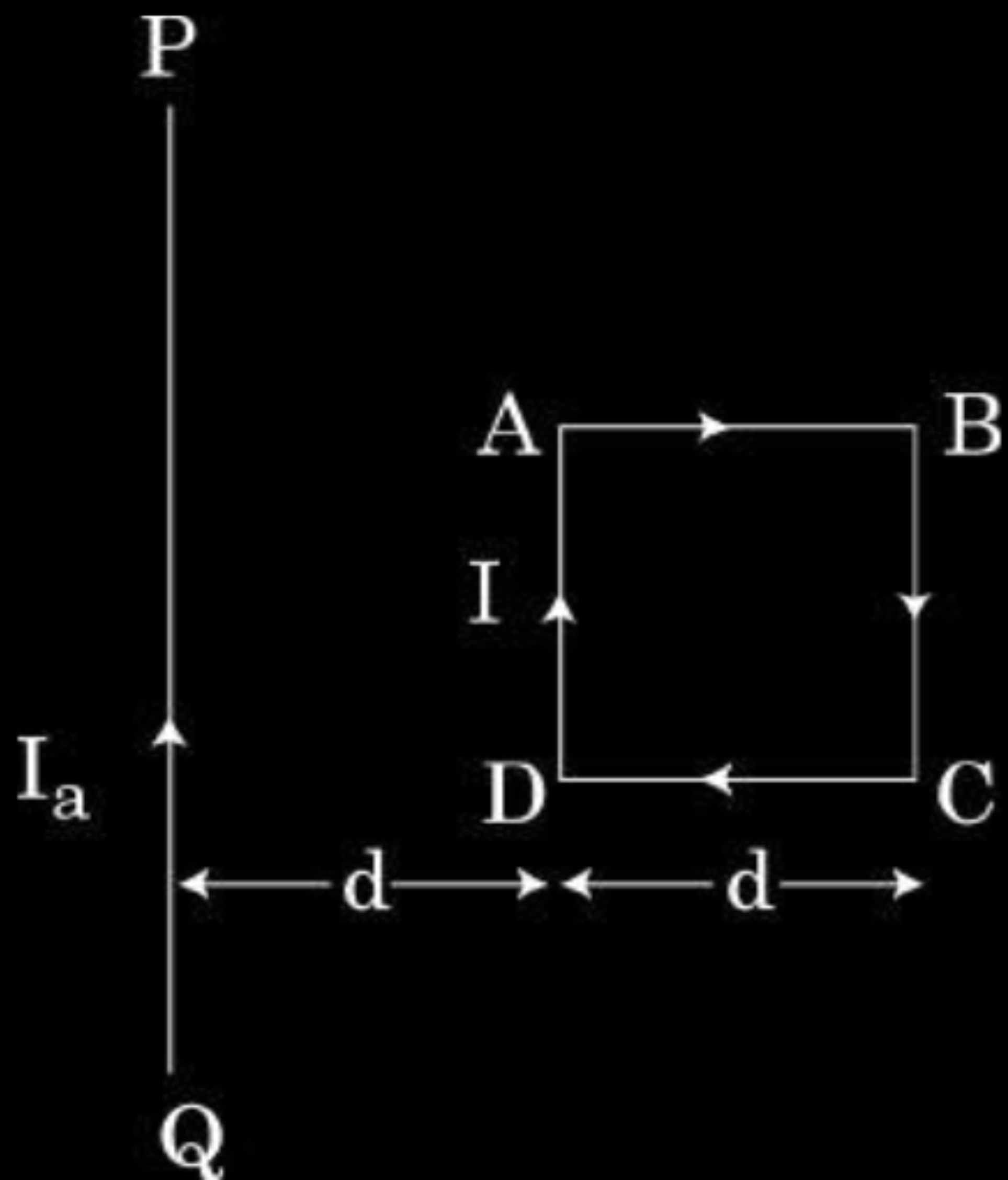


- (iii) An electron after being accelerated through a potential difference of 100 V enters a uniform magnetic field of 0.004 T perpendicular to its direction of motion. Calculate the radius of the path described by the electron.

- (iii) An electron after being accelerated through a potential difference of 100 V enters a uniform magnetic field of 0.004 T perpendicular to its direction of motion. Calculate the radius of the path described by the electron.

(ii)	$r = \frac{mv}{qB} = \frac{\sqrt{2mqV}}{qB}$	$\frac{1}{2}$
	$r = \frac{\sqrt{2 \times 9.1 \times 10^{-31} \times 1.6 \times 10^{-19} \times 100}}{1.6 \times 10^{-19} \times 0.004} \text{ m}$	$\frac{1}{2}$
	$r = \frac{5.4 \times 10^{-24}}{6.4 \times 10^{-22}} \text{ m}$	$\frac{1}{2}$
	$r = 8.4 \times 10^{-3} \text{ m}$	$\frac{1}{2}$

- (i) Depict magnetic field lines due to two straight, long, parallel conductors carrying steady currents I_1 and I_2 in the same direction.
- (ii) Write the expression for the magnetic field produced by one of the conductor over the other. Deduce an expression for the force per unit length.
- (iii) Determine the direction of this force.
- (iv) In figure given below, wire PQ is fixed while the square loop ABCD is free to move under the influence of currents flowing in them. State with reason, in which direction does the loop begin to move or rotate ?



(ii) $B_1 = \frac{\mu_0 I_1}{2\pi d}$ or $\frac{\mu_0 I_2}{2\pi d} = B_2$

$$F = F_{12} = F_{21} = I_1 B_2 L = I_2 B_1 L$$

$$= \left(\frac{\mu_0 I_1 I_2}{2\pi d} \right) L$$

Force per unit length

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$$

(iii) Attractive force

(iv) Loop ABCD will move towards wire PQ.

Current in wire PQ and Current in arm AD are in the same direction, so they attract each other.

Current in wire PQ and Current in arm BC are in opposite direction, so they repel each other.

Contribution due to current in AB and CD nullify each other.

Since arm AD is nearer than arm BC to arm PQ, so net force on the loop is attractive. Therefore, the loop will move towards the wire PQ.

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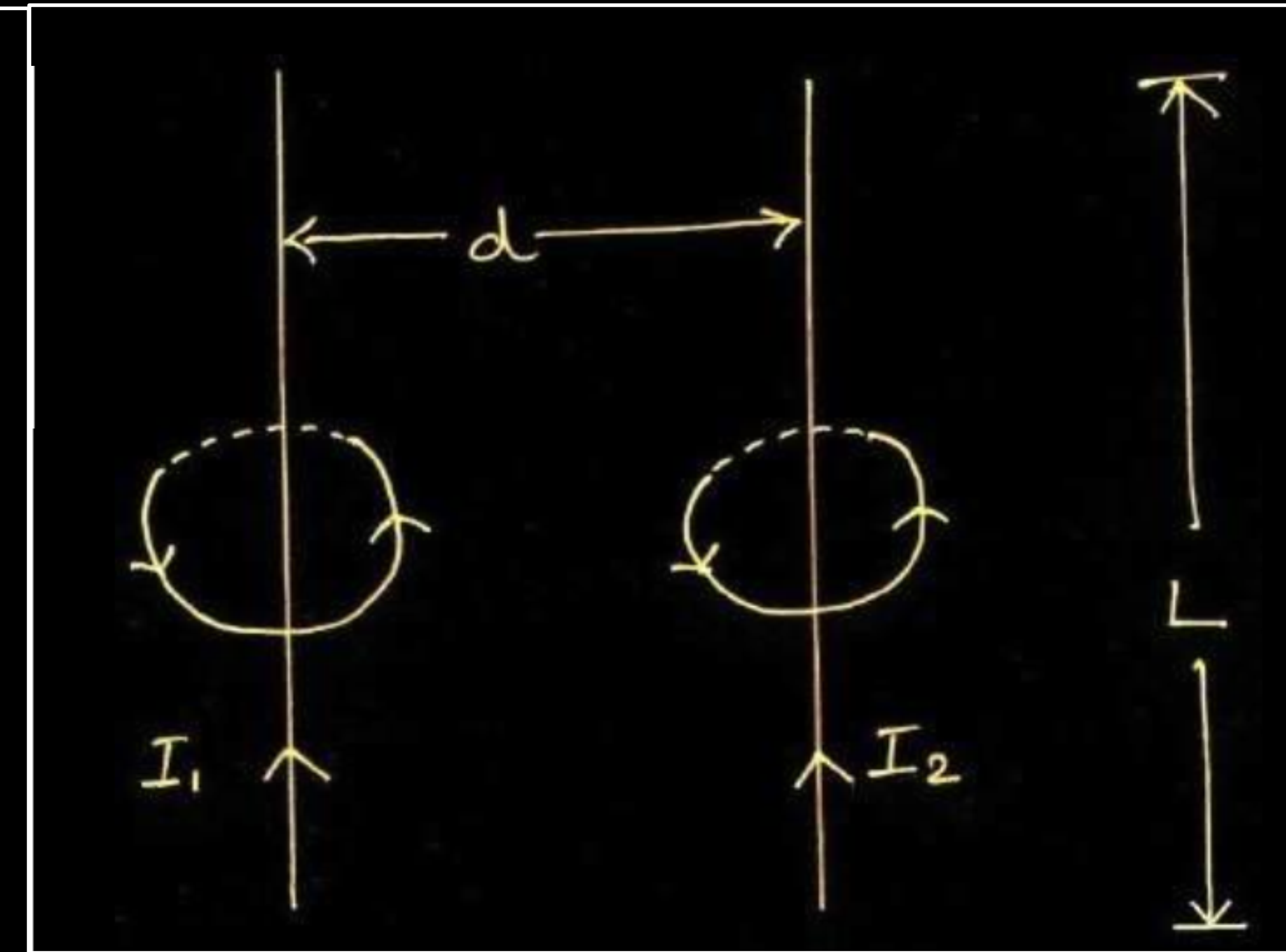
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Use Biot-Savart law to derive the expression for the magnetic field on the axis of a current carrying circular loop of radius R .

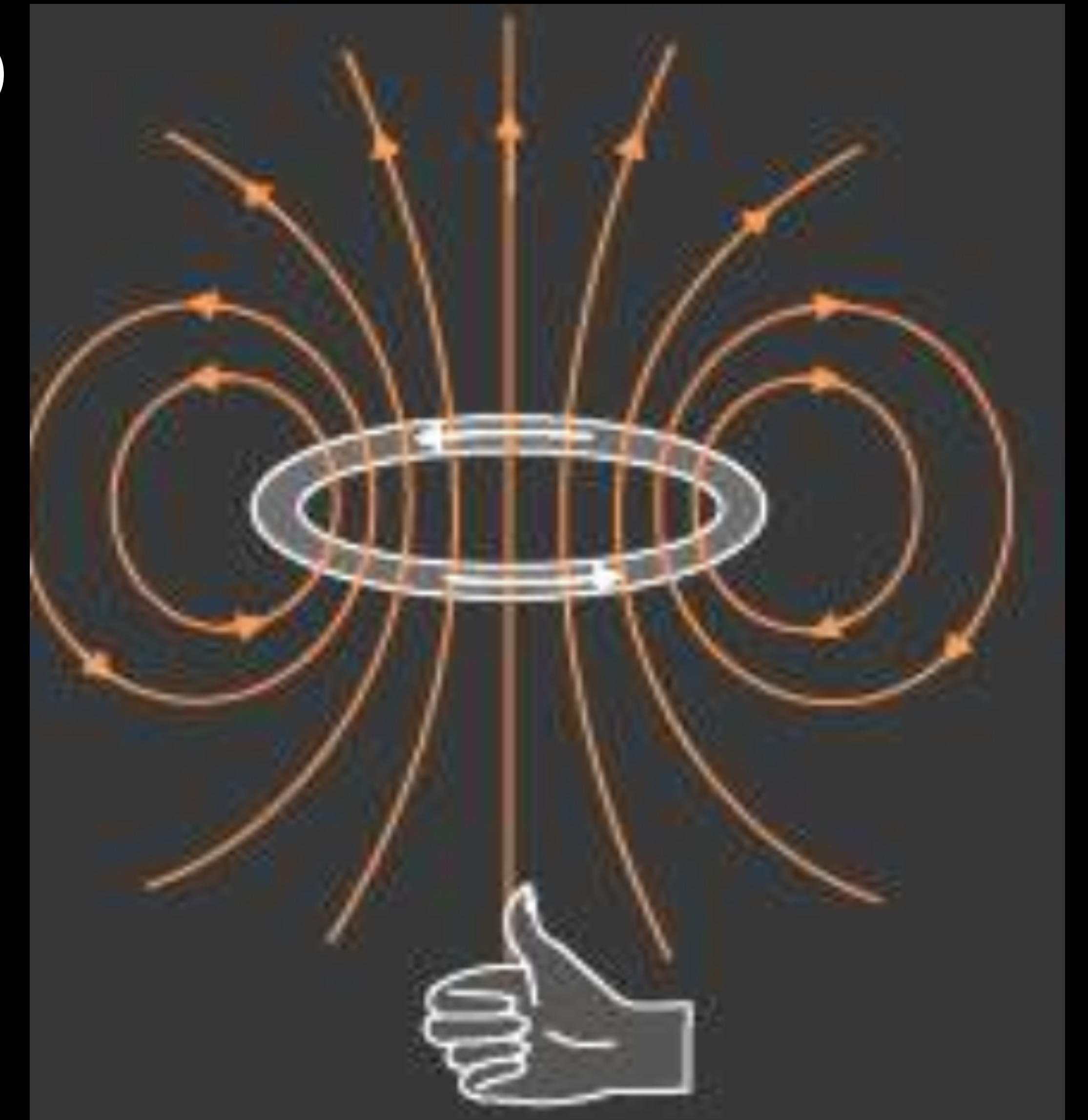
Draw the magnetic field lines due to a circular wire carrying current I .

Explanation for magnetic field on the axis of current loop	2
Drawing- magnetic field lines	1

Use Biot-Savart law to derive the expression for the magnetic field on the axis of a current carrying circular loop of radius R .

Draw the magnetic field lines due to a circular wire carrying current I .

(ii)



Explanation for magnetic field on the axis of current loop	2
Drawing- magnetic field lines	1